



SAMSUNG ELECTRONICS Co., Ltd.

2017 Business Report

For the year ended December 31, 2017

Certain statements in the document, other than purely historical information, including estimates, projections, statements relating to our business plans, objectives and expected operating results, and the assumptions upon which those statements are based, are "forward-looking statements." Forward-looking statements are based on current expectations and assumptions that are subject to risks and uncertainties which may cause actual results to differ materially from the forward-looking statements. A detailed discussion of risks and uncertainties that could cause actual results and events to differ materially from such forward-looking statements is included in our financial reports available on our website.

See, also, 『Note on Forward-Looking Statements』 in preamble of 『II. Business Overview』.

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Certification

Letter of Certification

We, Ki-Nam Kim and Hee-Chan Roh, as CEO and executive of the company in charge of reporting, certify that we have reviewed the annual business report of Samsung Electronics Co., Ltd. for the year ended December 31, 2017.

Based on our knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary that would be misleading with respect to the period covered by this report.

Based on our knowledge, the financial statements, and other financial information included in the report, fairly present in all materials respects the financial condition, results of operations and cash flows of the company as of, and for, the periods presented in this report.

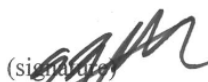
We confirm that Samsung Electronics operates an Internal Accounting Management System, responsible for establishing and maintaining disclosure controls and procedures and internal controls over financial reporting, as defined in Article 2.2 and 2.3 of the External Audit Act. We have disclosed our most recent evaluation of internal controls over financial reporting to the Company's auditors and to the audit committee of the Company's board of directors.

Date: 2018. 4. 2

Kinam Kim  (signature)

President and CEO,
Samsung Electronics Co., Ltd.

Executive responsible for reporting:

Hee Chan Roh  (signature)

President and CFO
Samsung Electronics Co., Ltd.

I. Corporate Overview

1. Overview

A. Legal, Commercial Name:

- Samsung Electronics Co., Ltd.

B. Date of Establishment:

- Samsung Electronics Co., Ltd. (“SEC”) was established as Samsung Electronics Industry Co., Ltd. on January 13, 1969, and held an initial public offering on June 11, 1975.
- SEC changed its name from Samsung Electronics Industry Co., Ltd. to Samsung Electronics Co., Ltd. following a resolution passed at the Annual General Meeting of Shareholders on February 28, 1984.

C. Address, Phone Number, and English Language Website of the Corporate Headquarter

- Address: 129, Samsung-ro, Yeongtong-gu, Suwon, Gyeonggi-do, Korea
- Phone Number: 82-31-200-1114
- Website: [http:// www.samsung.com/sec](http://www.samsung.com/sec)

D. The Company is not subject to Article 2 of the Framework Act on Small and Medium Enterprises.

E. Core Businesses

- The Company (defined below) comprises business divisions that operates independently (each a “**Division**”): CE (Consumer Electronics), IM (Information Technology & Mobile Communications), DS (Device Solutions), and Harman (Harman International Industries, Inc. and its subsidiaries), which produces automotive electronics components.

Products in each Division are presented below:

Division	Products
CE	TVs, monitors, refrigerators, washing machines, air conditioners, medical devices, etc.
IM	HHPs, network systems, computers, etc.
DS	DRAM, NAND flash, mobile APs, TFT-LCD panels, OLED panels, etc.
Harman	Head units, infotainment systems, telematics, speakers, etc.

SEC is a global electronics firm comprised of the headquarters in Korea and 270 subsidiaries (collectively, “Samsung Electronics” or the “Company”) across the world including nine (9) regional headquarters for the CE and IM Divisions, five (5) regional headquarters for the DS Division, and Harman’s subsidiaries.

[CE Division]

The CE Division is leading the global digital era by continuously offering new products with innovative technology, unique design, and enhanced value propositions.

TV is the core product of CE. The Company has maintained its position as the market leader for twelve (12) consecutive years by leveraging competitive advantages in hardware such as LCD/LED TVs as well as software driven product

features within our Smart TV product portfolio. We plan to solidify our market leadership by continue developing new technologies and making innovations.

[IM Division]

The IM Division is at the forefront of the mobile lifestyle, delivering consumer-friendly and innovative products such as the premium Galaxy Series, while also moving beyond mobile convergence to include a focus on software and services.

The Division is focused on meeting market demand by enhancing product differentiation via seamless integration of key features from various product categories.

The Division expects mobile phones, digital media devices and personal computers, cameras and other devices to increasingly converge on, and in some cases be replaced by, smartphones and tablets. As a result, it has invested and will continue to invest significantly in research and development in new technologies, products, and services.

In addition to handheld phone (“HHP”) products, which include smartphones, the IM Division is also leading technological development and standardization in the global network system market, including the one for Long Term Evolution (“LTE”).

[DS Division]

The DS Division comprises the Semiconductor sub-division and the Display Panel (“DP”) Business. The Semiconductor sub-division is further divided into the Memory Business and the System LSI Business. The Memory Business manufactures and sells DRAM and NAND products; the System LSI Business manufactures and sells mobile application processors (“mobile APs” or “APs”) and camera sensor chips; and the DP Business manufactures and sells display panels.

The Company is constantly working to increase its influence on end-product manufacturers and create new demand through technological breakthroughs and the development of high quality components as it has done with APs—which serve as the brains of mobile products—and with memory components like DRAM via enhanced capacity and density.

The Memory Business produces differentiated products and has maintained the top position in the global memory market by being the first to apply the latest advanced process technology for each technological iteration. The System LSI Unit has not been complacent, expanding its focus from mobile to include automotive chips, and will stay at the forefront of the market by offering differentiated products such as APs and CISs and strengthening competitiveness via its advanced manufacturing process technology.

Building on its 2016 achievement of launching the world’s first 10nm products, the Foundry Business seeks to maintain the leading status by being the first to develop the 7nm process

The Company’s OLED business is building a system that can respond swiftly to fluctuating market demand for flexible panels while maintaining a competitive edge by delivering differentiated products.

The TFT-LCD business is improving its sales mix via high-value-added high-definition and ultra-large panels for use in quantum dot and frameless products. It is also strengthening business competitiveness by continuously improving yield and reducing costs.

[Harman Division]

Harman designs and develops connected products and solutions for automakers, consumers, and companies worldwide, and is a global leader in the market for connected car systems, audio and visual products, professional solutions, and connected services.

Harman has established an internationally renowned brand and employs both internal development and strategic acquisitions to expand its lineup of quality products.

☞ See 『II. Businesses Overview』, for more details about each Division.

F. Affiliates

SEC is an affiliate of the Samsung group as defined under Korea's Monopoly Regulation and Fair Trade Act ("Samsung Group").

As of December 31, 2017, Samsung Group had a total of sixty-three (63) domestic affiliates, addition of five (5) affiliates (Samsung Active Asset Management, Samsung Hedge Asset Management, Harman International Korea, Red Bend Software Korea, SBTM) from December 31, 2016. Among the Samsung Group's sixty-three (63) domestic affiliates, sixteen (16) affiliates including Samsung Electronics are listed, and forty-seven (47) affiliates are unlisted.

[As of December 31, 2017]

	No. of affiliates	Name of affiliates
Listed	16	Samsung C&T, Samsung Electronics, Samsung SDI, SEMCO, Samsung Fire & Marine Insurance, Samsung Heavy Industries, Samsung Life Insurance, MULTICAMPUS, Samsung Securities, Samsung SDS, Samsung Card, Samsung Engineering, S1, Cheil Worldwide, Hotel Shilla, Samsung Biologics
Unlisted	47	Seoul Lakeside CC, Samwoo Architects & Engineers, CVnet Corporation, Samsung Bioepis, Samsung Display, Samsung Corning Advanced Glass, SU Materials, STECO, SEMES, Samsung Electronics Service, Samsung Electronics Sales, Samsung Electronics Logitech, Suwon Samsung Bluewings FC, Samsung Medison, Samsung Claim Adjustment Service, Samsung Fire & Marine Insurance Service, Samsung Fire & Marine Insurance Financial Service, Daejung Offshore Wind Power Co., Ltd., Samsung Futures, Samsung Asset Management, Saengbo, Samsung Life Service, Samsung SRA Asset Management, Samsung Life Financial Service, SD Flex, Cheil Fashion Retail Co., Ltd., Natural9, Samsung Welstory, S-Printing Solution, SECUI, STM, S-Core, OpenHands, Miracom, Samsung Card Customer Service, Human TSS, S-1CRM, Shilla Stay, HDC Shilla Duty Free LTD, SERI, Samsung Lions, Samsung Venture Investment Corporation, Samsung Active Asset Management, Samsung Hedge Asset Management, Harman International Korea, Red Bend Software Korea, SBTM
Total	63	

※ S-Printing Solution was sold in November, 2017 and was excluded from affiliates in February, 2018.

See 『IX. Affiliates and Subsidiaries』 for more details about domestic and overseas affiliates.

G. Subsidiaries Subject to Consolidated Financial Statements

As of December 31, 2017, Samsung Electronics' consolidated subsidiaries totaled 270 with the addition of one hundred and twenty (120) newly established or acquired subsidiaries and the subtraction of nineteen (19) subsidiaries when compared to December 31, 2016. Below is the list of consolidated subsidiaries.

(Unit: KRW million)

Name of Subsidiaries	Date of Establishment	Major business	Dec 31, 2017 Assets	% ownership	Classified as major subsidiary (Y/N)
Samsung Electronics America Inc.	1978.07	Sale of electronic goods	26,266,636	Over 50%	Y
NexusDX Inc.	2009.07	Medical equipment	6,046	Over 50%	N
NeuroLogica Corp.	2004.02	Medical equipment	180,138	Over 50%	Y
Dacor Holdings, Inc.	1998.12	Holding Company	45,265	Over 50%	N
Dacor	1965.03	Production and sale of home appliances	45,337	Over 50%	N
Dacor Canada Co.	2001.06	Sale of home appliances	52	Over 50%	N
EverythingDacor.com, Inc.	2006.06	Sale of home appliances	0	Over 50%	N
Distinctive Appliances of California, Inc.	2014.06	Sale of home appliances	0	Over 50%	N
Samsung HVAC America, LLC (formerly QuietSide LLC)	2001.07	Sale of Air-conditioner	36,562	Over 50%	N
SmartThings, Inc.	2012.04	Sale of smart-home devices	210,589	Over 50%	Y
Samsung Oak Holdings, Inc.	2016.06	Holding company	153,045	Over 50%	Y
Joyent, Inc.	2005.03	Cloud services	203,005	Over 50%	Y
Samsung Pay, Inc.	2006.03	Develop and service mobile payment	317,193	Over 50%	Y
Stellus Technologies, Inc.	2015.11	Production and sales of semiconductor systems	32,493	Over 50%	N
Prismview, LLC	2007.10	LED display panel production and sales	52,130	Over 50%	N
Samsung Semiconductor Inc.	1983.07	Sale of semiconductors and display panels	10,125,193	Over 50%	Y
Samsung Electronics Canada Inc.	1980.07	Sale of electronic goods	762,677	Over 50%	Y
AdGear Technologies Inc.	2010.08	Digital advertising platform	10,432	Over 50%	N
Viv Labs, Inc.	2012.09	Artificial Intelligence services	254,464	Over 50%	Y
SigMast Communications Inc. (formerly NewNet Communication Technologies (Canada), Inc.)	2009.07	Develop text messaging services	1,515	Over 50%	N
RT SV CO-INVEST, LP	2014.02	Investment in venture firms	11,764	Over 50%	N
Samsung Research America, Inc	1988.10	R&D	576,727	Over 50%	Y
Samsung Next LLC	2016.08	Holding company	53,584	Over 50%	N
Samsung Next Fund LLC	2016.08	Investment in venture firms and new technologies	54,185	Over 50%	N
Samsung International Inc.	1983.10	Production of TVs and monitors	792,117	Over 50%	Y
Samsung Mexicana S.A. de C.V	1988.03	Production of electronic goods	44,878	Over 50%	N
Samsung Austin Semiconductor LLC.	1996.02	Production of semiconductors	4,943,576	Over 50%	Y
Samsung Electronics Mexico S.A. De C.V.	1995.07	Sale of electronic goods	1,049,929	Over 50%	Y
SEMES America Inc.	1998.10	Semiconductor equipment	1,412	Over 50%	N
Samsung Electronics Digital Appliance Mexico, SA de CV	2012.12	Production of electronic goods	485,469	Over 50%	Y

Samsung Electronics Latinoamerica Miami, Inc.	1995.05	Sale of electronic goods	263,365	Over 50%	Y
Samsung Electronics Latinoamerica (Zona Libre)	1989.04	Sale of electronic goods	373,528	Over 50%	Y
Samsung Electronics Venezuela, C.A.	2010.05	Marketing and services	1,241	Over 50%	N
Samsung Electronica Colombia S.A.	1997.03	Sale of electronic goods	421,924	Over 50%	Y
Samsung Electronics Panama. S.A.	2012.07	Consulting	189	Over 50%	N
Samsung Electronica da Amazonia Ltda.	1995.01	Production and sale of electronic goods	5,838,533	Over 50%	Y
Samsung Electronics Argentina S.A.	1996.06	Marketing and services	55,595	Over 50%	N
Samsung Electronics Chile Limitada	2002.12	Sale of electronic goods	445,411	Over 50%	Y
Samsung Electronics Peru S.A.C.	2010.04	Sale of electronic goods	223,403	Over 50%	Y
Samsung Electronics Home Appliances America, LLC	2017.08	Production of home appliances	282,076	Over 50%	Y
AMX Holding Corporation	2005.12	Holding Company	0	Over 50%	N
AMX LLC	2005.11	Holding Company	73,777	Over 50%	N
Harman Becker Automotive Systems, Inc.	1981.06	Production and sale of audio products, R&D	1,942,159	Over 50%	Y
Harman Connected Services, Inc.	2002.02	Connected Service Provider	1,021,364	Over 50%	Y
Harman Connected Services Engineering Corp.	2004.09	Connected Service Provider	3,016	Over 50%	N
Harman Connected Services Holding Corp.	2012.03	Connected Service Provider	879,079	Over 50%	Y
Harman Connected Services South America S.R.L.	2015.04	Connected Service Provider	30	Over 50%	N
Harman da Amazonia Industria Electronica e Participacoes Ltda.	2005.07	Production and sale of audio products	48,331	Over 50%	N
Harman de Mexico S. de R.L. de C.V.	1997.02	Production of audio products	46,687	Over 50%	N
Harman do Brasil Industria Electronica e Participacoes Ltda.	1958.11	Sale of audio products, R&D	212,767	Over 50%	Y
Harman Financial Group, LLC	2004.06	Management Company	593,054	Over 50%	Y
Harman International Industries Canada Ltd.	2005.05	Sale of audio products	902	Over 50%	N
Harman International Industries, Inc.	1980.01	Holding Company	14,156,148	Over 50%	Y
Harman International Mexico, S. de R.L. de C.V.	2014.12	Sale of audio products	26,208	Over 50%	N
Harman Investment Group, LLC	2015.12	Financing Company	629,382	Over 50%	Y
Harman KG Holding, LLC	2009.03	Holding Company	0	Over 50%	N
Harman Professional, Inc.	2006.07	Sale of audio products, R&D	757,906	Over 50%	Y
Red Bend Software Inc.	2001.03	S/W design	14,379	Over 50%	N
S1NN USA, Inc.	2011.02	R&D	164	Over 50%	N
Southern Vision Systems, Inc	2003.04	Development of vision sensors	1,212	Over 50%	N
Triple Play Integration LLC	2003.11	Connected Service Provider	3,256	Over 50%	N
Beijing Integrated Circuit Industry International Fund, L.P	2014.12	Investment in venture firms	26,727	Over 50%	N
China Materialia New Materials 2016 Limited Partnership	2017.09	Investment in venture firms	1,544	Over 50%	N
Samsung Electronics (UK) Ltd.	1995.07	Sale of electronic goods	1,615,723	Over 50%	Y
Samsung Electronics(London) Ltd.	1999.01	Holding company	6,324	Over 50%	N
Samsung Electronics Holding GmbH	1982.02	Holding company	711,919	Over 50%	Y
Samsung Semiconductor Europe GmbH	1987.12	Sale of semiconductors and display panels	853,110	Over 50%	Y

Samsung Electronics GmbH	1984.12	Sale of electronic goods	1,743,138	Over 50%	Y
Samsung Electronics Iberia, S.A.	1989.01	Sale of electronic goods	734,838	Over 50%	Y
Samsung Electronics France S.A.S	1988.01	Sale of electronic goods	1,262,849	Over 50%	Y
Samsung Electronics Hungarian Private Co. Ltd.	1989.10	Production and sale of electronic goods	1,823,278	Over 50%	Y
Samsung Electronics Czech and Slovak s.r.o.	2010.01	Sale of electronic goods	140,176	Over 50%	Y
Samsung Electronics Italia S.P.A.	1991.04	Sale of electronic goods	822,200	Over 50%	Y
Samsung Electronics Europe Logistics B.V.	1991.05	Logistics	1,927,321	Over 50%	Y
Samsung Electronics Benelux B.V.	1995.07	Sale of electronic goods	1,484,026	Over 50%	Y
Samsung Display Slovakia s.r.o.	2007.03	Display panel processing	128,831	Over 50%	Y
Samsung Electronics Romania LLC	2007.09	Sale of electronic goods	211,181	Over 50%	Y
Samsung Electronics Polska, SP.Zo.o	1996.04	Sale of electronic goods	528,044	Over 50%	Y
Samsung Electronics Portuguesa S.A.	1982.09	Sale of electronic goods	159,301	Over 50%	Y
Samsung Electronics Nordic Aktiebolag	1992.03	Sale of electronic goods	701,024	Over 50%	Y
Samsung Semiconductor Europe Limited	1997.04	Sale of semiconductors and display panels	83,503	Over 50%	Y
Samsung Electronics Austria GmbH	2002.01	Sale of electronic goods	313,654	Over 50%	Y
Samsung Electronics Switzerland GmbH	2013.05	Sale of electronic goods	151,235	Over 50%	Y
Samsung Electronics Slovakia s.r.o	2002.06	Production of TVs and monitors	2,019,572	Over 50%	Y
Samsung Electronics Baltics SIA	2001.10	Sale of electronic goods	94,244	Over 50%	Y
Samsung Electronics Europe Holding Cooperatief U.A.	2008.10	Holding company	9,474,960	Over 50%	Y
Samsung Electronics Poland Manufacturing SP.Zo.o	2010.02	Production of home appliances	351,051	Over 50%	Y
Samsung Electronics Greece S.A.	2010.04	Sale of electronic goods	76,845	Over 50%	Y
Samsung Electronics Air Conditioner Europe B.V.	2017.04	Sale of Air-conditioner	73,016	Over 50%	N
Samsung Nanoradio Design Center	2004.02	R&D	23,863	Over 50%	N
Samsung Denmark Research Center ApS	2012.09	R&D	22,157	Over 50%	N
Samsung France Research Center SARL	2012.10	R&D	13,064	Over 50%	N
Samsung Cambridge Solution Centre Limited	2012.09	R&D	119,230	Over 50%	Y
Joyent Ltd.	2014.04	Cloud services	0	Over 50%	N
Samsung Electronics Overseas B.V.	1997.01	Sale of electronic goods	105,267	Over 50%	Y
Innoetics E.P.E.	2006.04	S/W development	536	Over 50%	N
Aditi Technologies Europe GmbH	2013.03	Sale of audio products, etc.	79	Over 50%	N
AKG Acoustics GmbH	1947.03	Production and sale of audio products	337,450	Over 50%	Y
AMX GmbH	2007.08	Sale of audio products	898	Over 50%	N
AMX UK Limited	1993.03	Sale of audio products	6,786	Over 50%	N
Arcam Limited	2004.07	Holding Company	5,073	Over 50%	N
A&R Cambridge Limited	1993.12	Sale of audio products	7,260	Over 50%	N
Duran Audio B.V.	1991.11	Sale of audio products, R&D	518,454	Over 50%	Y
Duran Audio Iberia Espana S.L.	2012.11	Sale of audio products	88	Over 50%	N

Endeleo Limited	2003.05	Sale of audio product, R&D	0	Over 50%	N
Harman Automotive UK Limited	2012.10	Production of audio products	449,850	Over 50%	Y
Harman Becker Automotive Systems GmbH	1990.07	Production and sale of audio products, R&D	4,105,781	Over 50%	Y
Harman Becker Automotive Systems Italy S.R.L.	2005.12	Sale of audio products	873	Over 50%	N
Harman Becker Automotive Systems Manufacturing Kft	1994.08	Production of audio products, R&D	2,789,251	Over 50%	Y
Harman Belgium SA	1967.04	Sale of audio products	978	Over 50%	N
Harman Connected Services AB.	1984.10	Connected Service Provider	51,834	Over 50%	N
Harman Connected Services Finland OY	1998.07	Connected Service Provider	1,159	Over 50%	N
Harman Connected Services GmbH	2005.12	Connected Service Provider	44,593	Over 50%	N
Harman Connected Services Limited	1992.12	Connected Service Provider	8,932	Over 50%	N
Harman Connected Services Poland Sp.zoo	2007.06	Connected Service Provider	4,387	Over 50%	N
Harman Connected Services UK Ltd.	2008.09	Connected Service Provider	79,516	Over 50%	Y
Harman Consumer Division Nordic A/S	1992.01	Sale of audio products	12,434	Over 50%	N
Harman Consumer Finland OY	2002.07	Sale of audio products	18	Over 50%	N
Harman Consumer Nederland B.V.	1995.12	Sale of audio products	392,974	Over 50%	Y
Harman Deutschland GmbH	1998.03	Sale of audio products	0	Over 50%	N
Harman Finance International GP S.a.r.l	2015.04	Holding Company	18	Over 50%	N
Harman France SNC	1995.11	Sale of audio products	136,765	Over 50%	Y
Harman Holding GmbH & Co. KG	2002.06	Management Company	4,762,822	Over 50%	Y
Harman Hungary Financing Ltd.	2012.06	Financing Company	795,547	Over 50%	Y
Harman Inc. & Co. KG	2012.06	Holding Company	3,831,950	Over 50%	Y
Harman International Estonia OU	2015.05	R&D	177	Over 50%	N
Harman International Industries Limited	1980.03	Sale of audio products, etc.	221,161	Over 50%	Y
Harman International Romania SRL	2015.02	R&D	11,255	Over 50%	N
Harman Finance International SCA	2015.04	Financing Company	465,108	Over 50%	Y
Harman International s.r.o	2015.02	Production of audio products	83	Over 50%	N
Harman International SNC	1989.02	Sale of audio products	2,161	Over 50%	N
Harman Management GmbH	2002.04	Holding Company	3	Over 50%	N
Harman Professional Kft	2014.12	Production of audio products, R&D	63,361	Over 50%	N
Inspiration Matters Limited	2002.06	Sale of audio products, R&D	0	Over 50%	N
Knight Image Limited	1989.09	Sale of audio products, R&D	0	Over 50%	N
Martin Manufacturing (UK) Ltd	1985.05	Production of audio products	4,133	Over 50%	N
Harman Professional Denmark ApS	1987.07	Sale of audio products, R&D	170,431	Over 50%	Y
Harman Professional France SAS	1990.01	Sale of audio products	631	Over 50%	N
Harman Professional Germany GmbH	1999.12	Sale of audio products	725	Over 50%	N
R&D International BVBA	1989.12	Production of audio products	121	Over 50%	N
Red Bend Software Ltd.	2004.08	S/W design	4,564	Over 50%	N

Red Bend Software SAS	2002.10	S/W design	3,817	Over 50%	N
Studer Professional Audio GmbH	2003.11	Sale of audio products, R&D	35,003	Over 50%	N
Samsung Electronics Rus Company LLC	2006.10	Sale of electronic goods	1,189,176	Over 50%	Y
Samsung Electronics Ukraine Company LLC	2008.09	Sale of electronic goods	135,052	Over 50%	Y
Samsung R&D Institute Rus LLC	2011.11	R&D	26,921	Over 50%	N
Samsung Electronics Central Eurasia LLP	2008.09	Sale of electronic goods	93,208	Over 50%	Y
Samsung Electronics Caucasus Co. Ltd.	2014.10	Marketing	1,746	Over 50%	N
Samsung Electronics Rus Kaluga LLC	2007.07	Production of TVs	960,288	Over 50%	Y
Harman Connected Services OOO	1998.11	Connected Service Provider	12,036	Over 50%	N
Harman RUS CIS LLC	2011.08	Sale of audio products	61,926	Over 50%	N
Samsung Electronics West Africa	2010.03	Marketing	36,712	Over 50%	N
Samsung Electronics East Africa	2011.12	Marketing	25,581	Over 50%	N
Samsung Gulf Electronics Co., Ltd.	1995.05	Sale of electronic goods	743,222	Over 50%	Y
Samsung Electronics Egypt S.A.E	2012.07	Production and sale of electronic goods	492,092	Over 50%	Y
Samsung Electronics Israel Ltd.	2012.09	Marketing	8,630	Over 50%	N
Samsung Electronics Tunisia S.A.R.L	2012.09	Marketing	3,458	Over 50%	N
Samsung Electronics Pakistan(Private) Ltd.	2012.11	Marketing	2,529	Over 50%	N
Samsung Electronics South Africa(Pty) Ltd.	1994.06	Sale of electronic goods	496,007	Over 50%	Y
Samsung Electronics South Africa Production (pty) Ltd.	2014.07	Production of TV and monitors	55,915	Over 50%	N
Samsung Electronics Turkey	1984.12	Sale of electronic goods	626,822	Over 50%	Y
Samsung Semiconductor Israel R&D Center, Ltd.	2007.10	R&D	48,431	Over 50%	N
Samsung Electronics Levant Co.,Ltd.	2009.07	Sale of electronic goods	382,616	Over 50%	Y
Samsung Electronics Maghreb Arab	2009.11	Sale of electronic goods	171,095	Over 50%	Y
Broadsense Ltd.	2007.03	Service	0	Over 50%	N
Global Symphony Technology Group Private Ltd.	2002.01	Holding Company	39,033	Over 50%	N
Harman Connected Services Morocco	2012.04	Connected Service Provider	3,321	Over 50%	N
Harman Industries Holdings Mauritius Ltd.	2009.10	Holding Company	79,330	Over 50%	Y
iOnRoad Ltd	2011.08	R&D	138	Over 50%	N
iOnRoad Technologies Ltd	2012.01	R&D	1,623	Over 50%	N
Red Bend Ltd.	1998.02	Production of audio products	68,744	Over 50%	N
Towersec Ltd.	2008.04	R&D	3,577	Over 50%	N
Samsung Japan Corporation	1975.12	Sale of electronic goods	934,227	Over 50%	Y
Samsung R&D Institute Japan Co. Ltd	1992.08	R&D	138,096	Over 50%	Y
Samsung Electronics Japan Co., Ltd.	2008.09	Sale of electronic goods	723,211	Over 50%	Y
Samsung Electronics Display (M) Sdn. Bhd.	1995.03	Production of electronic goods	24,794	Over 50%	N
Samsung Medison India Private Ltd.	2009.01	Medical equipment	0	Over 50%	N
Samsung Electronics (M) Sdn. Bhd.	1989.09	Production of home appliances	155,377	Over 50%	Y

Samsung Vina Electronics Co., Ltd.	1995.01	Sale of electronic goods	218,680	Over 50%	Y
Samsung Asia Private Ltd.	2006.07	Sale of electronic goods	6,138,652	Over 50%	Y
Samsung India Electronics Private Ltd.	1995.08	Production and sale of electronic goods	6,200,351	Over 50%	Y
Samsung R&D Institute India-Bangalore Private Ltd.	2005.05	R&D	222,450	Over 50%	Y
Samsung Nepal Services Pvt, Ltd	2017.11	Services	77	Over 50%	N
Samsung Electronics Australia Pty. Ltd.	1987.11	Sale of electronic goods	382,133	Over 50%	Y
Samsung Electronics New Zealand Limited	2013.09	Sale of electronic goods	74,917	Over 50%	N
PT Samsung Electronics Indonesia	1991.08	Production and sale of electronic goods	1,562,037	Over 50%	Y
PT Samsung Telecommunications Indonesia	2003.03	Sale and services of electronic goods	1,746	Over 50%	N
Thai Samsung Electronics Co., Ltd.	1988.10	Production and sale of electronic goods	2,191,519	Over 50%	Y
Laos Samsung Electronics Sole Co., Ltd	2016.09	Marketing	561	Over 50%	N
Samsung Electronics Philippines Corporation	1996.03	Sale of electronic goods	219,316	Over 50%	Y
Samsung Display Vietnam Co., Ltd	2014.07	Display panel production	10,460,755	Over 50%	Y
Samsung Malaysia Electronics (SME) Sdn. Bhd.	2003.05	Sale of electronic goods	337,274	Over 50%	Y
Samsung R&D Institute BanglaDesh	2010.08	R&D	7,288	Over 50%	N
Samsung Electronics Vietnam Co., Ltd.	2008.03	Production of electronic goods	9,592,074	Over 50%	Y
Samsung Electronics Vietnam THAINGUYEN Co., Ltd.	2013.03	Production of telecom products	10,044,337	Over 50%	Y
Samsung Electronics HCMC CE Complex Co. Ltd.,	2015.02	Production and sale of electronic goods	1,784,883	Over 50%	Y
AMX Products And Solutions Private Limited	2008.02	Sale of audio products	604	Over 50%	N
Harman Connected Services Technologies Pvt. Ltd.	1994.08	Connected Service Provider	73,264	Over 50%	N
Harman Connected Services Corp. India Pvt. Ltd.	2002.04	Connected Service Provider	454,392	Over 50%	Y
Harman Connected Services Japan Co. Ltd.	2009.12	Connected Service Provider	448	Over 50%	N
Harman International (India) Private Limited	2009.01	Sale of audio products, R&D	174,103	Over 50%	Y
Harman International Industries PTY, Ltd.	2014.12	Holding Company	0	Over 50%	N
Harman International Japan Co. Ltd.	1991.06	Sale of audio products, R&D	48,995	Over 50%	N
Harman International Singapore Pte. Ltd.	1993.12	Sale of audio products	4,326	Over 50%	N
Harman Malaysia Sdn. Bhd.	2008.07	Sale of audio products	3,909	Over 50%	N
Harman Professional Singapore Pte. Ltd	2007.08	Sale of audio products	5,640	Over 50%	N
INSP India Software Development Pvt. Ltd.	2007.06	S/W development and supply	0	Over 50%	N
Martin Professional Pte. Ltd.	1995.06	Sale of audio products	4,036	Over 50%	N
Red Bend Software Japan Co., Ltd.	2005.11	S/W design	10,689	Over 50%	N
Studer Japan, Ltd.	1982.12	Holding Company	6,077	Over 50%	N
Samsung Display Dongguan Co., Ltd.	2001.11	Display panel production	1,400,671	Over 50%	Y
Samsung Display TianJin Co., Ltd.	2004.06	Display panel production	1,036,566	Over 50%	Y
Samsung Electronics Hong Kong Co., Ltd.	1988.09	Sale of electronic goods	1,157,142	Over 50%	Y
Suzhou Samsung Electronics Co., Ltd.	1995.04	Production of home appliances	558,610	Over 50%	Y
Samsung Suzhou Electronics Export Co., Ltd.	1995.04	Production of home appliances	405,961	Over 50%	Y

Samsung (CHINA) Investment Co., Ltd.	1996.03	Sale of electronic goods	13,409,281	Over 50%	Y
Samsung Mobile R&D Center, China-Guangzhou	2010.01	R&D	62,796	Over 50%	N
Samsung Tianjin Mobile Development Center	2010.08	R&D	35,253	Over 50%	N
Samsung R&D Institute China-Shenzhen	2013.03	R&D	14,961	Over 50%	N
Samsung Electronics Suzhou Semiconductor Co., Ltd.	1994.12	Semiconductor processing	950,691	Over 50%	Y
SEMES (XIAN) Co., Ltd.	2013.07	Semiconductor equipment	1,107	Over 50%	N
Samsung Electronics Huizhou Co., Ltd.	1992.12	Production of electronic goods	6,440,564	Over 50%	Y
Tianjin Samsung Electronics Co., Ltd.	1993.04	Production of TV and monitors	672,687	Over 50%	Y
Samsung Electronics Taiwan Co., Ltd.	1994.11	Sale of electronic goods	1,447,907	Over 50%	Y
Beijing Samsung Telecom R&D Center	2000.09	R&D	87,170	Over 50%	Y
Tianjin Samsung Telecom Technology Co., Ltd.	2001.03	Production of telecom products	1,019,144	Over 50%	Y
Shanghai Samsung Semiconductor Co., Ltd.	2001.10	Sale of semiconductor and display panels	5,169,438	Over 50%	Y
Samsung Electronics Suzhou Computer Co., Ltd.	2002.09	Production of electronic goods	910,833	Over 50%	Y
Samsung Suzhou Module Co., Ltd	2002.09	Display panel processing	721,040	Over 50%	Y
Samsung Suzhou LCD Co., Ltd	2011.07	Display panel production	2,019,262	Over 50%	Y
Shenzhen Samsung Electronics Telecommunication Co., Ltd.	2002.02	Production of telecom products	129,943	Over 50%	Y
Samsung Semiconductor (China) R&D Co., Ltd.	2003.04	R&D	27,417	Over 50%	N
Samsung Electronics China R&D Center	2004.05	R&D	46,359	Over 50%	N
Samsung (China) Semiconductor Co., Ltd.	2012.09	Production of semiconductors	8,076,107	Over 50%	Y
Samsung SemiConductor Xian	2016.04	Sale of semiconductor and display panels	796,942	Over 50%	Y
Samsung Electronics (Beijing) Service Company Limited	2005.01	Services	138,193	Over 50%	Y
Tianjin Samsung LED Co., Ltd.	2009.05	LED production	416,578	Over 50%	Y
Harman (China) Technologies Co. Ltd.	2011.03	Production of audio products	132,594	Over 50%	Y
Harman (Suzhou) Audio and Infotainment Systems Co. Ltd.	2013.03	Sale of audio products	9,422	Over 50%	N
Harman Automotive Electronic Systems (Suzhou) Co. Ltd	2006.09	Production of audio products, R&D	307,506	Over 50%	Y
Harman Commercial (Shanghai) Co. Ltd.	2010.10	Sale of audio products	2,691	Over 50%	N
Harman Connected Services Solutions (Beijing) Co. Ltd.	2007.03	Connected Service Provider	332	Over 50%	N
Harman Connected Services Solutions (Chengdu) Co. Ltd.	2007.08	Connected Service Provider	18,211	Over 50%	N
Harman Connected Services Taiwan Inc.	2005.08	Connected Service Provider	263	Over 50%	N
Harman Holding Limited	2007.05	Sale of audio products	438,879	Over 50%	Y
Harman International (China) Holdings Co. Ltd	2009.06	Sale of audio products, R&D	553,526	Over 50%	Y
Harman Automotive Info Tech (Dalian) Co. Ltd	2013.12	S/W development and supply	543	Over 50%	N
Harman Technology (Shenzhen) Co. Ltd.	2004.09	Sale of audio products, R&D	27,115	Over 50%	N
Samsung Display	2012.04	Production and sale of display panels	50,723,199	Over 50%	Y
SU Materials	2011.08	Production of display panel component	31,730	Over 50%	N
STECO	1995.06	Production of semiconductor component	170,098	Over 50%	Y
SEMES	1993.01	Semiconductor/FPD manufacturing equipment	1,043,351	Over 50%	Y

Samsung Electronics Service	1998.10	Electronics goods repair services	347,411	Over 50%	Y
Samsung Electronics Sales	1996.07	Sales of electronic goods	590,110	Over 50%	Y
Samsung Electronics Logitech	1998.04	Total logistics services	155,878	Over 50%	Y
Samsung Medison	1985.07	Medical equipment	329,064	Over 50%	Y
Mirero System	1994.01	Semiconductor S/W	20,371	Over 50%	N
SVIC #21 Venture Capital Union	2011.11	Investment in venture firms and new technologies	85,857	Over 50%	Y
SVIC #22 Venture Capital Union	2011.11	Investment in venture firms and new technologies	126,264	Over 50%	Y
SVIC #23 Venture Capital Union	2012.10	Investment in venture firms and new technologies	24,340	Over 50%	N
SVIC #26 Venture Capital Union	2014.11	Investment in venture firms and new technologies	192,444	Over 50%	Y
SVIC #27 Venture Capital Union	2014.09	Investment in venture firms and new technologies	16,871	Over 50%	N
SVIC #28 Venture Capital Union	2015.02	Investment in venture firms and new technologies	186,164	Over 50%	Y
SVIC #29 Venture Capital Union	2015.04	Investment in venture firms and new technologies	48,251	Over 50%	N
SVIC #32 Venture Capital Union	2016.08	Investment in venture firms and new technologies	64,077	Over 50%	N
SVIC #33 Venture Capital Union	2016.11	Investment in venture firms and new technologies	64,873	Over 50%	N
SVIC #37 Venture Capital Union	2017.11	Investment in venture firms and new technologies	23,886	Over 50%	N
Private Equity Fund for semiconductor growth	2017.03	Investment in semiconductor	51,906	Over 50%	N
Harman International Korea	2005.01	S/W development and supply	17,868	Over 50%	N
Red Bend Software Korea	2007.02	S/W development and supply	960	Over 50%	N

※ Companies with over KRW 75 billion in total assets as of December 31, 2017, are classified as major subsidiaries.

※ See 『II. Businesses Overview』 for additional information about major business segments.

※ In 2017, Martin Professional ApS changed its name to Harman Professional Denmark ApS, and Martin Professional France SAS changed its name to Harman Professional France SAS. Also, Quietside LLC changed its name to Samsung HVAC America, LLC, Martin Professional GmbH changed its name to Harman Professional Germany GmbH, and Harman Neusoft Automotive Infotech (Dalian) Co. Ltd changed its name to Harman Automotive Infotech (Dalian) Co., Ltd. NewNet Communication Technologies(Canada), Inc. changed its name to SigMast Communications Inc.

(Changes in Subsidiaries)

	AMER	Europe/ MEA/ CIS	Asia	China	Domestic	Total	Increase	Decrease
Dec 31, 2012	30	61	23	32	20	166		
Dec 31, 2013	25	55	22	33	18	153	[Americas: 2] NeuroLogica Corp., Intellectual Keystone Technology LLC. [Europe/MEA/CIS: 1] Samsung Electronics Switzerland GmbH [Asia: 2] Samsung Electronics Vietnam THAINGUYEN Co., Ltd., Samsung Electronics New Zealand Ltd. [China: 3] Samsung Network R&D Center China-Shenzhen, Samsung R&D Institute China-Xian, SEMES (XIAN) Co., Ltd.	[Domestic: 2] Secron, GES [Americas: 7] Newton Sub. Corp., mSpot, Inc., Deltapoint Cardiac Diagnostics, Inc., Samsung Medison America, Inc., Intellectual Keystone Technology LLC., Samsung Medison Brasil Ltda., Samsung Electronics Corporativo, SA de CV [Europe/MEA/CIS: 7] Samsung Telecoms (UK) Ltd., Samsung LCD Netherlands R&D Center B.V, Samsung LCD Netherlands R&D Center (UK) Limited, General RF Modules AB, Samsung Medison France S.A.S., Samsung Opto-Electronics GmbH, Samsung Medison Italia S.r.l. [Asia: 3] Samsung Electronic Philippines Manufacturing Corp., Batino Realty Corporation, TNP Small/Medium Size & Venture Enterprises Growth Promotion Investment Limited Partnership(TSUNAMI) [China: 2] Samsung LCD Netherlands R&D Center (HK)Limited, Medison (shanghai) Co., Ltd.
Dec 31, 2014	30	56	22	32	18	158	[Domestic : 2] SVIC #26 Venture Capital Union SVIC #27 Venture Capital Union [America: 6] RT SV CO-INVEST, LP, Quietside LLC, SmartThings, Inc., PrinterOn Inc., PrinterOn America Corporation, 1397011 Ontario Ltd. [Europe/MEA/CIS: 3] PrinterOn Europe Limited, Samsung Electronics South Africa Production(pty) Ltd., Samsung Electronics Caucasus Co., Ltd. [Asia : 1] Samsung Display Bac Ninh	[Domestic: 2] Samsung Blue Wings, World Cyber Games [America: 1] 1397011 Ontario Ltd. [Europe: 2] Samsung Medison Europe B.V Nanogen Recognomics GmbH [Asia: 1] Medison Medical Systems India Private Ltd. [China: 1] Medison Medical Equipment Shanghai Co., Ltd..
Dec 31, 2015	33	55	24	30	17	159	[Domestic : 2] SVIC #28 Venture Capital Union SVIC #29 Venture Capital Union [America: 6] Simpres Comercio, Locacao e Servicos S.A., Samsung Pay, Inc. (formerly LoopPay, Inc.), Prismview LLC(formerly YESCO Electronics LLC), Paymate Global, Inc., Beijing Integrated Circuit Industry International Fund, Stellus Technologies [Asia: 2] Samsung Electronics HCMC CE Complex Co. Ltd. Future Technology & Service	[Domestic: 3] RAY High Pioneer Private Investment Trust #1, SVIC #6 Venture Capital Union [America: 3] Samsung Telecommunications America, Paymate Global, Nvelo [China: 2] Samsung Electronics Hainan Fiberoptics, Samsung Medison Shanghai Medical Instrument [Europe: 1] Nanoradio Hellas

Dec 31, 2016	44	53	23	29	20	169	[Domestic : 4] SVIC #32 Venture Capital Union SVIC #33 Venture Capital Union Mirero System, S-Printing Solution Co., Ltd. [America: 13] Samsung Oak Holdings, Inc., Joyent, Inc., Joyent Canada, Inc., AdGear Technologies Inc., Samsung Next LLC, Samsung Next Fund LLC, Dacor Holdings, Inc., Dacor, Dacor Canada Co., EverythingDacor.com, Inc., Distinctive Appliances of California, Inc., Viv Labs, Inc., NewNet Communication Technologies(Canada), Inc. [Europe: 1] Joyent Ltd. [Asia: 1] Laos Samsung Electronics Sole Co., Ltd [China: 1] Samsung SemiConductor Xian	[Domestic : 1] SVIC #14 Venture Capital Union [America: 2] Grandis, Inc., Joyent Canada, Inc. [Europe/CIS: 3] Samsung Russia Service Centre, SonoAce Deutschland GmbH Samsung Electronics Kazakhstan LLP [Asia: 2] Samsung Telecommunications Malaysia, Future Technology & Service [China: 2] Samsung R&D Institute China-Xian, Samsung Electronics Shanghai Telecommunication Co., Ltd.
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Dec 31, 2017	63	109	38	38	22	270	[Domestic : 4] Private Equity Fund for semiconductor growth Harman International Korea Red Bend Software Korea SVIC #37 Venture Capital Union [America: 25] Samsung Electronics Home Appliances America, LLC Kngine, Inc. AMX Holding Corporation AMX LLC Harman Becker Automotive Systems, Inc. Harman Connected Services Inc. Harman Connected Services Engineering Corp. Harman Connected Services Holding Corp. Harman Connected Services South America S.R.L. Harman da Amazonia Industria Electronica e Participacoes Ltda. Harman de Mexico S. de R.L. de C.V. Harman do Brasil Industria Electronica e Participacoes Ltda. Harman Financial Group, LLC Harman International Industries Canada Ltd. Harman International Industries, Inc. Harman International Mexico S de RL de CV Harman Investment Group, LLC Harman KG Holding, LLC Harman Professional, Inc. Red Bend Software Inc. S1NN USA, Inc. Southern Vision Systems, Inc TowerSec Inc. Triple Play Integration LLC China Materialia New Materials 2016 Limited Partnership [Europe/CIS/MEA: 62] Samsung Electronics Air Conditioner Europe B.V. Aditi Technologies Europe GmbH AKG Acoustics GmbH AMX GmbH AMX UK Limited Duran Audio B.V. Duran Audio Iberia Espana S.L. Endeleo Limited Harman Automotive UK Limited Harman Becker Automotive Systems GmbH Harman Becker Automotive Systems Italy S.R.L. Harman Becker Automotive Systems Manufacturing Kft Harman Belgium SA Harman Connected Services AB. Harman Connected Services Finland OY Harman Connected Services GmbH Harman Connected Services Limited Harman Connected Services Poland Sp.zoo Harman Connected Services UK Ltd. Harman Consumer Division Nordic A/S Harman Consumer Finland OY Harman Consumer Nederland B.V. Harman Deutschland GmbH Harman Finance International GP S.a.r.l Harman France SNC Harman Holding GmbH & Co. KG Harman Hungary Financing Ltd. Harman Inc. & Co. KG Harman International Estonia OU Harman International Industries Limited Harman International Romania SRL Harman Finance International SCA Harman International s.r.o Harman International SNC Harman Management GmbH Harman Professional Kft Inspiration Matters Limited Knight Image Limited Martin Manufacturing (UK) Ltd Harman Professional Denmark ApS Harman Professional France SAS. Harman Professional Germany GmbH	[America: 6] Samsung Receivables Corporation TowerSec Inc. Kngine, Inc. PrinterOn Inc. PrinterOn America Corporation Simpres Comercio, Locacao e Servicos S.A. [Europe/CIS: 6] Samsung Electronics Rus LLC Samsung Electronics Ukraine LLC Surfkitchen Limited AMX LLC (Russia) Martin Professional Ltd. PrinterOn Europe Limited [Asia: 2] VFX Systems PTY Ltd. I.P.S.G. International Product Solution Group FTY. LTD [China: 3] Samsung Electronics (Shandong) Digital Printing Co., Ltd. Tianjin Samsung Opto-Electronics Co., Ltd. Martin Trading Zhuhai Ltd. [Domestic : 2] SVIC #20 Venture Capital Union S-Printing Solution
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						Martin Professional Ltd. R&D International BVBA Red Bend Software Ltd. Red Bend Software SAS France., Studer Professional Audio GmbH Surfkitchen Limited AMX LLC (Russia) Harman Connected Services OOO Harman RUS CIS LLC Broadsense Ltd. Global Symphony Technology Group Private Ltd. Harman Connected Services Morocco Harman Industries Holdings Mauritius Ltd. iOnRoad Ltd. iOnRoad Technologies Ltd. Red Bend Ltd. TowerSec Ltd. Innoetics E.P.E. ARCAM Ltd A&R Cambridge Ltd [Asia: 17] AMX Products And Solutions Private Limited Harman Connected Services Technologies Pvt. Ltd. Harman Connected Services Corp. India Pvt. Ltd. Harman Connected Services Japan Co. Ltd. Harman International (India) Private Limited Harman International Industries PTY, Ltd. Harman International Japan Co. Ltd. Harman International Singapore Pte. Ltd. Harman Malaysia Sdn. Bhd. Harman Professional Singapore Pte. Ltd I.P.S.G. International Product Solution Group PTY. LTD. INSP India Software Development Pvt. Ltd. Martin Professional Pte. Ltd. Red Bend Software KK Studer Japan, Ltd. VFX Systems PTY Ltd. Samsung Nepal Services Pvt, Ltd [China: 12] Harman (China) Technologies Co. Ltd. Harman (Suzhou) Audio and Infotainment Systems Co. Ltd. Harman Automotive Electronic Systems (Suzhou) Co. Ltd Harman Commercial (Shanghai) Co. Ltd. Harman Connected Services (Beijing) Solutions Co. Ltd. Harman Connected Services (Chengdu) Solutions Co. Ltd. Harman Connected Services Taiwan Inc. Harman Holding Limited Harman International (China) Holdings Co. Ltd Harman Automotive Infotech (Dalian) Co. Ltd Harman Technology (Shenzhen) Co. Ltd. Martin Trading Zhuhai Ltd.	
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※ AMER = America; MEA = Middle East and Africa, CIS = Commonwealth of Independent States

H. Credit Rating

SEC's credit rating is assessed by two external credit ratings agencies. As of December 31, 2017, SEC's credit ratings are: "A1" and investment outlook is positive, as rated by Moody's; and "AA-" and investment outlook is stable, as rated by S&P.

Date	Securities	Ratings	Credit Rating Agency	Rating Range	Note
'14.08	Corporate bond	A+	S&P (USA)	(AAA ~ D)	Annual Review
'14.10	Corporate bond	A1	Moody's (USA)	(Aaa ~ C)	
'15.05	Corporate bond	A1	Moody's (USA)	(Aaa ~ C)	
'15.09	Corporate bond	A+	S&P (USA)	(AAA ~ D)	
'16.07	Corporate bond	A+	S&P (USA)	(AAA ~ D)	
'16.08	Corporate bond	A1	Moody's (USA)	(Aaa ~ C)	
'17.07	Corporate bond	AA-	S&P (USA)	(AAA ~ D)	
'17.08	Corporate bond	A1	Moody's (USA)	(Aaa ~ C)	

2. Company History

(The Information disclosed below outlines major changes to the Company such as asset transfers, M&A, and security offerings)

2011.01.01	Merged with Samsung Electronics Gwangju
2011.02.16	Acquired shares of Prosonic (100%) and Medison (43.5%)
2011.04.29	Acquired additional shares of Medison (22.3%)
2011.07.01	Transferred solar cell business to Samsung SDI
2011.07.22	Samsung Information System America ("SISA"), a subsidiary of SEC, acquired shares of Grandis (100%)
2012.01.19	Acquired remaining shares of S-LCD Corporation (50%) from SONY
2012.04.01	Separated LCD business (established Samsung Display)
2012.04.01	Merged Samsung LED into SEC
2012.04.10	Samsung Electronics America ("SEA"), a subsidiary of SEC, issued USD \$1 billion of corporate bonds.
2012.07.01	Samsung Mobile Display merged with S-LCD (a subsidiary of Samsung Display)
2012.09.01	Samsung Medison, a subsidiary of SEC, merged with Prosonic
2012.12.01	Merged SEHF Korea into SEC
2013.01.01	Merged SECRON with GES (a subsidiary of SEMES)
2013.01.28	SEA acquired shares of NeuroLogica (100%)
2014.01.15	Samsung Display disposed of stock of Samsung Corning Precision Materials and purchased convertible preferred shares of Corning Incorporated
2014.08.18	SEA acquired shares of SmartThings (100%)
2015.01.01	SEA merged with Samsung Telecommunications America LLC ("STA")
2015.02.23	SEA acquired shares of LoopPay (100%)
2016.01.28	Sold shares of Samsung Card (37.5%)
2016.06.24	SEA acquired shares of Joyent (100%)
2016.09.07	SEA acquired shares of Dacor (100%)

2016.10.07 SEA acquired shares of Viv Labs (100%)

2016.11.01 Spun-off printing solutions business (established S-Printing Solution Co., Ltd.)

2017.03.10 SEA acquired shares of Harman International Industries, Inc. (100%)

2017.11.01 Sold Printing Solution business

※ The Samsung Electronics Headquarters Address is: 129, Samsung-ro (Maetan-dong), Yeongtong-gu, Suwon-si, Gyeonggi-do

(Important Changes in Management Executives)

Following the annual general meeting of shareholders on March 16, 2012, of the three Executive Directors with expiring terms (Gee-Sung Choi, Yoon-Woo Lee, Ju-Hwa Yoon), Yoon-Woo Lee retired and Gee-Sung Choi and Ju-Hwa Yoon were re-appointed. Oh-Hyun Kwon was newly appointed as Executive Director. In addition, of the three Independent Directors with expiring terms (Dong-Min Yoon, Jae-Woong Lee, Oh-Soo Park), Jae-Woong Lee and Oh-Soo Park retired and Dong-Min Yoon was re-appointed. Han-Joong Kim and Byeong-Gi Lee were newly appointed as Independent Directors.

On June 8, 2012, Oh-Hyun Kwon succeeded Gee-Sung Choi as CEO.

On February 7, 2013, Independent Director Dong-Min Yoon retired (deceased).

On March 14, 2013, Executive Directors Gee-Sung Choi and Ju-Hwa Yoon resigned.

Following the shareholders' meeting on March 15, 2013, Boo-Keun Yoon, Jong-Kyun Shin, and Sang-Hoon Lee were newly appointed as Executive Directors. Independent Director In-Ho Lee was reappointed, and Kwang-Soo Song and Eun-Mee Kim were newly appointed as Independent Directors.

On March 15, 2013, SEC appointed Executive Directors Boo-Keun Yoon and Jong-Kyun Shin as CEOs to serve alongside incumbent CEO Oh-Hyun Kwon as co-CEOs.

On March 13, 2015, Executive Director Oh-Hyun Kwon and Independent Directors Han-Joong Kim and Byeong-Gi Lee were reappointed at the annual general meeting of shareholders.

On March 11, 2016, Executive Directors Boo-Keun Yoon, Jong-Kyun Shin and Sang-Hoon Lee were reappointed at the annual general meeting of shareholders. Of the three Independent Executive Directors with expiring terms (In-Ho Lee, Kwang-Soo Song and Eun-Mee Kim), Eun-Mee Kim retired, and In-Ho Lee, Kwang-Soo Song were reappointed. Jae-Wan Bahk was newly appointed as an Independent Director.

On October 27, 2016, Jae-Yong Lee was newly appointed as Executive Director at the Extraordinary General Meeting of Shareholders and Executive Director Sang-Hoon Lee resigned.

Following the annual meeting on March 23, 2018, Sang-Hoon Lee, Ki-Nam Kim, Hyun-Suk Kim, and Dong-Jin Koh were newly appointed as Executive Directors and Jeong Kim, Sun-Uk Kim, Byung-Gook Park were newly appointed as Independent Directors. In addition, Executive Director Oh-Hyun Kwon and Independent Directors Han-Joong Kim and Byeong-Gi Lee retired as terms expired. Executive Directors Boo-Keun Yoon and Jong-Kyun Shin resigned.

On March 23, 2018, Executive Directors Ki-Nam Kim, Hyun-Suk Kim, and Dong-Jin Koh were newly appointed as CEOs.

As of April 2, 2018, (the "Reporting Date"), SEC's BOD is comprised of five Executive Directors (Jae-Yong Lee, Sang-Hoon Lee, Ki-Nam Kim, Hyun-suk Kim, and Dong-Jon Koh) and six Independent Directors (In-Ho Lee, Kwang-Soo Song, Sun-Uk Kim, Jae-Wan Bahk, Byung-Gook Park, and Jeong Kim).

(Changes in Organizational Structure)

In December 2012, following an organizational change, CE and IM businesses were upgraded to individual Division status, and the IT Solution business was divided into Printing Solution and Computer businesses. The Printing Solution business was incorporated into the CE Division and the Computer business was converted to a team unit and integrated into the Mobile business under the IM Division. The Medical Device business team was upgraded to a business unit within the CE Division.

In December 2013, the Digital Imaging business was reorganized as a team unit, and then was integrated with the Mobile business unit. In December 2015, the LED business was reorganized as a team unit.

In November 2016, the Company spun off its printing solutions business and established S-Printing Solution Co., Ltd.

In June 2017, the System LSI business was divided into Foundry and System LSI businesses.

[As of December 2012]

	Before	After
Business organization	DMC Division (CE: Visual display, Digital appliances) (IM: Mobile, IT solution, Network, Digital imaging)	CE Division (Visual display, Digital appliances, Printing solution, Medical devices)
		IM Division (Mobile, Network, Digital imaging)
	DS Division (Memory, SYS.LSI, LCD, LED)	DS Division (Memory, SYS.LSI, DP, LED)
Regional headquarters	Korea, North America, Latin America, Europe, CIS, Southwest Asia, Southeast Asia, China, Middle East, Africa	Korea, North America, Latin America, Europe, CIS, Southwest Asia, Southeast Asia, China, Middle East, Africa
	Americas(DS), Europe(DS), China(DS), Southeast Asia(DS), Japan(DS)	Americas(DS), Europe(DS), China(DS), Southeast Asia(DS), Japan (DS)

[As of December 2013]

	Before	After
Business organization	CE Division (Visual display, Digital appliances, Printing solution, Medical devices)	CE Division (Visual display, Digital appliances, Printing solution, Medical devices)
	IM Division (Mobile, Network, Digital imaging)	IM Division (Mobile, Network)
	DS Division (Memory, SYS.LSI, DP, LED)	DS Division (Memory, SYS.LSI, DP, LED)
Regional headquarters	Korea, North America, Latin America, Europe, CIS, Southwest Asia, Southeast Asia, China, Middle East, Africa	Korea, North America, Latin America, Europe, CIS, Southwest Asia, Southeast Asia, China, Middle East, Africa
	Americas(DS), Europe(DS), China(DS), Southeast Asia(DS), Japan(DS)	Americas(DS), Europe(DS), China(DS), Southeast Asia(DS), Japan (DS)

[As of December 2015]

	Before	After
Business organization	CE Division (Visual display, Digital appliances, Printing solution, Medical devices)	CE Division (Visual display, Digital appliances, Printing solution, Medical devices)
	IM Division (Mobile, Network)	IM Division (Mobile, Network)
	DS Division (Memory, SYS.LSI, DP, LED)	DS Division (Memory, SYS.LSI, DP)
Regional headquarters	Korea, North America, Latin America, Europe, CIS, Southwest Asia, Southeast Asia, China, Middle East, Africa	Korea, North America, Latin America, Europe, CIS, Southwest Asia, Southeast Asia, China, Middle East, Africa
	Americas(DS), Europe(DS), China(DS), Southeast Asia(DS), Japan(DS)	Americas(DS), Europe(DS), China(DS), Southeast Asia(DS), Japan (DS)

[As of December 2016]

	Before	After
Business organization	CE Division (Visual display, Digital appliances, Printing solution, Medical devices)	CE Division (Visual display, Digital appliances, Medical devices)
	IM Division (Mobile, Network)	IM Division (Mobile, Network)
	DS Division (Memory, SYS.LSI, DP)	DS Division (Memory, SYS.LSI, DP)
Regional headquarters	Korea, North America, Latin America, Europe, CIS, Southwest Asia, Southeast Asia, China, Middle East, Africa	Korea, North America, Latin America, Europe, CIS, Southwest Asia, Southeast Asia, China, Middle East, Africa
	Americas(DS), Europe(DS), China(DS), Southeast Asia(DS), Japan(DS)	Americas(DS), Europe(DS), China(DS), Southeast Asia(DS), Japan (DS)

[As of June 2017]

	Before	After
Business organization	CE Division (Visual display, Digital appliances, Medical devices)	CE Division (Visual display, Digital appliances, Medical devices)
	IM Division (Mobile, Network)	IM Division (Mobile, Network)
	DS Division (Memory, SYS.LSI, DP)	DS Division (Memory, SYS.LSI, Foundry, DP)
	-	Harman Division
Regional headquarters	Korea, North America, Latin America, Europe, CIS, Southwest Asia, Southeast Asia, China, Middle East, Africa	Korea, North America, Latin America, Europe, CIS, Southwest Asia, Southeast Asia, China, Middle East, Africa
	Americas(DS), Europe(DS), China(DS), Southeast Asia(DS), Japan (DS)	Americas(DS), Europe(DS), China(DS), Southeast Asia(DS), Japan (DS)

3. Changes in Paid-in Capital

No changes were reported during the past five (5) consecutive years.

4. Stock Information

A. Total Number of Shares

The total number of authorized shares according to the articles of incorporation is 500,000,000 (par value per share: KRW 5,000). As of December 31, 2017, SEC has issued 129,098,494 shares of registered common stock and 18,250,580 shares of registered preferred stock without voting rights (excluding canceled shares). SEC has cumulatively canceled 26,510,843 shares of common stock and 5,642,847 shares of preferred stock in accordance with the resolution of the board of directors.

As of December 31, 2017, the number of floating common stock is 119,688,369 shares and the number of floating preferred stock is 16,530,409 shares, excluding treasury shares (of 9,410,125 shares of common stock and 1,720,171 shares of preferred stock).

[As of December 31, 2017]

(Unit: Shares)

Classification	Type of Stocks			Note
	Common	Preferred	Total	
I. Number of authorized shares	400,000,000	100,000,000	500,000,000	-
II. Number of shares issued	155,609,337	23,893,427	179,502,764	-
III. Number of shares decreased	26,510,843	5,642,847	32,153,690	-
1. Capital Reduction	-	-	-	-
2. Number of shares canceled	26,510,843	5,642,847	32,153,690	Cancellation of treasury shares
3. Redemption of redeemable shares	-	-	-	-
4. Others	-	-	-	-
IV. Number of outstanding shares (II-III)	129,098,494	18,250,580	147,349,074	-
V. Treasury shares	9,410,125	1,720,171	11,130,296	-
VI. Number of floating shares (IV-V)	119,688,369	16,530,409	136,218,778	-

※ SEC's common stocks carry voting rights but preferred stocks do not carry voting rights.

B. Treasury Shares

In 2017, Samsung Electronics acquired 3,009,282 shares of common stock and 753,325 shares of preferred stock and cancelled 11,580,843 shares of common stock and 2,262,847 shares of preferred stock, for enhancing shareholder return.

As of December 31, 2017, SEC holds 9,410,125 shares of common stock and 1,720,171 shares of preferred stock as treasury shares.

[As of December 31, 2017]

(Unit: Shares)

Acquisition method	Share type	Period-beginning shares	Change			Period-end shares
			Acquisition (+)	Disposal (-)	Cancellation (-)	
Intra-market direct acquisition (Within dividend related capital gains limit)	Common	17,981,686	3,009,282	-	11,580,843	9,410,125
	Preferred	3,229,693	753,325	-	2,262,847	1,720,171
Total	Common	17,981,686	3,009,282	-	11,580,843	9,410,125
	Preferred	3,229,693	753,325	-	2,262,847	1,720,171

C. Types of Registered Stock

SEC has two types of registered equity securities: 1) common stock; 2) non-voting and non-cumulative preferred stock.

The shareholders of preferred stock are entitled to dividend payments of an additional 1% of par value compared to shareholders of common stock. As of December 31, 2017, the total number of outstanding preferred stock is 18,250,580 shares.

5. Voting Shares

SEC has 129,098,494 shares of common stock outstanding, which represents 25.8% of the total number of authorized shares of 500 million (500,000,000). There are 18,250,580 shares of preferred stock outstanding (with no voting rights). SEC holds 9,410,125 shares of common stock (with no voting rights) in its treasury, and 12,506,577 shares of common stock have limited voting rights under applicable laws. Thus, the total number of shares with voting rights is 107,181,792.

[As of December 31, 2017]

(Unit: Shares)

Classification	Share type	Number of shares	Note
Number of outstanding shares (A)	Common	129,098,494	-
	Preferred	18,250,580	-
Shares with no voting rights (B)	Common	9,410,125	Treasury stock according to Korean Commercial Act
Shares with no voting rights according to Articles of Incorporation (C)	Preferred	18,250,580	1,720,171 shares of preferred treasury stock included
Shares with limited voting rights under relevant laws (D)	Common	12,479,184	Restricted by the Monopoly Regulation and Fair Trade Act: 10,622,814 shares held by Samsung Life Insurance & 1,856,370 shares held by Samsung Fire & Marine Insurance
	Common	27,393	Restricted by the Insurance Business Act: Shares held by Samsung Life Insurance in certain special accounts
Shares with voting rights (F = A - B - C - D)	Common	107,181,792	-
	Preferred	-	-

※ Pursuant to Korea's Monopoly Regulation and Fair Trade Act, shareholders with limited voting rights (D) can exercise certain rights in accordance with related laws, including the right to vote on agendas such as the appointment or dismissal of directors and revisions to the Articles of Incorporation.

6. Shareholder Return

As a part of the Company's shareholder return policy, a portion of the profits have been returned to shareholders by way of dividends and shareholder value has been increased through open market purchases of floating shares (share repurchase). The Company carefully considers strategic investments for sustainable growth, business performance, and cash flows in determining the level of total shareholder return.

Dividends paid in the most recent three (3) fiscal years are as follows:

(Unit: Shares / %)

Classification		2017	2016	2015
Par value per share (KRW)		5,000	5,000	5,000
Net profit (KRW million)		41,344,569	22,415,655	18,694,628
EPS (KRW)		299,868	157,967	126,305
Total cash dividend (KRW million)		5,826,302	3,991,892	3,068,737
Total stock dividend (KRW million)		-	-	-
Dividend payout ratio (%)		14.1	17.8	16.4
Cash dividend yield (%)	Common	1.7	1.6	1.6
	Preferred	2.1	2.0	1.9
Stock dividend yield (%)	-	-	-	-
	-	-	-	-
Cash dividend per share (KRW)	Common	42,500	28,500	21,000
	Preferred	42,550	28,550	21,050
Stock dividend per share (share)	Common	-	-	-
	Preferred	-	-	-

※ March quarterly dividend of 2017 is KRW 972,373 million (KRW 7,000 per share), and June quarterly dividend is KRW 965,348 million (KRW 7,000 per share). September quarterly dividend of 2017 is KRW 959,051 million (KRW 7,000 per share).

※ EPS denotes basic earnings per common share.

※ For further information relating to the calculation of basic EPS, see Earnings Per Share in 『2. Note to Consolidated Financial Statements』 in 『III. Financial Affairs』.

II. Businesses Overview

1. Overview

Note on Forward-Looking Statements

This report includes forward-looking statements that relate to future events and can be generally identified by phrases containing words such as “believes,” “expects,” “anticipates,” “foresees,” “forecasts,” “estimates” or other words or phrases of similar meaning. These forward-looking statements are not guarantees of future performance and may involve known and unknown risks, uncertainties and other factors that may affect the Company’s actual results, performance, achievements or financial position, making them materially different from the actual future results, performance, achievements or financial position expressed or implied by these forward-looking statements.

Uncertain events that could positively or negatively affect the Company’s management condition and financial performance include:

- *Trends of financial markets domestically and abroad, including changes in exchange rates and interest rates*
- *The Company’s strategic decision making, including disposals and purchases of businesses*
- *Unexpected sudden changes in core businesses such as CE, IM, Semiconductor, and DP*
- *Other changes domestically and abroad that can affect management condition and financial performance*

The Company assumes no obligation to revise or update this report to reflect risks or uncertainties that arise after the reporting period.

A. Business Overview by Division

In addition to our headquarters in Korea, Samsung Electronics is comprised of 270 subsidiaries across the world responsible for sales and production. There are nine (9) regional headquarters for CE and IM Divisions, five (5) regional headquarters for the DS Division, and Harman’s subsidiaries.

The Company’s business divisions are organized in a two-pronged framework consisting of set (brand products) and component businesses. The set business comprises the CE and IM Divisions, and the component business consists of the Semiconductor and DP Business Units. The CE Division is responsible for the production and sales of TVs, monitors, refrigerators, and washing machines; and the IM Division focuses on the production and sales of HHPs, which includes smartphones, as well as network systems, and computers. The DS Division comprises the Semiconductor business which manufactures and sells DRAMs, mobile APs, and other semiconductor-related products, and the DP Business, which manufactures and sells TFT-LCD and OLED panels for TVs, desktops, laptops, and mobile devices.

The Harman Division (acquired in the first half of 2017) manufactures and sells products such as Head units, infotainment systems, telematics, and speakers.

<Major Products, by Business Division>

Business Division		Major Products
CE Division		TVs, monitors, refrigerators, washing machines, air conditioners, medical devices, etc.
IM Division		HHPs, network systems, computers, etc.
DS Division	DRAM, NAND flash, mobile APs, etc.	DRAM, NAND Flash, Mobile AP, etc.
	TFT-LCD panels, OLED panels, etc.	TFT-LCD, OLED, etc.
Harman Division		Head units, infotainment systems, telematics, speakers, etc.

The Company maintains its corporate headquarters and twenty-two (22) consolidated subsidiaries in Korea.

The Company's corporate headquarters is divided along Divisions and/or Businesses, which are situated in the following 6 locations: Suwon (CE and R&D Center); Gumi (IM); Giheung, Hwasung, and Pyeongtaek (Semiconductor); and Gwangju (Home Appliance). The Company's unlisted domestic subsidiaries include Samsung Display for display panel production, Samsung Electronics Sales for domestic retail sales, Samsung Electronics Service for after-sales services, and Samsung Electronics Logitech for logistics.

We have 248 overseas subsidiaries for product manufacturing, sales, and R&D, with locations in the US, Europe, Asia, and Africa.

We have 63 subsidiaries in North America, including SEA (New Jersey; responsible for the sales of set products including HHPs), SAS (Austin; manufacturing semiconductor products), and Harman in the US and SAMEX (Tijuana; manufacturing TVs) in Mexico.

In Europe, we operate 79 subsidiaries. These include SEUK (UK), SEF (France), SEG (Germany), SEI (Italy) for sales of set products; SESK (Slovakia) and SEH (Hungary) for manufacturing TVs, and SEPM (Poland) for manufacturing home appliances such as refrigerators.

We have 38 subsidiaries in Asia, including SAPL (Singapore), SEAU (Australia), SEPCO (Philippines), and SME (Malaysia), which are responsible for regional sales. In addition, we operate numerous production sites including SEV and SEVT (Vietnam) for HHPs, SEHC (Vietnam) for TVs, and SIEL (India) for both TVs and HHPs.

We operate 38 subsidiaries in China, including SCIC (Beijing) and SEHK (Hong Kong) for the sales of set products in those regions. We also operate manufacturing facilities in that nation, including ones for set products in Tianjin (TSTC), and for home appliances (SSEC) and back-end processes for semiconductor & display (SESS) in Suzhou.

SEC also has 30 production and sales subsidiaries combined in Africa, the Middle East, and the CIS.

[CE Division]

□ Industry Overview

Since the first public demonstration of a true television system in 1926 and later mass production of color TVs, technological developments have led to the creation of products such as the Trinitron CRT (1967) and flat CRT (1996). As the penetration rate in major countries reached over 90%, the CRT TV business stagnated. The industry regained strong growth momentum following the launch of flat panel TVs (“FPTVs”) like LCDs and PDPs, especially with the expansion of digital broadcasting (UK/US 1998~). FPTVs replaced CRT sets on the back of their enhanced design, picture quality, etc., combined with a sharp decline in prices.

The year 2010 saw the launch of 3D TVs, and the rise of internet video services (e.g., over-the-top) along with increased consumer interest in smart devices from 2011 to 2012 led to the birth of the smart TV market. The UHD TV, an innovative product boasting significantly enhanced resolution and picture quality, launched in 2013 and a new form factor, the Curved TV, was released just a year later, indicating a constantly evolving TV market.

In 2016, overall TV demand was 222.7 million units, a 1.5% decline from the previous year. By product, LCD-TVs (including LED ones) sold 221.5 million units, a 1.3% decrease year-on-year, for a market share of over 99%, while OLED and CRT models sold a respective 700,000 and 500,000 units. In 2017, LCD TVs sold 220 million units to account for over 99% of the overall total of 221.9 million. Demand for OLED TVs rose to 1.38 million units but the OLED TV portion of total sales was just 0.6%. In 2018, we expect overall TV demand to increase 4-5% to around 227 million units.

Also, based on rising demand for high-definition and large-sized screens, UHD TV sales are expected to rise 25% year-over-year and reach 100 million units this year and account for about 44% of total TV demand, while sales of ultra large-sized TVs (60-inch and larger) are projected to hit around 19 million units, the total up at least 20% from the previous year. The increase in demand is likely to come from emerging countries—markets in Asia, Latin America, the Middle East, and Africa are expected to grow over 5%, driving overall growth in TV demand.

□ Market Condition

The market trend toward large, high-resolution screens is accelerating due to intensified competition between manufacturers. Accordingly, market shares of major players with high-quality products and brand power are rising. In addition, as consumer demand for high-resolution screens and slim designs increase, LED TVs, with eco-friendly LED back light units (“BLU”)—which increase brightness, contrast, and energy efficiency—have become the mainstream in the market.

The Company has focused on smart TV sales based on forecasts that demand to acquire information from the internet via TVs is rising. Our concentrated efforts resulted in the launch of multi-featured smart TVs in 2010 and a strong position as a market leader. In 2012, the Company introduced a new market trend with the introduction of Smart Interaction™—a system based on voice commands and command gestures to enable convenient interaction with Samsung Smart TVs. In 2013, the Company introduced a system that recommends TV programs based on a user’s viewing history, and in 2014, striving to further enhance user experiences, we focused on improving usability, expanding contents, and enhancing our Smart TV platform.

In 2015, as we saw a change in the delivery of content, we achieved an innovation in smart UX, enabling consumers to enjoy various contents via their TV. In 2016, we introduced smart TVs that automatically connect with other devices, allowing peripheral devices to be operated with a single remote control. Our Smart TVs enable consumers to more conveniently enjoy increasingly diverse entertainment services on one screen.

In 2017, we created an ecosystem and extended the connectivity of TVs to mobile devices, enabling easy use of the smart hub, control of the TV via mobile devices, and the ability to enjoy mobile phone contents on a TV screen, thus presenting a more integrated and consistent entertainment experience. In addition, users can take advantage of intelligent voice recognition technology to easily control the TV, including changing the volume, channels, or even the screen mode.

In 2018, we expect market competition to intensify as more companies shift their focus to premium products that offer

high-resolution and large-size screens, such as UHD TVs or those with 60-inch and larger screens.

< Market Share of the CE Division's Main Product >

Product	2017	2016	2015
TV	20.0%	21.6%	21.0%

※ Data for 2015 and 2016 is from research firm *IHS* (global market share in quantity). Data for 2017 is company's estimates

□ **Business Condition**

We have maintained the top position in the overall TV market, FPTVs, and LCD TVs for twelve straight years since 2006.

In 2009, we created the world's first LED TV (LED BLU, ultra-slim/ultra-light, eco-friendly); and in the first quarter of 2010, we launched the world's first 3D Total Solution by offering a 3D TV/Blue-ray player/3D glasses/Blu-ray titles in one package, taking the lead in the 3D TV market.

We launched Samsung Apps—the world's first app store for smart TVs—in 2010 and added various smart services in 2011, consolidating Samsung status as a leader in smart TVs. In 2012, we developed content that can be accessed via new input methods based on voice/gesture (Smart Interaction), thus improving services that every member of family can enjoy.

In 2013, we released UHD TVs, offering superior image quality and resolution compared to previous models. Also, by launching a new Curved TV form factor in 2014 and the first nano-crystal technology (also known as quantum dot) SUHD TV in 2015, we bolstered our leadership in the premium market. In 2016, by introducing second generation quantum dot and HDR 1000 technologies, we raised the bar for image quality even higher.

In 2017, the Company created a new category with QLED TVs, which offer 100% color volume and perfect color expression regardless of the level of brightness and show up to 2000 nits, allowing us to realize the ultimate image quality of TVs. Furthermore, we addressed a widespread consumer pain point—we removed the need for complicated wiring to connect peripheral devices. Consumers can now connect their devices (such as a set-top box and game console) via one seemingly transparent optical cable and operate them with a single remote control. As the market leader, this is just one example of how we are continuously working to introduce new concepts to benefit our consumers.

Another of the Company's vision, based on The Frame TV, is a concept called 'Screen Everywhere.' This idea envisions giving consumers the ability to hang their TV anywhere, as The Frame's innovative design can blend perfectly into any living space.

As the leading player in the market, we will continue to introduce innovative products—such as 8K ultra-definition QLED TVs that increase resolution fourfold from existing UHD standards—and expand the boundaries of TVs.

[IM Division]

□ Industry Overview

In 2018, the penetration rate of mobile phones will reach 78%, up from 77% in 2017 (*Strategy Analytics*, February 2018).

The mobile phone industry started with the first generation analogue phones in the early 1980s and evolved to second generation digital and third generation mobile communication standards such as WCDMA. Today, fourth generation LTE mobile communication technology with ultra-high-speed data transmission is becoming commonplace, and fifth generation mobile communication service is getting ready for commercialization. Also, as market demand continues to focus on smartphones, the importance of competitive software and services—such as content/services, applications, UX, mobile payment, AI, and AR—is rising alongside that of competitive hardware, which includes high-performance APs, AMOLED displays, high-resolution cameras, sensors, and waterproof & dustproof features..

□ Market Condition

Shipments of smartphones in 2018 are projected to reach 1.64 billion units, up 5% from 1.56 billion in 2017. Meanwhile, shipments of tablets are expected to remain flat year-over-year at 190 million units (*Strategy Analytics*, December 2017).

< Market share of the IM Division's Main Product >

Product	2017	2016	2015
HHP	19.5%	19.2%	20.7%

※ Data research firm *Strategy Analytics* (global market share in quantity)

□ Business Condition

SEC has always been at forefront of the HHP market, and in particular, maintained leadership in the global market for smartphones since 2011. To solidify our status in the overall mobile market, we are striving to improve our current products—including tablets, wearables, and accessories—while growing businesses in areas with high potential, such as the service, online, and B2B sectors.

For smartphones, we are establishing optimal portfolios for each region, accounting for varying market conditions and offering a wide lineup ranging from mass-market to premium devices.

We have been providing premium smartphones—namely the S and Note series—and delivering differentiated user experiences via our cutting-edge technology that includes large-screen AMOLED displays, Edge displays, the S-pen, water- and dust-proofing, high-speed and wireless charging, Samsung Pay, etc. For instance, the launch of the Galaxy S8 in 2017 brought users to a new level with its immersive Infinity Display, iris scanner, and smart assistant Bixby.

For low-end products, we have been focusing on boosting profitability by streamlining the lineup, enhancing yield, and migrating proven and popular functions from premium smartphones, including a metal design, water- and dust-proofing, high-resolution selfie cameras, Samsung Pay, etc.

Not content to dwell on our achievements in hardware, we have been focusing on delivering new and useful services, which include Samsung Pay, Samsung Health, and Samsung Cloud. In 2017, we rolled out Bixby, a platform that provides users with another way to increase convenience and personalize information and services.

Our goal is to develop IoT solutions and create an open ecosystem where all devices and services, even those of different companies, can connect, enabling consumers to enjoy a consistent and high-quality Multi-Device Experience.

As always, we will continue to invest in any field that has growth potential, with current targets including but not limited to 5G, AI, IoT, cloud, and mobile B2B. Utilizing our superior R&D capabilities, we will keep enhancing consumer experiences and expanding our leadership in the global arena.

[DS Division]

- Semiconductor

□ Industry Overview

In general, our Semiconductor business is divided into Memory (storage) and System LSI (logical processing). Memory semiconductor is divided into RAM products (ones that can be both written on and read) and ROM products (read only). RAM is called volatile memory as the information is deleted when the power is turned off. It is used for temporary loading and storage of application programs.

For System LSI products, there are many categories for various applications, with CPUs (central processing units) for PCs, mobile devices, and servers forming the largest market. System LSI products are used in myriad areas, such as household electronics, networks, and games. The Company manufactures mobile APs for smartphones and tablets, image sensors, and other customized products.

Growth in the semiconductor market is expected to be negatively impacted by slowing demand for mobile devices as the smartphone market slows and the tablet market contracts. The memory market, however, is projected to enjoy continuous growth, fueled by increasing shipments of high-density information storage devices. Also, we expect demand to grow rapidly in newly emerging areas such as IoT and automotive applications. Supply and demand volatility is expected to decrease as demand diversifies.

□ Market Condition

In the DRAM market, a supply shortage has persisted mainly due to rising demand for new mobile and server products and delays in tech migration at suppliers. Meanwhile, as other manufacturers are struggling with delays in stabilizing advanced manufacturing processes and developing high-performance and high-reliability products, demand is likely to continue to be concentrated on SEC. Demand for NAND is expected to grow as more memory is included in new smartphones and solid state drives (“SSDs”) are increasingly adopted, but the chip market is projected to experience a prolonged supply shortage as suppliers are facing difficulties in migrating to finer processes of planar NAND and competitors are slow in expanding vertical NAND production.

< Market share of the Semiconductor Business' Main Product >

Product	2017	2016	2015
DRAM	45.4%	48.0%	45.3%

※ Data for 2015 and 2016 is from research firm *DRAMeXchange* (global market share in revenue). Data for 2017 is company's estimates

□ Business Condition

In 2016, the Company launched a 10nm-class DRAM, a world's first, enabling us to secure a competitive position at least one year ahead of our competitors. Furthermore, the Company is increasing supply of HBM2, the second generation of high bandwidth memory, which supports AI or supercomputing. For NAND, the Company is mass producing both planar and vertical NAND and addressing customer demand in a timely manner. In the premium market in particular, we are using leading technology to mass-produce 4th generation V-NAND and applying the product to high-performance SSDs. At the same time, in 2017, the Company expanded the portable SSD market through a global launch of its T5 portable SSD. We will also work to push the boundaries of the premium memory market by mass producing eUFS for automobile applications—for the first time in the industry—and drive the growth of memory market for automobiles.

In 2018, the memory semiconductor market, as it did last year, is expected to grow. Therefore, the Company will expand into differentiated products based on advanced processes and address each market segment with diverse product lineups to maintain its position as the market's No.1 memory maker.

The System LSI industry is diversifying from mobile products into IoT, wearables, healthcare, etc. With the expansion

into new product categories, industry growth is forecast to accelerate. To respond to this market trend, the Company will continue to seek growth by developing cutting-edge processes and launching products based on differentiated technologies ahead of others. For SOCs, we have introduced industry-leading processes ahead of competitors, and are actively addressing both the premium and mid-range to low-end markets.

The Foundry Business forms partnerships with major fabless companies based on its advanced manufacturing processes. Building on our 2016 achievement of launching the world's first 10nm products, we seek to maintain our leading status by being the first to develop the 7nm process.

- DP

□ Industry Overview

Display includes TFT-LCD (thin film transistor liquid crystal display) and OLED (organic light emitting diode) panels.

OLED panels use organic materials, which offer sharp contrast and color, high color gamut, and fast response rates. Such differentiating features provide advantages for use in smartphones, and as a result, market demand for small- to medium-size OLED displays (i.e., 10-inch and under) is growing sharply. The use of OLED is now expanding to larger formats such as TVs and transparent displays.

A TFT-LCD panel is a liquid crystal based display that features a thinner and lighter profile while offering high resolution. It has a wide range of potential applications in devices of all sizes and functionality, from portable mobile devices to large-size TVs featuring bright, high-resolution images. The market for large-panel TFT-LCD grew rapidly as their use moved from laptops initially to monitors and then to TVs. Recently however, high saturation has caused growth to slow.

The TFT-LCD and OLED businesses have high entry-barriers as they are capital intensive and require large-scale production to realize economies of scale. These types of businesses are generally sensitive to business fluctuations, so even if demand continuously rises, supply and demand dynamics tend to change regularly due to competitive investments in facilities by producers. As such, when production capacity sharply increases, the average selling price ("ASP") of panels may decrease. In contrast, when supply cannot meet increasing demand, ASP may increase.

□ Market Condition

Most major display panel producers are based in Asia as indicated below:

- Korea: Samsung Display, LG Display
- Japan: Sharp, Japan Display
- Taiwan: AU Optronics, Innolux
- China: BOE, CSOT, Tianma, CEC Panda

The Company has not only led the large-sized TFT-LCD market, but also maintained unrivalled share in the mid to small-sized OLED market since 2007 when it successfully commercialized the world's first OLED product.

We expect that small- and mid-size OLED panels will become the mainstream in the market, but competition with LTPS LCD is heating up.

For large-sized LCD panels, supply and demand may continue to be imbalanced as companies are increasing production. Nevertheless, we expect the market for premium products such as ultra large-size and high-resolution TVs to keep growing.

< Market Share of the DP Business' Main Product >

Product	2017	2016	2015
Display Panel	14.8%	17.1%	21.1%

※ Data is from research firm *IHS* (global market share of large-size panels in terms of revenue)

□ **Business Condition**

Sales in the OLED panel business grew in the fourth quarter of 2017 on increasing demand from flagship smartphones. In the first quarter of 2018, however, we expect such demand to slow due to seasonality and continued competition with LTPS LCD.

Against this backdrop, we plan to secure profitability by expanding our consumer base, swiftly responding to changes in market conditions, and improving production yield.

For large-sized LCD panels, a drop in ASPs led to a decrease in sales, and weak seasonality may have led to an additional decline in the first quarter of 2018.

To remain competitive in such an environment, the Company will work to improve yield and cost effectiveness, while also increasing the high-end-product portion of sales.

[Harman Division]

□ Industry Overview

In 2018, global automobile production is projected to remain relatively flat, edging up just 1% (*JD Power Global Production*, December 2017).

In the automotive industry, demand for connectivity and entertainment systems has been rising in the past few years, and we expect it to keep rising as automakers seek to stay at the forefront of technology, particularly with the dual trends of autonomous vehicles and shared mobility shaping the industry.

In the audio industry (e.g., connected homes, headphones, smart audio, etc.), demand is projected to grow 5% for consumer audio (*Futuresource*, October 2017) and around 2% for professional audio (*Steirner Consulting*, June 2017).

Technological innovations (particularly in the connected home and smart audio spaces) have been a driving force behind a considerable hike in demand. We anticipate that this growth will continue as technology continues to advance amid a growing global economy.

□ Market Condition

The automotive and professional/consumer audio markets are highly competitive and growing rapidly.

In the connected car industry, there are numerous companies—e.g., Alpine, Aptive, Continental, Mitsubishi, and Panasonic—that are cooperating with automotive manufacturers. Competition is expected to remain fierce between dominant players and new entrants as consumers expect car makers to deliver the latest technology, including autonomous driving.

Car audio systems is another segment where several major players (e.g., Bose, Pioneer, and Panasonic) are competing fiercely, which is expected to drive technological advances as each player introduces unique sound-management solutions.

The professional/consumer audio market is rather fragmented and a handful of companies (Amazon, Beats, Bose, Ultimate Ears, etc.) are showing early signs of dominance. We anticipate competition to be intense as new players try to make inroads, considering the state of saturation in the market for connected homes and smart speakers.

< Market Share of Harman Business' Main Product >

Product	2017	2016	2015
Head units	25.4%	25.0%	21.2%

※ Head unit refers to a separate product besides telematics and backseat organizers

※ Company's estimates based on *IHS* data.

□ Business Condition

Harman is focused on continuing its strong performance in the automotive and professional/consumer audio markets. To attain this, it will continue to seek growth through innovation, M&A activities, and sales of its globally renowned brands.

Harman is positioned to stay atop the automotive market and plans to continue to leverage its wide variety of brands to outfit vehicles ranging from mass-market thorough to luxury models, while also maintaining the quality that is synonymous with Harman brands. Additionally, Harman's relentless pursuit of innovation in the audio and connectivity spaces will help ensure the development of thriving relationships with automakers.

Additionally, Harman has continued to develop its Connected Services business with industry leading OTA and software servicing products.

The same factors that Harman relies on within the automotive market will also be used in the professional/consumer audio

market. Having won three Grammy Awards and two Academy Awards, Harman's brands have earned a strong reputation amongst everyday consumers and audiophiles alike. New offerings in up-and-coming segments such as connected wireless and smart audio continue to help bring in new consumers and further enhance brand reputation. Through these and other efforts, the Company is confident that Harman will maintain its role as a preeminent player in all areas of its business.

B. Financial Summary by Division

(Unit: KRW million, %)

Division		Classification	2017		2016		2015	
			Amount	Portion	Amount	Portion	Amount	Portion
CE		Total sales	102,788,437	18.3%	104,870,428	21.8%	115,378,019	23.4%
		Internal sales	57,679,518	17.9%	59,767,666	21.3%	70,576,753	24.2%
		Net sales	45,108,919	18.8%	45,102,762	22.3%	44,801,266	22.3%
		Operating profit	1,653,300	3.1%	2,712,148	9.3%	1,448,568	5.5%
		Total asset	47,136,186	10.4%	47,202,301	12.0%	42,540,738	11.9%
IM		Total sales	226,004,156	40.2%	211,523,973	43.9%	222,023,600	45.0%
		Internal sales	119,335,838	37.0%	111,221,861	39.7%	118,469,345	40.5%
		Net sales	106,668,318	44.5%	100,302,112	49.7%	103,554,255	51.6%
		Operating profit	11,827,324	22.0%	10,807,569	37.0%	10,142,022	38.4%
		Total asset	111,441,835	24.5%	111,574,049	30.5%	98,463,323	27.4%
DS	Semi-conductor	Total sales	141,819,969	25.2%	99,527,926	20.6%	90,600,806	18.4%
		Internal sales	67,564,374	21.0%	48,370,924	17.3%	43,014,054	14.7%
		Net sales	74,255,595	31.0%	51,157,002	25.3%	47,586,752	23.7%
		Operating profit	35,204,143	65.6%	13,595,004	46.5%	12,787,297	48.4%
		Total asset	131,659,288	29.0%	102,251,069	28.0%	98,989,253	27.6%
	DP	Total sales	72,108,721	12.8%	55,884,739	11.6%	55,120,243	11.2%
		Internal sales	37,643,358	11.7%	28,956,095	10.3%	27,633,382	9.4%
		Net sales	34,465,363	14.4%	26,928,644	13.3%	27,486,861	13.7%
		Operating profit	5,398,401	10.1%	2,226,626	7.6%	2,295,367	8.7%
		Total asset	69,854,363	15.4%	57,240,065	15.6%	50,147,263	14.0%
	Total	Total sales	218,781,719	38.9%	159,473,455	33.1%	149,974,731	30.4%
		Internal sales	110,614,239	34.3%	81,325,252	29.0%	74,948,617	25.6%
		Net sales	108,167,480	45.1%	78,148,203	38.7%	75,026,114	37.4%
		Operating profit	40,327,933	75.2%	15,850,986	54.2%	14,887,262	56.4%
		Total asset	227,250,206	50.0%	183,951,625	50.3%	174,264,841	48.5%
Harman		Total sales	9,171,835	1.6%	-	-	-	-
		Internal sales	2,069,206	0.6%	-	-	-	-
		Net sales	7,102,629	3.0%	-	-	-	-
		Operating profit	57,421	0.1%	-	-	-	-
		Total asset	14,676,715	3.2%	-	-	-	-

※ Sales for each business unit includes intersegment sales.

- ※ For CE division, the financial information was restated to exclude Printing Solution business.
- ※ For Harman, the financial information is consolidated after the date of acquisition.

Cumulative net sales for 2017 were KRW 239,575 billion. By Division, CE reported net sales of KRW 45,109 billion (18.8%) and IM reported net sales of KRW 106,668 billion (44.5%). The DS Division contributed approximately 45.1% of net sales: KRW 74,256 billion (31.0%) by the Semiconductor Sub-Division and KRW 34,465 billion (14.4%) by the DP Business Unit. The Harman division reported net sales of KRW 7,103 billion (3.0%).

Cumulative operating profit for 2017 was KRW 53,645 billion. The CE Division accounted for 3.1% with KRW 1,653 billion, the IM Division accounted for 22.0% with KRW 11,827 billion and the DS Division accounted for 75.2% with KRW 40,328 billion. The Harman division posted operating profit of KRW 57.4 billion, contributing 0.1% of the cumulative operating profit for 2017.

Reasonable Allocation of Common SG&A (Selling General & Administrative) Expenses and Assets

(1) For common SG&A expenses, specific expenses that are allocable to a specific product/model are allocated to such product/model. However, common expenses that cannot be attributed to a specific product/model category are reasonably allocated throughout the Company based on an allocation standard (expense-to-sales ratio, number of personnel, etc.).

(2) For common assets, assets that can be directly allocated (inventory assets, fixed assets, investment assets, etc.) are allocated to the corresponding organizational unit. Assets that are commonly managed and cannot be directly allocated are allocated to each Division based on an allocation standard (expense-to-sales ratio, pre-tax profit, etc.).

2. Key Products and Services

A. Revenue

In 2017, the CE Division's revenue of KRW 45,109 billion accounted for 18.8% of the total net revenue; the IM Division accounted for 44.5% with KRW 106,668 billion; and the DS Division accounted for 45.1% with KRW 108,168 billion. The Harman division accounted for 3.0% with KRW 7,103 billion.

The Company's net revenue by Division as of December 31, 2017 is as follows:

(Unit: KRW 100 million, %)

Division		Major Products	Net Revenue	Portion
CE		TVs, monitors, refrigerators, washing machines, air conditioners, medical devices, etc.	451,089	18.8%
IM		HHPs, network systems, computers, etc.	1,066,683	44.5%
DS	Semiconductor	DRAM, NAND flash, mobile APs, etc.	742,556	31.0%
	DP	TFT-LCD panels, OLED panels, etc.	344,654	14.4%
	Sub-Total		1,081,675	45.1%
Harman		Head units, infotainment systems, telematics, speakers, etc.	71,026	3.0%
Others		-	-274,719	-11.5%
Total			2,395,754	100%

※ Includes sales between Divisions (on consolidated basis).

※ For Harman, the financial information was consolidated after the date of acquisition.

☞ See 『5. Sales and Distribution』 for sales by each product.

B. Average Selling Price (ASP) Changes

In 2017, the ASP of TVs increased by 9.7% compared to the previous year, and ASP of HHPs increased 5.5% from the previous year. Both the ASPs of memory products and display panels increased, by 25% and 3.7% respectively, compared to the previous year. The ASP of Head units decreased by 7.0% compared to the previous year.

3. Key Raw Materials

(On consolidated basis as of December 31, 2017)

(Unit: KRW 100 million, %)

Division	Type	Item	Specific usage	Purchase price	Portion	Note (supplier)
CE	Raw material	Display panel	Color picture signaler	57,938	27.1%	AUO, BOE, etc.
	Raw material	Others		156,022	72.9%	
	Division Total			213,960	100.0%	
IM	Raw material	Camera Module	Mobile phone camera	39,450	11.0%	Samsung Electro-Mechanics, Partron, etc.
	Raw material	Mobile display panel	Color picture signaler	30,611	8.5%	BOE, ELK, etc.
	Raw material	Baseband Chip	CPU	30,272	8.4%	Qualcomm, Media tek, etc.
	Raw material	Others		259,699	72.1%	
	Division Total			360,032	100.0%	
DS	Raw material	TSP	Touchscreen	26,058	10.6%	Apple, Dongwoo Fine-Chem, etc.
	Raw material	FPCA	Circuit	19,987	8.1%	BH, Apple, etc.
	Raw material	Window	Tempered glass	19,340	7.9%	Lens, Apple etc.
	Raw material	POL	Polarizer plate	18,488	7.5%	Nitto, Dongwoo Fine-Chem, etc.
	Raw material	Others		161,669	65.8%	
	Division Total			245,542	100.0%	
Harman	Raw material	Memory	Chips for vehicles	2,860	19.9%	Avnet, Microchip, etc.
	Raw material	System-on-Chip	Chips for vehicles	2,757	19.2%	NVIDIA, Renesas, etc.
	Raw material	Others		8,743	60.9%	
	Division Total			14,360	100.0%	
Others				8,048	-	
Total				841,942	-	

※ Includes sales between Divisions (on consolidated basis)

※ Samsung Electro-Mechanics, the Camera module supplier, is an affiliate of Samsung Group.

※ For Harman, the financial information was consolidated after the date of acquisition.

For the CE Division, key raw materials include display panel modules for TVs and monitors. For the IM Division, key raw materials include display panels, camera modules and baseband chips for mobile devices. For the DS Division, key raw materials include TSP, window, polarizer ("POL"), and FPCA.

Large display panels for TVs and monitors are supplied by AU Optronics (AUO) and BOE (among others), mobile display panels are supplied by Iljin Display and others, camera modules from Samsung Electro-Mechanics, and baseband chips are supplied by Qualcomm and others. The Company manufactures display panels with TSPs, windows, POLs, FPCAs supplied by Apple, Lens, Nitto, BH, etc.

For the Harman division, key raw materials include memory products for in-vehicle solutions and SOC (System-on-Chip)

products. Memory products are supplied by Avnet, Microchips, etc., and SOC products are supplied by NVIDIA, Renesas, etc.

(Raw Materials Price Trends)

The price of TV and monitor display panels, which are the major raw materials for the CE Division, declined 8% on average since 2016. For the IM Division, since 2016, the price of mobile display panels decreased by 7%, and the price of baseband chips increased by 33%. For the DS Division, the price of windows for tempered glass, a major raw material, declined by around 9% and TSPs for touch screens declined by 1% since the previous year. In addition, the price of FPCA for circuits and POL for polarizing plates rose 25% and 18% respectively. For the Harman division, the prices of memories and SOC products have decreased by 7% and 4% respectively, compared to the previous year.

4. Production and Facilities

A. Production Capacity, Output, Utilization Rate

(Capacity)

(Unit: 1,000)

Division	Item	2017	2016	2015
		Quantity	Quantity	Quantity
CE	TV	44,639	52,308	50,450
IM	HHP	415,200	447,200	497,050
DS	Memory	530,590,000	415,026,000	268,630,000
	Display Panel	8,723	10,028	9,459
Harman	Head units	5,483	-	-

※ Global production capacity for major product categories

※ For Harman, the financial information was consolidated after the date of acquisition.

The CE and IM Divisions' production capacity, by major product, is calculated as follows:

The average number of lines (x) the average output per hour (x) the average operation hours per day (x) the days of operation

Memory production capacity for the DS Division is calculated as follows:

Converted output (1GB equivalent) ÷ the utilization rate.

Display panel production capacity is calculated as follows:

The total producible panel surface area ÷ the dimensions of eighth generation glass (2200x2500mm)

Harman's production capacity for Head units is calculated as follows:

The number of production (assembly and test) cells for each customer/product (x) the average production capacity per hour by production cell (x) the average operation hours per day (x) the days of operation

(Output)

(Unit: 1,000)

Division	Item	2017	2016	2015
		Quantity	Quantity	Quantity
CE	TV	39,450	47,428	45,821
IM	HHP	393,693	389,838	423,058
DS	Memory	530,590,000	415,026,000	268,630,000
	Display Panel	7,798	8,307	8,284
Harman	Head units	4,221	-	-

※ Global output for major product categories

※ For Harman, the financial information was consolidated after the date of Acquisition.

In 2017, the CE Division's output of TVs was 39,450 thousand units (major production sites: Korea, China, Mexico, Brazil, and Hungary). The IM Division's output of HHPs was 393,693 thousand units (major production sites: Korea, China, Vietnam, and Brazil). The DS Division's memory output (1GB equivalent) was 530,590 million (major production sites: Korea and China). The DS Division's output of display panels was 7,798 thousand units (major production sites: Korea, China, etc.). The Harman Division's output was 4,221 thousand units.

(Utilization Rate)

(Unit: 1,000)

Division	Item	2017 Production capacity	2017 Actual output	Utilization Rate
CE	TV	44,639	39,450	88.4%
IM	HHP	415,200	393,693	94.8%

In 2017, CE and IM utilization rates were calculated as actual output relative to production capacity. The utilization rates were 88.4% for TVs and 94.8% for HHPs.

(Unit: hours)

Division	Item	2017 Potential Production Time	2017 Actual Production Time	Utilization Rate
DS	Memory	70,080	70,080	100.0%
	Display Panel	67,872	64,434	94.9%

The DS Division operates memory and display panel production in three shifts (24 hours a day). Cumulative operating days in 2017 including holidays were 365 days. The utilization rate was calculated as actual hours [365 days (x) number of production lines (x) 24 hours] relative to production capacity.

(Unit: 1,000)

Division	Item	2017 Production capacity	2017 Actual output	Utilization Rate
Harman	Head units	5,483	4,221	77.0%

In 2017, Harman's the utilization rate was 77.0% and was calculated as actual output relative to production capacity.

B. Production Facilities and Investment

(1) Key facilities for production and operation

The Company's operational activities include manufacturing, development, marketing, and sales in Korea—operations in Suwon, Gumi, Giheung, Onyang, Gwangju, 9 regional headquarters under the CE and IM Divisions, 5 under the DS Division in North America, Europe, and China, and Harman's subsidiaries.

[Operations]

Region	Operations	Location
Korea (12)	Seochon	Seoul, Korea
	Woomyeon	Seoul, Korea
	Suwon	Suwon, Korea
	Gumi1	Gumi, Korea
	Gumi2	Gumi, Korea
	Giheung	Yongin, Korea
	Onyang	Asan, Korea
	Hwaseong	Hwaseong, Korea
	Pyeongtaek	Pyeongtaek, Korea
	Gwangju	Gwangju, Korea
	Cheonan	Cheonan, Korea
	Asan	Asan, Korea
Overseas (9 Regional Headquarters for CE and IM Divisions)	North America	New Jersey, US
	Europe	London, UK
	China	Beijing, China
	Southeast Asia	Singapore, Singapore
	Southwest Asia	New Delhi, India
	CIS	Moscow, Russia
	Middle East	Dubai, UAE
	Africa	Johannesburg, Republic of South Africa
	Latin America	Sao Paulo, Brazil
Overseas (5 Regional Headquarters for DS Division)	Americas	San Jose, US
	Europe	Eschborn, Germany
	China	Shanghai, China
	Southeast Asia	Singapore, Singapore

	Japan	Tokyo, Japan
Harman	North America HQ	Stamford, US

※ Harman's Connected Car business is mainly based in the US (Novi), and its Lifestyle Audio business is located in Germany (Garching), etc. Professional Solutions business operates mainly in the US (Northridge), and Connected Services business is based in US (Mt. View), etc.

The Company's property, plant and equipment include land, buildings and structures, machinery and equipment, and construction in progress. As of December 31, 2017, their total book value is KRW 111,666 billion, which is an increase of KRW 20,193 billion from year-end 2016. In 2017, new acquisitions of KRW 43,417 billion and depreciation of KRW 20,594 billion were recorded.

(Unit: KRW million)

Classification		Land	Buildings and Structures	Machinery and Equipment	Construction in progress	Others	Total
At 1 January 2017	Book value	7,869,679	24,375,826	38,302,055	18,773,986	2,151,495	91,473,041
	Acquisition cost	7,869,679	36,474,462	155,285,378	18,773,986	6,769,149	225,172,654
	Accumulated depreciation (Including accumulated impairment loss)	-	-12,098,636	-116,983,323	-	-4,617,654	-133,699,613
Increase (Decrease)	General acquisition and capital expenditure	1,589,162	6,424,994	38,547,588	-4,396,126	1,251,342	43,416,960
	Acquisition as a result of business combination	13,643	184,845	419,927	90,958	149,425	858,798
	Depreciation	-	-1,836,073	-17,846,905	-	-910,638	-20,593,616
	Disposal/Discard	-33,850	-37,007	-150,722	-261	-89,786	-311,626
	Impairment	-	-	-146,930	-	-461	-147,391
	Others	-29,358	-1,054,429	-1,209,215	-557,808	-179,708	-3,030,518
Balance at 31 December 2017	Book value	9,409,276	28,058,156	57,915,798	13,910,749	2,371,669	111,665,648
	Acquisition cost	9,409,276	41,639,045	184,392,999	13,910,749	7,176,535	256,528,604
	Accumulated depreciation (Including accumulated impairment loss)	-	-13,580,889	-126,477,201	-	-4,804,866	-144,862,956

※ Acquisition as a result of business combination: Includes an increased amount as a result of SEA (Samsung Electronics America, a subsidiary of SEC)'s acquisition of 100% shares of Harman and its subsidiaries.

※ Others: Includes effects of changes in FX rates and effects of reduction in government subsidy.

※ Market value of major tangible assets is omitted as objective assessment is difficult.

※ Property, plant, and equipment above is presented on a consolidated basis.

(2) CAPEX

In 2017, the Company invested KRW 43.4 trillion in CAPEX, including upgrading production lines of the Semiconductor and the DP. The total CAPEX for 2018 is yet to be determined due to various factors that may affect business conditions.

[CAPEX by Division]

(Unit: KRW 100 million)

Business	Purpose	Period	Assets	Investment in 2017
Semiconductor	Establishment, addition, upgrade	Jan '17 ~ Dec '17	Buildings, facilities	273,456

DP	Establishment, addition, upgrade	Jan '17 ~ Dec '17	Buildings, facilities	135,456
Other	Other	Jan '17 ~ Dec '17	Buildings, facilities	25,258
Total				434,170

5. Sales and Distribution

A. Sales

In 2017, the Company recorded KRW 239,575 billion of total revenue, an increase of 18.7% compared to the previous year. By Division, when compared to 2016, CE Division's revenue increased by 0.0%, IM Division's revenue increased by 6.3% and DS Division's revenue increased by 38.4%.

(Unit: KRW 100 million)

Division		Sales types	Product categories	2017	2016	2015
CE		Goods Products Service Others	TV, monitor, air conditioner, refrigerator, washing machine, medical device, etc.	451,089	451,028	448,013
IM		Goods Products Service Others	HHP, Network System, Computer, etc.	1,066,683	1,003,021	1,035,543
DS	Semiconductor	Goods Products Service Others	DRAM, NAND Flash, Mobile AP, etc.	742,556	511,570	475,868
	DP	Goods Products Service Others	TFT-LCD, OLED, etc.	344,654	269,286	274,869
	Division total			1,081,675	781,482	750,261
Harman		Goods Products Service Others	Infotainment System, telematics, speakers, headphones, etc.	71,026	-	-
Others		Other revenue	-	- 274,719	-216,864	-227,282
Total				2,395,754	2,018,667	2,006,535

※ Includes internal sales between Divisions on a consolidated basis

※ CE division was restated to exclude Printing Solution business.

□ Sales by Major Product (on a consolidated basis)

(Unit: KRW 100 million)

Classification	2017	2016	2015
Image Devices	275,154	287,241	292,194
Mobile Devices	1,036,218	977,494	1,005,117
Memory	603,036	378,594	342,917
Display Panel	344,654	269,286	274,869

※ Including internal sales between Divisions on a consolidated basis.

☐ Sales by Type

(Unit: KRW 100 million)

Classification	2017	2016	2015
Products	2,368,198	2,006,326	1,988,452
Service and other Sales	27,556	12,341	18,083
Total	2,395,754	2,018,667	2,006,535

※ Other sales consists of royalty income and etc.

☐ Sales by Region (on a separate basis)

(Unit: KRW 100 million)

Classification	2017	2016	2015
Domestic	165,684	140,656	145,908
Americas	488,864	426,448	425,042
Europe	189,464	170,850	173,583
Asia and Africa	317,661	281,021	291,473
China	457,477	320,497	316,044
Total	1,619,150	1,339,472	1,352,050

B. Sales Channels

☐ Domestic

Seller	Sales Channel		Consumer
Manufacturer (Facility)	Retailer		Consumer
	Distributor (General Merchandise Store, Discount Store, Department Store, Home-Shopping, Internet)		
	Retailer	Sales Shop, Open Stores	
	Telecommunication Service Provider (SKT, KT, LG U+)		
	Direct Sales		

☐ Overseas

Seller	Sales Channel					Consumer
Production Subsidiaries	Regional Sales Office	Retailer				Consumer
		Dealer		Retailer		
		Distributor	Dealer	Retailer		
		Telecommunication Service Provider, Automotive OEM				
	Regional Distribution Office	Regional Sales Office	Retailer			
			Dealer	Retailer		
			Distributor	Dealer	Retailer	
	Direct Sales					

☐ Sales Ratio by Channel

Channel	Wholesale	Retail	Special/direct sale	Others
Ratio	23%	24%	46%	7%

※ On a global basis

C. Sales Methods and Conditions

☐ Domestic

Classification	Channel	Collection	Incidental expense sharing
Exclusive	Retailer	- Credit agreement (Cash, 30 day credit) (Credit applied within 100% of collateral)	- Case specific cost sharing as determined by mutual agreement
Distributor	General Merchandise Store, Discount Store, Department Store, Home Shopping, Internet	Case specific and contract specific	- Case specific cost sharing as determined by mutual agreement
Special/direct sale	Corporate Clients, etc.	Case specific and contract specific	N/A

☐ Overseas

Classification	Channel	Collection	Incidental expense sharing
Retailer	Retail store	Case specific and contract specific	- Case specific cost sharing as determined by mutual agreement
Dealer	General merchandise store, Discount store, Department store	Case specific and contract specific	- Case specific cost sharing as determined by mutual agreement
Distributor	Direct sales to local distributors	Case specific and contract specific	- Case specific cost sharing as determined by mutual agreement
B2B	Corporate Clients, etc.	Case specific and contract specific	N/A

D. Sales Strategy

- Expand market leadership based on premium products such as smart devices
- Provide differentiated value to customers through brand, products, and services
- Strengthen marketing activities to boost demand from customer/market

E. Major Customers

In 2017, major customers included Apple, BestBuy, Deutsche Telekom, Sprint and Verizon (in alphabetical order). Sales to our five major customers accounted for approximately 14% of total sales.

6. Long-term Contracts

As of December 31, 2017, there are no long-term contracts that have a significant impact on the Company's financial statement.

7. Financial Risk Management

The Company's financial risk management focuses on minimizing market risk, credit risk, and liquidity risk arising from operating activities. To mitigate these risks, the Company implements and operates a financial risk policy and program that closely monitors and manages such risks. In addition, the Company uses derivative financial instruments to hedge certain risk exposure.

The finance team mainly carries out the Company's financial risk management. With the cooperation of the Company's divisions, domestic and foreign subsidiaries, the finance team periodically measures, evaluates and hedges financial risk and also establishes and implements the global financial risk management policy.

Also, the Company operates local finance centers in major overseas regions (United States, United Kingdom, Singapore, China, Brazil and Russia) to manage global financial risks.

The Company's financial assets that are under financial risk management are comprised of cash and cash equivalents, short-term financial instruments, available-for-sale financial assets, trade and other receivables, and other financial assets. The Company's financial liabilities under financial risk management are comprised of trade and other payables, borrowings, debentures, and other financial liabilities.

A. Market Risk

(1) Foreign Exchange Risk

The Company experiences currency gains and losses based on the different functional currency of each entity due to global operations. As such, the Company is exposed to foreign currency volatility from exchange positions of currencies, especially the US Dollar, Euro, Japanese Yen and Indian Rupee.

To minimize foreign exchange risk arising from operating activities, the Company's foreign exchange management policy requires normal business transactions (including imports and exports) to be conducted in the local currency or for the cash-in currency to be matched with the cash-out currency. In addition, the Company makes use of derivative instruments to minimize uncertainty of cash flows and fluctuation of profit or loss occurring from foreign exchange risk. The Company's foreign exchange risk management policy also defines foreign exchange risk, measuring period, ownership responsibilities, management procedures, hedging period and hedge ratio.

The Company limits all speculative foreign exchange transactions and operates a system to manage receivables and payables denominated in foreign currency. It evaluates, manages and reports foreign currency exposures to receivables and payables.

As of December 31, 2017, the foreign currency exposure to financial assets and liabilities of a 5% currency rate change against the Korean won are presented below:

(In millions of Korean won)	December 31, 2017		December 31, 2016	
	Increase	Decrease	Increase	Decrease
USD	₩ 278,140	₩ (278,140)	₩ 222,149	₩ (222,149)
EUR	79,760	(79,760)	138,084	(138,084)
JPY	(100,127)	100,127	(61,294)	61,294
INR	77,963	(77,963)	38,858	(38,858)

(2) Price Risk

The Company's investment portfolio consists of direct and indirect investments in equity securities classified as available-for-sale, which is in line with the Company's strategy.

As of December 31, 2017 and December 31, 2016, a price fluctuation in relation to marketable equity securities by 1% would result in changes in other comprehensive income (before income tax) of KRW 29,086 million and KRW 23,622 million, respectively.

(3) Interest Rate Risk

Risk of changes in interest rates for floating interest rate financial instruments is defined as the risk that the fair value of components of the statement of financial position, and future cash flows of interest income (expenses) of a financial instrument, will fluctuate because of changes in market interest rates. The Group is exposed to interest rate risk mainly through interest bearing liabilities and assets. The Group's position with regard to interest rate risk exposure is mainly driven by its floating interest rate debt obligations and interest-bearing deposits. The Group implemented policies and operates to minimize uncertainty arising from changes in interest rates and finance costs.

In order to avoid interest rate risk, the Company maintains minimum external borrowings by facilitating cash pooling systems on a regional and global basis. The Company manages exposed interest rate risk via periodic monitoring and handles risk factors on a timely basis.

The sensitivity risk of the Company is determined based on the following assumption:

- Changes in market interest rates that could impact the interest income and expenses of floating interest rate financial instruments

Based on the assumption above, changes to profit and net equity as a result of a 1% change in interest rates on borrowings are presented below:

(In millions of Korean won)	December 31, 2017		December 31, 2016	
	Increase	Decrease	Increase	Decrease
Financial assets	₩ 49,117	₩ (49,117)	₩ 64,803	₩ (64,803)
Financial liabilities	(22,643)	22,643	(9,123)	9,123
Net effect	₩ 26,474	₩ (26,474)	₩ 55,680	₩ (55,680)

B. Credit Risk

Credit risk arises during the normal course of transactions and investing activities where clients or other parties fail to discharge an obligation. The Group monitors and sets the client's and counterparty's credit limit on a periodic basis based on the client's and counterparty's financial conditions, default history and other important factors. Adequate insurance coverage is maintained for accounts receivables related to trading partners situated in higher risk countries.

Credit risk can arise from transactions with financial institutions which include financial instrument transactions such as cash and cash equivalents, savings, and derivative instruments. To minimize such risk, the Company transacts only with banks which have strong international credit rating (S&P A and above), and all new transactions with financial institutions with no prior transaction history are approved, managed and monitored by the Company's finance team and the local finance center. The Company requires separate approval for contracts with restrictions.

As of December 31, 2017 and 2016 the Company estimates that its maximum exposure to credit risk is the carrying amount of its financial assets, net of impairment losses.

C. Liquidity Risk

Due to large investments made by the Company, maintaining adequate levels of liquidity is critical. The Company strives to achieve this goal by periodically forecasting its cash flow, estimating required cash levels, and managing income and expenses.

The Company manages its liquidity risk by periodically forecasting projected cash flows. If abnormal signs are identified, SEC works with the local finance center and provides liquidity support by utilizing its globally integrated finance structures such as cash pooling. In addition, the Company maintains a liquidity management process which provides additional financial support through the local finance center and the Company when necessary. The cash pooling program allows sharing of surplus funds among entities and contributes to minimizing liquidity risk and strengthening the Company's competitive position by reducing capital operating expenses and finance expenses.

In addition, the Company mitigates liquidity risk by contracting with financial institutions with respect to bank overdrafts and foreign trade finance and by providing payment guarantees to subsidiaries. For large scale facility investments, liquidity risk is minimized by utilizing internal reserves and long term borrowings according to the capital injection schedule.

As of December 31, 2017 and 2016, the following table is an undiscounted cash flow analysis for financial liabilities that are presented on the statements of financial position according to their remaining contractual maturity.

<i>(In millions of Korean won)</i>	December 31, 2017				
	Less than 3 months	4-6 months	7-12 months	1-5 years	More than 5 years
Financial liabilities	₩ 45,323,281	₩ 927,746	₩ 1,144,922	₩ 4,633,210	₩ 875,283

<i>(In millions of Korean won)</i>	December 31, 2016				
	Less than 3 months	4-6 months	7-12 months	1-5 years	More than 5 years
Financial liabilities	₩ 40,918,912	₩ 1,588,798	₩ 150,744	₩ 4,346,200	₩ 50,073

The table above shows the Company's financial liabilities based on the remaining period at the statement of financial position date until the contractual maturity date. The amounts disclosed in the table are the contractual undiscounted cash flows.

The Company's trading portfolio of derivative instruments has been included at its fair value of: KRW 47,478 million (December 31, 2016: KRW 74,697 million). These contracts are managed on a net-fair value basis rather than by maturity date. Net settled derivatives consist of forwards on currency rates used by the Company to manage the exchange rate profile.

Derivatives that are settled on a gross basis by delivery of the underlying item, including derivatives for hedging, will be settled within the next 53 months from the end of the reporting period. These derivatives are not included in the table above.

The maximum liquidity risk exposure from those other than the above financial liabilities (e.g., payment guarantees for affiliated companies and performance bonds) as at December 31, 2017 is: KRW 49,937 million (December 31, 2016: KRW 59,016 million).

D. Derivative Financial Instruments

The Company applies cash flow hedge accounting to hedge the foreign currency risk of forecasted transaction including hedging the price risk associated with forecast inventory purchases. Details of derivative financial instruments that qualify as cash flow hedges as of December 31, 2017 are as follows:

(in millions of Korean won)	December 31, 2017	
	Assets	Liabilities
Currency forward exchange contracts		
Current items	₩ 45,396	₩ 41,646
Non-current items	31,915	25,806
Total	₩ 13,481	₩ 15,840

As of December 31, 2016, the Company did not have derivative financial instruments designated as cash flow hedges.

During the year ended for December 31, 2017, The Company recognizes the gains or losses relating to the effective portion of changes in fair value of derivatives that are designated and qualify as cash flow hedges in other comprehensive income, which amount to KRW 37,121 million (after tax), and recognizes the gains or losses relating to the ineffective portion in profit or loss, which amount to KRW 4,940 million (before tax). Gains or losses reclassified directly from other comprehensive income to profit or loss amount to KRW 3,413 million (after tax), and gains or losses reclassified from other comprehensive income to the carrying amount of inventory amount to KRW 29,663 million (after tax).

E. Capital Risk Management

The purpose of capital management is to maintain a sound capital structure. The Company monitors capital on the basis of the ratio of total liabilities to total equity. This ratio is calculated by dividing total liabilities by total equity in the consolidated financial statements.

The Company's capital risk management policy has not changed since the previous fiscal year. The Company has maintained "AA-" and "A1" credit ratings from S&P and Moody's, respectively.

The total liabilities to equity ratio as of December 31, 2017 and December 31, 2016 are as follows:

(Unit: KRW million)

	December 31, 2017	December 31, 2016
Total liabilities	87,260,662	69,211,291
Total equity	214,491,428	192,963,033
Total liabilities to equity ratio	40.7%	35.9%

E. Fair Value Measurement

- Carrying amounts and fair values of financial instruments by category as of December 31, 2017 and December 31, 2016 are as follows:

(Unit: KRW million)

Classification	December 31, 2017		December 31, 2016	
	Carrying amount	Fair value	Carrying amount	Fair value
Financial assets				
Cash and cash equivalents	30,545,130	(1)	32,111,442	(1)
Short-term financial instruments	49,447,696	(1)	52,432,411	(1)
Short-term available-for-sale financial assets	3,191,375	3,191,375	3,638,460	3,638,460
Trade and other receivables	27,695,995	(1)	24,279,211	(1)
Long-term available-for-sale financial assets ⁽²⁾	7,752,180	6,561,155	6,804,276	5,826,507
Held-to-maturity financial assets	106,751	(1)	-	-
Other ⁽⁴⁾	6,325,825	113,098	3,459,863	919,071
Total Financial assets	125,064,952		122,725,663	
Financial liabilities				
Trade payables	9,083,907	(1)	6,485,039	(1)
Short-term borrowings	15,767,619	(1)	12,746,789	(1)
Other payables ⁽³⁾	12,106,609	316,928	10,225,271	(1)
Current portion of long-term borrowings	278,619	(1)	1,232,817	(1)
Debentures	953,361	978,643	58,542	76,129
Long-term borrowings	1,814,446	(1)	1,244,238	1,225,455
Long-term other payables	1,746,184	28,285	3,009,659	3,022,821
Other ⁽⁴⁾	10,954,513	222,012	11,942,469	74,697
Total Financial liabilities	52,705,258		46,944,824	

¹ Assets and liabilities whose carrying amount is reasonable approximations of their fair value are excluded from the fair value disclosures.

² Amount measured at cost of KRW 1,191,025 million (December 31, 2016: KRW 977,769 million) are excluded, as the range of reasonable fair value estimates are significant and the probabilities of the various estimates cannot be reasonably assessed.

³ Asset measured at cost of KRW 6,212,727 million (December 31, 2016: KRW 2,452,118 million) and liabilities measured at cost of KRW 24,240,081 million (December 31, 2016: KRW 11,867,772 million) are excluded as the carrying amount is a reasonable approximation of fair value.

- The following table presents the fair value of assets and liabilities, by Level (as defined below), measured as of the dates indicated:

(In millions of Korean won)	As of December 31, 2017			
	Level 1	Level 2	Level 3	Total balance
Assets				
Short-term available-for-sale financial assets	-	3,191,375	-	3,191,375
Long-term available-for-sale financial assets	2,908,581	-	3,652,574	6,561,155
Other	-	113,098	-	113,098
Liabilities				
Other payables	-	-	316,928	316,928
Debentures	-	978,643	-	978,643
Long-term other payables	-	-	28,285	28,285
Other	-	215,307	6,705	222,012

(In millions of Korean won)	As of December 31, 2016			
	Level 1	Level 2	Level 3	Total balance
Assets				
Short-term available-for-sale financial assets	-	3,638,460	-	3,638,460
Long-term available-for-sale financial assets	2,362,235	-	3,464,272	5,826,507
Other	-	919,071	-	919,071
Liabilities				
Debentures	-	76,129	-	76,129
Long-term borrowings	-	1,225,455	-	1,225,455
Long-term other payables	-	2,680,119	342,702	3,022,821
Other	-	74,697	-	74,697

The levels of the fair value hierarchy (based on characteristics of the input variables) and its application to financial assets and liabilities are described below.

- Level 1: Quoted market prices (unadjusted) for identical assets or liabilities
- Level 2: Fair valuation based on inputs that are observable in the market
(Other than quoted prices included within Level 1)
- Level 3: Fair valuation based on inputs that are unobservable in the market

The fair value of financial instruments traded in active markets is based on quoted market prices at the statement of financial position date. A market is regarded as active if quoted prices are readily and regularly available from an exchange, dealer, broker, industry group, pricing service, or regulatory agency, and those prices represent actual and regularly occurring market transactions on an arm's length basis. These instruments are included in Level 1. Instruments in Level 1 include listed equity investments classified as trading securities or available-for-sale financial assets.

The fair value of financial instruments that are not traded on an active market is determined using valuation models. These valuation models maximize the use of observable market data where it is available and rely as little as possible on entity specific information. If all significant inputs required to measure the fair value of an instrument are observable, the

instrument is included in Level 2.

If one or more of the significant inputs are not based on observable market data, the instrument is included in Level 3.

The Company performs fair value measurements required for financial reporting purposes, including Level 3 fair values, and discusses valuation processes and results at least once every quarter in line with SEC's quarterly reporting dates. The Company recognizes changes in Levels at the end of the reporting period, if corresponding events or changes in circumstances necessitating such reclassification have occurred.

Specific valuation methods used to value financial instruments include:

- Quoted market prices or dealer quotes for similar instruments.
- The fair value of a derivative is determined using, among other things, forward exchange rates at the statement of financial position date, with the resulting value discounted back to present value.

Other methods, such as discounted cash flow analysis, are used to determine fair value for the remaining financial instruments. For trade and other receivables, the book value approximates a reasonable estimate of fair value.

- Valuation Methods and Inputs

- The Company utilizes a present value method to discount future cash flows at a proper interest rate for corporate bonds, government and public bonds, and bank debentures that are classified as Level 2 in the fair value hierarchy.
- The following table presents the valuation methods and inputs used for major financial instruments classified as Level 3.

(In millions of Korean won)				
Classification	Fair Value	Valuation Technique	Level 3 Inputs	Input Range (Weighted Average)
Long-term available-for-sale financial assets				
Maltani	15,137	Discounted cash flow	Permanent growth rate	-1.00% ~ 1.00% (0%)
			Weighted average cost of capital	7.69% ~ 9.69% (8.69%)
Samsung Venture Investment	7,784	Discounted cash flow	Permanent growth rate	-1.00% ~ 1.00% (0%)
			Weighted average cost of capital	19.56% ~ 21.56% (20.56%)
Coming Incorporated convertible preferred shares	3,629,653	Trinomial model	Risk adjusted discount rate	4.98% ~ 6.98% (5.98%)
			Price volatility	26.50% ~ 32.50% (29.50%)
Other payables				
Contingent liabilities	316,928	Discounted cash flow	Discount rate	3.81% ~ 4.65% (4.23%)
Long-term others payable				
Contingent financial liabilities	28,285	Monte-Carlo Simulation	Discount rate	10.50%
			Risk free rate	0.97%
			Asset volatility	34.54%
			Credit spread	2.12%
Others				
Contingent liabilities	4,982	Monte-Carlo Simulation	Discount rate	17.57%
			Risk free rate	0.86%
			Operation leverage rate	60.00%
			Gross margin discount rate	6.68%
Contingent liabilities	1,723	Probability-weighted discounted cash flow	Weighted average cost of capital	8.60%
			Credit risk	2.12%

- Changes in Level 3 Instruments:

(In millions of Korean won)

Financial Assets	2017	2016
Balance as of January 1	3,464,272	3,488,344
Disposals	-	(14,805)
Amount recognized in profit or loss	-	(795)
Amount recognized in other comprehensive income	188,302	695,631
Other	-	(704,103)
Balance as of December 31	3,652,574	3,464,272

(In millions of Korean won)

Financial Liabilities	2017	2016
Balance as of January 1	342,702	312,738
Settled	(2,693)	-
Amount recognized in profit or loss	(24,372)	29,964
Acquisition as a result of business combination	39,083	-
Others	(2,802)	-
Balance as of December 31	351,918	342,702

- Sensitivity Analysis for Recurring Fair Value Measurements Categorized within Level 3

Sensitivity analysis of financial instruments is performed to measure favorable and unfavorable changes in the fair value of financial instruments which are affected by inputs that are unobservable in the market, using a statistical technique. When the fair value is affected by two or more input variables, such fair value is calculated using the most favorable or most unfavorable input values.

The results of the sensitivity analysis for the effect on profit or loss from changes in inputs for each type of financial instrument, which is categorized within Level 3 and subject to sensitivity analysis, are as follows:

(In millions of Korean won) Classification	Favorable Changes		Unfavorable Changes	
	Profit or Loss	Equity	Profit or Loss	Equity
Long-term available for sale financial assets ¹	-	189,757	-	(212,458)
Long-term other payables ²	445	445	(446)	(446)
Total	445	190,202	(446)	(212,904)

¹ For equity securities, changes in fair value are calculated with the correlation among growth rate (-1%~1%), volatility (26.5%~32.5%) and discount rate, which are significant unobservable inputs.

² The fair value of other payable is calculated by increasing or decreasing the discount rate by 10%, which is the significant unobservable input.

8. Derivative Instruments and Put Options

The value of derivative instruments of the Company and its subsidiaries as of December 31, 2017 is presented below:

(Unit: KRW million)

	Asset	Liabilities	Valuation Gain	Valuation Loss
Put options	-	-	-	114,036
Total	-	-	-	114,036

※ The option values were calculated using appropriate valuation methodologies, including the Black-Scholes model.

The Company and its subsidiaries hold convertible preferred shares of Corning Incorporated and its value as of December 31, 2017 is as follows.

(Unit: KRW million)

	Purchasing Price	Fair Value	Valuation Gain	Valuation Loss
Convertible Preferred Shares	2,434,320	3,629,653	1,195,333	-

※ Fair value is determined using Trinomial Tree model; valuation gain is reflected in equity (under other components of equity).

To manage foreign exchange risk, SEC's subsidiaries hedge their foreign currency positions by trading currency forward contracts of such currencies. Overseas subsidiaries buy or sell currency forwards through a bank in the purpose of trade.

As of December 31, 2017, the Company has 2,465 currency forward contracts involving 37 foreign currencies including USD/EUR/JPY. Currency forwards as of December 31, 2017 are as follows.

(Unit: KRW million)

	Asset	Liabilities	Valuation Gain	Valuation Loss
Currency forwards	100,346	310,608	80,804	184,608
Total	100,346	310,608	80,804	184,608

The Company applies cash flow hedge accounting to hedge the foreign currency risk of forecasted transaction including hedging the price risk associated with forecast inventory purchases. Details of derivative financial instruments that qualify as cash flow hedges as at December 31, 2017 are as follows:

December 31, 2017

(in millions of Korean won)

	Assets	Liabilities
Currency forward exchange contracts		
Current items	₩ 45,396	₩ 41,646
Non-current items	31,915	25,806
Total	₩ 13,481	₩ 15,840

As of December 31, 2017, the Company did not have derivative financial instruments designated as cash flow hedges. During the year ended for December 31, 2017, The Company recognizes the gains or losses relating to the effective portion of changes in fair value of derivatives that are designated and qualify as cash flow hedges in other comprehensive

income, which amount to KRW 37,121 million (after tax), and recognizes the gains or losses relating to the ineffective portion in profit or loss, which amount to KRW 4,940 million (before tax). Gains or losses reclassified directly from other comprehensive income to profit or loss amount to KRW 3,413 million (after tax), and gains or losses reclassified from other comprehensive income to the carrying amount of inventory amount to KRW 29,663 million (after tax).

9. Major Contracts

Account	Item	Contents
Ericsson	Contract type	Patent cross-license agreement
	Contract date	2014.01.25
	Purpose and contents	Secure operational advantage through mutual patent licensing
Google	Contract type	Patent cross-license agreement
	Contract date and period	2014.01.25 / indefinite term
	Purpose and contents	Secure operational advantage through mutual patent licensing
	Others	Permanent license contract (including patent applications in the next 10 years)
Cisco	Contract type	Patent cross-license agreement
	Contract date	2014.01.23
	Purpose and contents	Secure operational advantage through mutual patent licensing
Global Foundries Inc.	Contract type	Process technology license contract
	Contract date	2014.02.28
	Purpose and contents	Expand customer base of 14nm process
InterDigital	Contract type	Patent license agreement
	Contract date	2014.06.03
	Purpose and contents	Secure operational advantage through patent licensing
Sharp	Contract type	Patent cross-license agreement
	Contract date	2015.01.01
	Purpose and contents	Secure operational advantage through mutual patent licensing
Microsoft	Contract type	Settlement Agreement
	Contract date	-
	Purpose and contents	End dispute over royalty payment
Nokia	Contract type	Patent license agreement
	Contract date	2016.07.12
	Purpose and contents	Secure operational advantage through patent licensing

HP	Contract type	Sales of business unit
	Contract date	2016.09.12
	Purpose and contents	Concentrate on our core competencies and advance business structure
	Others	Transaction value: USD 1.05 billion

※ Information that may be referenced or used in other IP disputes, including contract amount, is not included.

10. Research and Development Activities

A. Summary of Activities and R&D Expenditures

The Company is leading the global market by continuously developing creative and innovative products and the future technology through shifting its way of thinking and understanding customer demand.

The Company is currently developing creative and innovative products and doing its utmost to cement its position in the global IT industry and become a worldwide leader in industrial technology by creating and securing next-generation technology.

[R&D expenses]

(Unit: KRW million)

		2017	2016	2015
Total R&D expenditure		16,805,637	14,794,318	14,850,534
(Government Subsidies)		-2,484	-1,975	-1,780
R&D expenditure		16,803,153	14,792,343	14,848,754
Accounting	Capitalization of development expenses (intangible asset)	-447,541	-680,962	-1,143,059
	R&D costs (expenses)	16,355,612	14,111,381	13,705,695
R&D expenses/sales ratio		7.0%	7.3%	7.4%

※ On a consolidated basis (in conformity with K-IFRS)

※ The percentage was calculated with the amount of the total R&D expenditure, which includes government subsidies

As of December 31, 2017, the Company's R&D expenses were KRW 16,806 billion. Among this, KRW 16,356 billion was recognized as expenditure, excluding KRW 2.5 billion of government subsidies and KRW 447.5 billion of capitalized expenses.

B. R&D Organization and Operations

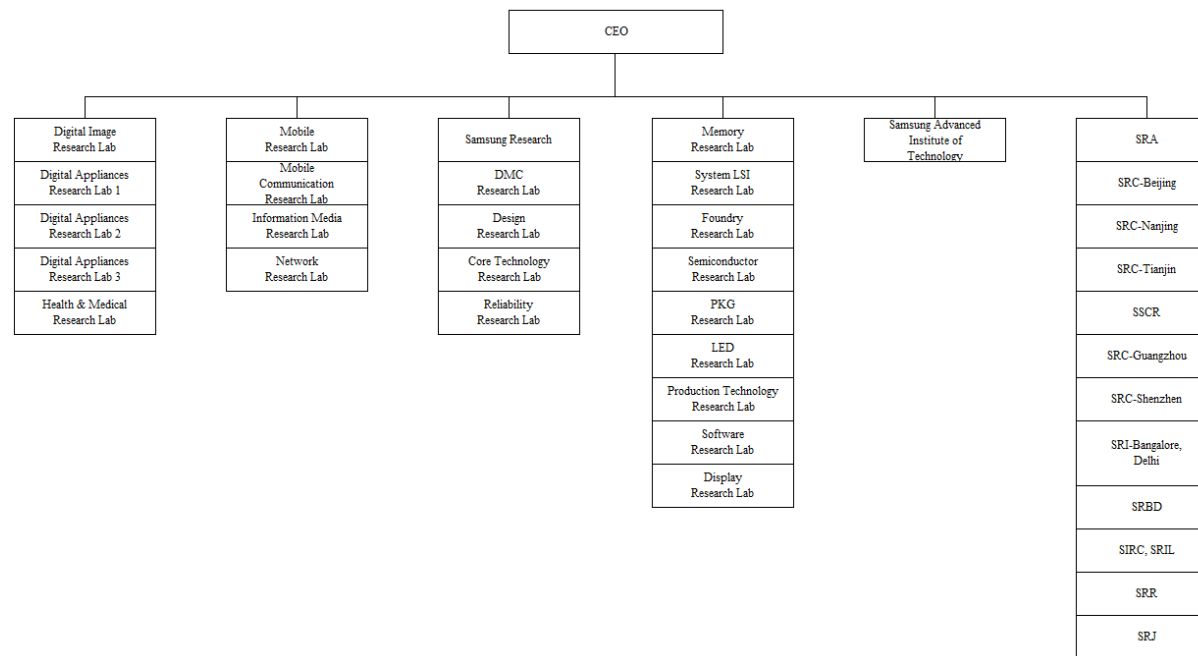
- Korea

The Company operates three levels of R&D organizations; a business unit development team under each Division that develops market ready technologies with a 1-2 year outlook, a research institute under each business unit that develops mid-to-long term technology with a 3-5 year outlook, and the Samsung Advanced Institute of Technology that develops core technology as seeds for future growth engines.

Samsung Advanced Institute of Technology is the central research institute of Samsung Electronics that has been established to lead the Company into the future and serve as an incubator of cutting-edge technology. It has a creative R&D system and provides R&D direction for promising growth sectors at the Company level, exploring future growth engines and strengthening technological competitiveness of core businesses.

- Overseas

The Company operates R&D organizations in the US (SRA), Russia (SRR), Israel (SRIL and SIRC), India (SRI-Bangalore and SRI-Delhi), Japan (SRJ), China (SSCR, SRC-Beijing, SRC-Nanjing, SRC-Tianjin, SRC-Guangzhou, and SRC-Shenzhen) to carry out research activities for product development and basic technological research.



※ As of December 31, 2017

※ Refer to 『G. Subsidiaries subject to consolidation』 in 『I. Corporate Overview』 for more details about overseas R&D organizations.

C. R&D Results

Research Project	Research Results and Expected Effect	In case applied to a product, name of the product and application
Develop new smart card product	- Launched NFC with internal flash memory (e-flash) - First to Use 45nm e-flash logic process, which improves density and power efficiency compared to 90nm products - Miniaturized products by supporting smallest antenna solution - Enabled mobile POS through NFC chip as the first in the industry ※ POS (Point of Sales): point of sale information management system	- Region : Global - Launch : Jan '14 - Name : S3FWRN5
High-resolution low power consumption Tablet product technology	- Mass produce high-resolution Tablet panel - Achieved low power consumption of tablet products and higher rate of production by enhancing transmissivity	- Region : Global - Launch : Feb '14
Mass produce 20nm 4GB DDR3 DRAM for server	- First in the world to mass produce ultra-fine 20nm 4GB DDR3 DRAM - Applied independently developed high-density/high-speed/low-power consumption technology - Enhanced productivity by 30% compare to 25nm - PC and server full line-up to be launched with 20nm 4GB DRAM - Enhance competitiveness by launching the whole line-up including mobile in the future	- Region : Global - Launch : Feb '14 - Name : 4GB DDR3 DRAM
Develop new ISOCELL image sensor	- Launched new 1600 megapixel ISOCELL image sensor - Clear image even in dark places 16 megapixel / 30fps per second (first in the industry) 16:9 aspect ratio FullHD resolution	- Region : Global - Launch : Mar '14 - Name : S5K2P2
Develop new Mobile AP product with HMP solution	- Launched new Exynos 5 Octa (5422) product - Enhance power efficiency by combining and utilizing 8 high-capacity/low-power consumption cores through applying HMP solution - Support WQHD and WQXGA ultra-high resolution - Strengthen low-power consumption capacity by using automatic conversion to power save mode and mobile video compression technology - Launched new Exynos 5 Hexa (5250) product - Enhance power efficiency by combining and utilizing 6 high-capacity/low power consumption cores through applying HMP solution - Support WQXGA ultra-high resolution	- Region : Global - Launch : Jan/Mar '14 - Name : Exynos5250/Exynos5422
UHD TV HU9000	- Ultimate Curved UHD TV that provides perfect immersive experience - Adopted ergonomic Curved LED Panel - Clear differentiation values compare to competitors - Smart Hub function with Multi-Link Screen added - The only Evolutionary UHD TV - Premium design - Ergonomic Curved Design - eEnhance premium image through Immersive Viewing Experience	- Region : Global - Launch : Mar '14 - Inch : 55"/65"
Mass produce 2 nd generation V-NAND based 1TB SSD for PC	- First in the world to mass produce 2nd generation 3D V-NAND based 1TB SSD - Opened up 'the era of popularized 3D memory' with 2nd generation V-NAND - Target PC market after creating server market last year - Launched high-reliability, high-capacity, low-power consumption V-NAND SSD line-up - Increased life by twofold and reduced power consumption by 20% compare to existing SSD - Launch 'V-NAND SSD' in 53 nations around the world starting in July	- Region : Global - Launch : May '14 - Name : 1TB, V-NAND SSD
Develop new ISOCELL Image sensor	- Develop 13 megapixel stacked ISOCELL image sensor - Smaller chip size and wider circuit range by applying stacked structure - Diversification of exposure with smart WDR function improves color in backlight	- Region : Global - Launch : Jun '14 - Name : S5K3M2
Develop new Mobile AP product	- Launched new Exynos 5 Octa (5430) product - Increased power efficiency by applying 20nm low-power consumption HKMG process - Enhanced performance by applying HMP solution and can work individually depending on low power core needs - Enhanced low power consumption property by using automatic conversion to power save mode and mobile video compression technology - Support WQHD and WQXGA ultra high-resolution	- Region : Global - Launch : Jun '14 - Name : Exynos 5430

Research Project	Research Results and Expected Effect	In case applied to a product, name of the product and application
UHD TV (UN105S9WAF)	□ Product Concept - Wide Curved(21:9) 105" UHD TV - Need to solidify market leadership by introducing ultra-large size 105" UHD TV as the first in the world □ Spec and effects - Adopted SDC 4200R 105" Wide Curved Panel - Applied Golf-AP/MP Platform - Provide the ultimate immersive and realistic viewing experience	□ Region : Global □ Launch : Jul '14 □ Inch: 105"
LFD QMD	□ Introduced Premium UHD Line Up - Introduced high resolution UHD LFD lineup - Applied SE13U Platform - Applied spec to respond to LFD Usage - Reduced additional investment costs by sharing TV HU7K design □ Main functions - Support full HD screen split in 4 (2/3/4 splits) - DP 1.2 and SBB applicable	□ Region : Global □ Launch : Aug '14 □ Inch: 85"
Mass produce PC/server 32 layer 3bit 3D V-NAND	□ Mass produced the world's first 3bit 3D V-NAND based 1TB SSD - Opened up the era of "popularized V-NAND SSD" with 3bit V-NAND - Target the standard SSD market with V-NAND that has 50% higher productivity □ Reinforced highly reliable, high-performance, low energy consuming V-NAND SSD lineup - Launched mSATA and M.2 SSD lineups in addition to the existing 2.5" - Launched '850 EVO SSD' in 53 nations around the world in Dec	□ Region : Global □ Launch : Oct '14 □ Inch: 128Gb 3bit 3D V-NAND (850 EVO SSD)
Bendable TV Nov (UN78S9BAF)	□ World's first Bendable UHD TV - Secured technology leadership □ Design - Adopted Timeless Gallery Design - Becomes an aesthetic object when power is off	□ Region : Domestic □ Launch : Nov '14 □ Inch: 78"
Mass produce mobile 20nm 8Gb LPDDR4 DRAM	□ Mass produced the world's first ultrafine 20nm 8Gb LPDDR4 DRAM - Applied "highly integrated/ultrahigh speed/high-density" solution developed by the Company - Created the world's first 4GB market and took the lead □ Led growth of the DRAM market by expanding 20nm DRAM lineup - Plan to take the lead in the market by expanding next-generation lineups(8/6/4Gb) in the future	□ Region : Global □ Launch : Dec '14 □ Inch: 8Gb LPDDR4 DRAM
Launch new Mobile AP product	□ Industry's first 14nm FinFET process-based Mobile AP - Compared to 20nm process, 14nm process has enhanced performance by 20%, reduced power consumption by 30%, and improved productivity by 35%	□ Region : Global □ Launch : Jan '15 □ Name: Exynos 7 Octa
Launch new NFC product	□ 4th generation NFC solution with enhanced RF performance - Compared to 3rd generation products, doubled card mode and enhanced reader mode by 20% □ Support mobile POS in smartphone environment □ Applied 45nm embedded flash process - Reduced customers' product development and certification period	□ Region : Global □ Launch : Jan '15 □ Name: S3FWRN5P
Mass produce 10nm-class 128GB UFS for mobile	□ Mass produced the world's first 128GB UFS memory for mobile - Performance was enhanced by 12 times compared to memory card and 2.7 times compared to eMMC □ Lead the growth of the premium market by expanding the high-density memory market - Launched high-density line up with twice the density (128/64/32GB)	□ Region : Global □ Launch : Feb '15 □ Name: 128GB UFS (Internal memory card)Lau
Launch SUHD TV	□ Curved SUHD TV : Provide the ultimate immersive experience ※ S: Spectacular, Smart, Stylish, Superb □ Design : Chamfer Design (Real Metal)	□ Region : Global □ Launch : Mar '15 □ Inch: 65"/78"/88"

Research Project	Research Results and Expected Effect	In case applied to a product, name of the product and application
Mass produce NAND-based 10nm-class 512GB M.2 NVMe SSD for PC	- Mass produced world's first 512GB M.2 NVMe SSD for the next generation PC - Read speed of 2,260MB/s, 4 times faster than SATA SSD - Took the lead in the mobile workstation and slim PC markets - Led the premium market by expanding the ultra-high speed, high-density SSD market - Plan to target PC market in earnest by launching V- NAND based line ups in the future	- Region : Global - Launch : Apr '15 - Name: 512GB M.2 NVMe SSD (SM951 SSD)
HAV(WAA) WAM5500/ZA	- Ambient Audio - Defined a new audio category with differentiated sound quality and sensible design (Blending, conventional design) - Provide the same sound quality through Wireless Audio 360 and Ring Radiator Tech - Expand multi-room audio usability through compatibility with TV/AV products - Provide sensible user experience by applying Analog Wheel - Easy to understand setting information through Voice AUI - Provide visual experience by applying OLED	- Region : Americas - Launch : Aug '15
Launch new CMOS image sensor	- Industry's first 1.0um 16 mega-pixel mobile image sensor - Provide the same image quality with 1.12um pixels by applying ISOCELL - With camera module of under 5mm in height, mobile device design became slimmer - Compared to 1.12um pixel sensor, module height decreased by 20%	- Region : Global - Launch : Jul '15 - Name: S5K3P3
Mass produce 3rd generation(48-layer) 256Gb 3D V-NAND for SSD	- Mass produced world's first 3rd generation(48-layer) 3D V-NAND - Began mass producing 256Gb in earnest, which has twice the degree of integration than 128Gb - Took the lead in the PC, enterprise server, and datacenter SSD markets - Led 'popularization of Tera SSD' by expanding ultra-high-speed SSD line ups - Target the market by expanding 3rd generation V-NAND based line ups	- Region : Global - Launch : Aug '15 - Name: 256Gb 3D V-NAND (3G 48-layer)
Mass produce 20nm DRAM-based 12Gb LPDDR4 mobile DRAM for mobile	- Mass produced world's first next generation 12Gb LPDDR4 DRAM for mobile - First to achieve 4,266Mbps, which is twice the speed of PC DRAM - Took the lead in not only mobile but also PC, consumer electronics, and automobile markets - Led the premium DRAM market by taking the lead in the ultra-high-speed DRAM market - Plan to target the next generation DRAM market in the future by launching 6GB line ups	- Region : Global - Launch : Sep '15 - Name: 12Gb LPDDR4 Mobile DRAM (20nm DRAM)
LFD OHD	- Product Concept - Satisfied market demand with accelerated digitization of outdoor Signage. - Varied its sizes from 46" to 46"/55" - Improved brightness to 2,500nit from 1,500nit and applied new panels which withstand high-temperature - Specification - Outdoor Kit product with Power Box - Can operate in a wide range of temperature, from -30°C to +50°C (fans and heating films are applied)	- Region : Global - Launch : Oct '15 - Inch: 46"/55"
Mass produce 20nm DRAM-based 12GB 3D TSV DDR4 RDIMM for server	- First to mass produce next-generation 128GB 3D TSV RDIMM for server 3D TSV technology doubled up the speed while reducing electricity consumption by 50% - Surpassed the limit of DRAM module density for server (RDIMM/LRDIMM) - Leading the premium DRAM market with the next-generation line-up - Will create new market by launching HBM2, following the TSV LRDIMM	- Region : Global - Launch : Nov '15 - Name: 12GB 3D TSV DDR4 RDIMM (20nm DRAM)
System Air-Conditioner (CAC) AC9000K	- Product features - Developed the world-first Bladeless 360 Cassette-type air-conditioner - Provides even distribution of air and minimized untouched area with circular air wave - Increased space coverage to 83% from 44% - Cold Draft Free (reduced sensory temperature gap to 0.2°C from 7.9°C)	- Region : Global - Launch : Dec '15
System Air-Conditioner (DVM) AM7500K	- Product features - Has the biggest capacity at the same size level - Provided in Compact Size & improved installation and service (the width is 940mm, the smallest in the market) - Provides highest efficiency (increased China's 12HP IPLV by 17%)	- Region : Global - Launch : Dec '15
Mass produce all-in-one Bio-Processor Chip	- Mass produced the industry's first all-in-one Bio-Processor Chip - Its all-in-one chip took up every steps from capturing signals to processing, by including AEF, MCU, DSP, eFlash and PMIC - Equipped with five AFEs and measures five different biometric signals: BIA (body fat), PPG (heartbeat rate), ECG (electrocardiogram), skin temperature, and GSR (stress level)	- Region : Global - Launch : Dec '15 - Name: S3FBP5A

Research Project	Research Results and Expected Effect	In case applied to a product, name of the product and application
Mass produce 20nm DRAM-based 4GB HBM2 DRAM for HPC	□ Mass produced the world's first next-generation 4GB HBM2 DRAM for HPC - Improved system speed by 3.6x, board space savings of up to 97% - Exceeded speed limitation through TSV technology with 37x higher density □ Lead the growth of premium memory market with the launch of 8GB HBM2 - Continue to stay ahead in the network market, as well as graphic and HPC markets	□ Region : Global □ Launch : Jan '16 □ Name: 4GB HBM2 DRAM (20nm DRAM)
LFD DCE	□ Specification and Effects - Slim design though common use of mechanical engineering and panels of DBE model / Provide light MagicInfo-E Solution by applying NT14 platform - Slim Design: Bezel 9.5mm, Depth 49.9mm - Provide USB Contents Player through MagicInfo-E Solution - Strengthened controlling functions such as RJ45/RS232C	□ Region : Global □ Launch : Jan '16 □ Inch: 32"/40"/48"/55"
LM TC2	□ Specification and Effects - Enhanced competitiveness by upgrading CPU specifications ※ 1.0GHZ Dual Core → 2.2GHz Dual Core - Provides more OS options by introducing 64bit WES7 - Added USB port (Serial Port → USB x 2) ※ Replaced the existing serial port with USB to Serial Adapter	□ Region : Global □ Launch : Jan '16 □ Inch: 22"/24"
DVM System Air-Conditioner AM9100K	□ Product features - Made an entrance into high-efficiency Inverter chiller market - Efficiency (Europe): EER : 3.4 / ESEER: 5.7 (40% higher than competitor) - Installment: 1.38m³ (39% smaller than competitors) - Operate in a wider range of temperature, from -25°C to +48°C (competitor: -15°C to +43°C)	□ Region : Global □ Launch : Jan '16
Mass produce 3 rd Generation (48-layer) V-NAND based 256GB UFS	□ Mass produced the world's first next-generation 256GB UFS for smartphone - High-performance: up to 9 times faster than uSD card, twice as fast as SSD - Provides high-speed, high-density solution with sub-micro size □ Strengthen memory competitiveness though 256GB UFS market expansion - Constantly lead high growth of UFS in internal storage market	□ Region : Global □ Launch : Feb '16 □ Name: 256GB UFS (3rd generation 48-layer 256Gb V-NAND)
Mass produce 3 rd Generation (48-layer) V-NAND based 15.36TB SAS SSD	□ Mass produced the world's first next-generation 15.36TB SAS SSD for server - Provide 15.36TB SSD for the first time in the world through application of 256Gb V-NAND - The highest density among single form factor storage devices □ Focus on SAS market by constantly expanding high-density line-ups - Expand enterprise market significantly following data center market	□ Region : Global □ Launch : Feb '16 □ Name: 15.36TB SAS SSD (3rd generation 48-layer 256Gb V-NAND)
Mass produce 10nm-class (1x) 8Gb DDR4 DRAM for PC/server	□ Mass produced the world's first 10nm-class 8Gb DDR4 DRAM - Enhanced productivity and speed by above 30% compared to 20nm, save electricity by up to 20% - Exceeded limitation of migration, with our three distinctive innovative technology □ Lead the market growth through high-density DRAM line-up expansion - Continue to strengthen its dominance in the mobile market as well as PC and server markets	□ Region : Global □ Launch : Feb '16 □ Name: 8Gb DDR4 DRAM (10nm-class DRAM)
LM CF59	□ Specifications and Effects - Circuits: Provides triple Interface (1 D-Sub/1HDMI/1DP) - Mechanical Engineering: The 3-side bezel less technique gave birth to distinctive design and provides seamless usage through multi-display - Panel: Curved panel with a curvature of 1800R (4000R in 2015)	□ Region : Global □ Launch : Feb '16 □ Inch: 27"
FDR Refrigerator RF9500KF	□ Product features - Create new demand and take leadership in the smart home appliances market by providing new user experience which meets new consumer needs in IOT era - Applied 21.5" LCD - Provides Smart Things and Sticki Shopping features that all family members can share to achieve a Smart Home.	□ Region : Global □ Launch : Feb '16
Exynos 8 Octa	□ 14nm 2nd generation process-based premium mobile SOC - Enhanced performance and energy-saving effect - Applied distinctive custom CPU core technology - The first integrated one-chip solution with highest-specification LTE modem	□ Region : Global □ Launch : Mar '16 □ Name: S5E8990

Research Project	Research Results and Expected Effect	In case applied to a product, name of the product and application
Exynos 7	☐ Industry's first 14nm based mid to low-end mobile SOC - Applied 14nm derivative process - Enhanced energy-efficiency by 30%, compared to the existing 28nm products with same performance level	☐ Region : Global ☐ Launch : Mar '16 ☐ Name: S5E7870
Mobile Image Sensor	☐ 1.2 megapixel mobile image sensor with dual pixel technology - Phase-detection AF in all resolutions at high speed - Auto-focus feature that is quick and accurate even in darker environments - Maximized functions with Samsung isocell technology	☐ Region : Global ☐ Launch : Mar '16 ☐ Name: S5K2L1
LFD OHE	☐ Specification and Effects - Applied the existing outdoor specifications - 110°C TNI Panel made outdoor usage possible - Internalized Quad-Core SOC to provide PC-less Solution - Provides optimized brightness level through automatic illumination sensor, saving energy-consumption and maximizing product life	☐ Region : Global ☐ Launch : Mar '16 ☐ Inch: 24"
LM CF39	☐ Specification and Effects - 16:9 ratio, the most optimal curvature of 1800R - Offers distinctive design by applying round-type stand base and simple single hinge - Simple and sensational design even on the back, maximizing the beauty of curved design	☐ Region : Global ☐ Launch : Mar '16 ☐ Inch: 24"/27"
SUHD TV (KS9000)	☐ Curved SUHD TV ☐ Design: Dignity, Bezel-less, Screw-less, Axis Stand ☐ Specification and Effects - Platform (H/W, S/W): Jazz-M, Tizen OS - Image quality: UHD 120Hz, Curved, QD, Local Dimming - Feature: Live and OTT combined Home helps easy access to TV content/service and improves consumer experience	☐ Region : Global ☐ Launch : Mar '16 ☐ Inch: 49"/55"/65"
SUHD TV (KS7500/KS7000)	☐ Curved SUHD TV ☐ Design : Triumph, Bezel-less, Screw-less, Branch Stand ☐ Specification and Effects - Platform (H/W,S/W): Jazz-M, Tizen OS - Image quality: UHD 120Hz, Curved, QD, Local Dimming - Feature: Live and OTT combined Home helps easy access to TV content/service and improves consumer experience	☐ Region : Global ☐ Launch : Mar '16 ☐ Inch: 43"/49"/55"/65"
UHD TV (KU6500/KU6400)	☐ UHD Curved TV ☐ Design : Metal Design, Bolt-less Clean Back, Ultimate Slim Design ☐ Specification and Effects - Platform (H/W,S/W): Jazz-M, Tizen OS - Image quality: Wide Color Gamut - Feature: Live and OTT combined Home helps easy access to TV content/service and improves consumer experience	☐ Region : Global ☐ Launch : Mar '16 ☐ Inch: 40"/43"/49"/55"/65"
UHD TV (KU6000)	☐ UHD Flat TV ☐ Design : Minimalism Design, V-Shape stand ☐ Specification and Effects - Platform (H/W,S/W): Jazz-M, Tizen OS - Image quality: UHD Upscaling, Auto Contrast Enhancer, PurColor - Feature: New Smart TV feature provides easier access to TV content/service and improved consumer experience	☐ Region : Global ☐ Launch : Mar '16 ☐ Inch: 40"/43"/50"/55"/60"/65"
FHD TV (K5100) (K4100)	☐ FHD Flat TV ☐ Design : Louvre Design, Semi Edge Slim ☐ Specification and Effects - Platform (H/W,S/W): XL1, NT14L, NT16L(Non-Smart) - Image quality: FHD image quality enable consumers to watch various contents - Feature: Sport mode, USB 2.0	☐ Region : Global ☐ Launch : Apr '16 ☐ Inch: 32"/40"/43"/49"/55"
HD-TV (HE690)	☐ Product Concept - To penetrate the market by providing solutions and customized products for each market segment - Premium Smart TV for four to five-starred hotels ☐ Specification and Effects - HMS, SINC, REACH, H.Browser, Bluetooth Music Player, Ethernet Bridge(32"↑), Swivel Stand	☐ Region : Global ☐ Launch : Aug '16 ☐ Inch: 22"/24"/28"/32"/40"/49"/55"

Research Project	Research Results and Expected Effect	In case applied to a product, name of the product and application
Exynos 7 Quad	□ Connectivity integrated mobile SOC - Increased performance by 70% and power efficiency by 30% vs. the previous 28nm products - Reduced the size by 20% by integrating major features such as the modem, connectivity and PMIC.	□ Region : Global □ Launch : Aug '16 □ Name: S5E7570
Dishwasher (DW9900M)	□ Product Concept - 2nd Generation WaterWall technology - Improved Zone booster - Wi-Fi application and differentiated Rack feature - Sump & New Filter System - Hidden control, Touch Type display	□ Region : North America □ Launch : Dec '16
Residential Air-conditioner (AR5500M)	□ Product Concept - Adopt environment-friendly (Low-GWP) R32 refrigerant to respond to market trends - Applied S-Inv to strengthened the competitiveness of Inv-type products - N-PFC, R32 8-pole compressor	□ Region : Global □ Launch : Dec '16
Exynos 9	□ 10nm FinFET process based Premium AP - Cutting-edge 10nm process based model, low power consumption, high performance - Gygabit-class LTE modem-integrated chip - Provide unrivalled performance such as 2nd generation custom CPU and next-generation GPU	□ Region : Global □ Launch : 1Q '17
UHD TV (MU7000)	□ Flat UHD TV □ Design: Slim Unibody, Real 360, Bezel-less, Screw-less, Branch Stand □ Specification and Effects - Platform (H/W,S/W): Kant-M, Tizen OS - Image quality: UHD 120Hz, Flat, Local Dimming - Feature: Provide optimal convenience in terms of connectivity though One Connect mini	□ Region : Global □ Launch : Feb '17 □ Inch: 49"/55"/65"
UHD TV (MU6500)	□ UHD Curved TV □ Design: Metal Design, Bolt-less Clean Back, Ultimate Slim Design, Y-Shape Stand □ Specification and Effects - Platform (H/W,S/W): Kant-M, Tizen OS - Image quality: Active Crystal Color (DCI 92%) - Feature: New Smart features (Eden 2.0, Eden Mobile) provides users with easier and more convenient content experiences	□ Region : Global □ Launch : Mar '17 □ Inch: 49"/55"/65"
UHD TV (MU6400)	□ UHD Curved TV □ Design: Metal Design, Bolt-less Clean Back, Ultimate Slim Design, Y-Shape Stand □ Specification and Effects - Platform (H/W,S/W): Kant-M, Tizen OS - Image quality: Active Crystal Color (DCI 92%) - Feature: New Smart features (Eden 2.0, Eden Mobile) provides users with easier and more convenient content experiences	□ Region : Global □ Launch : Mar '17 □ Inch: 40"/49"/55"/65"
UHD TV (MU6300 /MU6100)	□ UHD Curved TV □ Design: Minimalism Design, V-Shape Stand □ Specification and Effects - Platform (H/W,S/W): Kant-M, Tizen OS - Image quality: Active Crystal Color (DCI 92%) - Feature: New Smart features (Eden 2.0, Eden Mobile) provides users with easier and more convenient content experiences	□ Region : Global □ Launch : Mar '17 □ Inch: 40"/43"/49"/50"/55"/65"/75"
QLED TV (Q8C)	□ Curved QLED TV □ Design: 4 Sides Bezel-less, Simple Edge, Metal Back Stainless Silver stand (Type-U), Slim Wall Mount □ Specification and Effects - Platform (H/W,S/W): Kant-M, Tizen OS - Concept: 4 Sides Bezel-less Design QLED TV - Image quality: Highest-level of brightness, expanded color volume, accurate image, better viewing angles and deeper black levels - Feature: With Optical Cable and One connect, provides users with optimal convenience in terms of connectivity	□ Region : Global □ Launch : Apr '17 □ Inch: 75"

Research Project	Research Results and Expected Effect	In case applied to a product, name of the product and application
UHD TV (MU6100)	□ UHD Flat TV □ Design: Minimalism Design, V-Shape Stand □ Specification and Effects - Platform (H/W,S/W): Kant-M, Tizen OS - Image quality: Active Crystal Color (DCI 92%) - Feature: New Smart features (Eden 2.0, Eden Mobile) provides users with easier and more convenient content experiences	□ Region : Global □ Launch : Apr '17 □ Inch: 49"/55"/58"/75"
UHD TV (LS003)	□ Lifestyle TV □ Design: Frame Edge Design, Zero Gap Wall Mount □ Specification and Effects - Platform (H/W,S/W): Kant-M, Tizen OS - Image quality: Active Crystal Color (DCI 92%) Paper Look - Feature: Art Mode on Paper like PQ, Mobile Control, Samsung Collection, Invisible Connection, Replaceable Deco Frame	□ Region : Global □ Launch : May '17 □ Inch: 55"/65"
Exynos i (T200)	□ SOC optimized for IoT - 28nm HKMG process-based, multi-core performance - Encode/Decode features, enhanced security function with hardware block system which prevents illegal copy - Certified by WiFi Alliance, supports IoT protocol standard 'IoTivity'	□ Region : Global □ Launch : May '17 □ Inch: S5JT200
UHD TV (MU6303)	□ Economical UHD Curved TV □ Design: Minimalism Design, V-Shape Stand □ Specification and Effects - Platform (H/W,S/W): KANT-M, Tizen OS - Image quality: Active Crystal Color (DCI 92%) - Features: New Smart features (Eden 2.0, Eden Mobile) provides users with easier and more convenient content experiences - Not supporting BT vs. MU6300 (Not supporting Smart Controller)	□ Region : Global □ Launch : Jul '17 □ Inch: 49"/55"/65"
UHD TV (MU6103)	□ Economical UHD Flat TV □ Design: Minimalism Design, V-Shape Stand □ Specification and Effects - Platform(H/W,S/W) : KANT-M, Tizen OS - Image quality: Active Crystal Color (DCI 92%) - Features: New Smart features (Eden 2.0, Eden Mobile) provides users with easier and more convenient content experiences - Not supporting BT vs. MU6100 (Not supporting Smart Controller)	□ Region : Global □ Launch : Jul '17 □ Inch: 40"/43"/49"/50"/55"/65"/75"
HD-TV HF690	□ Product Concept - Offer customized products and solutions by market segment - New Premium Smart Line-up (HF590/690) for 3~4 star hotels - Platform: KANT-M (Tizen3.0) □ Specification and Effects - Panel: Same specifications as M5500, 32"/43"/49"/55" - Circuits platform : Kant-M (Tizen3.0) ※ SMPS: Shares M5500 SMPS design ※ HDMIx3, USBx2, RJ12, LAN, Opt, Comp/AV, RJP, BathSPK, LAN OUT - Device design : Shares M5500 series design ※ Swivel Stand/Box: Same features with 690 series of 2016	□ Region : Global □ Launch : Aug '17 □ Inch: 32"/43"/49"/55"
LFD QMH	□ Product Concept - Introduced Premium UHD Line Up ※ UHD (High resolution) Signage Line-up ※ Provides Signage solutions through SOC application (Major enhancement compared to QMF of 2016) - Improved size competitiveness of UHD line-ups ※ Respond to the market's needs for high-definition products by building full line-ups with diverse sizes - Non-Glare screen, 24/7 usage, same as QMF of 2016 ※ Applied required specifications for LFD usage □ Specification and Effects - Platform : Orsay → Tizen 3.0, removed DTS and DIVX - Circuits: Integrated with platform for FHD/UHD models → Kant-M SOC ※ Improved specification/features: SE13U (Scaler) → Kant-M (SOC) - Panel: 24/7, V-PID, Non-Glare (Haze 25%)	□ Region : Global □ Launch : Aug '17 □ Inch: 49"/55"/65"

Research Project	Research Results and Expected Effect	In case applied to a product, name of the product and application
Vaccum (VS8000ML)	□ Product features - World's most powerful suction vacuum "POWERstick" - Suction Power 150W, lasts for up to 40 minutes - Flex Body, Dual action brush (consists of two brushes each with a separate drum) - EzClean Dustbin & Brush - Hygienic Solution: HEPA Filter	□ Region : Global □ Launch : Aug '17 □ Name: POWERstick
UHD TV (LS003)	□ Lifestyle TV □ Design: Frame Edge Design, Zero Gap WMT □ Specification and Functions - Platform(H/W,S/W) : KANT-M, Tizen OS - Picture Quality: Active Crystal Color (DCI 92%) Paper Look - Features: Art Mode on Paper like PQ, Mobile Control, Samsung Collection, Invisible Connection, Replaceable Deco Frame	□ Region : Global □ Launch : Dec '17 □ Inch: 43"
Mass production of 512GB eUFS based on 4th generation V-NAND (64 layers) for mobiles	□ The world's first mass production of 512GB eUFS SSD based on 64 layer V-NAND - Lead the high density memory market by mass producing 512Gb V-NAND - Provide the fastest solution with the biggest capacity □ Drive market growth by expanding the production of 4th generation 512Gb VNAND - Expand market leadership from smartphones and tablets to the SSDs	□ Region : Global □ Launch : Nov '17 □ Name: 512 GB eUFS for mobiles (4th generation V-NAND)
Mass production of 10nm-class (1y) 8Gb DDR4 DRAM for PCs and servers	□ The world's first 2nd generation 10nm-class 8Gb DDR4 DRAM - 30% better productivity, 10% faster speed, and 15% improved energy efficiency compared to the 1x- nano DRAM - Development of 'Three innovative Technologies' boosting product competitiveness □ Lead the market with the capability to mass-produce next-generation DRAMs - Established a mass production system for DDR5/ LPDDR5/ HBM3/ GDDR6	□ Region : Global □ Launch : Nov '17 □ Name: 8Gb DDR4 DRAM (1y-nano DRAM)
Mobile Image sensor	□ Ultra-small high-resolution image sensor, ISOCELL Fast 2L9 - Autofocus, dual pixel technologies - Micro-manufacturing process reduced the size of the pixel to 1.2μm from 1.4μm - A single image sensor is capable of out-focusing	□ Region : Global □ Launch : 4Q17 □ Name: S5K2X7
Mobile Image sensor	□ Ultra-small high-resolution image sensor, ISOCELL Slim 2X7 - Vivid and bright pictures enabled by TetraCell technology, which allows pixel self-adjustment - The ultra-small image sensor with the smallest ever pixel size (0.9μm - DTI processing applied physical walls between pixels.	□ Region : Global □ Launch : 4Q17 □ Name: S5K2L9SX

11. Other Information

In 2017, the Company invested KRW 16.8 trillion and registered 5,394 domestic and 14,279 overseas patents, including approval of 6,072 US patents. In 2017, the Company registered 6,072 patents in the US, ranking second following IBM for twelve consecutive years.

<Number of Patents Registered in Each Country (as of December 31, 2017, cumulative)>

(Unit: # of Patents)

	Korea	US	Europe	China	Japan	Others
<i>Number</i>	24,860	46,150	20,509	11,005	7,086	9,727

After registering its first US patent in 1984, the Company registered and now holds 119,337 patents around the world. The Company holds the highest number of patents (cumulative number) in the US and aims to address conflicts in the US more effectively.

<Number of Patent Registrations Per Year>

(Unit: # of Patents)

	'17	'16	'15	'14	'13	'12	'11	'10
<i>Korea</i>	2,703	3,462	3,002	3,985	2,775	2,024	1,616	1,620
<i>US</i>	6,072	5,683	5,220	5,085	4,802	5,194	4,991	4,653

These patents are mostly related to smartphone, smart TV, memory, and system LSI products for the Company's strategic business products or for future use. These patents not only protect the Company's business but also play a role in keeping similar technology and patents, as well as competitors, in check. Additionally, the Company is focusing on securing early patents in new technologies ahead of others, in order to protect opportunities and have the freedom to operate when entering new businesses.

The Company has signed cross-licensing agreements with Google (on January 2014, a permanent license), Nokia (on July 2016), and WD (on December 2016, till 2024), through which the Company has secured an extensive patent protection in mainstay businesses as well as in the fields of new business.

The Company has also been focusing on securing design patents to protect its original design applied to smartphones and LED TVs. In 2017, the Company acquired 1,063 US patents, in part for the aforementioned purposes.

B. Environmental regulations

The Company strictly abides by environmental regulations on products and in the workplace, as prescribed by law. In addition, in accordance with "the low carbon green growth policy" of the Korean government, the Company reports "the amount of CO2 emission and energy use" to the government and provides related information to stakeholders by providing various reports, including the Samsung Sustainability Report.

(See 『9. Green Management』 of 『XI. Other Information』 for more details about Green Technology Certification.)

(Environmental Regulation of Products)

Environmental regulation of products are becoming stricter, reflecting the concerns of governments and regulators about potential direct and indirect impact of products to consumers' health and safety as consumers. Accordingly, the Company is working to minimize the environmental impact throughout the entire life cycle of products from the development stage of components and products to manufacturing, distribution, use, and disposal. The Company offers "Eco-Partner Certification" to suppliers for components free of harmful substances, and runs an "eco-design evaluation" system to reflect eco-friendly elements on products (reduced use of resources, energy and harmful substances, and the use of eco-friendly materials) at the development stage, and operates a "waste electronics collection recycle system" in Europe, North America, Korea, and India for collecting and recycling waste electronics. These activities are in line with domestic and foreign environmental laws on electronics and are a differentiation factor for the Company and its products.

Relevant laws are as follows:

1. law on collection and recycle of waste electronics (e.g., EU WEEE Directive)
2. limit on the use of harmful substances (e.g., EU RoHS Directive, REACH Regulation)
3. regulation on energy efficiency (e.g., EU ErP Directive)

(Environmental Regulations in the Workplace)

The Company operates environmental pollution prevention facilities to reduce air pollution, water pollution, and waste disposal, and to minimize the discharge of pollutants, thereby minimizing impact on the surrounding environment.

Environmental management of workplace is supervised by the relevant government authorities. All production facilities, domestic and overseas, have acquired the International Occupational Health and Safety Management System Certification (ISO 14001, OHSAS18001) to strengthen compliance.

Major Relevant Domestic and International Laws are as follows:

1. Regulations related to emission of pollutants: Water Quality and Ecosystem Conservation Act, Clean Air Conservation Act, Wastes Control Act, Noise and Vibration Control Act, Environmental Impact Assessment Act
2. Management of greenhouse gas emission: Act on Allocation and Trading of Greenhouse Gas Emission, Framework Act on Low Carbon, Green Growth
3. Others: Toxic Chemicals Control Act, Act on the Registration and Evaluation, etc. of Chemical Substances, Odor Control Law, Soil Environment Conservation Act

(Greenhouse gas emission and energy consumption management)

The Company is a “controlled entity” according to Article 42 of Korea’s “Framework Act on Low Carbon, Green Growth.” Thus, the Company has been reporting the amount of greenhouse gas emission and energy consumption, verified by a third-party, to the authorities and disclosing it to stakeholders according to Article 44 of the same Act since May 2011.

The reported amount of greenhouse gas emission and energy use are provided below:

	2017	2016	2015
Greenhouse gas (Unit: tCO ₂ e)	8,411,674	6,885,300	6,729,419
Energy (Unit: TJ)	130,836	107,740	111,166

- ※ Domestic manufacturing facilities, office buildings, buildings owned by the Company, leased buildings, etc.
- ※ Reported Greenhouse gas emission excludes ozone depletion substances (ODS).
- ※ DP business was excluded from the calculation as Samsung Display was spun off from Samsung Electronics in 2012.

From 2015, in accordance with the Article 8 of the Act on the Allocation and Trading of Greenhouse-Gas Emission Permits, the Company is an eligible business entity under the Act.

III. Financial Affairs

1. Consolidated Financial Statements

CONSOLIDATED STATEMENTS OF FINANCIAL POSITION

(In millions of Korean won, in thousands of US dollars (Note 2.3))

	Notes	December 31, 2017 KRW	December 31, 2016 KRW	December 31, 2017 USD	December 31, 2016 USD
Assets					
Current assets					
Cash and cash equivalents	4, 6, 7, 31	30,545,130	32,111,442	27,005,370	28,390,168
Short-term financial instruments	5, 6, 7, 31	49,447,696	52,432,411	43,717,389	46,356,216
Short-term available-for-sale financial assets	6, 9, 31	3,191,375	3,638,460	2,821,539	3,216,813
Trade receivables	6, 7, 10, 31	27,695,995	24,279,211	24,486,410	21,465,585
Non-trade receivables	6, 10	4,108,961	3,521,197	3,632,789	3,113,139
Advance payments		1,753,673	1,439,938	1,550,446	1,273,069
Prepaid expenses		3,835,219	3,502,083	3,390,770	3,096,240
Inventories	11	24,983,355	18,353,503	22,088,128	16,226,585
Other current assets	6	1,421,060	1,315,653	1,256,378	1,163,185
Assets held-for-sale	36	-	835,806	-	738,948
Total current assets		146,982,464	141,429,704	129,949,219	125,039,948
Non-current assets					
Long-term available-for-sale financial assets	6, 9, 31	7,752,180	6,804,276	6,853,809	6,015,754
Held-to-maturity financial assets	6	106,751	-	94,380	-
Investment in associates and joint ventures	12	6,802,351	5,837,884	6,014,052	5,161,354
Property, plant and equipment	13	111,665,648	91,473,041	98,725,136	80,872,574
Intangible assets	14	14,760,483	5,344,020	13,049,946	4,724,722
Long-term prepaid expenses		3,434,375	3,834,831	3,036,378	3,390,427
Net defined benefit assets	17	825,892	557,091	730,182	492,532
Deferred income tax assets	28	5,061,687	5,321,450	4,475,107	4,704,767
Other non-current assets	5, 6	4,360,259	1,572,027	3,854,967	1,389,850
Total assets		301,752,090	262,174,324	266,783,176	231,791,928

CONSOLIDATED STATEMENTS OF FINANCIAL POSITION

(In millions of Korean won, in thousands of US dollars (Note 2.3))

	Notes	December 31, 2017 KRW	December 31, 2016 KRW	December 31, 2017 USD	December 31, 2016 USD
Liabilities and Equity					
Current liabilities					
Trade payables	6, 31	9,083,907	6,485,039	8,031,207	5,733,512
Short-term borrowings	6, 8, 15, 31	15,767,619	12,746,789	13,940,369	11,269,612
Other payables	6, 31	13,899,633	11,525,910	12,288,857	10,190,216
Advances received		1,249,174	1,358,878	1,104,412	1,201,403
Withholdings	6	793,582	685,028	701,617	605,643
Accrued expenses	6	13,996,273	12,527,300	12,374,298	11,075,558
Income tax payable		7,408,348	2,837,353	6,549,822	2,508,543
Current portion of long-term liabilities	6, 15, 16, 31	278,619	1,232,817	246,331	1,089,950
Provisions	18	4,294,820	4,597,417	3,797,109	4,064,640
Other current liabilities	6	403,139	351,176	356,421	310,478
Liabilities held-for-sale	36	-	356,388	-	315,088
Total current liabilities		67,175,114	54,704,095	59,390,443	48,364,643
Non-current liabilities					
Debentures	6, 16, 31	953,361	58,542	842,880	51,758
Long-term borrowings	6, 15, 31	1,814,446	1,244,238	1,604,177	1,100,048
Long-term other payables	6, 31	2,043,729	3,317,054	1,806,889	2,932,653
Net defined benefit liabilities	17	389,922	173,656	344,735	153,532
Deferred income tax liabilities	28	11,710,781	7,293,514	10,353,663	6,448,296
Provisions	18	464,324	358,126	410,515	316,624
Other non-current liabilities	6	2,708,985	2,062,066	2,395,050	1,823,101
Total liabilities		87,260,662	69,211,291	77,148,352	61,190,655

CONSOLIDATED STATEMENTS OF FINANCIAL POSITION

(In millions of Korean won, in thousands of US dollars (Note 2.3))

	Notes	December 31, 2017 KRW	December 31, 2016 KRW	December 31, 2017 USD	December 31, 2016 USD
Equity attributable to owners of the parent					
Preference shares	20	119,467	119,467	105,622	105,622
Ordinary shares	20	778,047	778,047	687,882	687,882
Share premium		4,403,893	4,403,893	3,893,542	3,893,542
Retained earnings	21	215,811,200	193,086,317	190,801,652	170,710,271
Other components of equity	23	(13,899,191)	(11,934,586)	(12,288,465)	(10,551,532)
Accumulated other comprehensive income attributable to assets held-for-sale	36	-	(28,810)	-	(25,471)
		207,213,416	186,424,328	183,200,233	164,820,314
Non-controlling interests		7,278,012	6,538,705	6,434,591	5,780,959
Total equity		214,491,428	192,963,033	189,634,824	170,601,273
Total liabilities and equity					
		301,752,090	262,174,324	266,783,176	231,791,928

CONSOLIDATED STATEMENTS OF PROFIT OR LOSS

(In millions of Korean won, in thousands of US dollars (Note 2.3))

	Notes	For the year ended December 31,			
		2017	2016	2017	2016
		KRW	KRW	USD	USD
Revenue	32	239,575,376	201,866,745	211,811,887	178,473,168
Cost of sales	24	129,290,661	120,277,715	114,307,653	106,339,183
Gross profit		110,284,715	81,589,030	97,504,234	72,133,985
Selling and administrative expenses	24, 25	56,639,677	52,348,358	50,075,918	46,281,904
Operating profit		53,645,038	29,240,672	47,428,316	25,852,081
Other non-operating income	26	3,010,657	3,238,261	2,661,763	2,862,991
Other non-operating expense	26	1,419,648	2,463,814	1,255,130	2,178,292
Share of profit of associates and joint ventures	12	201,442	19,501	178,098	17,241
Financial income	27	9,737,391	11,385,645	8,608,961	10,066,205
Financial expense	27	8,978,913	10,706,613	7,938,380	9,465,864
Profit before income tax		56,195,967	30,713,652	49,683,628	27,154,362
Income tax expense	28	14,009,220	7,987,560	12,385,744	7,061,911
Profit for the period		42,186,747	22,726,092	37,297,884	20,092,451
Profit attributable to owners of the parent		41,344,569	22,415,655	36,553,302	19,817,989
Profit attributable to non-controlling interests		842,178	310,437	744,582	274,462
Earnings per share (in Korean Won, in US dollars)	29				
- Basic		299,868	157,967	265.12	139.66
- Diluted		299,868	157,967	265.12	139.66

CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME

(In millions of Korean won, in thousands of US dollars (Note 2.3))

	Notes	For the year ended December 31,			
		2017	2016	2017	2016
		KRW	KRW	USD	USD
Profit for the period		42,186,747	22,726,092	37,297,884	20,092,451
Other comprehensive income (loss)					
Items not to be reclassified to profit or loss subsequently:					
Remeasurement of net defined benefit liabilities, net of tax	17, 23	414,247	963,602	366,241	851,934
Share of other comprehensive income (loss) of associates and joint ventures, net of tax	12, 23	(6,347)	50,438	(5,611)	44,593
Items to be reclassified to profit or loss subsequently:					
Changes in value of available-for-sale financial assets, net of tax	9, 23	511,207	(23,839)	451,965	(21,076)
Share of other comprehensive income (loss) of associates and joint ventures, net of tax	12, 23	(49,256)	(130,337)	(43,548)	(115,233)
Foreign currency translation, net of tax	23	(6,334,987)	1,131,536	(5,600,849)	1,000,406
Gain (loss) on valuation of derivatives		(37,121)	-	(32,820)	-
Other comprehensive income (loss) for the period, net of tax		(5,502,257)	1,991,400	(4,864,622)	1,760,624
Total comprehensive income for the period		36,684,490	24,717,492	32,433,262	21,853,075
Comprehensive income attributable to:					
Owners of the parent		35,887,505	24,310,814	31,728,637	21,493,525
Non-controlling interests		796,985	406,678	704,625	359,550

CONSOLIDATED STATEMENTS OF CHANGES IN EQUITY

(In millions of Korean won)

2016 KRW	Notes	Preference shares	Ordinary shares	Share premium	Retained earnings	Other components of equity	Accumulated other comprehensive income attributable to assets held-for-sale	Equity attributable to owners of the parent	Non- controlling interests	Total
Balance as at January 1, 2016		119,467	778,047	4,403,893	185,132,014	(17,580,451)	23,797	172,876,767	6,183,038	179,059,805
Profit for the period		-	-	-	22,415,655	-	-	22,415,655	310,437	22,726,092
Changes in value of available-for-sale financial assets, net of tax	9, 23	-	-	-	-	(87,706)	(23,797)	(111,503)	87,664	(23,839)
Share of other comprehensive income (loss) of associates and joint ventures, net of tax	12, 23	-	-	-	-	(80,146)	212	(79,934)	35	(79,899)
Foreign currency translation, net of tax	23	-	-	-	-	1,160,316	-	1,160,316	(28,780)	1,131,536
Remeasurement of net defined benefit liabilities, net of tax	17, 23	-	-	-	-	926,280	-	926,280	37,322	963,602
Classified as held-for-sale	36	-	-	-	-	29,022	(29,022)	-	-	-
Total comprehensive income (loss)		-	-	-	22,415,655	1,947,766	(52,607)	24,310,814	406,678	24,717,492
Dividends	22	-	-	-	(3,061,361)	-	-	(3,061,361)	(65,161)	(3,126,522)
Capital transaction under common control		-	-	-	-	(37)	-	(37)	12,272	12,235
Changes in consolidated entities		-	-	-	-	-	-	-	1,790	1,790
Acquisition of treasury stock	23	-	-	-	-	(7,707,938)	-	(7,707,938)	-	(7,707,938)
Retirement of treasury stock	23	-	-	-	(11,399,991)	11,399,991	-	-	-	-
Others		-	-	-	-	6,083	-	6,083	88	6,171
Total transactions with owners		-	-	-	(14,461,352)	3,698,099	-	(10,763,253)	(51,011)	(10,814,264)
Balance as at December 31, 2016		119,467	778,047	4,403,893	193,086,317	(11,934,586)	(28,810)	186,424,328	6,538,705	192,963,033

CONSOLIDATED STATEMENTS OF CHANGES IN EQUITY

(In thousands of US dollars (Note 2.3))

2016 USD	Notes	Preference shares	Ordinary shares	Share premium	Retained earnings	Other components of equity	Accumulated other comprehensive income attributable to assets held-for- sale	Equity attributable to owners of the parent	Non- controlling interests	Total
Balance as at January 1, 2016		105,622	687,882	3,893,542	163,677,762	(15,543,119)	21,040	152,842,729	5,466,509	158,309,238
Profit for the period		-	-	-	19,817,989	-	-	19,817,989	274,462	20,092,451
Changes in value of available-for-sale financial assets, net of tax	9, 23	-	-	-	-	(77,542)	(21,039)	(98,581)	77,505	(21,076)
Share of other comprehensive income (loss) of associates and joint ventures, net of tax	12, 23	-	-	-	-	(70,858)	187	(70,671)	31	(70,640)
Foreign currency translation, net of tax	23	-	-	-	-	1,025,851	-	1,025,851	(25,445)	1,000,406
Remeasurement of net defined benefit liabilities, net of tax	17, 23	-	-	-	-	818,937	-	818,937	32,997	851,934
Classified as held-for-sale	36	-	-	-	-	25,659	(25,659)	-	-	-
Total comprehensive income (loss)		-	-	-	19,817,989	1,722,047	(46,511)	21,493,525	359,550	21,853,075
Dividends	22	-	-	-	(2,706,591)	-	-	(2,706,591)	(57,610)	(2,764,201)
Capital transaction under common control		-	-	-	-	(33)	-	(33)	10,850	10,817
Changes in consolidated entities		-	-	-	-	-	-	-	1,583	1,583
Acquisition of treasury stock	23	-	-	-	-	(6,814,695)	-	(6,814,695)	-	(6,814,695)
Retirement of treasury stock	23	-	-	-	(10,078,889)	10,078,889	-	-	-	-
Others		-	-	-	-	5,379	-	5,379	77	5,456
Total transactions with owners		-	-	-	(12,785,480)	3,269,540	-	(9,515,940)	(45,100)	(9,561,040)
Balance as at December 31, 2016		105,622	687,882	3,893,542	170,710,271	(10,551,532)	(25,471)	164,820,314	5,780,959	170,601,273

CONSOLIDATED STATEMENTS OF CHANGES IN EQUITY

(In millions of Korean won)

2017 KRW	Notes	Preference shares	Ordinary shares	Share premium	Retained earnings	Other components of equity	Accumulated other comprehensive income attributable to assets held-for- sale	Equity attributable to owners of the parent	Non- controlling interests	Total
Balance as at January 1, 2017		119,467	778,047	4,403,893	193,086,317	(11,934,586)	(28,810)	186,424,328	6,538,705	192,963,033
Profit for the period		-	-	-	41,344,569	-	-	41,344,569	842,178	42,186,747
Changes in value of available-for-sale financial assets, net of tax	9, 23	-	-	-	-	489,150	-	489,150	22,057	511,207
Share of other comprehensive income (loss) of associates and joint ventures, net of tax	12, 23	-	-	-	-	(54,300)	-	(54,300)	(1,303)	(55,603)
Foreign currency translation, net of tax	23	-	-	-	-	(6,289,926)	28,810	(6,261,116)	(73,871)	(6,334,987)
Remeasurement of net defined benefit liabilities, net of tax	17, 23	-	-	-	-	406,323	-	406,323	7,924	414,247
Gain (loss) on valuation of derivatives		-	-	-	-	(37,121)	-	(37,121)	-	(37,121)
Total comprehensive income (loss)		-	-	-	41,344,569	(5,485,874)	28,810	35,887,505	796,985	36,684,490
Dividends	22	-	-	-	(6,747,123)	-	-	(6,747,123)	(64,277)	(6,811,400)
Capital transaction under common control		-	-	-	-	(2,992)	-	(2,992)	15,114	12,122
Changes in consolidated entities		-	-	-	-	(2,699)	-	(2,699)	(9,352)	(12,051)
Acquisition of treasury stock	23	-	-	-	-	(8,350,424)	-	(8,350,424)	-	(8,350,424)
Retirement of treasury stock	23	-	-	-	(11,872,563)	11,872,563	-	-	-	-
Others		-	-	-	-	4,821	-	4,821	837	5,658
Total transactions with owners		-	-	-	(18,619,686)	3,521,269	-	(15,098,417)	(57,678)	(15,156,095)
Balance as at December 31, 2017		119,467	778,047	4,403,893	215,811,200	(13,899,191)	-	207,213,416	7,278,012	214,491,428

CONSOLIDATED STATEMENTS OF CHANGES IN EQUITY

(In thousands of US dollars (Note 2.3))

2017 USD	Notes	Preference shares	Ordinary shares	Share premium	Retained earnings	Other components of equity	Accumulated other comprehensive income attributable to assets held-for- sale	Equity attributable to owners of the parent	Non- controlling interests	Total
Balance as at January 1, 2017		105,622	687,882	3,893,542	170,710,271	(10,551,532)	(25,471)	164,820,314	5,780,959	170,601,273
Profit for the period		-	-	-	36,553,302	-	-	36,553,302	744,582	37,297,884
Changes in value of available-for-sale financial assets, net of tax	9, 23	-	-	-	-	432,464	-	432,464	19,501	451,965
Share of other comprehensive income (loss) of associates and joint ventures, net of tax	12, 23	-	-	-	-	(48,007)	-	(48,007)	(1,152)	(49,159)
Foreign currency translation, net of tax	23	-	-	-	-	(5,561,010)	25,471	(5,535,539)	(65,310)	(5,600,849)
Remeasurement of net defined benefit liabilities, net of tax	17, 23	-	-	-	-	359,237	-	359,237	7,004	366,241
Gain (loss) on valuation of derivatives		-	-	-	-	(32,820)	-	(32,820)	-	(32,820)
Total comprehensive income (loss)		-	-	-	36,553,302	(4,850,136)	25,471	31,728,637	704,625	32,433,262
Dividends	22	-	-	-	(5,965,224)	-	-	(5,965,224)	(56,828)	(6,022,052)
Capital transaction under common control		-	-	-	-	(2,645)	-	(2,645)	13,362	10,717
Changes in consolidated entities		-	-	-	-	(2,386)	-	(2,386)	(8,268)	(10,654)
Acquisition of treasury stock	23	-	-	-	-	(7,382,726)	-	(7,382,726)	-	(7,382,726)
Retirement of treasury stock	23	-	-	-	(10,496,697)	10,496,697	-	-	-	-
Others		-	-	-	-	4,263	-	4,263	741	5,004
Total transactions with owners		-	-	-	(16,461,921)	3,113,203	-	(13,348,718)	(50,993)	(13,399,711)
Balance as at December 31, 2017		105,622	687,882	3,893,542	190,801,652	(12,288,465)	-	183,200,233	6,434,591	189,634,824

CONSOLIDATED STATEMENTS OF CASH FLOWS

(In millions of Korean won, in thousands of US dollars (Note 2.3))

	Notes	For the year ended December 31,			
		2017	2016	2017	2016
		KRW	KRW	USD	USD
Cash flows from operating activities					
Profit for the period		42,186,747	22,726,092	37,297,884	20,092,451
Adjustments	30	36,211,232	30,754,471	32,014,849	27,190,451
Changes in assets and liabilities arising from operating activities	30	(10,620,547)	(1,180,953)	(9,389,772)	(1,044,097)
Cash generated from operations		67,777,432	52,299,610	59,922,961	46,238,805
Interest received		1,581,117	1,405,085	1,397,887	1,242,255
Interest paid		(542,715)	(443,838)	(479,822)	(392,403)
Dividends received		173,305	256,851	153,221	227,086
Income tax paid		(6,827,098)	(6,132,064)	(6,035,931)	(5,421,443)
Net cash inflow from operating activities		62,162,041	47,385,644	54,958,316	41,894,300
Cash flows from investing activities					
Net decrease (increase) in short-term financial instruments		387,627	(6,780,610)	342,706	(5,994,831)
Disposal of short-term available-for-sale financial assets		499,856	3,010,003	441,930	2,661,185
Acquisition of short-term available-for-sale financial assets		-	(2,129,551)	-	(1,882,765)
Disposal of long-term financial instruments		1,750,221	789,862	1,547,394	698,328
Acquisition of long-term financial instruments		(1,079,355)	(1,741,547)	(954,273)	(1,539,726)
Disposal of long-term available-for-sale financial assets		191,826	2,010,356	169,596	1,777,383
Acquisition of long-term available-for-sale financial assets		(358,497)	(1,498,148)	(316,952)	(1,324,533)
Acquisition of held-to-maturity financial assets		191,826	2,010,356	169,596	1,777,383
Disposal of investment in associates and joint ventures		355,926	2,280,203	314,679	2,015,959
Acquisition of investment in associates and joint ventures		(25,293)	(84,306)	(22,362)	(74,536)
Disposal of property, plant and equipment		308,354	270,874	272,620	239,483
Acquisition of property, plant and equipment		(42,792,234)	(24,142,973)	(37,833,203)	(21,345,135)
Disposal of intangible assets		733	6,944	648	6139
Acquisition of intangible assets		(983,740)	(1,047,668)	(869,738)	(926,258)
Cash outflow from business combinations		(8,754,268)	(622,050)	(7,739,769)	(549,963)
Cash inflow from business transfers		1,248,834	-	1,104,111	-
Others		(28,455)	19,936	(25,156)	17,627
Net cash used in investing activities		(49,385,216)	(29,658,675)	(43,662,149)	(26,221,643)

CONSOLIDATED STATEMENTS OF CASH FLOWS

(In millions of Korean won, in thousands of US dollars (Note 2.3))

	Notes	For the year ended December 31,			
		2017	2016	2017	2016
		KRW	KRW	USD	USD
Cash flows from financing activities					
Net increase in short-term borrowings		2,730,676	1,351,037	2,414,228	1,194,470
Acquisition of treasury stock		(8,350,424)	(7,707,938)	(7,382,726)	(6,814,695)
Proceeds from long-term borrowings and debentures		998,311	1,041,743	882,620	921,019
Repayment of long-term borrowings and debentures		(1,140,803)	(252,846)	(1,008,600)	(223,545)
Dividends paid		(6,804,297)	(3,114,742)	(6,015,773)	(2,753,786)
Net increase in non-controlling interests		5,670	13,232	5,016	11,700
Net cash outflow from financing activities		(12,560,867)	(8,669,514)	(11,105,235)	(7,664,837)
Effect of exchange rate changes on cash and cash equivalents		(1,782,270)	417,243	(1,575,730)	368,891
Net increase (decrease) in cash and cash equivalents		(1,566,312)	9,474,698	(1,384,798)	8,376,711
Cash and cash equivalents					
Beginning of the period		32,111,442	22,636,744	28,390,168	20,013,457
End of the period		30,545,130	32,111,442	27,005,370	28,390,168

2. Notes to Consolidated Financial Statements

1. General Information

1.1 Company Overview

Samsung Electronics Co., Ltd. (the “Company”) was incorporated under the laws of the Republic of Korea in 1969 and listed its shares on the Korea Stock Exchange in 1975. SEC and its subsidiaries (collectively referred to as the “Company”) operate four business divisions: Consumer Electronics (“CE”), Information technology & Mobile communications (“IM”), Device Solutions (“DS”) and Harman. The CE division includes digital TVs, monitors, air conditioners and refrigerators and the IM division includes mobile phones, communication systems, and computers. The DS division includes products such as Memory, Foundry and System LSI in the semiconductor business (“Semiconductor”), and LCD and OLED panels in the display business (“DP”). The Harman division includes connected car systems, audio and visual products, enterprise automation solutions and connected services. The Company is domiciled in the Republic of Korea and the address of its registered office is Suwon, the Republic of Korea.

These consolidated financial statements have been prepared in accordance with Korean International Financial Reporting Standards (“Korean IFRS”) 1110, *Consolidated Financial Statements*. SEC, as the controlling company, consolidates its 270 subsidiaries including Samsung Display and Samsung Electronics America. The Company also applies the equity method of accounting for its 41 associates, including Samsung Electro-Mechanics.

1.2 Consolidated Subsidiaries

(A) The consolidated subsidiaries as at December 31, 2017 are as follows:

Area	Subsidiaries	Industry	Percentage of ownership ¹
Domestic	Samsung Display (SDC)	Manufacture and sale of display panels	84.8
	SU Materials	Manufacture of display components	50.0
	STECO	Manufacture of semiconductor components	70.0
	SEMES	Manufacture of semiconductor/FPD	91.5
	Samsung Electronics Service	Repair services for electronic devices	99.3
	Samsung Electronics Sales	Sale of electronic devices	100.0
	Samsung Electronics Logitech	General logistics agency	100.0
	Samsung Medison	Manufacture and sale of medical equipment	68.5
	Samsung Venture Capital Union #21	Technology business, Venture capital investments	99.0
	Samsung Venture Capital Union #22	Technology business, Venture capital investments	99.0
	Samsung Venture Capital Union #23	Technology business, Venture capital investments	99.0
	Samsung Venture Capital Union #26	Technology business, Venture capital investments	99.0
	Samsung Venture Capital Union #27	Technology business, Venture capital investments	99.0
	Samsung Venture Capital Union #28	Technology business, Venture capital investments	99.0
	Samsung Venture Capital Union #29	Technology business, Venture capital investments	99.0
	Samsung Venture Capital Union #32	Technology business, Venture capital investments	99.0
	Samsung Venture Capital Union #33	Technology business, Venture capital investments	99.0
	Samsung Venture Capital Union #37	Technology business, Venture capital investments	99.0
	Mirero System	Development and supply of semiconductor process defect and quality control software	99.9
	Growth type private equity trust specialized in semiconductors	Investment on semiconductor industry	66.7
	Harman International Korea	Software development and supply, etc.	100.0
	Red Bend Software Korea	Software development and supply	100.0

¹ Ownership represents the Company’s ownership of voting rights in each entity.

Area	Subsidiaries	Industry	Percentage of ownership ¹
America	Samsung Electronics America (SEA)	Sale of electronic devices	100.0
	NexusDX (Nexus)	Medical equipment	100.0
	NeuroLogica	Medical equipment	100.0
	Samsung Semiconductor (SSI)	Sale of semiconductor/display panels	100.0
	Samsung Electronics Canada (SECA)	Sale of electronic devices	100.0
	Samsung Research America (SRA)	R&D	100.0
	Samsung Mexicana (SAMEX)	Manufacture of electronic devices	100.0
	Samsung International (SII)	Manufacture of TV/monitors	100.0
	Samsung Austin Semiconductor (SAS)	Manufacture of semiconductor	100.0
	Samsung Electronics Mexico (SEM)	Sale of electronic devices	99.9
	SEMES America (SEMESA)	Semiconductor equipment	100.0
	Samsung Electronics Digital Appliance Mexico (SEDAM)	Manufacture of electronic devices	99.9
	Samsung Electronics Latinoamerica Miami (SEMI)	Sale of electronic devices	100.0
	Samsung Electronics Latinoamerica (SELA)	Sale of electronic devices	100.0
	Samsung Electronics Venezuela (SEVEN)	Marketing and services	100.0
	Samsung Electronica Colombia (SAMCOL)	Sale of electronic devices	100.0
	Samsung Electronics Panama (SEPA)	Consulting	100.0
	Samsung Electronica da Amazonia (SEDA)	Manufacture and sale of electronic devices	100.0
	Samsung Electronics Argentina (SEASA)	Marketing and services	100.0
	Samsung Electronics Chile (SECH)	Sale of electronic devices	100.0
	Samsung Electronics Peru (SEPR)	Sale of electronic devices	100.0
	RT SV CO-INVEST (RT-SV)	Venture capital investments	99.9
	Samsung HVAC (formerly Quietside)	Sale of heating and cooling products	100.0
	SmartThings	Sale of smart home electronics	100.0
	Samsung Pay	Develop and provide mobile payment service	100.0
	Prismview	Manufacture and sale of LED displays	100.0
	Beijing Integrated Circuit Industry International Fund (Beijing Fund)	Venture capital investments	61.4
	Stellus Technologies	Manufacture and sale of server semiconductor storage system	100.0
	Samsung Oak Holdings (SHI)	Holding company	100.0
	AdGear Technologies	Digital advertising platforms	100.0
	Joyent	Cloud Services	100.0
	Samsung Next	Holding Company	100.0
	Samsung Next Fund	Technology business, Venture capital investments	100.0
	Dacor Holdings	Holding Company	100.0
	Dacor	Manufacture and sale of Home appliances	100.0
	Dacor Canada	Sale of Home appliances	100.0
	EverythingDacor.com	Sale of Home appliances	100.0
	Distinctive Appliances of California	Sale of Home appliances	100.0
	Viv Labs	Research of AI technology	100.0

¹ Ownership represents the Company's ownership of the voting rights in each entity, including subsidiaries' ownerships.

Area	Subsidiaries	Industry	Percentage of ownership ¹
America	SigMast Communications (formerly NewNet Communication Technologies (Canada))	RCS (Rich Communication Service)	100.0
	AMX Holding Corporation	Holding Company	100.0
	AMX LLC	Holding Company	100.0
	Harman Becker Automotive Systems	Manufacture and sale of audio products, R&D	100.0
	Harman Connected Services Engineering	Connected Service Provider	100.0
	Harman Connected Services Holding	Connected Service Provider	100.0
	Harman Connected Services, Inc.	Connected Service Provider	100.0
	Harman Connected Services South America	Connected Service Provider	100.0
	Harman da Amazonia Industria Electronica e Participacoes	Manufacture and sale of audio products	100.0
	Harman de Mexico S. de R.L. de C.V.	Manufacture of audio products	100.0
	Harman do Brasil Industria Electronica e Participacoes.	Sale of audio products, R&D	100.0
	Harman Financial Group	Management Company	100.0
	Harman International Industries Canada	Sale of audio products	100.0
	Harman International Industries, Inc.	Holding Company	100.0
	Harman International Mexico S de RL de CV	Sale of audio products	100.0
	Harman Investment Group, LLC	Financing Company	100.0
	Harman KG Holding, LLC	Holding Company	100.0
	Harman Professional	Sale of audio products, R&D	100.0
	Red Bend Software	Software design	100.0
	S1NN USA	R&D	100.0
	Southern Vision Systems	Development of image sensing device	100.0
	Triple Play Integration	Connected Service Provider	100.0
	Samsung Electronics Home Appliances America(SEHA)	Manufacture of home appliances	100.0
	China Materialia	Venture capital investments	98.0

¹ Ownership represents the Company's ownership of the voting rights in each entity, including subsidiaries' ownerships.

Area	Subsidiaries	Industry	Percentage of ownership ¹
Europe/CIS	Samsung Electronics (UK) (SEUK)	Sale of electronic devices	100.0
	Samsung Electronics Holding (SEHG)	Holding Company	100.0
	Samsung Semiconductor Europe GmbH (SSEG)	Sale of semiconductor/display panels	100.0
	Samsung Electronics GmbH (SEG)	Sale of electronic devices	100.0
	Samsung Electronics Iberia (SESA)	Sale of electronic devices	100.0
	Samsung Electronics France (SEF)	Sale of electronic devices	100.0
	Samsung Electronics Hungarian (SEH)	Manufacture and sale of electronic devices	100.0
	Samsung Electronics Czech and Slovak (SECZ)	Sale of electronic devices	100.0
	Samsung Electronics Italia (SEI)	Sale of electronic devices	100.0
	Samsung Electronics Europe Logistics (SELS)	Logistics	100.0
	Samsung Electronics Benelux (SEBN)	Sale of electronic devices	100.0
	Samsung Display Slovakia (SDSK)	Toll processing of display panels	100.0

¹ Ownership represents the Company's ownership of the voting rights in each entity, including subsidiaries' ownerships.

Area	Subsidiaries	Industry	Percentage of ownership ¹
Europe/CIS	Samsung Electronics Romania (SEROM)	Sale of electronic devices	100.0
	Samsung Electronics Overseas (SEO)	Sale of electronic devices	100.0
	Samsung Electronics Polska (SEPOL)	Sale of electronic devices	100.0
	Samsung Electronics Portuguesa (SEP)	Sale of electronic devices	100.0
	Samsung Electronics Nordic (SENA)	Sale of electronic devices	100.0
	Samsung Semiconductor Europe (SSEL)	Sale of semiconductor/display panels	100.0
	Samsung Electronics Austria (SEAG)	Sale of electronic devices	100.0
	Samsung Electronics Slovakia (SESK)	Manufacture of TV/monitors	100.0
	Samsung Electronics Europe Holding (SEEH)	Holding Company	100.0
	Samsung Electronics Poland Manufacturing (SEPM)	Manufacture of home appliances	100.0
	Samsung Electronics Greece (SEGR)	Sale of electronic devices	100.0
	Samsung Nanoradio Design Center (SNDK)	R&D	100.0
	Samsung Electronics Air Conditioner Europe B.V.(SEACE)	Sale of heating and cooling products	100.0
	Samsung Electronics Rus Company (SERC)	Sale of electronic devices	100.0
	Samsung Electronics Baltics (SEB)	Sale of electronic devices	100.0
	Samsung Electronics Ukraine Company (SEUC)	Sale of electronic devices	100.0
	Samsung R&D Institute Rus (SRR)	R&D	100.0
	Samsung Electronics Central Eurasia (SECE)	Sale of electronic devices	100.0
	Samsung Electronics Rus Kaluga (SERK)	Manufacture of TV	100.0
	Samsung Electronics (London) Limited (SEL)	Holding Company	100.0
	Samsung Denmark Research Center (SDRC)	R&D	100.0
	Samsung France Research Center (SFRC)	R&D	100.0
	Samsung Cambridge Solution Centre (SCSC)	R&D	100.0
	Samsung Electronics Switzerland GmbH (SESG)	Sale of electronic devices	100.0
	Samsung Electronics Caucasus (SECC)	Marketing	100.0
	Joyent (UK)	Cloud services	100.0
	Harman Connected Services OOO	Connected Service Provider	100.0
	Harman RUS CIS LLC	Sale of audio products	100.0
	Aditi Technologies Europe	Sale of audio products, etc.	100.0
	AKG Acoustics	Manufacture and sale of audio products	100.0
	AMX (Germany)	Sale of audio products	100.0
	AMX UK	Sale of audio products	100.0
	Duran Audio B.V.	Sale of audio products, R&D	100.0
	Duran Audio Iberia Espana	Sale of audio products	100.0
	Endeleo	Sale of audio products, R&D	100.0
	Harman Automotive UK	Manufacture of audio products	100.0
	Harman Becker Automotive Systems (Germany)	Manufacture and sale of audio products, R&D	100.0

¹ Ownership represents the Company's ownership of the voting rights in each entity, including subsidiaries' ownerships.

Area	Subsidiaries	Industry	Percentage of ownership ¹
Europe/CIS	Harman Becker Automotive Systems Italy	Sale of audio products	100.0
	Harman Becker Automotive Systems Manufacturing Kft	Manufacture of audio products, R&D	100.0
	Harman Belgium	Sale of audio products	100.0
	Harman Connected Services AB.	Connected Service Provider	100.0
	Harman Connected Services Finland OY	Connected Service Provider	100.0
	Harman Connected Services (Germany)	Connected Service Provider	100.0
	Harman Connected Services	Connected Service Provider	100.0
	Harman Connected Services Poland Sp.zoo	Connected Service Provider	100.0
	Harman Connected Services UK	Connected Service Provider	100.0
	Harman Consumer Division Nordic A/S	Sale of audio products	100.0
	Harman Consumer Finland OY	Sale of audio products	100.0
	Harman Consumer Nederland B.V.	Sale of audio products	100.0
	Harman Deutschland	Sale of audio products	100.0
	Harman Finance International GP S.a.r.l	Holding Company	100.0
	Harman Finance International SCA	Financing Company	100.0
	Harman France SNC	Sale of audio products	100.0
	Harman Holding & Co. KG	Management Company	100.0
	Harman Hungary Financing	Financing Company	100.0
	Harman Inc. & Co. KG	Holding Company	100.0
	Harman International Estonia OU	R&D	100.0
	Harman International Industries (UK)	Sale of audio products, etc.	100.0
	Harman International Romania SRL	R&D	100.0
	Harman International s.r.o	Manufacture of audio products	100.0
	Harman International SNC	Sale of audio products	100.0
	Harman Management	Holding Company	100.0
	Harman Professional Kft	Manufacture of audio products, R&D	100.0
	Inspiration Matters	Sale of audio products, R&D	100.0
	Knight Image	Sale of audio products, R&D	100.0
	Martin Manufacturing (UK)	Manufacture of audio products	100.0
	Harman Professional Denmark ApS (formerly Martin Professional ApS)	Sale of audio products, R&D	100.0
	Harman Professional France SAS (formerly Martin Professional France)	Sale of audio products	100.0
	Harman Professional Germany GmbH (formerly Martin Professional GmbH)	Sale of audio products	100.0
	R&D International	Manufacture of audio products	100.0
	Red Bend Software (UK)	Software design	100.0
	Red Bend Software SAS	Software design	100.0
	Studer Professional Audio	Sale of audio products, R&D	100.0
	Innoetics E.P.E.	Software development	100.0
	ARCAM	Holding Company	100.0
	A&R Cambridge	Sale of audio products	100.0

¹ Ownership represents the Company's ownership of the voting rights in each entity, including subsidiaries' ownerships.

Area	Subsidiaries	Industry	Percentage of ownership ¹
Middle East and Africa	Samsung Electronics West Africa (SEWA)	Marketing	100.0
	Samsung Electronics East Africa (SEEA)	Marketing	100.0
	Samsung Gulf Electronics (SGE)	Sale of electronic devices	100.0
	Samsung Electronics Egypt (SEEG)	Manufacture and sale of electronic devices	100.0
	Samsung Electronics Israel (SEIL)	Marketing	100.0
	Samsung Electronics Tunisia (SETN)	Marketing	100.0
	Samsung Electronics Pakistan (SEPAK)	Marketing	100.0
	Samsung Electronics South Africa (SSA)	Sale of electronic devices	100.0
	Samsung Electronics Turkey (SETK)	Sale of electronic devices	100.0
	Samsung Semiconductor Israel R&D Center (SIRC)	R&D	100.0
	Samsung Electronics Levant (SELV)	Sale of electronic devices	100.0
	Samsung Electronics Maghreb Arab (SEMAG)	Sale of electronic devices	100.0
	Samsung Electronics South Africa Production (SSAP)	Manufacture of TV/monitors	100.0
	Broadsense	Service	100.0
	Global Symphony Technology Group	Holding Company	100.0
	Harman Connected Services Morocco	Connected Service Provider	100.0
	Harman Industries Holdings Mauritius	Holding Company	100.0
	iOnRoad	R&D	100.0
	iOnRoad Technologies	R&D	100.0
	Red Bend	Manufacture of audio products	100.0
	Towersec (Israel)	R&D	100.0

¹ Ownership represents the Company's ownership of the voting rights in each entity, including subsidiaries' ownerships.

Area	Subsidiaries	Industry	Percentage of ownership ¹
Asia (Except China)	Samsung Japan (SJC)	Sale of electronic devices	100.0
	Samsung R&D Institute Japan (SRJ)	R&D	100.0
	Samsung Electronics Japan (SEJ)	Sale of electronic devices	100.0
	Samsung Electronics Display (M) (SDMA)	Manufacture of electronic devices	100.0
	Samsung Electronics (M) (SEMA)	Manufacture of home appliances	100.0
	Samsung Vina Electronics (SAVINA)	Sale of electronic devices	100.0
	Samsung Asia Private (SAPL)	Sale of electronic devices	100.0
	Samsung India Electronics (SIEL)	Manufacture and sale of electronic devices	100.0
	Samsung R&D Institute India-Bangalore (SRI-B)	R&D	100.0
	Samsung Nepal Services(SNSL)	Service	100.0
	Samsung Electronics Australia (SEAU)	Sale of electronic devices	100.0
	Samsung Electronics Indonesia (SEIN)	Manufacture and sale of electronic devices	100.0
	Samsung Telecommunications Indonesia (STIN)	Sale and service of electronic devices	100.0
	Thai Samsung Electronics (TSE)	Manufacture and sale of electronic devices	91.8
	Samsung Electronics Philippines (SEPCO)	Sale of electronic devices	100.0
	Samsung Malaysia Electronics (SME)	Sale of electronic devices	100.0
	Samsung R&D Institute Bangladesh (SRBD)	R&D	100.0
	Samsung Electronics Vietnam (SEV)	Manufacture of electronic devices	100.0
	Samsung Electronics Vietnam THAINGUYEN (SEVT)	Manufacture and sale of communication equipment	100.0
	Samsung Medison India (SMIN)	Medical equipment	100.0
	Samsung Electronics New Zealand (SENZ)	Sale of electronic devices	100.0
	Samsung Display Vietnam (SDV)	Manufacture of display panels	100.0
	Samsung Electronics HCMC CE Complex (SEHC)	Manufacture and sale of electronic devices	100.0
	Laos Samsung Electronics Sole (LSE)	Marketing	100.0
	AMX Products and Solutions Private	Sale of audio products	100.0
	Harman Connected Services India	Connected Service Provider	100.0
	Harman Connected Services Technologies	Connected Service Provider	100.0
	Harman International (India) Private	Sale of audio products, R&D	100.0
	Harman International Industries PTY	Holding Company	100.0
	Harman International Singapore	Sale of audio products	100.0
	Harman Malaysia Sdn. Bhd.	Sale of audio products	100.0
	Harman Professional Singapore Pte.	Sale of audio products	100.0
	INSP India Software Development Pvt.	Software development and supply	100.0
	Martin Professional Pte.	Sale of audio products	100.0
	Harman Connected Services Japan	Connected Service Provider	100.0
	Harman International Japan	Sale of audio products, R&D	100.0
	Red Bend Software Japan	Software design	100.0
	Studer Japan	Holding Company	100.0

¹ Ownership represents the Company's ownership of the voting rights in each entity, including subsidiaries' ownerships.

Area	Subsidiaries	Industry	Percentage of ownership ¹
China	Samsung Display Dongguan (SDD)	Manufacture of display panels	100.0
	Samsung Display Tianjin (SDT)	Manufacture of display panels	95.0
	Samsung Electronics Hong Kong (SEHK)	Sale of electronic devices	100.0
	Suzhou Samsung Electronics (SSEC)	Manufacture of home appliances	88.3
	Samsung Suzhou Electronics Export (SSEC-E)	Manufacture of home appliances	100.0
	Samsung (China) Investment (SCIC)	Sale of electronic devices	100.0
	Samsung Mobile R&D Center China-Guangzhou (SRC-Guangzhou)	R&D	100.0
	Samsung Tianjin Mobile Development Center (STMC)	R&D	100.0
	Samsung R&D Institute China-Shenzhen(SRC-Shenzhen)	R&D	100.0
	Samsung Electronics Suzhou Semiconductor (SESS)	Toll processing of semiconductor	100.0
	Samsung Electronics Huizhou (SEHZ)	Manufacture of electronic devices	99.9
	Tianjin Samsung Electronics (TSEC)	Manufacture of TV/monitors	91.2
	Samsung Electronics Taiwan (SET)	Sale of electronic devices	100.0
	Beijing Samsung Telecom R&D Center (BST)	R&D	100.0
	Tianjin Samsung Telecom Technology (TSTC)	Manufacture of communication equipment	90.0
	Shanghai Samsung Semiconductor (SSS)	Sale of semiconductor/display panels	100.0
	Samsung Electronics Suzhou Computer (SESC)	Manufacture of electronic devices	100.0
	Samsung Suzhou Module (SSM)	Toll processing of display panels	100.0
	Samsung Suzhou LCD (SSL)	Manufacture of display panels	60.0
	Shenzhen Samsung Electronics Telecommunication (SSET)	Manufacture of communication equipment	95.0
	Samsung Semiconductor (China) R&D (SSCR)	R&D	100.0
	Samsung Electronics China R&D Center (SCRC)	R&D	100.0
	Samsung (China) Semiconductor (SCS)	Manufacture of semiconductor	100.0
	Samsung Electronics (Beijing) Service (SBSC)	Services	100.0
	Tianjin Samsung LED (TSLED)	Manufacture of LED	100.0
	SEMES (Xian)	Semiconductor equipment	100.0
	Samsung Semiconductor Xian (SSCX)	Sale of semiconductor/display panels	100.0
	Harman (China) Technologies	Manufacture of audio products	100.0
	Harman (Suzhou) Audio and Infotainment Systems	Sale of audio products	100.0
	Harman Automotive Electronic Systems (Suzhou)	Manufacture of audio products, R&D	100.0
	Harman Commercial (Shanghai)	Sale of audio products	100.0
	Harman Connected Services Taiwan	Connected Service Provider	100.0
	Harman Holding	Sale of audio products	100.0
	Harman International (China) Holdings	Sale of audio products, R&D	100.0
	Harman Automotive InfoTech (Dalian) (formerly Harman Neusoft Automotive Infotech (Dalian))	Software development and supply	100.0
	Harman Technology (Shenzhen)	Sale of audio products, R&D	100.0
	Harman Connected Services Solutions (Beijing)	Connected Service Provider	100.0
	Harman Connected Services Solutions (Chengdu)	Connected Service Provider	100.0

¹ Ownership represents the Company's ownership of the voting rights in each entity, including subsidiaries' ownerships.

(B) A summary of financial data of major consolidated subsidiaries is as follows:

(1) 2017

	2017			
<i>(In millions of Korean won)</i>	Assets	Liabilities	Sales	Net Income (Loss)
Samsung Display (SDC)	50,723,199	12,243,688	30,111,427	3,352,802
Samsung Electronics America (SEA)	26,266,636	8,980,828	33,329,340	(70,198)
Harman and its subsidiaries ¹	14,676,715	5,391,669	7,103,437	209,015
Samsung (China) Investment (SCIC)	13,409,281	11,297,307	5,133,676	268,481
Samsung Display Vietnam (SDV)	10,460,755	9,495,803	18,193,257	1,257,300
Samsung Semiconductor (SSI)	10,125,193	5,376,696	28,870,987	83,113
Samsung Electronics Vietnam THAINGUYEN (SEVT)	10,044,337	2,325,221	28,323,302	3,043,210
Samsung Electronics Vietnam (SEV)	9,592,074	1,875,105	19,344,045	2,007,876
Samsung Electronics Europe Holding (SEEH)	9,474,960	7,241,753	-	18,294
Samsung (China) Semiconductor (SCS)	8,076,107	1,499,220	4,513,291	1,359,007
Samsung Electronics Huizhou (SEHZ)	6,440,564	980,064	13,521,077	689,594
Samsung India Electronics (SIEL)	6,200,351	3,406,484	10,393,877	654,408
Samsung Asia Private (SAPL)	6,138,652	481,546	1,848,905	1,293,924
Samsung Electronica da Amazonia (SEDA)	5,838,533	1,514,891	7,108,691	1,449,426
Shanghai Samsung Semiconductor (SSS)	5,169,438	4,182,832	24,198,255	236,410
Samsung Austin Semiconductor (SAS)	4,943,576	448,772	3,473,545	197,729
Thai Samsung Electronics (TSE)	2,191,519	371,379	4,594,861	164,627
Samsung Electronics Slovakia (SESK)	2,019,572	1,051,361	3,237,384	100,617
Samsung Suzhou LCD (SSL)	2,019,262	852,618	1,802,543	107,677
Samsung Electronics Europe Logistics (SELS)	1,927,321	1,810,014	14,060,511	6,718
Samsung Electronics Hungarian (SEH)	1,823,278	238,383	2,768,150	459,848
Samsung Electronics HCMC CE Complex (SEHC)	1,784,883	1,282,489	3,940,762	302,674
Samsung Electronics GmbH (SEG)	1,743,138	1,685,252	6,536,684	530
Samsung Electronics (UK) (SEUK)	1,615,723	1,094,995	4,843,150	99,024
Samsung Electronics Indonesia (SEIN)	1,562,037	1,002,213	3,025,611	77,149

¹ Consolidated financial data of an intermediate company that includes Harman International Industries, Inc. and its subsidiaries.

(2) 2016

	2016			
<i>(In millions of Korean won)</i>	Assets	Liabilities	Sales	Net Income (Loss)
Samsung Display (SDC)	43,305,405	8,361,256	24,658,814	1,498,628
Samsung Electronics America (SEA)	21,810,492	9,496,649	34,521,654	246,141
Samsung (China) Investment (SCIC)	13,632,938	11,672,755	8,792,750	298,373
Samsung (China) Semiconductor (SCS)	9,749,448	4,564,793	4,152,137	1,113,218
Samsung Electronics Vietnam (SEV)	9,134,023	1,258,948	19,426,334	2,046,280
Samsung Electronics Europe Holding (SEEH)	8,643,308	6,661,092	-	350,974
Samsung Semiconductor (SSI)	7,804,698	3,746,687	19,911,135	30,247
Samsung Electronics Vietnam THAINGUYEN (SEVT)	7,646,828	2,179,023	23,563,736	2,641,418
Samsung Electronics Huizhou (SEHZ)	6,174,579	1,106,633	12,971,475	764,426
Shanghai Samsung Semiconductor (SSS)	5,862,409	5,166,385	20,983,314	181,041
Samsung Asia Private (SAPL)	5,528,472	592,320	1,458,176	1,056,956
Samsung Electronica da Amazonia (SEDA)	5,200,799	1,510,972	6,092,245	966,821
Samsung Austin Semiconductor (SAS)	4,940,748	1,293,458	3,586,127	104,747
Samsung India Electronics (SIEL)	4,563,407	2,256,194	8,827,028	753,164
Samsung Display Vietnam (SDV)	3,165,239	3,437,791	5,230,581	(98,102)
Samsung Electronics Europe Logistics (SELS)	2,887,230	2,779,296	13,157,455	36,768
Samsung Suzhou LCD (SSL)	2,499,917	1,376,439	1,494,787	1,091
Thai Samsung Electronics (TSE)	2,079,865	364,518	4,069,078	198,980
Samsung Electronics Slovakia (SESK)	2,053,467	440,402	3,634,166	115,387
Samsung Electronics Taiwan (SET)	1,857,017	1,540,478	3,533,924	(36,178)
Samsung Electronics HCMC CE Complex (SEHC)	1,814,566	1,572,982	2,010,442	118,091
Samsung Electronics Hungarian (SEH)	1,743,979	633,975	2,441,881	89,712
Samsung Electronics GmbH (SEG)	1,621,827	1,618,305	6,257,480	2,187
Samsung Display Dongguan (SDD)	1,584,504	518,511	5,187,954	199,922
Samsung Electronics (UK) (SEUK)	1,526,879	1,103,579	4,731,464	107,243

(C) Changes in scope for consolidation

(1) Subsidiaries excluded from the consolidation for the year ended December 31, 2017:

Area	Subsidiary	Description
Domestic	Samsung Venture Capital Union #20	Liquidation
	S-Printing Solution	Sale of Business
America	Samsung Receivables(SRC)	Liquidation
	TowerSec	Merger ¹
	Kngine	Merger ²
	PrinterOn	Sale of Business
	PrinterOn America	Sale of Business
	Simpres	Sale of Business
Europe/CIS	Samsung Electronics Rus(SER)	Merger ³
	Samsung Electronics Ukraine(SEU)	Liquidation
	AMX LLC (Russia)	Liquidation
	Surfkitchen	Liquidation
	Martin Professional	Liquidation
	PrinterOn Europe	Sale of Business
Asia	I.P.S.G. International Product Solution Group PTY.	Liquidation
	VFX Systems PTY	Liquidation
China	Samsung Electronics (Shandong) Digital Printing(SSDP)	Sale of Business
	Tianjin Samsung Opto-Electronics(TSOE)	Liquidation
	Martin Trading Zhuhai	Liquidation

¹ TowerSec merged into Harman International Industries, Inc., a subsidiary of the Company, in May 2017.

² Kngine merged into Samsung Research America (SRA), a subsidiary of the Company, in October 2017.

³ Samsung Electronics Rus (SER) merged into Samsung R&D Institute Rus (SRR), a subsidiary of the Company, in January 2017.

(2) Subsidiaries newly included in the consolidation for the year ended December 31, 2017:

Area	Subsidiary	Description
Domestic	Growth type private equity trust specialized in semiconductors	Incorporation
	Harman International Korea	Acquisition of shares
	Red Bend Software Korea	Acquisition of shares
	Samsung Venture Capital Union #37	Incorporation
America	Kngine	Acquisition of shares
	AMX Holding Corporation	Acquisition of shares
	AMX LLC	Acquisition of shares
	Harman Becker Automotive Systems	Acquisition of shares
	Harman Connected Services Engineering	Acquisition of shares
	Harman Connected Services Holding	Acquisition of shares
	Harman Connected Services	Acquisition of shares
	Harman Connected Services South America	Acquisition of shares
	Harman da Amazonia Industria Electronica e Participacoes	Acquisition of shares
	Harman de Mexico S. de R.L. de C.V.	Acquisition of shares
	Harman do Brasil Industria Electronica e Participacoes.	Acquisition of shares
	Harman Financial Group	Acquisition of shares

Area	Subsidiary	Description
America	Harman International Industries Canada	Acquisition of shares
	Harman International Industries, Inc.	Acquisition of shares
	Harman International Mexico S de RL de CV	Acquisition of shares
	Harman Investment Group, LLC	Acquisition of shares
	Harman KG Holding, LLC	Acquisition of shares
	Harman Professional	Acquisition of shares
	Red Bend Software	Acquisition of shares
	S1NN USA	Acquisition of shares
	Southern Vision Systems	Acquisition of shares
	TowerSec	Acquisition of shares
	Triple Play Integration	Acquisition of shares
	Samsung Electronics Home Appliances America(SEHA)	Incorporation
	China Materialia	Incorporation
Europe/CIS	AMX LLC (Russia)	Acquisition of shares
	Harman Connected Services OOO	Acquisition of shares
	Harman RUS CIS LLC	Acquisition of shares
	Aditi Technologies Europe	Acquisition of shares
	AKG Acoustics	Acquisition of shares
	AMX (Germany)	Acquisition of shares
	AMX UK	Acquisition of shares
	Duran Audio B.V.	Acquisition of shares
	Duran Audio Iberia Espana	Acquisition of shares
	Endeleo	Acquisition of shares
	Harman Automotive UK	Acquisition of shares
	Harman Becker Automotive Systems (Germany)	Acquisition of shares
	Harman Becker Automotive Systems Italy	Acquisition of shares
	Harman Becker Automotive Systems Manufacturing Kft	Acquisition of shares
	Harman Belgium	Acquisition of shares
	Harman Connected Services AB.	Acquisition of shares
	Harman Connected Services Finland OY	Acquisition of shares
	Harman Connected Services (Germany)	Acquisition of shares
	Harman Connected Services	Acquisition of shares
	Harman Connected Services Poland Sp.zoo	Acquisition of shares
	Harman Connected Services UK	Acquisition of shares
	Harman Consumer Division Nordic A/S	Acquisition of shares
	Harman Consumer Finland OY	Acquisition of shares
	Harman Consumer Nederland B.V.	Acquisition of shares
	Harman Deutschland	Acquisition of shares
	Harman Finance International GP S.a.r.l	Acquisition of shares
	Harman Finance International SCA	Acquisition of shares
	Harman France SNC	Acquisition of shares
	Harman Holding & Co. KG	Acquisition of shares
	Harman Hungary Financing	Acquisition of shares
	Harman Inc. & Co. KG	Acquisition of shares
	Harman International Estonia OU	Acquisition of shares
	Harman International Industries (UK)	Acquisition of shares
	Harman International Romania SRL	Acquisition of shares
	Harman International s.r.o	Acquisition of shares
	Harman International SNC	Acquisition of shares

Area	Subsidiary	Description
Europe/CIS	Harman Management	Acquisition of shares
	Harman Professional Kft	Acquisition of shares
	Inspiration Matters	Acquisition of shares
	Knight Image	Acquisition of shares
	Martin Manufacturing (UK)	Acquisition of shares
	Harman Professional Denmark ApS (formerly Martin Professional ApS)	Acquisition of shares
	Harman Professional France SAS (formerly Martin Professional France)	Acquisition of shares
	Harman Professional Germany GmbH (formerly Martin Professional GmbH)	Acquisition of shares
	Martin Professional	Acquisition of shares
	R&D International	Acquisition of shares
	Red Bend Software (UK)	Acquisition of shares
	Red Bend Software SAS	Acquisition of shares
	Samsung Electronics Air Conditioner Europe B.V. (SEACE)	Incorporation
	Studer Professional Audio	Acquisition of shares
Middle East and Africa	Surfkitchen	Acquisition of shares
	Innoetics E.P.E.	Acquisition of shares
	ARCAM	Acquisition of shares
	A&R Cambridge	Acquisition of shares
	Broadsense	Acquisition of shares
	Global Symphony Technology Group	Acquisition of shares
	Harman Connected Services Morocco	Acquisition of shares
	Harman Industries Holdings Mauritius	Acquisition of shares
	iOnRoad	Acquisition of shares
	iOnRoad Technologies	Acquisition of shares
Asia (Except China)	Red Bend	Acquisition of shares
	TowerSec (Israel)	Acquisition of shares
	AMX Products and Solutions Private	Acquisition of shares
	Harman Connected Services India	Acquisition of shares
	Harman Connected Services Technologies	Acquisition of shares
	Harman International (India) Private	Acquisition of shares
	Harman International Industries PTY	Acquisition of shares
	Harman International Singapore	Acquisition of shares
	Harman Malaysia Sdn. Bhd.	Acquisition of shares
	Harman Professional Singapore Pte.	Acquisition of shares
	I.P.S.G. International Product Solution Group PTY.	Acquisition of shares
	INSP India Software Development Pvt.	Acquisition of shares
	Martin Professional Pte.	Acquisition of shares
	VFX Systems PTY	Acquisition of shares
	Harman Connected Services Japan	Acquisition of shares
	Harman International Japan	Acquisition of shares
	Red Bend Software Japan	Acquisition of shares
	Studer Japan	Acquisition of shares
China	Samsung Nepal Services (SNSL)	Incorporation
	Harman (China) Technologies	Acquisition of shares
	Harman (Suzhou) Audio and Infotainment Systems	Acquisition of shares
	Harman Automotive Electronic Systems (Suzhou)	Acquisition of shares
	Harman Commercial (Shanghai)	Acquisition of shares
	Harman Connected Services Taiwan	Acquisition of shares

Area	Subsidiary	Description
China	Harman Holding	Acquisition of shares
	Harman International (China) Holdings	Acquisition of shares
	Harman Automotive InfoTech (Dalian) (formerly Harman Neusoft automotive infotech (Dalian))	Acquisition of shares
	Harman Technology (Shenzhen)	Acquisition of shares
	Martin Trading Zhuhai	Acquisition of shares
	Harman Connected Services Solutions (Beijing)	Acquisition of shares
	Harman Connected Services Solutions (Chengdu)	Acquisition of shares

2. Summary of Significant Accounting Policies

The principal accounting policies applied in the preparation of these consolidated financial statements are set out below. These policies have been consistently applied to all the years presented, unless otherwise stated.

2.1 Basis of Presentation

The Company maintains its accounting records in Korean won and prepares statutory financial statements in the Korean language (Hangul) in accordance with *Korean IFRS*. The accompanying consolidated financial statements have been condensed, restructured and translated into English from the Korean language financial statements.

Certain information attached to the Korean language financial statements, but not required for a fair presentation of the Company's financial position, financial performance or cash flows, is not presented in the accompanying consolidated financial statements.

The consolidated financial statements of the Company presented have been prepared in accordance with *Korean IFRS*. International Financial Reporting Standards ("IFRS") have been adopted by the Korean Accounting Standards Board as *Korean IFRS* based on standards and interpretations published by the International Accounting Standards Board.

Korean IFRS permits the use of critical accounting estimates in the preparation of the financial statements and requires management judgments in applying accounting policies. The areas involving a higher degree of judgment or complexity, or areas where assumptions and estimates are significant to the consolidated financial statements, are disclosed in Note 3.

2.2 Changes in Accounting Policy and Disclosures

(A) New and amended standards adopted by the Company

The Company applied the following amended and enacted standards for the annual period beginning on January 1, 2017:

Amendment to Korean IFRS 1007, Statement of Cash Flows

Amendments to *Korean IFRS 1007 Statement of Cash flows* require disclosures that enable users of financial statements to evaluate changes in liabilities arising from financing activities, including both changes arising from cash flows and non-cash changes.

(B) New and amended standards not adopted by the Company

New standards, amendments and interpretations issued but not effective for the annual period beginning on January 1, 2017, and not early adopted by the Company, are as follows:

Korean IFRIC 2122, Foreign Currency Transaction and Advance Consideration

According to these enactments, the date of the transaction for the purpose of determining the exchange rate to use on initial recognition of the related asset, expense or income (or part of it) is the date on which an entity initially recognizes the non-monetary asset or non-monetary liability arising from the payment or receipt of advance consideration. If there are multiple payments or receipts in advance, the entity shall determine a date of the transaction for each payment or receipt of advance consideration. These enactments will be effective for annual periods beginning on or after January 1, 2018, with early adoption permitted. The Company does not expect the enactments to have a significant impact on the financial statements.

Korean IFRS 1109, Financial Instruments

The new standard for financial instruments issued on September 25, 2015 is effective for annual periods beginning on or after January 1, 2018 with early application permitted. This standard will replace *Korean IFRS 1039 Financial Instruments: Recognition and Measurement*. The Company will apply the standard for the annual periods beginning on or after January 1, 2018.

The standard requires retrospective application with some exceptions. For example, the entity is not required to restate prior periods

in relation to classification, measurement and impairment of financial instruments. The standard requires prospective application of its hedge accounting requirements for all hedging relationships except the accounting for time value of options and other exceptions.

Korean IFRS 1109 Financial Instruments requires all financial assets to be classified and measured on the basis of the entity's business model for managing financial assets and the contractual cash flow characteristics of the financial assets. A new impairment model, an expected credit loss model, is introduced and any subsequent changes in expected credit losses will be recognized in profit or loss. Also, hedge accounting rules were amended to extend the hedging relationship, which consists only of eligible hedging instruments and hedged items, for determining what qualifies for hedge accounting.

An effective implementation of *Korean IFRS 1109* requires preparation processes including financial impact assessment, accounting policy establishment, accounting system development and system stabilization. The impact on the Company's financial statements due to the application of the standard is dependent on judgements made in applying the standard, financial instruments held by the Company and macroeconomic variables.

With the implementation of *Korea IFRS 1109*, the Company prepared for changes to internal management processes and completed modification of its accounting system for financial instruments reporting. The Company performed an impact assessment to identify potential financial impacts of applying Korean IFRS 1109. The assessment was performed based on available information as at December 31, 2017, and the results of the assessment are explained as below.

(a) Classification and Measurement of Financial Assets

When implementing *Korean IFRS 1109*, the classification of financial assets will be driven by the Company's business model for managing the financial assets and contractual terms of cash flow. The following table shows the classification of financial assets measured subsequently at amortized cost, at fair value through other comprehensive income and at fair value through profit or loss. If a hybrid contract contains a host that is a financial asset, the classification of the hybrid contract shall be determined for the entire contract without separating the embedded derivative.

Business model for the contractual cash flows characteristics	Solely represent payments of principal and interest	All other
Hold the financial asset for the collection of the contractual cash flows	Measured at amortized cost ¹	
Hold the financial asset for the collection of the contractual cash flows and trading	Recognized at fair value through other comprehensive income ¹	Recognized at fair value through profit or loss ²
Hold the financial asset for trading	Recognized at fair value through profit or loss	

¹ A designation at fair value through profit or loss is allowed only if such designation mitigates an accounting mismatch (irrevocable).

² A designation at fair value through other comprehensive income is allowed only if the financial instrument is an equity investment not held for trading (irrevocable).

With the implementation of *Korean IFRS 1109*, the criteria to classify the financial assets at amortized cost or at fair value through other comprehensive income are more strictly applied than the criteria applied with *Korean IFRS 1039*. Accordingly, the financial assets at fair value through profit or loss may increase by implementing *Korean IFRS 1109* and may result an extended fluctuation in profit or loss.

As at December 31, 2017, the Company owns loan and trade receivables amounting to ₩113,901,548 million, financial assets held-to-maturity amounting to ₩106,751 million, and financial assets available-for-sale amounting to ₩10,943,555 million.

According to *Korean IFRS 1109*, debt investments are measured at amortized cost if: a) the objective of the business model is to hold the financial asset for the collection of the contractual cash flows, and b) the contractual terms of cash flows solely represent payments of principal and interest. As at December 31, 2017, the Company measured loan and trade receivables amounting to ₩113,901,548 million and financial assets held-to-maturity amounting to ₩106,751 million at amortized costs.

When applying *Korean IFRS 1109* to the above financial assets as at December 31, 2017, as most of the financial assets are held within a business model with the objective to hold the financial assets in order to collect contractual cash flows and the contractual terms of the financial asset give rise on specified dates to cash flows that are solely payments of principal and interest on the principal amount outstanding, the Company shall classify the financial assets as subsequently measured at amortized cost. It is expected that the financial impact of this change on the financial statements will be immaterial.

Korean IFRS 1109 measures debt investments at fair value through other comprehensive income for which terms of cash flows solely represent payment of the principal and interest on a due date, where the purpose of holding the debt investment is to collect contractual cash flows and trade. As at December 31, 2017, the Company has debt investments classified as available-for-sale financial assets amounting to ₩3,305,850 million

As at December 31, 2017, based on the result from the impact assessment of *Korean IFRS 1109*, it is expected that the financial impact on the financial statements will be immaterial, because debt instruments classified as available-for-sale financial assets will be classified as amortized cost.

According to *Korean IFRS 1109*, equity investments not being held for trading may be given an irrevocable election to be classified as financial instruments measured at fair value through other comprehensive income at initial recognition and the cumulative gain or loss previously recognized in other comprehensive income is not subsequently recycled from equity to profit or loss as a reclassification adjustment. As at December 31, 2017, the Company's equity investments classified as available-for-sale financial assets amount to ₩7,637,705 million and the cumulative unrealized profit or loss on available-for-sale equity investments

amounting to ₩3,199 million was recycled from equity to profit or loss as a reclassification adjustment for the year ended December 31, 2017.

The Company plans to designate long-term equity instruments held for long-term investment purposes, which account for most of the available-for-sale equity instruments, as financial instruments measured at fair value through other comprehensive income. Therefore, the Company expects the application of *Korean IFRS 1109* on these financial assets will not have a material impact on the financial statements. However, the Company has identified that some available-for-sale equity instruments will have increased profit or loss variability, as they are classified as fair value through profit or loss.

According to *Korean IFRS 1109*, debt investments for which the contractual terms of cash flows are not solely representing payments of principal and interest, or which are held for trading, are classified as fair value through profit or loss. Equity investments not designated at fair value through comprehensive income are also measured at fair value through profit or loss. As at December 31, 2017, the Company did not have debt or equity investments classified as financial instruments at fair value through profit or loss.

(b) Classification and Measurement of Financial Liabilities

Korean IFRS 1109 requires the change in the liability's fair value attributable to changes in the credit risk to be recognized in other comprehensive income, unless this treatment of the credit risk component creates or enlarges a measurement mismatch. Amounts presented in other comprehensive income are not subsequently transferred to profit or loss.

Under *Korean IFRS 1039*, all the fair value movements for financial liabilities designated to be measured at fair value through profit or loss, were recognized through profit or loss. However, under *Korean IFRS 1109*, certain fair value movements will be recognized in other comprehensive income thus profit or loss from fair value movements may decrease.

As at December 31, 2017, total financial liabilities account for ₩52,705,258 million of which ₩525,579 million are designated to be measured at fair value through profit or loss.

According to the result of the impact assessment, financial liabilities measured at fair value through profit and loss for the current terms as at December 31, 2017 have mostly short maturities and the credit risk fluctuation of financial liabilities is insignificant. Therefore, it is expected that the impact of adopting *Korean IFRS 1109* will not be significant.

(c) Impairment: Financial Assets and Contract Assets

The new impairment model requires the recognition of impairment provisions based on expected credit losses (ECL) rather than only incurred credit losses as is the case under *Korean IFRS 1039*. It applies to financial assets classified at amortized cost, debt instruments measured at fair value through other comprehensive income, lease receivables, contract assets, loan commitments and certain financial guarantee contracts.

Under *Korean IFRS 1109*, a credit event (or impairment 'trigger') no longer has to occur before credit losses are recognized. The Company will always recognize (at a minimum) 12-month expected credit losses in profit or loss. Lifetime expected credit losses will be recognized on assets for which there is a significant increase in credit risk after initial recognition.

Stage ¹		Loss allowance
1	No significant increase in credit risk after initial recognition ²	12-month expected credit losses (expected credit losses that result from those default events on the financial instruments that are possible within 12 months after the reporting date)
2	Significant increase in credit risk after initial recognition	Lifetime expected credit losses (expected credit losses that result from all possible default events over the life of the financial instruments)
3	Objective evidence of credit-impaired	

¹ The Company shall measure the loss allowance at an amount equal to Lifetime expected credit losses for contract assets or trade receivables under the standard, *Korean IFRS 1115 Revenue from Contracts with Customers*, which do not contain a significant financing component. However, the Company can elect to measure the loss allowance at an amount equal to Lifetime expected credit losses for all contract assets or all trade receivables which contain a significant financing component in accordance with *Korean IFRS 1115*. The Company can also elect to measure the loss allowance at an amount equal to Lifetime expected credit losses for lease receivables.

² If the financial instrument has low credit risk at the reporting date, the Company may assume that the credit risk has not increased significantly since initial recognition.

Under *Korean IFRS 1109*, an asset that is credit-impaired at initial recognition would recognize all changes in lifetime expected credit losses since the initial recognition as a loss allowance with any changes recognized in profit or loss.

As at December 31, 2017, the Company owns debt investments measured at amortized cost of ₩114,673,402 million (loans and receivables of ₩114,566,651 million, financial assets held-to-maturity of ₩106,751 million), which is classified as available-for-sale and measured at fair value through other comprehensive income, of ₩3,305,850 million. The Company has recognized a loss allowance of ₩665,103 million for these assets.

When applying *Korean IFRS 1109*, the Company will measure the loss allowance at an amount equal to lifetime expected credit losses for trade receivables which contain a significant financing component, contract assets, and lease receivables, and the Company may use a simplified approach which assumes the credit risk on a financial instrument has not increased significantly since initial recognition if the financial instrument is determined to have low credit risk at the reporting date. It is expected that the impact of adopting *Korean IFRS 1109* will not be significant.

(d) Hedge Accounting

Hedge accounting mechanics (fair value hedges, cash flow hedges and hedges of net investments in foreign operations) required by *Korean IFRS 1039* remains unchanged in *Korean IFRS 1109*, however, the new hedge accounting rules will align the accounting for hedging instruments more closely with the Company's risk management practices. As a general rule, more hedge relationships may be eligible for hedge accounting, as the standard introduces a more principles-based approach. *Korean IFRS 1109* allows more hedging instruments and hedged items to qualify for hedge accounting, and relaxes the hedge accounting requirement by removing two hedge effectiveness tests: the prospective test to ensure that the hedging relationship is expected to be highly effective and the quantitative retrospective test (within range of 80~125%) to ensure that the hedging relationship has been highly effective throughout the reporting period.

With the implementation of *Korean IFRS 1109*, volatility in profit or loss may be reduced as some items that were not eligible as hedged items or hedging instruments under *Korean IFRS 1039* are now eligible under *Korean IFRS 1109*.

When the Company first applies *Korean IFRS 1109*, it may choose its accounting policy to continue to apply all of the hedge accounting requirements of *Korean IFRS 1039* instead of the requirements of *Korean IFRS 1109*.

With the implementation of *Korean IFRS 1109*, the Company has aligned the application of hedge accounting requirements more closely with risk management practices. It is expected that the impact of adopting *Korean IFRS 1109* will not be significant.

Korean IFRS 1115, Revenue from Contracts with Customers

Korean IFRS 1115, Revenue from Contracts with Customers issued on November 6, 2015 is effective for annual periods beginning on or after January 1, 2018 with early adoption permitted. This standard replaces *Korean IFRS 1018 Revenue*, *Korean IFRS 1011 Construction Contracts*, *Interpretation 2031 Revenue-Barter Transactions Involving Advertising Services*, *Interpretation 2113 Customer Loyalty Programs*, *Interpretation 2115 Agreements for the Construction of Real Estate* and *Interpretation 2118 Transfers of assets from customers*.

The Company will apply *Korean IFRS 1115* to the annual period beginning January 1, 2018 and will recognize the cumulative

impact of initially applying the standard as an adjustment to retained earnings (or other component of equity, as appropriate) as at January 1, 2018, the period of initial application. Under this transition method, the Company will elect to apply this standard retrospectively only to contracts that are not completed at the date of initial application.

Korean IFRS 1018 and other current revenue standards identify revenue as income that arises in the course of ordinary activities of an entity and provides guidance on a variety of different types of revenue, such as sale of goods, rendering of services, interest, dividends, royalties and construction contracts. However, the Company shall recognize revenue in accordance with the new standard by applying the following 5 steps: ①Identify the contract with the customer, ②Identify the separate performance obligations, ③Determine the transaction price of the contract, ④Allocate the transaction price to each of the separate performance obligations, and ⑤Recognize the revenue as each performance obligation is satisfied.

As at December 31, 2017, for the preparation of implementing *Korean IFRS 1115*, the Company formed a task force team which consists of members of the accounting department, the accounting firm which serves as an external advisor, and also other practical departments such as the IT department, if necessary. An assessment was performed, by revenue stream, and the Company has subsequently aligned certain internal management processes and modified part of its accounting system based on the results of the assessment.

Korean IFRS 1115 will affect overall business practices including accounting treatment, sales strategy and operations. Therefore, the Company provides training for changes due to the application of the new standard to employees, and periodically reports the plans and progress of the implementation to the management team.

(a) Identification of performance obligations

The Company exports various products and merchandise in accordance with Incoterms Group C trading conditions (Incoterms CIF, etc.). According to *Korean IFRS 1115*, since the seller provides the shipping service after control of the product or merchandise has been transferred to the customer, the shipping service (including insurance) must be recognized as a separate performance obligation.

As a result of analyzing Incoterms Group C trading condition (Incoterms CIF, etc.) contracts, the Company considers it appropriate to recognize the transportation service after the control has been transferred as a separate performance obligation apart from the sale of the goods. The adoption of *Korean IFRS 1115* for these arrangements is not expected to have a material impact on the financial statements.

(b) A performance obligation is satisfied over time: CE Operating Segment

The CE segment of the Company delivers products to customers and installs them in accordance with the system air conditioner contracts awarded by the Public Procurement Service. According to *Korean IFRS 1115*, the Company can recognize revenue over time if the Company's performance creates or enhances an asset (for example, work in progress) that the customer controls as the asset is created or enhanced

As a result of the analysis of the contract terms, the Company considers that it should recognize the revenue from system air conditioner installation over time as the customer controls the outcome of the service. The adoption of *Korean IFRS 1115* for these arrangements is not expected to have a material impact on the financial statements.

(c) A performance obligation is satisfied over time - custom-order semiconductor supply contracts: DS Operating Segment

The DS segment of the Company supplies semiconductor products according to customer orders. According to *Korean IFRS 1115*, the Company can recognize revenue over time if the Company's performance does not create an asset with an alternative use to the entity and the entity has an enforceable right to payment for performance completed to date.

As a result of analyzing the agreement terms, the Company considers the revenue from custom-order semiconductor supply contracts with no alternative use and an enforceable right to payment should be recognized over time. The adoption of *Korean IFRS 1115* is not expected to have a material impact on the financial statements.

(d) Variable consideration

According to *Korean IFRS 1115*, the Company shall recognize a refund liability if the entity receives consideration from a customer and expects to refund some or all of the consideration to the customer. The Company recognizes an asset (refund asset) for the Company's right to recover the products from the customer when the refund liability is paid.

The Company considers that the refund asset should be classified and presented separately from the refund liability. The adoption of *Korean IFRS 1115* is not expected to have a material impact on the financial statements.

Enactment of Korean IFRS 1116, Leases

Korean IFRS 1116 Leases issued on May 22, 2017 is effective for annual periods beginning on or after January 1, 2019, with early adoption permitted. This standard will replace *Korean IFRS 1017 Leases*, *Interpretation 2104 Determining whether an Arrangement contains a Lease*, *Interpretation 2015 Operating Leases-Incentives*, and *Interpretation 2027 Evaluating the Substance of Transactions Involving the Legal Form of a Lease*.

At inception of a contract, the Company shall assess whether the contract is, or contains, a lease. Also, at the date of initial application, the Company shall assess whether the contract is, or contains, a lease in accordance with the standard. However, the Company will not need to reassess all contracts with applying the practical expedient because the Company elected to apply the practical expedient only to contracts entered before the date of initial application.

For a contract that is, or contains, a lease, the Company shall account for each lease component within the contract as a lease separately from non-lease components of the contract. A lessee is required to recognize a right-of-use asset representing its right to use the underlying leased asset and a lease liability representing its obligation to make lease payments. The lessee may elect not to apply the requirements to short-term leases (a lease term of 12 months or less at the commencement date) and low value assets (e.g. underlying assets below \$5,000). In addition, as a practical expedient, the lessee may elect, by class of underlying asset, not to separate non-lease components from lease components, and instead account for each lease component and any associated non-lease components as a single lease component.

The accounting standard as a lessor did not change significantly from *Korean IFRS 1017 Leases*.

(a) Lessee accounting

A lessee shall apply this standard to its leases either: a) retrospectively to each prior reporting period presented applying *Korean IFRS 1008 Accounting Policies*, *Changes in Accounting Estimates and Errors* (Full retrospective application) or b) retrospectively with the cumulative effect of initially applying the standard recognized at the date of initial application. The Company has not yet elected an adoption method.

The Company is in the process of performing a preliminary assessment of potential financial impact of applying *Korean IFRS 1116* to the 2017 financial statements.

(b) Lessor accounting

When applying *Korean IFRS 1116*, the Company does not expect the enactments to have a significant impact on the financial statements as the accounting standard as a lessor did not change significantly from *IFRS 1017*.

2.3 Consolidation

The Company prepares the consolidated financial statements in accordance with *Korean IFRS 1110, Consolidated Financial Statements*.

(A) Subsidiaries

Subsidiaries are all entities (including special purpose entities) over which the Company has control. The Company controls the corresponding investee when it is exposed, or has rights, to variable returns from its involvement with the investee and has the ability to affect those returns through its power over the investee. Consolidation of a subsidiary begins from the date the Company obtains control of a subsidiary and ceases when the Company loses control of the subsidiary.

The Company applies the acquisition method to account for business combinations. The consideration transferred is measured at the fair values of the assets transferred, and identifiable assets acquired and liabilities and contingent liabilities assumed in a business combination are initially measured at their fair values at the acquisition date. The Company recognizes any non-controlling interest in the acquiree on an acquisition-by-acquisition basis in the event of liquidation at the non-controlling interest's proportionate share of the recognized amounts of acquiree's identifiable net assets. Acquisition-related costs are expensed as incurred.

Goodwill is recognized as the excess of (1) the aggregate of i) the consideration transferred, ii) the amount of any non-controlling interest in the acquiree and iii) the acquisition-date fair value of the Company's previously held equity interest in the acquiree over (2) the net identifiable assets acquired. If this consideration (1) is lower than the fair value of the acquiree's net assets in (2), the difference is recognized in profit or loss.

Balances of receivables and payables, income and expenses and unrealized gains or losses on transactions between the Company subsidiaries are eliminated. Accounting policies of subsidiaries are changed where necessary to ensure consistency with the policies adopted by the Company.

(B) Changes in ownership interests in subsidiaries without change of control

Transactions with non-controlling interests that do not result in loss of control are accounted for as equity transactions – that is, as transactions with the owners in their capacity as owners. The difference between fair value of any consideration paid and the relevant share acquired of the carrying value of net assets of the subsidiary is recorded in equity. Gains or losses on disposals of non-controlling interests are also recorded in equity.

(C) Disposal of subsidiaries

If the Company loses control of a subsidiary, any investment continuously retained in the subsidiary is re-measured at its fair value at the date when control is lost and any resulting differences are recognized in profit or loss. Such fair value becomes the initial carrying amount for the subsequent measurement of the retained interest accounted for as an associate, joint venture, or financial asset. In addition, any amounts previously recognized in other comprehensive income in respect of such entity are accounted for as if the Company had directly disposed of the related assets or liabilities. As a result, the previously recognized other comprehensive income are reclassified into profit or loss.

(D) Non-controlling interests

Each component of profit or loss and other comprehensive income is attributed to owners of the parent and to non-controlling interests. Total comprehensive income is attributed to owners of the parent and to non-controlling interests even if this results in a negative balance of non-controlling interests.

(E) Associates

Associates are all entities over which the Company has significant influence but does not have control, generally investees of which from 20% to 50% of voting stock is owned by the Company. Investments in associates are initially recognized at acquisition cost

using the equity method. Unrealized gains on transactions between the Company and its associates are eliminated to the extent of the Company's interest in the associates. If there is any objective evidence that the investment in the associate is impaired, the Company recognizes the difference between the recoverable amount of the associate and its book value as impairment loss.

(F) Joint arrangements

A joint arrangement of which two or more parties have joint control is classified as either a joint operation or a joint venture. A joint operator has rights to the assets, and obligations for the liabilities, relating to the joint operation and recognizes the assets, liabilities, revenues and expenses relating to its interest in a joint operation. A joint venturer has rights to the net assets relating to the joint venture and accounts for that investment using the equity method.

2.4 Foreign Currency Translation

(A) Functional and presentation currency

Items included in the financial statements of each of the Company's entities are measured using the currency of the primary economic environment in which each entity operates (the "functional currency"). The consolidated financial statements are presented in Korean won, which is the parent company's functional and presentation currency.

(B) Transactions and balances

Foreign currency transactions are translated into the functional currency using the exchange rates prevailing at the dates of the transactions or valuation where items are re-measured. Foreign exchange gains and losses resulting from the settlement of such transactions and from the translation at year-end exchange rates of monetary assets and liabilities denominated in foreign currencies are recognized in profit or loss.

Exchange differences arising on non-monetary financial assets and liabilities such as equity instruments at fair value through profit or loss and available-for-sale equity instruments are recognized in profit or loss and other comprehensive income, respectively, as part of the fair value gain or loss.

(C) Translation into the presentation currency

The results and financial position of all the foreign entities that have a functional currency different from the presentation currency of the Company are translated into the presentation currency as follows:

- Assets and liabilities for each statement of financial position presented are translated at the closing rate at the end of the reporting date.
- Income and expenses for each statement of income are translated at average exchange rates, unless this average is not a reasonable approximation of the cumulative effect of the rates prevailing on the transaction dates, in which case income and expenses are translated at the rate on the dates of the transactions.
- All resulting exchange differences are recognized in other comprehensive income.

2.5 Cash and Cash Equivalents

Cash and cash equivalents include cash on hand, deposits held at call with banks, and other short-term highly liquid investments that are readily convertible to a known amount of cash and are subject to an insignificant risk of change in value.

2.6 Financial Assets

(A) Classification

The Company classifies its financial assets in the following categories: financial assets at fair value through profit or loss, loans and receivables, available-for-sale financial assets, and held-to-maturity financial assets. The classification depends on the terms of the instruments and purpose for which the financial assets were acquired. Management determines the classification of its financial assets at initial recognition.

(1) Financial assets at fair value through profit or loss

Financial assets at fair value through profit or loss are financial assets held for trading. A financial asset is classified in this category if acquired principally for the purpose of selling in the short-term. Derivatives not subject to hedge accounting and derivatives separated from financial instruments, such as embedded derivatives, are also categorized as held for trading. Assets in this category are classified as current assets.

(2) Loans and receivables

Loans and receivables are non-derivative financial assets with fixed or determinable payments that are not quoted in an active market. They are included in current assets, except for those with maturities greater than 12 months after the end of the reporting period which are classified as non-current assets.

(3) Available-for-sale financial assets

Available-for-sale financial assets are non-derivative financial assets that are either designated in this category or not classified in any of the other categories. They are included in non-current assets unless an investment matures or management intends to dispose of it within 12 months of the end of the reporting period.

(B) Recognition and measurement

Regular purchases and sales of financial assets are recognized on the trade date. At initial recognition, financial assets are measured at fair value plus, in the case of financial assets not carried at fair value through profit or loss, transaction costs. Transaction costs of financial assets carried at fair value through profit or loss are expensed in the statement of income. After the initial recognition, available-for-sale financial assets and financial assets at fair value through profit or loss are subsequently carried at fair value. Loans and receivables and held-to-maturity investments are subsequently carried at amortized cost using the effective interest method.

Changes in fair value of financial assets at fair value through profit or loss are recognized in profit or loss and changes in fair value of available-for-sale financial assets are recognized in other comprehensive income. When the available-for-sale financial assets are sold or impaired, the fair value adjustments recorded in equity are reclassified into profit or loss.

Interest on available-for-sale financial assets and held-to-maturity financial assets calculated using the effective interest method is recognized in the statement of income as part of financial income. Dividends on available-for-sale financial assets are recognized in the statement of income as part of other non-operating income when the Company's right to receive payments is established.

(C) Offsetting financial instruments

Financial assets and liabilities are offset, and the net amount reported in the statement of financial position, when there is a legally enforceable right to offset the recognized amounts and there is an intention to settle on a net basis, or realize the asset and settle the liability simultaneously. The legally enforceable right must not be contingent on future events and must be enforceable in the normal course of business and in the event of default, insolvency or bankruptcy of the Company or the counterparty.

(D) Derecognition of financial assets

If the Company transfers a financial asset and the transfer does not result in derecognition because the Company has retained substantially all of the risks and rewards of ownership of the transferred asset due to a recourse in the event the debtor defaults, the Company continues to recognize the transferred asset in its entirety and recognizes a financial liability for the consideration received. The related financial liability is classified as 'borrowings' in the statement of financial position.

2.7 Impairment of Financial Assets

The Company assesses at the end of each reporting period whether there is objective evidence that a financial asset or group of financial assets is impaired. A financial asset or a group of financial assets is impaired and impairment loss is recognized only if there is objective evidence and that loss event (or events) has an impact on the estimated future cash flows of the financial asset or group of financial assets that can be reliably estimated.

Impairment of loans and receivables is presented as a deduction in an allowance account. Impairment of other financial assets is directly deducted from their carrying amount. The Company writes off financial assets when the assets are determined to no longer be recoverable.

The objective evidence that a financial asset is impaired includes significant financial difficulty of the issuer or obligor, a delinquency in interest or principal payments, or the disappearance of an active market for that financial asset because of financial difficulties. A significant and prolonged decline below its cost in the fair value of an available-for-sale equity instrument is also objective evidence of impairment.

2.8 Trade Receivables

Trade receivables are amounts due from customers for merchandise sold or services performed in the ordinary course of business. If collection is expected in one year or less, they are classified as current assets. If collection is expected beyond one year, they are presented as non-current assets. Trade receivables are recognized initially at fair value and subsequently measured at amortized cost using the effective interest method, less provision for impairment.

2.9 Inventories

Inventories are stated at the lower of cost and net realizable value. Cost is determined using the average cost method, except for materials in transit. The cost of finished goods and work in progress comprises raw materials, direct labor, other direct costs and related production overheads (based on normal operating capacity). It excludes costs of idle plant and abnormal waste. Net realizable value is the estimated selling price in the ordinary course of business, less applicable variable selling expenses.

Inventories are reduced for the estimated losses arising from excess, obsolescence, and decline in value. This reduction is determined by estimating market value based on future customer demand. The losses on inventory obsolescence are recorded as a part of cost of sales.

2.10 Disposal Group Held-for-Sale

Non-current assets (or disposal groups) are classified as assets held-for-sale when their carrying amount is to be recovered principally through a sale transaction and a sale is considered highly probable. The assets are measured at the lower of their carrying amount and the fair value less costs to sell.

2.11 Property, Plant and Equipment

Property, plant and equipment are stated at cost less accumulated depreciation and accumulated impairment losses. Historical cost includes expenditures that are directly attributable to the acquisition of the items. Subsequent costs are included in the asset’s carrying amount or recognized as a separate asset, as appropriate, only when it is probable that future economic benefits associated with the item will flow to the Company and the cost of the item can be measured reliably. The carrying amount of those parts that are replaced is derecognized and repairs and maintenance expenses are recognized in profit or loss in the period they are incurred.

Depreciation on tangible assets is calculated using the straight-line method to allocate the difference between their cost and their residual values over their estimated useful lives. Land is not depreciated. Costs that are directly attributable to the acquisition, construction or production of a qualifying asset, including capitalized interest costs, form part of the cost of that asset and are amortized over the estimated useful lives.

The Company’s policy is that property, plant and equipment should be depreciated over the following estimated useful lives:

	<u>Estimated useful lives</u>
Buildings and structures	15, 30 years
Machinery and equipment	5 years
Other	5 years

The depreciation method, residual values and useful lives of property, plant and equipment are reviewed, and adjusted if appropriate, at the end of each reporting period. An asset’s carrying amount is written down immediately to its recoverable amount if the asset’s carrying amount is greater than its estimated recoverable amount. Gains and losses on disposals are determined by comparing the proceeds with the carrying amount and are recognized within the statement of income as part of other non-operating income and expenses.

2.12 Borrowing Costs

General and specific borrowing costs that are directly attributable to the acquisition, construction or production of a qualifying asset are capitalized during the period of time that is required to complete and prepare the asset for its intended use or sale. Investment

income earned on the temporary investment of specific borrowings pending their expenditure on qualifying assets is deducted from the borrowing costs eligible for capitalization. Other borrowing costs are expensed in the period in which they are incurred.

2.13 Intangible Assets

Goodwill represents the excess of the cost of an acquisition over the fair value of the Company’s share of the net identifiable assets of the acquired subsidiary, associates, joint ventures and businesses at the date of acquisition. Goodwill on acquisitions of subsidiaries and businesses is included in intangible assets and goodwill on acquisition of associates and joint ventures is included in the investments in associates and joint ventures.

Intangible assets, except for goodwill, are initially recognized at their historical cost and carried at cost less accumulated amortization and accumulated impairment losses.

Internally generated development costs are the aggregate costs recognized after meeting the asset recognition criteria, including technical feasibility, and determined to have future economic benefits. Membership rights and certain trademarks are regarded as intangible assets with an indefinite useful life and not amortized because there is no foreseeable limit to the period over which the assets are expected to be utilized. Intangible assets with definite useful lives such as trademarks and licenses, are amortized using the straight-line method over their estimated useful lives.

The Company’s policy is that intangible assets should be amortized over the following estimated useful lives:

	<u>Estimated useful lives</u>
Development costs	2 years
Trademarks, licenses and other intangible assets	5 - 10 years

2.14 Impairment of Non-Financial Assets

Goodwill or intangible assets with indefinite useful lives are not subject to amortization and are tested annually for impairment. Assets that are subject to amortization are reviewed for impairment whenever events or changes in circumstances indicate that the carrying amount may not be recoverable. An impairment loss is recognized for the amount by which the asset’s carrying amount exceeds its recoverable amount. The recoverable amount is the higher of an asset’s fair value less costs to sell and value in use. For the purposes of assessing impairment, assets are grouped at the lowest levels for which there are separately identifiable cash flows (cash-generating units). Non-financial assets other than goodwill for which an impairment charge was previously recorded are reviewed for possible reversal of the impairment at each reporting date.

2.15 Financial Liabilities

(A) Classification and measurement

Financial liabilities at fair value through profit or loss are financial instruments held for trading. Financial liabilities are classified in this category if incurred principally for the purpose of repurchasing them in the near term. Derivatives that are not designated as hedges or bifurcated from financial instruments containing embedded derivatives are also categorized as held-for-trading.

The Company classifies non-derivative financial liabilities, except for financial liabilities at fair value through profit or loss, financial guarantee contracts and financial liabilities that arise when a transfer of financial assets does not qualify for derecognition, as financial liabilities carried at amortized cost and presented as 'trade payables', 'borrowings', and 'other financial liabilities' in the statement of financial position.

(B) Derecognition

Financial liabilities are removed from the statement of financial position when they are extinguished, for example, when the obligation specified in the contract is discharged, cancelled or expires or when the terms of an existing financial liability are substantially modified.

2.16 Trade Payables

Trade payables are amounts due to suppliers for merchandise purchased or services received in the ordinary course of business. If payment is expected in one year or less, they are classified as current liabilities. If payment is expected beyond one year, they are presented as non-current liabilities. Non-current trade payables are recognized initially at fair value and subsequently measured at amortized cost using the effective interest method.

2.17 Borrowings

Borrowings are recognized initially at fair value, net of transaction costs, and are subsequently measured at amortized cost. Any difference between cost and the redemption value is recognized in the statement of income over the period of the borrowings using the effective interest method. If the Company has an indefinite right to defer payment for a period longer than 12 months after the end of the reporting date, such liabilities are recorded as non-current liabilities, otherwise, they are recorded as current liabilities.

2.18 Provisions

A provision is recognized when the Company has a present legal or constructive obligation as a result of a past event, it is probable that an outflow of resources embodying economic benefits will be required to settle the obligation, and a reliable estimate can be made of the amount of the obligation. Provisions are not recognized for future operating losses.

Provisions are measured at the present value of the expenditures expected to be required to settle the obligation using a pre-tax rate that reflects current market assessments of the time value of money and the risks specific to the obligation. The increase in the provision due to passage of time is recognized as interest expense.

When it is probable that an outflow of economic benefits will occur due to a present obligation resulting from a past event, and the amount is reasonably estimable, a corresponding provision is recognized in the financial statements. However, when such outflow is dependent upon a future event that is not certain to occur, or cannot be reliably estimated, a disclosure regarding the contingent liability is made in the notes to the financial statements.

2.19 Employee Benefits

The Company has a variety of retirement pension plans including defined benefit and defined contribution plans. A defined contribution plan is a pension plan under which the Company pays fixed contributions into a separate entity. The Company has no legal or constructive obligations to pay further contributions if the fund does not hold sufficient assets to pay all employees the benefits relating to employee service in the current and prior periods. For defined contribution plans, the Company pays contributions to annuity plans that are managed either publicly or privately on a mandatory, contractual or voluntary basis. The Company has no further future payment obligations once the contributions have been paid. The contributions are recognized as employee benefit expense when they are due. Prepaid contributions are recognized as an asset to the extent that a cash refund or a reduction in the future payments is available.

A defined benefit plan is a pension plan that is not a defined contribution plan. Typically defined benefit plans define an amount of pension benefit that an employee will receive on retirement, usually dependent on one or more factors such as age, years of service and compensation. The liability recognized in the statement of financial position in respect to defined benefit pension plans is the present value of the defined benefit obligation at the end of the reporting period less the fair value of plan assets. The defined benefit obligation is calculated annually by independent actuaries using the projected unit credit method. The present value of the defined benefit obligation is determined by discounting the estimated future cash outflows using interest rates of high-quality corporate bonds that are denominated in the currency in which the benefits will be paid and that have terms to maturity approximating to the terms of the related pension obligation.

Actuarial gains and losses resulting from the changes in actuarial assumptions, and the differences between the previous actuarial assumptions and what has actually occurred, are recognized in other comprehensive income in the period in which they were incurred. Past service costs are immediately recognized in profit or loss.

2.20 Financial Guarantee Contract

Financial guarantee contracts are contracts that require the issuer to make specified payments to reimburse the holder for a loss it incurs because a specified debtor fails to make payments when due. Financial guarantees are initially recognized in the financial statements at fair value on the date the guarantee was given. If the amount measured in subsequent periods exceeds the unamortized balance of the amount initially recognized, the excess is classified as other financial liability.

2.21 Current and Deferred Tax

The tax expense for the period comprises current and deferred tax. Tax is recognized on the profit for the period in the statement of income, except to the extent that it relates to items recognized in other comprehensive income or directly in equity, in which case the tax is also recognized in other comprehensive income or directly in equity, respectively. The tax expense is calculated on the basis of the tax laws enacted or substantively enacted at the end of the reporting period.

Deferred tax is recognized for temporary differences arising between the tax bases of assets and liabilities and their carrying amounts as expected tax consequences at the recovery or settlement of the carrying amounts of the assets and liabilities. However, deferred tax assets and liabilities are not recognized if they arise from initial recognition of an asset or liability in a transaction other than a business combination that at the time of the transaction affects neither accounting nor taxable profit or loss. Deferred tax assets are recognized only to the extent that it is probable that future taxable profit will be available against which the temporary differences can be utilized.

A deferred tax liability is recognized for taxable temporary differences associated with investments in subsidiaries, associates, and interests in joint ventures, except to the extent that the Company is able to control the timing of the reversal of the temporary differences and it is probable that the temporary difference will not reverse in the foreseeable future. In addition, a deferred tax asset is recognized for deductible temporary differences arising from such investments to the extent that it is probable the temporary difference will reverse in the foreseeable future and taxable profit will be available against which the temporary difference can be utilized.

Deferred tax assets and liabilities are offset when there is a legally enforceable right to offset current tax assets against current tax liabilities and when the deferred income taxes assets and liabilities relate to income taxes levied by the same taxation authority on either the same taxable entity or different taxable entities where there is an intention to settle the balances on a net basis.

2.22 Derivative Instruments

Derivatives are initially recognized at fair value on the date when a derivative contract is entered into and are subsequently remeasured at their fair value at the end of each reporting period. Changes in the fair value of any derivative instrument that does not qualify for hedge accounting are recognized immediately in profit or loss.

The Company applies cash flow hedge accounting to hedge the foreign currency risk of forecasted transactions including hedging the price risk associated with forecast inventory purchase. The effective portion of changes in fair value of derivatives that are designated and qualify as cash flow hedges is recognized in other comprehensive income, and the ineffective portion is recognized in financial income or cost.

2.23 Dividend

Dividend distribution to the Company's shareholders is recognized as a liability when the dividends are approved.

2.24 Share Capital

Ordinary shares and preference shares with no repayment obligations are classified as equity. When the Company purchases its ordinary shares, the acquisition costs, including direct transaction costs, are deducted from equity until the redemption or reissuance as treasury shares. Consideration received on the subsequent sale or issuance of treasury shares is credited to equity.

2.25 Revenue Recognition

Revenue mainly comprises the fair value of the consideration received or receivable for the sale of goods in the ordinary course of the Company's activities. Revenue is shown net of value-added tax, returns, sales incentives and discounts and after eliminating intercompany transactions.

The Company recognizes revenue when the amount of revenue can be reliably measured; when it is probable that future economic benefits will flow to the entity; and when specific criteria have been met for each of the Company's activities, as described below. The Company measures revenue by reliably estimating the contingencies associated with revenue based on historical results, taking into consideration the type of customer, the type of transaction and the specifics of each arrangement.

Where multiple-element arrangements exist, the fair values of each element are determined based on the current market price of each of the elements when sold separately. When the fair values of each element are indeterminable, the fair values of deliverables which have already been provided are calculated in such way that the fair values of elements which are yet to be provided are subtracted from total contract value of the arrangement.

(A) Sales of goods

Sales of products and merchandise are recognized upon delivery when the significant risks and rewards of ownership of goods have transferred to the buyer. Revenue is recognized net of discounts and returns, estimated at the time of sale based on past experience.

(B) Sales of services

Revenues from rendering services are generally recognized using the percentage-of-completion method based on the percentage of costs to date compared to the total estimated costs, contractual milestones or performance.

(C) Interest income

Interest income is recognized using the effective interest method according to the time passed. When a loan or receivable is impaired, the Company reduces the carrying amount to its recoverable amount and continues unwinding the discount as interest income. Interest income on impaired loans and receivables is recognized using the original effective interest rate.

(D) Royalty income

Royalty income is recognized on an accrual basis in accordance with the substance of the relevant agreements.

(E) Dividend income

Dividend income is recognized when the right to receive payment is established.

2.26 Leases

A lease is an agreement, whereby the lessor conveys to the lessee, in return for a payment or series of payments, the right to use an asset for an agreed period of time. Leases in which a significant portion of the risks and rewards of ownership are not transferred to the Company are classified as operating leases. Payments made under operating leases are charge to profit or loss on a straight-line basis over the period of lease.

Leases where the Company, as lessee, has substantially all the risks and rewards of ownership are classified as finance leases. Finance leases are capitalized at the lease's inception at the fair value of the leased property or, if lower, the present value of the minimum lease payments. The corresponding rental obligations, net of finance charges, are included in other short-term and long-term payables. Each lease payment is allocated between the liability and financial expense.

If the Company is a lessor, a lease is classified as a finance lease if it transfers substantially all the risks and rewards incidental to ownership at the inception of the lease. A lease other than a finance lease is classified as an operating lease. Lease income from operating leases is recognized in income on a straight-line basis over the lease term. Initial direct costs incurred by the lessor in negotiating and arranging an operating lease is added to the carrying amount of the leased asset and recognized as an expense over the lease term on the same basis as the lease income.

2.27 Government Grants

Government grants are recognized at their fair values when there is reasonable assurance that the grant will be received and the Company will comply with the conditions attached to it. Government grants relating to costs are deferred and recognized in the income statement over the period necessary to match them with the costs that they are intended to compensate. Government grants relating to assets are recognized in liabilities as deferred income government grants and are credited to the income statement on a straight-line basis over the expected lives of the related assets.

2.28 Earnings per Share

Basic earnings per share is calculated by dividing net profit for the period available to ordinary shareholders by the weighted-average number of ordinary shares outstanding during the year. Diluted earnings per share is calculated using the weighted-average number of ordinary shares outstanding adjusted to include the potentially dilutive effect of common equivalent shares outstanding.

2.29 Operating Segments

Operating segments are disclosed in the manner reported to the chief operating decision-maker. The chief operating decision-maker is responsible for making strategic decisions on resource allocation and performance assessment of the operating segments. The Management Committee, which makes strategic decisions, is regarded as the chief operating decision-maker.

2.30 Convenience Translation into United States Dollar Amounts

The Company operates primarily in Korean won and its official accounting records are maintained in Korean won. The US dollar amounts provided in the financial statements represent supplementary information solely for the convenience of the reader. All Korean won amounts are expressed in U.S. dollars at the rate of ₩1131.08 to US \$1, the average exchange rate for the year ended December 31, 2017. Such presentation is not in accordance with generally accepted accounting principles, and should not be construed as a representation that the Korean won amounts shown could be readily converted, realized or settled in US dollars at this or at any other rate.

2.31 Approval of the Consolidated Financial Statements

These consolidated financial statements were approved by the Board of Directors on January 31, 2018.

3. Critical Accounting Estimates and Assumptions

The Company makes estimates and assumptions concerning the future. The estimates and assumptions are continuously assessed, considering historical experience and other factors, including expectations of future events that are believed to be reasonable under the circumstances. The resulting accounting estimates will, by definition, seldom equal the related actual results. The estimates and assumptions that have a significant risk of causing a material adjustment to the carrying amounts of assets and liabilities within the next financial year are addressed below.

(A) Revenue recognition

The Company uses the percentage-of-completion method in accounting for its contracts to deliver installation services. Use of the percentage-of-completion method requires the Company to estimate the services performed to date as a proportion of the total services to be performed. Revenues and earnings are subject to significant change, effected by early steps in a long-term projects, change in scope of a project, cost, period, and plans of the customers.

(B) Provision for warranty

The Company recognizes provision for warranty on products sold. The Company accrues provision for warranty based on the best estimate of amounts necessary to settle future and existing claims. The amounts are estimated based on past experience.

(C) Fair value of financial instruments

The fair value of financial instruments that are not traded in an active market is determined by using a variety of methods and assumptions that are mainly based on market conditions existing at the end of each reporting period.

(D) Net defined benefit liabilities (assets)

The net defined benefit liabilities (assets) depend on a number of factors that are determined on an actuarial basis using a number of assumptions including the discount rate. Any changes in these assumptions will impact the carrying amount of the net defined benefit liability. The Company, in consideration of the interest rates of high-quality corporate bonds, determines the appropriate discount rate at the end of each year. This is the interest rate that is used to determine the present value of estimated future cash outflows expected to be required to settle the net defined benefit liabilities (assets). The principal actuarial assumptions associated with the net defined benefit liabilities (assets) are based on the current market expectations.

(E) Impairment of goodwill

At the end of each reporting period, the Company tests whether goodwill has become impaired by comparing the carrying amounts of cash-generating units to the recoverable amounts. The recoverable amounts of cash-generating units have been determined based on value-in-use calculations, and these calculations are based on estimates.

(F) Income taxes

Income taxes on the Company's taxable income from operating activities are subject to various tax laws and determinations of each tax authority across various countries throughout the world. There is uncertainty in determining the eventual tax effects on the taxable income from operating activities. The Company has recognized current tax and deferred tax at the end of the fiscal year based on the best estimation of future taxes payable as a result of operating activities. However, the resulting deferred income tax assets and liabilities may not equal the actual future taxes payable and such difference may impact the current tax and deferred income tax assets and liabilities upon the determination of eventual tax effects.

4. Cash and Cash Equivalents

Cash and cash equivalents include cash on hand, deposits held at call with banks, and other short-term highly liquid investments that are readily convertible to a known amount of cash and are subject to an insignificant risk of change in value.

Cash and cash equivalents as at December 31, 2017 and 2016 consist of the following:

<i>(In millions of Korean won)</i>	December 31, 2017		December 31, 2016	
Cash on hand	₩	37,521	₩	51,770
Bank deposits and others		30,507,609		32,059,672
Total	₩	30,545,130	₩	32,111,442

5. Financial Assets Subject to Withdrawal Restrictions

Financial instruments subject to withdrawal restrictions as at December 31, 2017 and 2016 consist of the following:

<i>(In millions of Korean won)</i>	December 31, 2017		December 31, 2016	
Short-term financial instruments	₩	30,377	₩	21,541
Other non-current assets		3,507		28,828
Total	₩	33,884	₩	50,369

6. Financial Instruments by Category

(A) Categorizations of financial assets and liabilities as at December 31, 2017 and 2016 are as follows:

(1) As at December 31, 2017

<i>(In millions of Korean won)</i>	Assets at fair value through profit or loss		Loans and receivables		Available-for-sale financial assets		Held-to-maturity financial assets		Other financial assets¹		Total
Financial Assets											
Cash and cash equivalents	₩	-	₩ 30,545,130	₩	-	₩	-	₩	-	₩ 30,545,130	
Short-term financial instruments		-	49,447,696		-		-		-	49,447,696	
Short-term available-for-sale financial assets		-	-		3,191,375		-		-	3,191,375	
Trade receivables		-	27,695,995		-		-		-	27,695,995	
Long-term available-for-sale financial assets		-	-		7,752,180		-		-	7,752,180	
Held-to-maturity financial assets		-	-		-		106,751		-	106,751	
Others		67,702	6,212,727		-		-		45,396	6,325,825	
Total	₩	67,702	₩113,901,548	₩	10,943,555	₩	106,751	₩	45,396	₩125,064,952	

¹Other financial assets include derivatives designated as hedging instruments, which are not subject to categorizations.

<i>(In millions of Korean won)</i>	Liabilities at fair value through profit or loss		Financial liabilities measured at amortized cost		Other financial liabilities¹		Total
Financial Liabilities							
Trade payables	₩	-	₩ 9,083,907	₩	-	₩ 9,083,907	
Short-term borrowings		-	1,497,417		14,270,202	15,767,619	
Other payables		316,928	11,789,681		-	12,106,609	
Current portion of long-term liabilities		-	278,619		-	278,619	
Debentures		-	953,361		-	953,361	
Long-term borrowings		-	1,814,446		-	1,814,446	
Long-term other payables		28,285	1,717,899		-	1,746,184	
Others		180,366	10,732,501		41,646	10,954,513	
Total	₩	525,579	₩ 37,867,831	₩	14,311,848	₩ 52,705,258	

¹Other financial liabilities include collateralized borrowings and derivatives designated as hedging instruments, which are not subject to categorizations.

(2) As at December 31, 2016

<i>(In millions of Korean won)</i>	Assets at fair value through profit or loss		Loans and receivables	Available-for-sale financial assets	Total
Financial Assets					
Cash and cash equivalents	₩	-	₩ 32,111,442	₩ -	₩ 32,111,442
Short-term financial instruments		-	52,432,411	-	52,432,411
Short-term available-for-sale financial assets		-	-	3,638,460	3,638,460
Trade receivables		-	24,279,211	-	24,279,211
Long-term available-for-sale financial assets		-	-	6,804,276	6,804,276
Others		63,208	3,396,655	-	3,459,863
Total	₩	63,208	₩112,219,719	₩ 10,442,736	₩122,725,663

<i>(In millions of Korean won)</i>	Liabilities at fair value through profit or loss	Financial liabilities measured at amortized cost	Other financial liabilities ¹	Total
Financial Liabilities				
Trade payables	₩ -	₩ 6,485,039	₩ -	₩ 6,485,039
Short-term borrowings	-	1,817,021	10,929,768	12,746,789
Other payables	-	10,225,271	-	10,225,271
Current portion of long-term liabilities	-	1,232,817	-	1,232,817
Debentures	-	58,542	-	58,542
Long-term borrowings	-	1,244,238	-	1,244,238
Long-term other payables	342,702	2,666,957	-	3,009,659
Others	74,697	11,867,772	-	11,942,469
Total	₩ 417,399	₩ 35,597,657	₩ 10,929,768	₩ 46,944,824

¹ Other financial liabilities include collateralized borrowings, which are not subject to categorizations.

(B) Net gains or net losses on each category of financial assets and liabilities for the years ended December 31, 2017 and 2016 are as follows:

A. For the year ended December 31, 2017

<i>(In millions of Korean won)</i>	Assets at fair value through profit or loss		Loans and receivables		Available-for-sale financial assets		Held-to-maturity financial assets		Other financial assets		Total	
Financial Assets												
Gain (loss) on valuation (other comprehensive income)	₩	-	₩	-	₩	513,434	₩	-	₩	(47,048)	₩	466,386
Gain (loss) on valuation/disposal (profit or loss)		(40,218)		(20,163)		161,848		-		(3,022)		98,445
Reclassification from other comprehensive income to profit or loss		-		-		(2,227)		-		15,612		13,385
Interest income		-		1,613,010		1,157		56		-		1,614,223
Foreign exchange differences (profit or loss)		-		(913,989)		-		-		-		(913,989)
Dividend income		-		-		128,946		-		-		128,946
Impairment/reversal (profit or loss)		-		(215,349)		(13,538)		-		-		(228,887)

<i>(In millions of Korean won)</i>	Liabilities at fair value through profit or loss	Financial liabilities measured at amortized cost	Other financial liabilities	Total
Financial Liabilities				
Gain on valuation (other comprehensive income)	₩ -	₩ -	₩ (16,751)	₩ (16,751)
Gain (loss) on valuation/disposal (profit or loss)	(130,684)	-	(1,918)	(132,602)
Reclassification from other comprehensive income to profit or loss	-	-	5,559	5,559
Interest expense	-	358,911	296,491	655,402
Foreign exchange differences (profit or loss)	-	772,975	137,168	910,143

B. For the year ended December 31, 2016

<i>(In millions of Korean won)</i>	Assets at fair value through profit or loss		Loans and receivables	Available-for-sale financial assets		Total		
Financial Assets								
Gain on valuation (other comprehensive income)	₩	-	₩	-	₩	607,762	₩	607,762
Gain (loss) on valuation/disposal (profit or loss)		(24,573)		(5,648)		953,416		923,195
Reclassification from other comprehensive income to profit or loss		-		-		(631,601)		(631,601)
Interest income		-		1,475,357		28,961		1,504,318
Foreign exchange differences (profit or loss)		-		772,552		-		772,552
Dividend income		-		-		239,899		239,899
Impairment/reversal (profit or loss)		-		(135,046)		(341,790)		(476,836)

<i>(In millions of Korean won)</i>	Liabilities at fair value through profit or loss		Financial liabilities measured at amortized cost		Other financial liabilities		Total
Financial Liabilities							
Gain (loss) on valuation/disposal (profit or loss)	₩	(61,221)	₩	-	₩	-	₩ (61,221)
Interest expense		-		351,009		236,822	587,831
Foreign exchange differences (profit or loss)		-		(623,777)		(212,788)	(836,565)

7. Credit Quality of Financial Assets

The credit quality of financial assets that are neither past due nor impaired is assessed by reference to external credit ratings as at December 31, 2017 and 2016 as follows:

- Superior ability to repay: Aaa~Aa (Moody's), AAA~AA (S&P, Fitch), A1 (Credit rating agencies in Korea)
- Strong ability to repay: A (Moody's, S&P, Fitch), A2 (Credit rating agencies in Korea)
- Acceptable ability to repay: Baa (Moody's), BBB (S&P, Fitch), A3 (Credit rating agencies in Korea)
- Currently having the ability to repay: Ba or below (Moody's), BB or below (S&P, Fitch), B or below (Credit rating agencies in Korea)
- Group 1: Customers not having experienced capital erosion or default risk
- Group 2: Customers having experienced capital erosion or default risk, where all default risk is relieved as the trade payables are guaranteed by credit insurance or collateral.

(A) Trade receivables

<i>(In millions of Korean won)</i>	December 31, 2017		December 31, 2016	
Counterparties managed by external credit rating:				
Superior ability to repay	₩	4,772,359	₩	2,743,633
Strong ability to repay		2,642,998		3,170,573
Acceptable ability to repay		3,810,177		3,839,674
Currently having the ability to repay		1,812,583		2,744,783
Subtotal		13,038,117		12,498,663
Counterparties managed by internal credit rating:				
Group 1		13,990,823		8,465,341
Group 2		178,452		239,626
Subtotal		14,169,275		8,704,967
Total	₩	27,207,392	₩	21,203,630

(B) Cash equivalents and short-term financial instruments

<i>(In millions of Korean won)</i>	December 31, 2017		December 31, 2016	
Superior ability to repay	₩	5,596,487	₩	9,319,341
Strong ability to repay		72,204,793		70,916,126
Acceptable ability to repay		1,983,351		4,040,000
Currently having the ability to repay		66,139		171,393
Other ¹		104,534		45,223
Total	₩	79,955,304	₩	84,492,083

¹ Short-term financial instruments held at financial institutions (such as credit unions) without an external credit rating.

8. Transfer of Financial Assets

Trade receivables of the Company have been discounted through factoring agreements with banks in 2017 and 2016. Collateral (trade receivables and other) provided in such factoring transactions do not meet the requirements for asset derecognition as risks and rewards are not substantially transferred in the event the debtor defaults. Financial liabilities recognized in relation to these transactions are included as short-term borrowings on the statement of financial position (Note 15).

The following table presents a breakdown of discounted trade receivables as at December 31, 2017 and 2016:

<i>(In millions of Korean won)</i>	December 31, 2017		December 31, 2016	
Carrying amount of the discounted trade receivables ¹	₩	14,270,202	₩	10,929,768
Carrying amount of the related borrowings		14,270,202		10,929,768

¹ The discounted trade receivables include intercompany balances.

9. Available-for-Sale Financial Assets

Changes in available-for-sale financial assets for the years ended December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	2017		2016	
Balance as at January 1	₩	10,442,736	₩	12,960,010
Acquisitions		15,529,275		17,804,905
Disposals		(15,562,233)		(18,896,371)
Valuation of available-for-sale financial assets		721,730		798,698
Impairment		(8,083)		(326,672)
Foreign exchange differences		(254,693)		(156,050)
Other ¹		74,823		(1,741,784)
Balance as at December 31	₩	10,943,555	₩	10,442,736
(A) Current portion		3,191,375		3,638,460
(B) Non-current portion		7,752,180		6,804,276

¹ Due to increase in shareholding ratio for the year ended December 31, 2016, Samsung SDI and Cheil Worldwide were reclassified from available-for-sale financial assets to investments in associates.

As at the end of the reporting period, the above available-for-sale financial assets are measured at fair value, and changes in valuation gains (losses) recognized in equity (other comprehensive income) on available-for-sale financial assets for the years ended December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	2017		2016	
Balance as at January 1	₩	1,969,833	₩	2,008,223
Fair value gains		721,730		798,698
Net gains transferred from equity		(3,199)		(837,088)
Balance as at December 31		2,688,364		1,969,833
Deferred income tax and non-controlling interests		(808,590)		(579,209)
Total	₩	1,879,774	₩	1,390,624

(A) Short-term available-for-sale financial assets

Details of short-term available-for-sale financial assets as at December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	December 31, 2017		December 31, 2016	
Beneficiary certificates ¹	₩	3,191,375	₩	3,638,460

¹ Beneficiary certificates are comprised of time deposits and others.

(B) Long-term available-for-sale financial assets

Details of long-term available-for-sale financial assets as at December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	December 31, 2017		December 31, 2016	
Equity securities - Listed	₩	2,908,581	₩	2,362,235
Equity securities - Non-listed		4,729,124		4,337,328
Debt securities ¹		114,475		104,713
Total	₩	7,752,180	₩	6,804,276

¹ The maximum exposure to credit risk of available-for-sale debt securities is the carrying amount at the reporting date.

(1) Equity securities - Listed

Details of listed equity securities as at December 31, 2017 and 2016 are as follows:

(In millions of Korean won, number of shares and percentage)

	December 31, 2017			December 31, 2016	
	Number of Shares Owned	Percentage of Ownership (%)¹	Acquisition Cost	Book Value (Market Value)	Book Value (Market Value)
Samsung Heavy Industries	65,930,982	16.9	473,727	483,274	609,862
Hotel Shilla	2,004,717	5.1	13,957	170,200	96,527
iMarket Korea	647,320	1.8	324	5,832	6,732
SFA ²	3,644,000	10.2	38,262	141,205	117,519
Wonik Holdings	3,518,342	4.6	30,821	27,760	23,714
Wonik IPS	3,701,872	9.0	32,428	123,643	96,989
ASML	6,297,787	1.4	363,012	1,169,393	851,395
Wacom	8,398,400	5.0	62,013	48,631	26,647
BYD	52,264,808	1.9	528,665	556,381	449,872
Other			158,688	182,262	82,978
Total			₩ 1,701,897	₩ 2,908,581	₩ 2,362,235

¹ Ownership represents the Company's ownership of ordinary shares issued in each entity.

² As SFA executed a capital increase without consideration, the Company's number of SFA shares owned increased.

Acquisition cost includes impairment loss on available-for-sale financial assets recognized due to the decline in realizable value below acquisition cost. The difference between the acquisition cost, net of impairment loss and the current fair value is recorded within other components of equity, net of tax effects (unrealized gains or losses on available-for-sale financial assets).

(2) Equity securities – Non-listed

Details of non-listed equity securities as at December 31, 2017 and 2016 are as follows:

	December 31, 2017			December 31, 2016	
	Number of Shares Owned	Percentage of Ownership (%)	Acquisition Cost	Book Value ¹	Book Value
Kihyup Technology	1,000,000	17.2	₩ 5,000	₩ 5,000	₩ 5,000
Samsung Venture Investment	980,000	16.3	4,900	7,784	7,515
Maltani	45,000	15.0	16,544	15,137	16,270
Corning Inc.	2,300	7.4	2,434,320	3,629,653	3,440,487
CSOSDT	-	9.8	357,315	357,315	357,315
Nanosys	22,500,681	17.5	35,814	37,475	28,985
Other			721,252	676,760	481,756
Total			₩ 3,575,145	₩ 4,729,124	₩ 4,337,328

¹ Nonmarketable shares are measured at cost as the variability of estimated cash flow is significant and the probability of various estimates, including discount rate, cannot be reasonably assessed.

Acquisition cost includes impairment loss on available-for-sale financial assets recognized due to the decline in realizable value below acquisition cost. The difference between the acquisition cost, net of impairment loss and the current fair value is recorded within other components of equity, net of tax effects (unrealized gains or losses on available-for-sale financial assets).

(3) Debt securities

Details of debt securities as at December 31, 2017 and 2016 are as follows:

(In millions of Korean won)	December 31, 2017		December 31, 2016	
Corporate bonds	₩	114,475	₩	104,713
Total	₩	114,475	₩	104,713

10. Trade and Other Receivables

(A) Trade and other receivables as at December 31, 2017 and 2016 are as follows:

(In millions of Korean won)	December 31, 2017		December 31, 2016	
	Trade	Non-Trade	Trade	Non-Trade
Receivables	₩ 30,316,958	₩ 4,275,407	₩ 24,704,524	₩ 3,582,229
Less: Provisions for impairment	(639,223)	(27,305)	(420,889)	(25,503)
Receivables, net	29,677,735	4,248,102	24,283,635	3,556,726
Less: Non-current portion	(1,981,740)	(139,141)	(4,424)	(35,529)
Current portion	₩ 27,695,995	₩ 4,108,961	₩ 24,279,211	₩ 3,521,197

(B) Movements in the provisions for impairment of receivables for the years ended December 31, 2017 and 2016 are as follows:

(In millions of Korean won)	2017		2016	
	Trade	Non-Trade	Trade	Non-Trade
Balance as at January 1	₩ 420,889	₩ 25,503	₩ 326,861	₩ 49,291
Provisions for impaired receivables / (reversals of unused amounts)	206,561	8,277	117,207	(19,209)
Receivables written off during the year as uncollectible	(14,367)	(1,133)	(20,421)	(3,978)
Other	26,140	(5,342)	(2,758)	(601)
Balance as at December 31	₩ 639,223	₩ 27,305	₩ 420,889	₩ 25,503

(C) The aging analysis of trade and other receivables as at December 31, 2017 and 2016 is as follows:

(In millions of Korean won)	December 31, 2017		December 31, 2016	
Receivables not past due	₩	31,113,883	₩	24,598,074
Past due but not impaired ¹ :				
Less than 31 days overdue		2,435,133		2,281,693
Impaired:				
31 days to 90 days overdue		473,009		881,736
Over 90 days overdue		570,340		525,250
Total	₩	34,592,365	₩	28,286,753

¹ The Company does not consider receivables that are overdue for less than or equal to 31 days as impaired.

(D) The maximum exposure to current credit risk is equivalent to the carrying amount of receivables as at December 31, 2017. The Company has in place insurance contracts covering the Company's major receivables, and has accrued provisions against receivables in accordance with the overdue payment history for those receivables not covered by insurance contracts.

11. Inventories

Inventories as at December 31, 2017 and 2016 are as follows:

(In millions of Korean won)	December 31, 2017			December 31, 2016		
	Gross Amount	Valuation Allowance ¹	Carrying Amount	Gross Amount	Valuation Allowance ¹	Carrying Amount
Finished goods	₩ 8,201,526	₩ (897,089)	₩ 7,304,437	₩ 7,982,850	₩(2,077,511)	₩ 5,905,339
Work in process	7,331,394	(217,493)	7,113,901	5,334,607	(317,223)	5,017,384
Raw materials and supplies	10,196,123	(782,906)	9,413,217	7,526,608	(1,032,442)	6,494,166
Materials in transit	1,151,800	-	1,151,800	936,614	-	936,614
Total	₩26,880,843	₩(1,897,488)	₩24,983,355	₩21,780,679	₩(3,427,176)	₩18,353,503

¹ Inventories for which the Company has suspended sales or production, are evaluated based on net realizable value. Valuation allowance is recorded if the net realizable value is less than the carrying amount.

The cost of inventories recognized as expense and included in 'cost of sales' for the year ended December 31, 2017, amounts to ₩128,823,788 million (2016: ₩119,611,006 million). The amount includes inventory valuation losses.

12. Investments in Associates and Joint Ventures

(A) Changes in investments in associates and joint ventures for the years ended December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	2017	2016
Balance as at January 1	₩ 5,837,884	₩ 5,276,348
Acquisition	25,293	84,306
Disposal ¹	(53,187)	(1,343,936)
Share of profit	201,442	19,501
Others ²	790,919	1,801,665
Balance as at December 31	₩ 6,802,351	₩ 5,837,884

¹ The Company sold its entire stake in Samsung Card for the year ended December 31, 2016.

² Others consist of dividends, impairment, reversal of impairment and reclassification.

(B) Major investments in associates and joint ventures as at December 31, 2017 are as follows:

(1) Investments in associates

Investee	Nature of Relationship with Associate	Percentage of Ownership ¹ (%)	Principal Business Location	The End of Reporting Period
Samsung Electro-Mechanics	Manufacture and supply electronic components including passive components, circuit boards, and modules	23.7	Korea	December
Samsung SDS	Provide IT services including computer programming, system integration and management and logistical services	22.6	Korea	December
Samsung Biologics	New business investment	31.5	Korea	December
Samsung SDI ²	Manufacture and supply electronics including secondary cell batteries	19.6	Korea	December
Cheil Worldwide	Advertising agency	25.2	Korea	December

¹ Ownership represents the Company's ownership of ordinary shares issued by each entity.

² The Company's ownership of ordinary shares outstanding is 20.6%.

(2) Investments in joint ventures

Investee	Nature of Relationship with Joint Venture	Percentage of Ownership ¹ (%)	Principal Business Location	The End of Reporting Period
Samsung Corning Advanced Glass	Manufacture and supply industrial glass devices	50.0	Korea	December

¹ Ownership represents the Company's ownership of ordinary shares issued by each entity.

(C) Details of investments in associates and joint ventures as at December 31, 2017, 2016 are as follows:

(1) Investments in associates

(In millions of Korean won)

(In millions of Korean won)			December 31, 2017			
Investee	Acquisition cost		Net asset value of equity		Carrying amount	
			shares ¹			
Samsung Electro-Mechanics	₩	359,237	₩	989,924	₩	991,579
Samsung SDS		147,963		1,256,881		1,282,205
Samsung Biologics		443,193		1,251,292		1,254,937
Samsung SDI		1,242,605		2,266,451		2,126,244
Cheil Worldwide		506,162		241,335		540,114
Other		593,080		215,025		376,349
Total	₩	3,292,240	₩	6,220,908	₩	6,571,428

¹ The Company's portion of net asset value of associates is based on the Company's ownership percentage.

(In millions of Korean won)

(In millions of Korean won)			December 31, 2016			
Investee	Acquisition cost		Net asset value of equity		Carrying amount	
			shares ¹			
Samsung Electro-Mechanics	₩	359,237	₩	993,031	₩	997,022
Samsung SDS		147,963		1,161,197		1,185,703
Samsung Biologics		443,193		1,285,706		1,289,351
Samsung SDI		1,242,605		2,131,718		1,232,986
Cheil Worldwide		506,162		192,594		517,885
Other		642,536		182,279		390,438
Total	₩	3,341,696	₩	5,946,525	₩	5,613,385

¹ The Company's portion of net asset value of associates is based on the Company's ownership percentage.

(2) Investments in joint ventures

(In millions of Korean won)

		December 31, 2017			
Investee	Acquisition cost	Net asset value of equity shares ¹		Carrying amount	
Samsung Corning Advanced Glass	₩ 215,000	₩	170,440	₩	170,425
Others	259,994		65,106		60,498
Total	₩ 474,994	₩	235,546	₩	230,923

¹ The Company's portion of net asset value of joint ventures is based on the Company's ownership percentage.

(In millions of Korean won)

		December 31, 2016			
Investee	Acquisition cost	Net asset value of equity shares ¹		Carrying amount	
Samsung Corning Advanced Glass	₩ 215,000	₩	169,521	₩	169,485
Others	259,977		59,342		55,014
Total	₩ 474,977	₩	228,863	₩	224,499

¹ The Company's portion of net asset value of joint ventures is based on the Company's ownership percentage.

(D) Details of the valuation of investments in associates and joint ventures under the equity method are as follows:

(1) For the year ended December 31, 2017

<i>(In millions of Korean won)</i>	Balance as at January 1		Gain/loss on valuation		Other comprehensive income/loss		Others¹	Balance as at December 31
Samsung Electro-Mechanics	₩	997,022	₩	34,353	₩	(30,949)	₩ (8,847)	₩ 991,579
Samsung SDS		1,185,703		121,968		(12,363)	(13,103)	1,282,205
Samsung Biologics		1,289,351		(31,546)		(2,868)	-	1,254,937
Samsung SDI		1,232,986		(50)		8,038	885,270	2,126,244
Cheil Worldwide		517,885		36,083		(5,142)	(8,712)	540,114
Samsung Corning Advanced Glass		169,485		1,039		(97)	(2)	170,425
Others		445,452		39,595		(11,043)	(37,157)	436,847
Total	₩	5,837,884	₩	201,442	₩	(54,424)	₩ 817,449	₩ 6,802,351

¹ Others consist of acquisitions, disposals, dividends, impairment, reversal of impairment and reclassification.

(2) For the year ended December 31, 2016

<i>(In millions of Korean won)</i>	Balance as at January 1		Gain/loss on valuation		Other comprehensive income/loss		Others¹	Balance as at December 31
Samsung Card	₩	1,338,679	₩	-	₩	-	₩ (1,338,679)	₩ -
Samsung Electro-Mechanics		994,489		1,738		9,641	(8,846)	997,022
Samsung SDS		1,060,396		105,424		28,620	(8,737)	1,185,703
Samsung Biologics		1,310,202		(66,212)		(1,189)	46,550	1,289,351
Samsung SDI		-		(8,017)		(1,602)	1,242,605	1,232,986
Cheil Worldwide		-		4,375		7,348	506,162	517,885
Samsung Corning Advanced Glass		188,371		(18,742)		(144)	-	169,485
Others		384,211		935		(6,522)	66,828	445,452
Total	₩	5,276,348	₩	19,501	₩	36,152	₩ 505,883	₩ 5,837,884

¹ Others consist of acquisitions, disposals, dividends, impairment and reclassification.

(E) Summary of condensed financial information of major associates and joint ventures

(1) A summary of condensed financial information of major associates and dividends received from associates as at December 31, 2017 and 2016, and for the years ended December 31, 2017 and 2016 is as follows:

(In millions of Korean won)	2017				
	Samsung Electro- Mechanics	Samsung SDS	Samsung Biologics	Samsung SDI	Cheil Worldwide
1. Condensed financial information					
Condensed statements of financial position:					
Current assets	₩ 2,478,798	₩ 5,117,499	₩ 620,995	₩ 3,605,076	₩ 1,889,096
Non-current assets	5,288,605	2,160,262	6,562,096	12,146,401	349,197
Current liabilities	2,454,110	1,324,862	2,287,442	2,670,360	1,281,293
Non-current liabilities	981,802	233,469	919,198	1,629,117	101,012
Non-controlling interests	99,848	161,151	-	194,698	9,927
Condensed statements of comprehensive income:					
Revenue	₩ 6,838,474	₩ 9,299,206	₩ 464,629	₩ 6,321,561	₩ 3,374,987
Profit or loss from continuing operations ¹	161,739	530,348	(96,972)	657,236	127,153
Other comprehensive income(loss) ¹	(132,298)	(63,585)	(8,957)	27,870	(25,115)
Total comprehensive income(loss) ¹	29,441	466,763	(105,929)	685,106	102,038
2. Details of adjustments from the book value of investments in associates					
Net assets (a)	4,231,643	5,558,279	3,976,451	11,257,302	846,061
Ownership percentage (b) ²	23.4%	22.6%	31.5%	20.1%	28.7%
Net assets of equity shares (a x b)	989,924	1,256,881	1,251,292	2,266,451	241,335
Goodwill	7,081	26,801	3,645	-	298,779
Intercompany transactions and others ³	(5,426)	(1,477)	-	(140,207)	-
Book value of associates	991,579	1,282,205	1,254,937	2,126,244	540,114
3. Dividends from associates					
Dividends	₩ 8,847	₩ 13,104	₩ -	₩ 13,463	₩ 8,711

¹ Income (loss) attributable to owners of the parent.

² Ownership percentage includes ordinary and preference shares.

³ Consists of unrealized gains and losses and other differences.

2016

(In millions of Korean won)	2016				
	Samsung Electro- Mechanics	Samsung SDS	Samsung Biologics	Samsung SDI	Cheil Worldwide
1. Condensed financial information					
Condensed statements of financial position:					
Current assets	₩ 2,812,409	₩ 4,548,448	₩ 1,461,425	₩ 3,958,266	₩ 1,794,812
Non-current assets	4,850,220	2,293,556	6,071,580	10,942,046	356,619
Current liabilities	2,043,155	1,347,385	2,476,545	2,212,796	1,244,899
Non-current liabilities	1,281,889	203,495	974,089	1,723,405	130,248
Non-controlling interests	97,467	150,429	-	241,980	13,277
Condensed statements of comprehensive income:					
Revenue	₩ 6,033,040	₩ 8,180,187	₩ 294,622	₩ 5,200,823	₩ 3,232,594
Profit or loss from continuing operations ¹	14,707	463,858	(176,832)	(878,504)	88,263
Post-tax profit or loss from discontinued operations ¹	-	-	-	1,089,615	-
Other comprehensive income(loss) ¹	41,212	35,664	108	(222,175)	(13,711)
Total comprehensive income(loss) ¹	55,919	499,522	(176,724)	(11,064)	74,552
2. Details of adjustments from the book value of investments in associates					
Net assets (a)	₩ 4,240,118	₩ 5,140,695	₩ 4,082,371	₩10,722,131	₩ 763,007
Ownership percentage (b) ²	23.4%	22.6%	31.5%	19.9%	25.2%
Net assets of equity shares (a x b)	993,031	1,161,197	1,285,706	2,131,718	192,594
Goodwill	7,081	26,801	3,645	-	325,291
Intercompany transactions and others ³	(3,090)	(2,295)	-	(898,732)	-
Book value of associates	997,022	1,185,703	1,289,351	1,232,986	517,885
3. Dividends from associates					
Dividends	₩ 8,847	₩ 8,736	-	-	-

¹ Income (loss) attributable to owners of the parent.

² Ownership percentage includes ordinary and preference shares.

³ Consists of unrealized gains and losses and other differences.

(2) A summary of the condensed financial information of major joint ventures and dividends received from joint ventures as at December 31, 2017 and 2016, and for the years ended December 31, 2017 and 2016 is as follows:

	Samsung Corning Advanced Glass			
(In millions of Korean won)	2017		2016	
1. Condensed financial information				
Condensed statements of financial position:				
Current assets	₩	128,499	₩	170,614
- Cash and cash equivalent		11,035		16,021
Non-current assets		253,840		209,881
Current liabilities		41,325		41,076
- Current financial liabilities ¹		18,900		14,779
Non-current liabilities		134		377
Condensed statements of comprehensive income ² :				
Revenue	₩	242,452	₩	257,041
Depreciation and amortization		1,356		2,202
Interest income		738		1,433
Income tax expense		129		(8,841)
Profit or loss from continuing operations ²		2,034		(37,531)
Other comprehensive income(loss) ²		-		-
Total comprehensive loss ²		2,034		(37,531)

¹ Trade payables, other payables, and provisions are excluded.

² Income(loss) attributable to owners of the parent.

	Samsung Corning Advanced Glass			
(In millions of Korean won)	2017		2016	
2. Details of adjustments from the book value of investments in joint ventures				
Net assets(a)	₩	340,880	₩	339,042
Ownership percentage(b)		50.0%		50.0%
Net assets of equity shares(a x b)		170,440		169,521
Intercompany transactions and others ¹		(15)		(36)
Book value of joint ventures		170,425		169,485
3. Dividends from joint ventures				
Dividends	₩	-	₩	-

¹ Consists of unrealized gains and losses and other differences.

(3) Income (loss) amounts attributable to owners of the parent from associates and joint ventures which are not individually material for the years ended December 31, 2017 and 2016 are as follows:

(In millions of Korean won)	2017				2016			
	Associates		Joint ventures		Associates		Joint ventures	
Profit or loss from continuing operations	₩	33,544	₩	6,219	₩	4,993	₩	(3,567)
Other comprehensive income (loss)		(9,386)		(1,661)		(2,042)		(4,505)
Total comprehensive income (loss)	₩	24,158	₩	4,558	₩	2,951	₩	(8,072)

(F) Details of marketable investments in associates as at December 31, 2017 and 2016 is as follows:

(In millions of Korean won and number of shares)	December 31, 2017		December 31, 2016	
	Number of shares held	Market value	Market value	
Samsung Electro-Mechanics	17,693,084	₩ 1,769,308	₩	898,809
Samsung SDS	17,472,110	3,494,422		2,437,359
Samsung Biologics	20,836,832	7,730,465		3,146,362
Samsung SDI	13,462,673	2,753,117		1,467,431
Cheil Worldwide	29,038,075	615,607		457,350

13. Property, Plant and Equipment

(A) Changes in property, plant and equipment for the years ended December 31, 2017 and 2016 are as follows:

(In millions of Korean won)	2017					
	Land	Buildings and Structures	Machinery and Equipment	Construction In Progress	Other	Total
Balance as at January 1	₩ 7,869,679	₩ 24,375,826	₩ 38,302,055	₩ 18,773,986	₩ 2,151,495	₩ 91,473,041
Acquisition cost	7,869,679	36,474,462	155,285,378	18,773,986	6,769,149	225,172,654
Accumulated depreciation and impairment	-	(12,098,636)	(116,983,323)	-	(4,617,654)	(133,699,613)
Acquisitions and capital expenditures ¹	1,589,162	6,424,994	38,547,588	(4,396,126)	1,251,342	43,416,960
Business combinations	13,643	184,845	419,927	90,958	149,425	858,798
Depreciation	-	(1,836,073)	(17,846,905)	-	(910,638)	(20,593,616)
Disposals/Scrap	(33,850)	(37,007)	(150,722)	(261)	(89,786)	(311,626)
Impairment	-	-	(146,930)	-	(461)	(147,391)
Other ²	(29,358)	(1,054,429)	(1,209,215)	(557,808)	(179,708)	(3,030,518)
Balance as at December 31	₩ 9,409,276	₩ 28,058,156	₩ 57,915,798	₩ 13,910,749	₩ 2,371,669	₩111,665,648
Acquisition cost	9,409,276	41,639,045	184,392,999	13,910,749	7,176,535	256,528,604
Accumulated depreciation and impairment	-	(13,580,889)	(126,477,201)	-	(4,804,866)	(144,862,956)

¹ The capitalized borrowing costs are ₩19,711million and the interest rate used to calculate the borrowing costs eligible for capitalization is 0.71%.

² Other includes effects of changes in foreign currency exchange rates and effects of the offset related to government assistance.

2016

<i>(In millions of Korean won)</i>		Land	Buildings and Structures	Machinery and Equipment	Construction In Progress	Other	Total
Balance as at January 1	₩	7,848,432	₩ 22,453,296	₩ 43,077,879	₩ 10,970,052	₩2,127,451	₩ 86,477,110
Acquisition cost		7,848,432	32,850,110	147,315,096	10,970,052	6,303,834	205,287,524
Accumulated depreciation and impairment		-	(10,396,814)	(104,237,217)	-	(4,176,383)	(118,810,414)
Acquisitions and capital expenditures ¹		37,735	3,482,228	12,769,230	8,230,900	974,275	25,494,368
Business combinations		-	-	4,492	240	2,271	7,003
Depreciation		-	(1,631,089)	(16,814,751)	-	(866,680)	(19,312,520)
Disposals/Scrap		(28,331)	(26,384)	(80,552)	(5)	(66,684)	(201,956)
Impairment		-	(2,805)	(370,574)	-	(1,731)	(375,110)
Reclassification of assets held-for-sale		-	(11,922)	(20,131)	(7,660)	(45,156)	(84,869)
Other ²		11,843	112,502	(263,538)	(419,541)	27,749	(530,985)
Balance as at December 31	₩	7,869,679	₩ 24,375,826	₩ 38,302,055	₩ 18,773,986	₩2,151,495	₩ 91,473,041
Acquisition cost		7,869,679	36,474,462	155,285,378	18,773,986	6,769,149	225,172,654
Accumulated depreciation and impairment		-	(12,098,636)	(116,983,323)	-	(4,617,654)	(133,699,613)

¹ The capitalized borrowing costs are ₩17,644 million and the interest rate used to calculate the borrowing costs eligible for capitalization is 0.95%.

² Other includes effects of changes in foreign currency exchange rates and effects of the offset related to government assistance.

(B) Details of depreciation of property, plant and equipment for the years ended December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	2017	2016
Cost of sales	₩ 18,440,750	₩ 17,348,302
Selling and administrative expenses and others	2,152,866	1,964,218
Total	₩ 20,593,616	₩ 19,312,520

14. Intangible Assets

(A) Changes in intangible assets for the years ended December 31, 2017 and 2016 are as follows:

(In millions of Korean won)	2017					
	Intellectual property rights	Development cost	Membership	Goodwill	Other	Total
Balance as at January 1	₩ 1,349,764	₩ 1,180,618	₩ 184,399	₩ 1,343,580	₩ 1,285,659	₩ 5,344,020
Internally generated (development costs)	-	447,541	-	-	-	447,541
External acquisitions	236,918	-	57,751	-	241,530	536,199
Business combinations	3,011,641	-	-	4,453,214	2,552,673	10,017,528
Amortization	(252,191)	(687,366)	-	-	(584,219)	(1,523,776)
Disposals/scrap	(27,123)	-	(1,162)	-	(587)	(28,872)
Impairment	-	(110,409)	(23,617)	-	(9,085)	(143,111)
Other ¹	(184,554)	-	5,598	(93,656)	383,567	110,955
Balance as at December 31	₩ 4,134,455	₩ 830,384	₩ 222,969	₩ 5,703,138	₩ 3,869,537	₩ 14,760,483

¹ Other includes effects of changes in foreign currency exchange rates and others.

(In millions of Korean won)	2016					
	Intellectual property rights	Development cost	Membership	Goodwill	Other	Total
Balance as at January 1	₩ 1,342,104	₩ 1,697,545	₩ 184,915	₩ 910,539	₩ 1,261,208	₩ 5,396,311
Internally generated (development costs)	-	680,962	-	-	-	680,962
External acquisitions	275,288	-	802	4,922	85,694	366,706
Business combinations	70,199	-	-	503,045	63,674	636,918
Amortization	(234,666)	(748,573)	-	-	(417,206)	(1,400,445)
Disposals/scrap	(49,700)	-	(1,005)	-	(2,257)	(52,962)
Impairment	-	(449,297)	-	(15,143)	(9,054)	(473,494)
Reclassification of assets held-for-sale	(41,032)	-	(89)	(41,650)	(41,800)	(124,571)
Other ¹	(12,429)	(19)	(224)	(18,133)	345,400	314,595
Balance as at December 31	₩ 1,349,764	₩ 1,180,618	₩ 184,399	₩ 1,343,580	₩ 1,285,659	₩ 5,344,020

¹ Other includes effects of changes in foreign currency exchange rates, and others.

(B) Goodwill

Goodwill is allocated to cash-generating units at the end of the reporting period. Details of goodwill as at December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	December 31, 2017		December 31, 2016	
CE	₩	500,611	₩	532,669
IM		600,633		644,468
Semiconductor		81,142		82,400
DP		80,299		80,299
Harman		4,436,715		-
Other		3,738		3,744
Total	₩	5,703,138	₩	1,343,580

Goodwill impairment reviews are undertaken annually, and the recoverable amounts of cash-generating units have been determined based on value-in-use calculations. These calculations use pre-tax cash flow projections based on financial budgets approved by management covering a five-year period. Cash flows beyond the five-year period are extrapolated using the estimated growth rates stated below. The growth rate does not exceed the long-term average growth rate for the industry. In addition, a constant growth rate assumption is used for perpetual cash flow calculation.

(C) Details of amortization of intangible assets for the years ended December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	2017		2016	
Cost of sales	₩	894,529	₩	959,545
Selling and administrative expenses and others		629,247		440,900
Total	₩	1,523,776	₩	1,400,445

15. Borrowings

(A) Details of the carrying amounts of borrowings as at December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	Financial Institutions	Annual Interest Rates (%) as at December 31, 2017		December 31, 2017		December 31, 2016
Short-term borrowings						
Collateralized borrowings ¹	Woori Bank and others	0.1 ~ 11.1	₩	14,270,202	₩	10,929,768
Non-collateralized borrowings	Citibank and others	0.1 ~ 15.9		1,497,417		1,817,021
Total			₩	15,767,619	₩	12,746,789
Current portion of long-term borrowings						
Bank borrowings	BTMU	LIBOR+0.5	₩	262,493		₩ 684
Financial lease liabilities ²	CSSD and others	1.1 ~ 15.7		10,925		18,599
Total			₩	273,418	₩	19,283
Long-term borrowings						
Bank borrowings	Citibank and others	LIBOR+0.4 ~ 3.5	₩	1,756,908	₩	1,179,111
Financial lease liabilities ²	CSSD and others	1.1 ~ 15.7		57,538		65,127
Total			₩	1,814,446	₩	1,244,238

¹ Collateralized borrowings are secured by trade receivables (refer to Note 8).

² Leased property, plant and equipment were pledged as collateral (refer to Note 19).

(B) Maturities of long-term borrowings outstanding as at December 31, 2017 are as follows:

<i>(In millions of Korean won)</i>	Long-term borrowings	
Repayment terms		
2018	₩	273,418
2019		1,718,616
2020		57,812
2021		6,089
2022 and thereafter		31,929
Total	₩	2,087,864

16. Debentures

(A) Details of foreign currency denominated debentures as at December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean Won)</i>	Issue Date	Due Date	Annual Interest Rates (%) as at December 31, 2017	2017	2016
US dollar denominated straight bonds ¹	1997.10.2	2027.10.1	7.7	₩ 53,570 (US\$50 million)	₩ 66,468 (US\$55 million)
US dollar denominated unsecured bonds ²	2012.4.10	2017.4.10	-	-	1,208,500 (US\$1,000 million)
US dollar denominated debenture bonds ³	2015.5.6	2025.5.15	4.2	428,560 (US\$400 million)	-
EURO denominated debenture bonds ⁴	2015.5.20	2022.5.27	2.0	447,739 (EUR€350 million)	-
Less: Current Portion				(5,357)	(1,214,543)
Less: Discounts				(1,405)	(1,883)
More: Premium				30,254	-
Total				₩ 953,361	₩ 58,542

¹ US dollar denominated straight bonds are repaid annually for twenty years after a ten-year grace period from the date of issuance. Interest is paid semi-annually.

² Samsung Electronics America issued dollar denominated unsecured bonds. The amount was repaid during 2017.

³ Harman International Industries issued dollar denominated debenture bonds. Repayment of these debentures is due on the date of maturity and interest is paid semi-annually.

⁴ Harman Finance International SCA issued euro denominated debenture bonds. Repayment of these debentures is due on the date of maturity and interest is paid annually.

(B) Maturities of debentures outstanding as at December 31, 2017 are as follows:

<i>(In millions of Korean won)</i>	Debentures	
Repayment Term		
2018	₩	5,357
2019		5,357
2020		5,357
2021		5,357
2022 and thereafter		908,441
Total	₩	929,869

17. Net Defined Benefit Liabilities (Assets)

(A) Details of net defined benefit liabilities(assets) recognized on the statements of financial position as at December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	December 31, 2017		December 31, 2016	
Present value of funded defined benefit obligations	₩	7,302,621	₩	7,167,929
Present value of unfunded defined benefit obligations		234,315		110,885
Subtotal		7,536,936		7,278,814
Fair value of plan assets		(7,972,906)		(7,662,249)
Total	₩	(435,970)	₩	(383,435)

(B) The amounts recognized in the statements of profit or loss for the years ended December 31, 2017 and 2016 related to defined benefit plans are as follows:

<i>(In millions of Korean won)</i>	2017		2016	
Current service cost	₩	885,101	₩	1,077,511
Net interest cost (income)		(13,381)		11,221
Others		(2,206)		2,319
Total	₩	869,514	₩	1,091,051

(C) The amounts recognized as expense of defined contribution plans for the years ended December 31, 2017 and 2016 are ₩106,814 million and ₩105,971 million, respectively.

(D) The pension expenses related to defined benefit plans recognized on the statements of profit or loss for the years ended December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	2017		2016	
Cost of sales	₩	353,539	₩	469,172
Selling and administrative expenses and others		515,975		621,879
Total	₩	869,514	₩	1,091,051

(E) Changes in the defined benefit obligations for the years ended December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	2017		2016	
Balance as at January 1	₩	7,278,814	₩	7,832,779
Current service cost		885,101		1,077,511
Interest cost		278,029		298,003
Remeasurement:				
Actuarial gains or losses arising from changes in demographic assumptions		(16,602)		(41,608)
Actuarial gains or losses arising from changes in financial assumptions		(494,523)		(1,062,656)
Other		(131,126)		(295,125)
Benefits paid		(505,689)		(474,112)
Foreign exchange differences		6,243		1,846
Other ¹		236,689		(57,824)
Balance as at December 31	₩	7,536,936	₩	7,278,814

¹ Other includes effects of business combinations.

(F) Changes in the fair value of plan assets for the years ended December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	2017		2016	
Balance as at January 1	₩	7,662,249	₩	7,473,959
Expected return on plan assets		291,410		286,782
Remeasurement factor of plan assets		(152,821)		(123,794)
Contributions by employer		422,680		498,504
Benefits paid		(250,341)		(422,249)
Other ¹		(271)		(50,953)
Balance as at December 31	₩	7,972,906	₩	7,662,249

¹ Other includes effects of changes in foreign currency exchange rates and business combinations.

Expected contributions to post-employment benefit plans for the year ending December 31, 2018 are ₩883,058 million.

(G) Plan assets as at December 31, 2017 and 2016 consist of the following:

<i>(In millions of Korean won)</i>	December 31, 2017		December 31, 2016	
Debt instruments	₩	7,945,391	₩	7,635,150
Other		27,515		27,099
Total	₩	7,972,906	₩	7,662,249

Plan assets are mostly invested in instruments which have a quoted price in active markets.

(H) The principal actuarial assumptions as at December 31, 2017 and 2016 are as follows:

	December 31, 2017	December 31, 2016
Discount rate	0.4 ~ 8.2 %	0.4 ~ 8.2 %
Salary growth rate (including the effects of inflation)	1.5 ~ 10.0 %	1.5 ~ 10.0 %

(I) The sensitivity of the defined benefit obligations as at December 31, 2017 and 2016 to changes in the weighted principal assumptions is as follows:

	December 31, 2017	December 31, 2016
Discount rate		
1% increase	92%	90%
1% decrease	110%	111%
Salary growth rate		
1% increase	110%	111%
1% decrease	92%	90%

(J) The weighted average maturity of the defined benefit obligations is 8.95 years.

18. Provisions

Changes in provisions for the year ended December 31, 2017 are as follows:

(In millions of Korean won)	Warranty (A)	Royalty expenses (B)	Long-term incentives (C)	Others (D, E)	Total
Balance as at January 1	₩ 1,747,857	₩ 1,588,037	₩ 716,252	₩ 903,397	₩ 4,955,543
Charged (credited) to the statement of profit or loss	2,032,311	741,497	478,507	61,745	3,314,060
Payment	(1,920,926)	(430,195)	(526,615)	(730,023)	(3,607,759)
Others ¹	152,336	(140,271)	4,509	80,726	97,300
Balance as at December 31	₩ 2,011,578	₩ 1,759,068	₩ 672,653	₩ 315,845	₩ 4,759,144

¹ Others include effects of changes in foreign currency exchange rates and business combinations.

- (A) The Company accrues warranty reserves for estimated costs of future service, repairs and recalls, based on historical experience and terms of warranty programs.
- (B) The Company recognizes provisions for the estimated royalty expenses that are under negotiation with counterparties. The timing of payment depends on the settlement of the negotiation.
- (C) The Company has a long-term incentive plan for its executives based on a three-year management performance criteria and recognizes provisions for the estimated incentive cost for the accrued period.
- (D) The Company records provisions for the estimated expenses occurring from discontinuing production and sale of products.
- (E) The Company makes provisions for the carrying amounts of emission rights held by the Company and the emission in excess of the emission rights for the applicable years. Details of emission rights and liabilities as at December 31, 2017 are as follows:

(1) Allocated amount of emission permits and estimated amount of emission as at December 31, 2017 are as follows:

<i>(In ten thousand metric tons)</i>	December 31, 2017
Allocated emission permits	1,225
Estimated volume of emission	1,271

(2) Changes in the emission permits rights for the year ended December 31, 2017 are as follows:

<i>(In millions of Korean won)</i>		2017
Balance as at January 1	₩	15,067
Acquisition		15,926
Disposal		(5,934)
Balance as at December 31	₩	25,059

(3) Changes in the provisions for emissions liabilities for the year ended December 31, 2017 are as follows:

<i>(In millions of Korean won)</i>		2017
Balance as at January 1	₩	12,692
Charged (credited) to the statement of profit or loss		424
Balance as at December 31	₩	13,116

19. Commitments and Contingencies

(A) Guarantees

Details of guarantees of debt provided by the Company as at December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	December 31, 2017		December 31, 2016	
Guarantees of debt for housing rental ¹	₩	49,937	₩	56,752

¹ As at December 31, the maximum amount of debt guarantee, which was provided for employees who took debt from financial institutions in order to finance employee housing rental, is ₩167,470 million.

(B) Leases

(1) Finance leases

The Company leases certain property, plant and equipment under various finance lease arrangements and recognizes the related amounts as lease assets or liabilities. Assets with a net book value of ₩51,663 million (2016: ₩87,106 million) are treated as finance lease agreements and are included in property, plant and equipment. Depreciation expense for the finance lease assets amounted to ₩13,506 million for the year ended December 31, 2017 (2016: ₩13,495 million). Leased property, plant and equipment were pledged as collateral (Note 15).

The minimum lease payments under finance lease agreements and their present value as at December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	2017				2016			
	Minimum Lease payments		Present values		Minimum Lease payments		Present values	
Within one year	₩	18,348	₩	10,925	₩	25,928	₩	18,645
Later than one year but not later than five years		51,861		28,801		56,732		31,461
More than five years		45,303		28,737		56,249		33,620
Total	₩	115,512	₩	68,463	₩	138,909	₩	83,726
Present value adjustment		(47,049)				(55,183)		
Finance lease payable	₩	68,463			₩	83,726		

(2) Operating leases as lessee

As at December 31, 2017, the Company is engaged in non-cancellable lease agreements of which future minimum lease payments amount to ₩412,787 million within one year, ₩814,608 million later than one year but not later than five years, and ₩168,515 million more than five years. The aggregate of future minimum lease payments under non-cancellable leases amounts to ₩1,395,910 million. For the year ended December 31, 2017, the lease payments recognized as expense resulting from operating leases amount to ₩485,695 million.

(3) Operating leases as lessor

As at December 31, 2017, the Company is engaged in non-cancellable lease agreements related to rentals of properties for which future minimum lease receipts amount to ₩60,793 million within one year, ₩16,428 million later than one year but not later than five years and ₩4,870 million more than five years. The aggregate of future minimum lease receipts under non-cancellable leases amounts to ₩82,091 million. For the year ended December 31, 2017, the lease receipts recognized as income resulting from operating leases amount to ₩114,312 million.

(C) Litigation

The litigation with Apple Inc. ("Apple") is ongoing in the United States as at the reporting date. On August 24, 2012, the jury at the District Court for the Northern District of California determined that the Company partially infringed Apple's design and utility patent and should pay damages to Apple. However, on March 1, 2013, the Judge ordered a new trial for a certain portion of the damages, ruling that it was originally miscalculated. On November 21, 2013 a jury verdict was rendered on the recalculated damages amount, and on March 6, 2014, the Judge made a final judgement to confirm the total damages and deny Apple's bid for a permanent injunction against the Company. The Company appealed the decision on the damages amount on March 7, 2014, and a hearing on the appeal was held on December 4, 2014. On May 18, 2015, the appeals court affirmed in part and reversed in part a previous decision, and remanded it. On June 17, 2015, the Company petitioned for an en banc rehearing regarding the design infringement, and on August 13, 2015, the federal court dismissed the Company's request. After the remand procedure, the Court of First Trial announced a partial final judgment on the appeals on September 18, 2015. On October 13, 2015, the immediate appeal was dismissed and on November 19, 2015, the Federal Circuit Court denied an en banc rehearing request. On December 11, 2015, the Company made payment for the damages. On December 14, 2015, the Company filed an appeal to the Supreme Court regarding the design patent infringement ruling. Thereafter, the two parties have submitted in writing to the District court details of supplemental damages incurred in connection with the ruling. On March 21, 2016, the Supreme Court granted the design-related appeals filed by the Company and on March 22, 2016, the Court of First Trial ordered all proceedings for review of damages scheduled to commence March 28, 2016 suspended until the sentence rendered by the Supreme Court was confirmed. On June 1, 2016, the Company submitted the draft document in the design-related appeal, and on June 8, 2016, several

companies and organizations presented the document in support of the Company. Apple filed a dissenting document on July 29, 2016, and on August 5, 2016, several companies and organizations presented an advocative document. On August 29, 2016, the Company submitted a rebuttal letter. Oral statements for the appeal were held at the Supreme Court on October 11, 2016. On December 6, 2016, the Supreme Court issued a ruling citing the Company appeal and returned the case to the Court of Appeals. On February 7, 2017, the Court of Appeals reversed the case to the Court of First Trial. On October 12, 2017, the Court of First Instance held a court hearing on the necessity of retrial and decided to proceed with a design retrial on October 22, 2017.

Additionally, on May 5, 2014, the jury at the District Court for the Northern District of California in another ongoing patent lawsuit determined that the Company partially infringed Apple's utility patent and should pay damages to Apple. On November 25, 2014, the first trial judgment was pronounced to confirm the jury's verdict. The Company appealed on November 25, 2014 and the rehearing was held on January 5, 2016. On August 27, 2014, the Judge denied Apple's request for a permanent injunction on the Company's product. However, on September 17, 2015, the appellate court reversed and remanded a previous decision and on December 16, 2015, the Federal Circuit Court denied an en banc hearing request of the Company. On January 18, 2016, the Court of First instance ordered a permanent injunction on the Company's product. On February 26, 2016, a panel of appellate court reversed the first trial judgment of November 25, 2014 and ruled that the Company had not infringed Apple's some patents and its remaining patents were invalid. On March 30, 2016, Apple filed a petition for an en banc rehearing about the rulings of the panel. On October 7, 2016, the en banc appellate court reversed the panel rulings, affirming the first trial judgment that the Company had infringed Apple's patent. On March 10, 2017, the Company appealed to the Supreme Court against the rulings of the en banc appellate Court. On October 4, 2017, the US Department of Justice filed with the Supreme Court a legal advice letter against the company's appeal. On November 6, 2017, the Supreme Court dismissed the appeal. On January 11, 2018, the Court of First Trial held a court hearing on whether to recognize additional damages.

The final conclusion and the effect of the patent lawsuits with Apple are uncertain as at the reporting date.

In August 2014, the Company and Apple reached an agreement to withdraw from ongoing litigation in all regions other than the United States, and the Company has withdrawn all non-United States based lawsuits.

- (1) The Company is involved in claims, disputes, and investigations conducted by regulatory bodies at the reporting date, including civil claims from some overseas buyers for price-fixing related to the sale of TFT-LCD. Although the outflow of resources and timing of these matters are uncertain, the Company believes the outcome will not have a material impact on the financial condition of the Company.
- (2) In addition, during the normal course of business with numerous companies, the Company has been involved in various claims, disputes, and investigations conducted by regulatory bodies. Although the outflow of resources and timing of these matters are uncertain, the Company believes the outcome will not have a material impact on the financial condition of the Company.

(D) Other commitments

As at December 31, 2017, the Company has trade financing agreements, trade notes receivable discounting facilities, and loan facilities with accounts receivable pledged as collateral with Woori Bank and 23 other financial institutions, with a combined limit of up to ₩19,810,202 million. In addition, the Company has a trade financing agreement (up to ₩13,514,808 million) with Shinhan Bank and 23 other financial institutions and loan facilities with accounts receivable pledged as collateral and other financial agreements (up to ₩6,833,388 million).

Samsung Display Co., Ltd entered into a collaboration agreement with Corning Incorporated on October 23, 2013 that includes a condition relating to mutual loss preservation which can cause inflows or outflows of future economic benefits and the Company has recorded an estimated liability as a result of this commitment as at December 31, 2016 (Notes 6 and 31).

20. Share Capital

The Company's total number of authorized shares is 500,000,000 shares (₩5,000 per share). The Company has issued 129,098,494 ordinary shares and 18,250,580 preference shares as at December 31, 2017, excluding retired shares. Due to the retirement of shares, the total par value of the shares issued is ₩736,745 million (ordinary shares ₩645,492 million and preference shares ₩91,253 million), which does not agree with paid-in capital of ₩897,514 million.

Changes in the number of shares outstanding for the years ended December 31, 2017 and 2016 are as follows:

<i>(number of shares)</i>	Preference shares	Ordinary shares
Balance as at January 1, 2016	18,547,833	127,396,848
Acquisition of treasury stock	(1,264,099)	(4,699,197)
Balance as at December 31, 2016	17,283,734	122,697,651
Acquisition of treasury stock	(753,325)	(3,009,282)
Balance as at December 31, 2017	16,530,409	119,688,369

During the year ended December 31, 2017, the Company retired 11,580,843 ordinary shares and 2,262,847 preference shares of which acquisition cost is ₩11,872,563 million in total on the basis of the Board of Directors' approval on January 24, 2017, April 27, 2017, and July 27, 2017.

21. Retained Earnings

(A) Retained earnings as at December 31, 2017 and 2016 consist of the following:

<i>(In millions of Korean won)</i>	December 31, 2017	December 31, 2016
Appropriated	₩ 139,629,195	₩ 143,007,192
Unappropriated	76,182,005	50,079,125
Total	₩ 215,811,200	₩ 193,086,317

22. Dividends

Details of interim and year-end dividends are as follows:

(A) Interim dividends (Record date: March 31, 2017, June 30, 2017, September 30, 2017 and June 30, 2016)

(In millions of Korean won, number of shares)

		2017		2016	
1 st quarter	Number of shares eligible for dividends	Ordinary shares	121,840,851	-	-
		Preference shares	17,069,534	-	-
	Dividend rate (based on par value)		140%	-	-
	Dividend amount	Ordinary shares	₩ 852,886	-	-
		Preference shares	119,487	-	-
	Total	₩	972,373	-	-
2 nd quarter	Number of shares eligible for dividends	Ordinary shares	121,038,051	123,958,561	
		Preference shares	16,868,834	17,580,920	
	Dividend rate (based on par value)		140%	20%	
	Dividend amount	Ordinary shares	₩ 847,266	₩ 123,958	
		Preference shares	118,082	17,581	
	Total	₩	965,348	₩ 141,539	
3 rd quarter	Number of shares eligible for dividends	Ordinary shares	120,316,189	-	-
		Preference shares	16,691,095	-	-
	Dividend rate (based on par value)		140%	-	-
	Dividend amount	Ordinary shares	₩ 842,213	-	-
		Preference shares	116,838	-	-
	Total	₩	959,051	-	-

(B) Year-end dividends (Record date: December 31, 2017 and 2016)

(In millions of Korean won and number of shares)

		2017		2016	
Number of shares eligible for dividends	Ordinary shares	119,688,369 shares		122,697,651 shares	
	Preference shares	16,530,409 shares		17,283,734 shares	
Dividend rate(based on par value)	Ordinary shares	430%		550%	
	Preference shares	431%		551%	
Dividend amount	Ordinary shares	₩ 2,573,300		₩ 3,374,185	
	Preference shares	356,230		476,167	
	Total	₩ 2,929,530		₩ 3,850,352	

23. Other Components of Equity

(A) Other components of equity as at December 31, 2017 and 2016 consist of the following:

<i>(In millions of Korean won)</i>	December 31, 2017		December 31, 2016	
Treasury stock	₩	(6,228,187)	₩	(9,750,326)
Unrealized gains on available-for-sale financial assets		1,879,774		1,390,624
Share of other comprehensive income of associates and joint ventures		40,394		94,694
Foreign currency translation		(9,192,002)		(2,902,076)
Remeasurement of net defined benefit liabilities		(405,206)		(811,529)
Others		6,036		44,027
Total	₩	(13,899,191)	₩	(11,934,586)

(B) The Company repurchases registered ordinary shares and non-voting preference shares for the purpose of stock price stability and increase in shareholder value. The Company recognizes the repurchase amount in other components of equity. Treasury stock as at December 31, 2017 and 2016 consists of the following:

<i>(Number of shares and in millions of Korean won)</i>	December 31, 2017		December 31, 2016	
	Ordinary Shares	Preference Shares	Ordinary Shares	Preference Shares
Number of shares	9,410,125	1,720,171	17,981,686	3,229,693
Acquisition cost	₩ 5,560,506	₩ 667,681	₩ 8,871,509	₩ 878,817

24. Expenses by Nature

Expenses by nature for the years ended December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	2017		2016	
Changes in finished goods and work in process	₩	(3,495,615)	₩	626,040
Raw materials used and merchandise purchased		79,647,552		73,512,658
Wages and salaries		22,380,595		19,269,035
Pension		976,328		1,197,022
Depreciation		20,593,616		19,312,520
Amortization		1,523,776		1,400,445
Welfare		3,812,521		3,495,336
Commission and service charges		7,703,111		8,002,513
Advertising		5,350,839		4,432,109
Sales promotion expenses		7,262,078		7,080,554
Other		40,175,537		34,297,841
Total¹	₩	185,930,338	₩	172,626,073

¹ Equal to the sum of cost of sales and selling and administrative expenses on the consolidated statements of profit or loss.

25. Selling and Administrative Expenses

Selling and administrative expenses for the years ended December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	2017		2016	
1) Selling and Administrative Expenses				
Wages and salaries	₩	6,540,715	₩	5,687,494
Pension		256,448		288,767
Commissions and service charges		7,703,111		8,002,513
Depreciation		948,393		780,223
Amortization		388,227		221,593
Advertising		5,350,839		4,432,109
Sales promotion expenses		7,262,078		7,080,554
Transportation		3,602,838		3,334,693
Warranty		3,051,538		3,752,603
Other		5,179,878		4,656,428
2) Research and development expenses				
Total expenses		16,803,153		14,792,343
Capitalized expenses		(447,541)		(680,962)
Total	₩	56,639,677	₩	52,348,358

26. Other Non-Operating Income and Expense

Details of other non-operating income for the years ended December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	2017		2016	
Dividend income	₩	128,946	₩	239,899
Rental income		143,482		107,664
Gain on disposal of investments		199,311		2,053,744
Gain on disposal of property, plant and equipment		115,001		193,020
Other		2,423,917		643,934
Total	₩	3,010,657	₩	3,238,261

Details of other non-operating expense for the years ended December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	2017		2016	
Loss on disposal of property, plant and equipment	₩	122,548	₩	126,516
Donations		309,763		407,087
Impairment losses on investments		13,538		341,790
Impairment losses on intangible assets		143,401		473,494
Other		830,398		1,114,927
Total	₩	1,419,648	₩	2,463,814

27. Financial Income and Expenses

(A) Details of financial income and expenses for the years ended December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	2017		2016	
Financial income				
Interest income:				
Interest income from loans and receivables	₩	1,613,010	₩	1,475,357
Interest income from available-for-sale financial assets		1,157		28,961
Interest income from held-to-maturity financial assets		56		-
Foreign exchange differences		7,278,888		9,052,485
Gains from derivatives		844,280		828,832
Total	₩	9,737,391	₩	11,385,645

<i>(In millions of Korean won)</i>	2017		2016	
Financial expenses				
Interest expenses:				
Interest expense from financial liabilities measured at amortized cost	₩	358,911	₩	351,009
Other financial liabilities		296,491		236,822
Foreign exchange differences		7,269,465		9,232,249
Losses from derivatives		1,054,046		886,533
Total	₩	8,978,913	₩	10,706,613

(B) The Company recognizes foreign exchange gains and losses arising from foreign currency transactions and translation as financial income and expenses.

28. Income Tax Expense

(A) Income tax expense for the years ended December 31, 2017 and 2016, consists of the following:

<i>(In millions of Korean won)</i>	2017	2016
Current taxes:		
Current tax on profits for the year	₩ 10,642,009	₩ 6,161,609
Adjustments in respect to prior years	301,924	(244,791)
Deferred taxes:		
Changes in carryforward of unused tax credits	(50,413)	(166,206)
Changes in temporary differences	3,417,698	1,869,700
Changes in carryforward of unused tax losses	(309,604)	397,329
Other	7,606	(30,081)
Items charged directly to equity	-	-
Income tax expense	₩ 14,009,220	₩ 7,987,560

(B) The tax on the Company's profit before tax differs from the amount that would arise using the weighted average statutory tax rate applicable to profits of the Company as follows:

<i>(In millions of Korean won)</i>	2017	2016
Income before tax	₩ 56,195,967	₩ 30,713,652
Tax calculated at weighted average of applicable tax rates ¹	14,425,196	7,009,003
Tax effects of:		
Permanent differences	(239,050)	(687,901)
Temporary differences for which no deferred income tax was recognized	(24,156)	(12,825)
Tax credits	(3,194,533)	(608,218)
Results of interest in subsidiaries, associates and joint ventures	3,281,740	2,288,893
Impact of changes in tax rates	(532,165)	1,280
Other	292,188	(2,672)
Income tax expense	₩ 14,009,220	₩ 7,987,560

¹ The weighted average of statutory tax rates are applied to the respective profits of the Company applicable to each tax authority as at December 31, 2017 and 2016.

(C) Changes in deferred income tax assets and liabilities resulting from the tax effect of temporary differences for the years ended December 31, 2017 and 2016 are as follows:

(1) 2017

(In millions of Korean won)	Temporary Differences			Deferred Income Tax Assets (Liabilities)		
	Balance as at January 1	Increase (Decrease)	Balance as at December 31	Balance as at January 1	Increase (Decrease)	Balance as at December 31
Deferred tax arising from temporary differences						
Special reserves appropriated for tax purposes	₩ (368)	₩ -	₩ (368)	₩ (89)	₩ (12)	₩ (101)
Revaluation of land	(3,444,254)	19,021	(3,425,233)	(833,510)	(108,429)	(941,939)
Investments in subsidiaries, associates and joint ventures ¹	(59,270,306)	(12,634,105)	(71,904,411)	(7,536,737)	(3,621,634)	(11,158,371)
Depreciation	2,908,546	(6,462,748)	(3,554,202)	644,126	(1,419,254)	(775,128)
Accrued income	(258,310)	66,156	(192,154)	(78,971)	32,186	(46,785)
Provisions and accrued expenses	13,458,781	(746,637)	12,712,144	3,232,788	168,637	3,401,425
Foreign currency translation	165,558	(401,645)	(236,087)	44,607	(107,490)	(62,883)
Asset impairment losses	1,227,353	537,407	1,764,760	301,206	185,953	487,159
Other	1,479,671	2,077,446	3,557,117	650,132	115,501	765,633
Subtotal	(43,733,329)	(17,545,105)	(61,278,434)	(3,576,448)	(4,754,542)	(8,330,990)
Deferred tax arising from carryforwards						
Unused tax losses	₩ 4,691,360	₩ 582,852	₩ 5,274,212	₩ 1,098,647	₩ 309,604	₩ 1,408,251
Unused tax credits	873,455	27,182	900,637	721,247	50,413	771,660
Deferred tax recognized in other comprehensive income						
Valuation of available-for-sale financial instruments	₩ (1,969,833)	₩ (718,531)	₩ (2,688,364)	₩ (446,756)	₩ (207,322)	₩ (654,078)
Actuarial valuation	1,023,559	(489,430)	534,129	231,246	(75,183)	156,063
Subtotal	₩ (946,274)	₩ (1,207,961)	₩ (2,154,235)	₩ (215,510)	₩ (282,505)	₩ (498,015)
Deferred tax assets						₩ 5,061,687
Deferred tax liabilities						(11,710,781)
Total						₩(6,649,094)

¹ Deferred tax assets were not recognized if it is probable that the temporary differences will not reverse in the foreseeable future for investments in subsidiaries, associates and joint ventures.

(2) 2016

(In millions of Korean won)	Temporary Differences			Deferred Income Tax Assets (Liabilities)		
	Balance as at January 1	Increase (Decrease)	Balance as at December 31	Balance as at January 1	Increase (Decrease)	Balance as at December 31
Deferred tax arising from temporary differences						
Special reserves appropriated for tax purposes	₩ (7,368)	₩ 7,000	₩ (368)	₩ (1,783)	₩ 1,694	₩ (89)
Revaluation of land	(3,455,958)	11,704	(3,444,254)	(836,342)	2,832	(833,510)
Investments in subsidiaries, associates and joint ventures ¹	(41,729,872)	(17,540,434)	(59,270,306)	(5,173,897)	(2,362,840)	(7,536,737)
Depreciation	2,627,073	281,473	2,908,546	651,988	(7,862)	644,126
Accrued income	(309,545)	51,235	(258,310)	(71,086)	(7,885)	(78,971)
Provisions and accrued expenses	13,322,516	136,265	13,458,781	3,519,564	(286,776)	3,232,788
Foreign currency translation	221,793	(56,235)	165,558	58,567	(13,960)	44,607
Asset impairment losses	652,768	574,585	1,227,353	160,975	140,231	301,206
Other	(193,612)	1,673,283	1,479,671	(14,734)	664,866	650,132
Subtotal	(28,872,205)	(14,861,124)	(43,733,329)	(1,706,748)	(1,869,700)	(3,576,448)
Deferred tax arising from carryforwards						
Unused tax losses	₩ 6,178,327	₩(1,486,967)	₩ 4,691,360	₩ 1,495,976	₩ (397,329)	₩ 1,098,647
Unused tax credits	631,362	242,093	873,455	555,041	166,206	721,247
Deferred tax recognized in other comprehensive income						
Valuation of available-for-sale financial instruments	₩(1,976,829)	₩ 6,996	₩(1,969,833)	₩ (453,192)	₩ 6,436	₩ (446,756)
Actuarial valuation	2,299,154	(1,275,595)	1,023,559	543,239	(311,993)	231,246
Subtotal	322,325	(1,268,599)	(946,274)	90,047	(305,557)	(215,510)
Deferred tax assets						₩ 5,321,450
Deferred tax liabilities						(7,293,514)
Total						₩ (1,972,064)

¹ Deferred tax assets were not recognized if it is probable that the temporary differences will not reverse in the foreseeable future for investments in subsidiaries, associates and joint ventures.

The Company periodically assesses its ability to recover deferred tax assets. In the event of a significant uncertainty regarding the Company's ultimate ability to recover such assets, deferred tax assets are recognized only to the extent that it is probable that future taxable profit will be available against which the temporary differences can be utilized.

Temporary differences whose deferred tax effects were not recognized due to uncertainty regarding the ultimate realizability of such assets as at December 31, 2017 and 2016, are as follows:

(In millions of Korean won)	2017	2016
Unused tax losses ¹	₩ 364,236	₩ 58,969
Unused tax credits ¹	69,800	31,205

¹ Expiry dates of unused tax losses and unused tax credits for which no deferred tax asset is recognized in the balance sheet are as follows:

<i>(In millions of Korean won)</i>	2018		2019		2020		2021 and after	
Undisposed accumulated deficit	₩	233	₩	85	₩	85	₩	363,833
Tax credit carryforwards		13,019		10,656		20,784		25,341

(D) The liquidity analysis of deferred tax assets and deferred tax liabilities for the years ended December 31, 2017 and 2016 is as follows:

<i>(In millions of Korean won)</i>	2017		2016	
Deferred tax assets				
Deferred tax assets to be recovered within 12 months	₩	2,279,651	₩	3,066,577
Deferred tax assets to be recovered after more than 12 months		2,782,036		2,254,873
Subtotal		5,061,687		5,321,450
Deferred tax liabilities				
Deferred tax liabilities to be recovered after more than 12 months		(11,710,781)		(7,293,514)
Subtotal		(11,710,781)		(7,293,514)
Total	₩	(6,649,094)	₩	(1,972,064)

29. Earnings per Share

(A) Basic earnings per share

Basic earnings per share for the years ended December 31, 2017 and 2016 are calculated as follows:

(1) Ordinary shares

<i>(In millions of Korean won, thousands of number of shares)</i>	2017		2016	
Profit or loss attributable to owners of the Parent company	₩	41,344,569	₩	22,415,655
Profit or loss available for ordinary shares		36,323,723		19,647,199
Weighted-average number of ordinary shares outstanding		121,132		124,375
Basic earnings per share(Korean won)	₩	299,868	₩	157,967

(2) Preference shares

<i>(In millions of Korean won, thousands of number of shares)</i>	2017		2016	
Profit or loss attributable to owners of the Parent company	₩	41,344,569	₩	22,415,655
Profit or loss available for preference shares		5,020,846		2,768,456
Weighted-average number of preference shares outstanding		16,893		17,692
Basic earnings per preference share(Korean won)	₩	297,217	₩	156,480

(B) Diluted earnings per share

The Company does not have potential ordinary shares and as a result, basic earnings per share and diluted earnings per share are the same for the years ended December 31, 2017 and 2016.

30. Cash Flows

(A) Adjustments and changes in assets and liabilities arising from operating activities for the years ended December 31, 2017 and 2016 are as follows:

- Adjustments

<i>(In millions of Korean won)</i>	2017	2016
Adjustments for:		
Income tax expense	₩ 14,009,220	₩ 7,987,560
Financial income	(3,447,029)	(3,521,050)
Financial expenses	2,590,683	2,500,020
Post-employment benefits	976,328	1,197,022
Depreciation	20,593,616	19,312,520
Amortization	1,523,776	1,400,445
Bad debt expenses (reversal)	206,561	117,207
Dividend income	(128,946)	(239,899)
Gain on valuation of equity method	(201,442)	(19,501)
Gain on disposal of property, plant and equipment	(115,001)	(193,020)
Loss on disposal of property, plant and equipment	122,548	126,516
Obsolescence and scrapping of inventories	1,296,366	2,959,042
Gain on disposal of investments	(199,311)	(2,053,744)
Impairment losses on investments	13,538	341,790
Impairment losses on intangible assets	143,401	473,494
Other income/expense	(1,173,076)	366,069
Total	₩ 36,211,232	₩ 30,754,471

- Changes in assets and liabilities arising from operating activities

<i>(In millions of Korean won)</i>	2017	2016
Changes in assets and liabilities :		
(Increase) decrease in trade receivables	₩ (6,597,339)	₩ 1,473,776
Increase in other receivables	(1,078,836)	(160,500)
(Increase) decrease in advance payments	(244,027)	145,053
Decrease (increase) in prepaid expenses	580,977	(46,026)
Increase in inventories	(8,444,506)	(2,830,602)
Increase in trade payables	4,006,410	200,240
Increase in other payables	1,096,019	1,144,756
Decrease in advances received	(73,214)	(105,460)
Increase (decrease) in withholdings	651,350	(302,901)
Increase in accrued expenses	1,292,420	1,136,440
Decrease in provisions	(106,064)	(1,604,824)
Payment of post-employment benefits	(582,711)	(474,112)
Increase in plan assets	(172,339)	(76,255)
Others	(948,687)	319,462
Total	₩ (10,620,547)	₩ (1,180,953)

(B) The Company's statements of cash flows are prepared using the indirect method. Significant transactions not affecting cash flows for the years ended December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	2017	2016
Valuation of available-for-sale financial assets	₩ 721,730	₩ 798,698
Valuation of investments in associates, joint ventures	(54,424)	36,152
Reclassification of construction in progress and machinery in transit to property, plant and equipment	47,499,881	17,131,444
Reclassification of current maturities of long-term borrowings	273,418	19,283
Reclassification of current maturities of bonds	5,357	1,214,543
Reclassification of available-for-sale financial assets to investments in associates	-	1,742,242
Reclassification of Printing Solutions division to held-for-sale	-	508,228

(C) The Company reported cash receipts and payments arising from transactions occurring frequently and short-term financial instruments and borrowings on a net basis.

(D) Changes in liabilities arising from financial activities for the year ended December 31, 2017 are as follows:

(In millions of Korean won)	As at January 1, 2017	Cash flows	Non-cash changes				As at December 31, 2017
			Business Combination	Amortization	Others ¹		
Short-Term Borrowings	₩ 12,746,789	₩ 2,730,676	₩ 558,396	₩ -	₩ (268,242)	₩ 15,767,619	
Debentures & Long-Term Borrowings	2,535,596	(142,492)	1,001,537	332	(348,547)	3,046,426	
Total	₩ 15,282,385	₩ 2,588,184	1,559,933	₩ 332	₩ (616,789)	₩ 18,814,045	

¹ Others include effects of changes in foreign currency exchange rates.

31. Financial Risk Management

The Company's financial risk management focuses on minimizing market risk, credit risk, and liquidity risk arising from operating activities. To mitigate these risks, the Company implements and operates a financial risk policy and program that closely monitors and manages such risks. In addition, the Company uses derivatives to hedge certain risk exposure.

The finance team mainly carries out the Company's financial risk management. With the cooperation of the Company's divisions, domestic and foreign subsidiaries, the finance team periodically measures, evaluates and hedges financial risk and also establishes and implements the global financial risk management policy.

The Company also operates local finance centers in accordance with global financial risk management across major regions including the United States, United Kingdom, Singapore, China, Brazil and Russia.

The Company's financial assets that are under financial risk management are comprised of cash and cash equivalents, short-term financial instruments, available-for-sale financial assets, trade and other receivables and other financial assets. The Company's financial liabilities under financial risk management are comprised of trade and other payables, borrowings, debentures, and other financial liabilities.

(A) Market risk

(1) Foreign exchange risk

Revenues and expenses arise from foreign currency transactions and exchange positions, and the most widely used currencies are the US Dollar, EU's Euro, Japanese Yen and Indian Rupee.

To minimize foreign exchange risk arising from operating activities, the Company's foreign exchange management policy requires normal business transactions to be in local currency or for the cash-in currency to be matched up with the cash-out currency. In addition, the Company makes use of derivatives to minimize uncertainty of cash flows and fluctuation of profit or loss occurring from foreign exchange risk. The Company's foreign exchange risk management policy also defines foreign exchange risk, measuring period, controlling responsibilities, management procedures, hedging period and hedge ratio.

The Company limits all speculative foreign exchange transactions and operates a system to manage receivables and payables denominated in foreign currency. It evaluates, manages and reports foreign currency exposures to receivables and payables.

As at December 31, 2017 and 2016, the foreign currency exposure to financial assets and liabilities of a 5% currency rate change against the Korean won are presented below:

(In millions of Korean won)	December 31, 2017		December 31, 2016	
	Increase	Decrease	Increase	Decrease
USD	₩ 278,140	₩ (278,140)	₩ 222,149	₩ (222,149)
EUR	79,760	(79,760)	138,084	(138,084)
JPY	(100,127)	100,127	(61,294)	61,294
INR	77,963	(77,963)	38,858	(38,858)

(2) Price risk

The Company's investment portfolio consists of direct and indirect investments in equity securities classified as available-for-sale, which is in line with the Company's strategy.

As at December 31, 2017 and 2016, a price fluctuation in relation to marketable equity securities by 1% would result in changes in other comprehensive income (before income tax) of ₩29,086 million and ₩23,622 million, respectively.

(3) Interest rate risk

Risk of changes in interest rates for floating interest rate financial instruments is defined as the risk that the fair value of components of the statement of financial position, and future cash flows of interest income (expenses) of a financial instrument, will fluctuate because of changes in market interest rates. The Company's position with regard to interest rate risk exposure is mainly driven by its floating interest rate debt obligations and interest-bearing deposits. The Company implemented policies and operates to minimize uncertainty arising from changes in interest rates and financial expenses.

In order to avoid interest rate risk, the Company maintains minimum external borrowings by facilitating cash pooling systems on a regional and global basis. The Company manages exposed interest rate risk via periodic monitoring and handling risk factors on a timely basis.

The sensitivity risk of the Company is determined based on the following assumption:

- Changes in market interest rates that could impact the interest income and expenses of floating interest rate financial instruments

Based on the above assumption, changes to profit and net equity as a result of a 1% change in interest rates on borrowings are presented below:

(In millions of Korean won)	December 31, 2017		December 31, 2016	
	Increase	Decrease	Increase	Decrease
Financial assets	₩ 49,117	₩ (49,117)	₩ 64,803	₩ (64,803)
Financial liabilities	(22,643)	22,643	(9,123)	9,123
Net effect	₩ 26,474	₩ (26,474)	₩ 55,680	₩ (55,680)

(B) Credit risk

Credit risk arises during the normal course of transactions and investing activities where clients or other parties fail to discharge an obligation. The Company monitors and sets the client's and counterparty's credit limit on a periodic basis based on the client's and counterparty's financial conditions, default history and other important factors. Adequate insurance coverage is maintained for accounts receivables related to trading partners situated in higher risk countries.

Credit risk can arise from transactions with financial institutions which include financial instrument transactions such as cash and cash equivalents, savings, and derivative instruments. To minimize such risk, the Company transacts only with banks which have strong international credit rating (S&P A and above), and all new transactions with financial institutions with no prior transaction history are approved, managed and monitored by the Company's finance team and the local finance center. The Company requires separate approval for contracts with restrictions.

As at December 31, 2017 and 2016 the Company estimates that its maximum exposure to credit risk is the carrying amount of its financial assets, net of impairment losses.

(C) Liquidity risk

Due to large investments made by the Company, maintaining adequate levels of liquidity risk is critical. The Company strives to achieve this goal by periodically forecasting its capital balance, estimating required cash levels, and managing income and expenses.

The Company manages its liquidity risk by periodically forecasting projected cash flows. If abnormal signs are identified, the Company works with the local finance center and provides liquidity support by utilizing a globally integrated finance structure, such as Cash Pooling. In addition, the Company maintains a liquidity management process which provides additional financial support by the local finance center and the Company. The Cash Pooling program allows sharing of surplus funds among entities and contributes to minimizing liquidity risk and strengthening the Company's competitive position by reducing capital operation expenses and financial expenses.

In addition, the Company mitigates liquidity risk by contracting with financial institutions with respect to bank overdrafts and foreign trade finance, and by providing payment guarantees to subsidiaries. For large scale facility investments, liquidity risk is minimized by utilizing internal reserves and long term borrowings according to the capital injection schedule.

As at December 31, 2017 and 2016, the following table is an undiscounted cash flow analysis for financial liabilities that are presented on the statements of financial position according to their remaining contractual maturity.

(In millions of Korean won)	December 31, 2017				
	Less than 3 months	4-6 months	7-12 months	1-5 years	More than 5 years
Financial liabilities	₩ 45,323,281	₩ 927,746	₩ 1,144,922	₩ 4,633,210	₩ 875,283

(In millions of Korean won)	December 31, 2016				
	Less than 3 months	4-6 months	7-12 months	1-5 years	More than 5 years
Financial liabilities	₩ 40,918,912	₩ 1,588,798	₩ 150,744	₩ 4,346,200	₩ 50,073

The table above shows the Company's financial liabilities based on the remaining period at the statement of financial position date until the contractual maturity date. The amounts disclosed in the table are the contractual undiscounted cash flows.

The Company's trading portfolio of derivative instruments has been included at its fair value of ₩47,478 million (December 31, 2016: ₩74,697 million). These contracts are managed on a net-fair value basis rather than by maturity date. Net settled derivatives consist of forwards on currency rates used by the Company to manage the exchange rate profile.

Derivatives that are settled on a gross basis by delivery of the underlying item, including derivatives for hedging, will be settled within the next 53 months from the end of the reporting period. These derivative are not included in the table above.

The maximum liquidity risk exposure from those other than the above financial liabilities (e.g., payment guarantees for affiliated companies and performance bonds) as at December 31, 2017 is ₩49,937 million (December 31, 2016: ₩59,016 million).

(D) Derivative Financial Instruments

The Company applies cash flow hedge accounting to hedge the foreign currency risk of forecasted transaction including hedging the price risk associated with forecast inventory purchases. Details of derivative financial instruments that qualify as cash flow hedges as at December 31, 2017 are as follows:

December 31, 2017

<i>(in millions of Korean won)</i>	Assets	Liabilities
Currency forward exchange contracts		
Current items	₩ 45,396	₩ 41,646
Non-current items	31,915	25,806
Total	₩ 13,481	₩ 15,840

As at December 31, 2016, the Company did not have derivative financial instruments designated as cash flow hedges.

During the year ended for December 31, 2017, The Company recognizes the gains or losses relating to the effective portion of changes in fair value of derivatives that are designated and qualify as cash flow hedges in other comprehensive income, which amount to ₩37,121 million (after tax), and recognizes the gains or losses relating to the ineffective portion in profit or loss, which amount to ₩4,940 million (before tax). Gains or losses reclassified directly from other comprehensive income to profit or loss amount to ₩3,413 million (after tax), and gains or losses reclassified from other comprehensive income to the carrying amount of inventory amount to ₩29,663 million (after tax).

(E) Capital risk management

The purpose of capital management is to maintain a sound capital structure. The Company monitors capital on the basis of the ratio of total liabilities to total equity. This ratio is calculated by dividing total liabilities by total equity in the financial statements.

The Company's capital risk management policy has not changed since the previous financial year. The Company has maintained an AA- and A1 credit rating from S&P and Moody's, respectively.

The total liabilities to equity ratios as at December 31, 2017 and 2016 are as follows:

<i>(In millions of Korean won)</i>	December 31, 2017	December 31, 2016
Total liabilities	₩ 87,260,662	₩ 69,211,291
Total equity	214,491,428	192,963,033
Total liabilities to equity ratio	40.7%	35.9%

(F) Fair value estimation

(1) Carrying amounts and fair values of financial instruments by category as at December 31, 2017 and 2016 are as follows:

(In millions of Korean won)	December 31, 2017		December 31, 2016	
	Carrying amount	Fair value	Carrying amount	Fair value
Financial assets				
Cash and cash equivalents	₩ 30,545,130	₩ ¹	₩ 32,111,442	₩ ¹
Short-term financial instruments	49,447,696	¹	52,432,411	¹
Short-term available-for-sale financial assets	3,191,375	3,191,375	3,638,460	3,638,460
Trade receivables	27,695,995	¹	24,279,211	¹
Long-term available-for-sale financial assets ²	7,752,180	6,561,155	6,804,276	5,826,507
Held-to-maturity financial assets	106,751	¹	-	-
Others ³	6,325,825	113,098	3,459,863	919,071
Total financial assets	₩ 125,064,952		₩ 122,725,663	
Financial liabilities				
Trade payables	₩ 9,083,907	₩ ¹	₩ 6,485,039	₩ ¹
Short-term borrowings	15,767,619	¹	12,746,789	¹
Other payables ³	12,106,609	316,928	10,225,271	¹
Current portion of long-term liabilities	278,619	¹	1,232,817	¹
Debentures	953,361	978,643	58,542	76,129
Long-term borrowings	1,814,446	¹	1,244,238	1,225,455
Long-term other payables ³	1,746,184	28,285	3,009,659	3,022,821
Others ³	10,954,513	222,012	11,942,469	74,697
Total financial liabilities	₩ 52,705,258		₩ 46,944,824	

¹ Assets and liabilities whose carrying amount is a reasonable approximation of fair value are excluded from the fair value disclosures.

² Amounts measured at cost of ₩1,191,025 million (December 31, 2016: ₩977,769 million) are excluded as the range of reasonable fair value estimates is significant and the probabilities of the various estimates cannot be reasonably assessed.

³ Assets measured at cost of ₩6,212,727 million (December 31, 2016: ₩2,452,118 million) and liabilities measured at cost of ₩24,240,081 million (December 31, 2016: ₩11,867,772) are excluded as the carrying amount is a reasonable approximation of fair value.

(2) Fair value hierarchy classifications of the financial instruments that are measured at fair value or its fair value is disclosed as at December 31, 2017 and 2016 are as follows :

December 31, 2017					
(In millions of Korean won)	Level 1	Level 2	Level 3	Total	
1) Assets					
Short-term available-for-sale financial assets	₩ -	₩ 3,191,375	₩ -	₩	3,191,375
Long-term available-for-sale financial assets	2,908,581	-	3,652,574		6,561,155
Others	-	113,098	-		113,098
2) Liabilities					
Other payables	₩ -	₩ -	₩ 316,928	₩	316,928
Debentures	-	978,643	-		978,643
Long-term other payables	-	-	28,285		28,285
Others	-	215,307	6,705		222,012

December 31, 2016					
(In millions of Korean won)	Level 1	Level 2	Level 3	Total	
1) Assets					
Short-term available-for-sale financial assets	₩ -	₩ 3,638,460	₩ -	₩	3,638,460
Long-term available-for-sale financial assets	2,362,235	-	3,464,272		5,826,507
Others	-	919,071	-		919,071
2) Liabilities					
Debentures	₩ -	₩ 76,129	₩ -	₩	76,129
Long-term borrowings	-	1,225,455	-		1,225,455
Long-term other payables	-	2,680,119	342,702		3,022,821
Others	-	74,697	-		74,697

The levels of the fair value hierarchy and its application to financial assets and liabilities are described below.

- Level 1: Quoted prices (unadjusted) in active markets for identical assets or liabilities
- Level 2: Inputs other than quoted prices included within Level 1 that are observable for the asset or liability, either directly or indirectly
- Level 3: Inputs for the asset or liability that are not based on observable market data (that is, unobservable inputs)

The fair value of financial instruments traded in active markets is based on quoted market prices at the statement of financial position date. A market is regarded as active if quoted prices are readily and regularly available from an exchange, dealer, broker, industry group, pricing service, or regulatory agency, and those prices represent actual and regularly occurring market transactions on an arm's length basis. The quoted market price used for financial assets held by the Company is the current bid price. These instruments are included in Level 1. Instruments included in Level 1 are listed equity investments classified as trading securities or available-for-sale financial assets.

The fair value of financial instruments that are not traded in an active market (for example, over-the-counter derivatives) is determined by using valuation techniques. These valuation techniques maximize the use of observable market data where it is available and rely as little as possible on entity specific estimates. If all significant inputs required to fair value an instrument are observable, the instrument is included in Level 2.

If one or more of the significant inputs are not based on observable market data, the instrument is included in Level 3.

The Company performs the fair value measurements required for financial reporting purposes, including Level 3 fair values and discusses valuation processes and results at least once every quarter in line with the Company's quarterly reporting dates. The Company's policy is to recognize transfers between levels at the end of the reporting period, if corresponding events or changes in circumstances have occurred.

Specific valuation techniques used to value financial instruments include:

- Quoted market prices or dealer quotes for similar instruments
- The fair value of derivatives is determined using forward exchange rates at the statement of financial position date, with the resulting value discounted back to present value

Other techniques, such as discounted cash flow analysis, are used to determine fair value for the remaining financial instruments. For trade and other receivables, the carrying amount approximates a reasonable estimate of fair value.

(3) Valuation technique and the inputs

The Company utilizes a present value technique to discount future cash flows using a proper interest rate for corporate bonds, government and public bonds, and bank debentures that are classified as Level 2 in the fair value hierarchy.

The following table presents the valuation technique and the inputs used for major financial instruments classified as Level 3.

(In millions of Korean won)

(In millions of Korean won)				
Classification	Fair Value	Valuation Technique	Level 3 Inputs	Input Range (Weighted Average)
Long-term available-for-sale financial assets				
Maltani	₩ 15,137	Discounted cash flow	Permanent growth rate	-1.00% ~ 1.00% (0%)
			Weighted average cost of capital	7.69%~9.69% (8.69%)
Samsung Venture Investment	7,784	Discounted cash flow	Permanent growth rate	-1.00% ~ 1.00% (0%)
			Weighted average cost of capital	19.56%~21.56% (20.56%)
Corning Incorporated convertible preference shares	3,629,653	Trinomial model	Risk adjusted discount rate	4.98%~6.98% (5.98%)
			Price volatility	26.5%~32.5% (29.5%)
Other payables				
Contingent financial liability	316,928	Discounted cash flow	Discount rate	3.81%~4.65% (4.23%)
Long-term others payables				
Contingent financial liability	28,285	Monte Carlo simulation	Discount rate	10.50%
			Risk free rate	0.97%
			Asset volatility	34.54%
			Credit spread	2.12%
Others				
Contingent financial liability	4,982	Monte Carlo simulation	Discount rate	17.57%
			Risk free rate	0.86%
			Operational leverage ratio	60.00%
			Gross margin discount rate	6.68%
Contingent financial liability	1,723	Probability-weighted discounted cash flow	Weighted average cost of capital	8.60%
			Credit risk	2.12%

(4) Changes in Level 3 instruments:

(In millions of Korean won)

(In millions of Korean won)	2017		2016	
Financial Assets				
Balance as at January 1	₩	3,464,272	₩	3,488,344
Disposals		-		(14,805)
Amount recognized in profit or loss		-		(795)
Amount recognized in other comprehensive income		188,302		695,631
Others		-		(704,103)
Balance as at December 31	₩	3,652,574	₩	3,464,272

(In millions of Korean won)

(In millions of Korean won)	2017		2016	
Financial Liabilities				
Balance as at January 1	₩	342,702	₩	312,738
Settled		(2,693)		-
Amount recognized in profit or loss		(24,372)		29,964
Acquisition in the business combination		39,083		-
Others		(2,802)		-
Balance as at December 31	₩	351,918	₩	342,702

(5) Sensitivity analysis for recurring fair value measurements categorized within Level 3

Sensitivity analysis of financial instruments is performed to measure favorable and unfavorable changes in the fair value of financial instruments which are affected by unobservable parameters, using a statistical technique. When the fair value is affected by more than two input parameters, the amounts represent the most favorable or most unfavorable.

The results of the sensitivity analysis for the effect on profit or loss (before tax amount for other comprehensive income or loss) from changes in inputs for major financial instruments which are categorized within Level 3 and subject to sensitivity analysis, are as follows:

(In millions of Korean won)

Classification	Favorable Changes		Unfavorable Changes	
	Profit or Loss	Equity	Profit or Loss	Equity
Long-term available-for-sale financial assets ¹	₩ -	₩ 189,757	₩ -	₩ (212,458)
Other payables ²	445	445	(446)	(446)
Total	₩ 445	₩ 190,202	₩ (446)	₩ (212,904)

¹ For equity securities, changes in fair value are calculated with the correlation among growth rate (-1%~1%), volatility (26.5%~32.5%) and discount rate, which are significant unobservable inputs.

² The fair value of other payables is calculated by increasing or decreasing the discount rate by 10%, which is the significant unobservable input.

32. Segment Information

(A) Operating segment information

The chief operating decision maker has been identified as the Management Committee. The Company determines operating segments based on the units reported to the Management Committee. The Management Committee reviews operating profits of each operating segment in order to assess performance and to make strategic decisions regarding allocation of resources to the segment.

The operating segments are product-based and are identified based on the internal organization and revenue streams. As at the reporting date, the operating segments are comprised of CE, IM, Semiconductor, DP, Harman and others.

The segment information for each reporting period such as depreciation, amortization of intangible assets, and operating profit is prepared after the allocation of internal transaction adjustments. Total assets and liabilities of each operating segment are excluded from the disclosure as these have not been provided regularly to the Management Committee.

(1) For the year ended December 31, 2017

(In millions of Korean won)	CE	IM	DS				Total ¹	Intercompany elimination within the Company	Consolidated
			Total ¹	Semi- conductor	DP	Harman			
Total segment revenue	102,788,437	226,004,156	218,781,719	141,819,969	72,108,721	9,171,835	561,861,438	(322,286,062)	239,575,376
Intercompany revenue	(57,679,518)	(119,335,838)	(110,614,239)	(67,564,374)	(37,643,358)	(2,069,206)	(322,286,062)	322,286,062	-
Net revenue ²	45,108,919	106,668,318	108,167,480	74,255,595	34,465,363	7,102,629	239,575,376	-	239,575,376
Depreciation	600,487	1,264,596	18,175,388	12,920,501	5,080,610	184,458	20,593,616	-	20,593,616
Amortization	62,935	152,939	935,022	797,630	122,300	172,324	1,523,776	-	1,523,776
Operating profit	1,653,300	11,827,324	40,327,933	35,204,143	5,398,401	57,421	53,645,038	-	53,645,038

¹ Includes other amounts not included in specific operating segments.

² Segment net revenue includes intersegment revenue.

(2) For the year ended December 31, 2016

(In millions of Korean won)	CE ³	IM	DS				Total ¹	Intercompany elimination within the Company	Consolidated
			Total ¹	Semi- conductor	DP	Harman			
Total segment revenue	104,870,428	211,523,973	159,473,455	99,527,926	55,884,739		482,046,125	(280,179,380)	201,866,745
Intercompany revenue	(59,767,666)	(111,221,861)	(81,325,252)	(48,370,924)	(28,956,095)		(280,179,380)	280,179,380	-
Net revenue ²	45,102,762	100,302,112	78,148,203	51,157,002	26,928,644		201,866,745	-	201,866,745
Depreciation	547,956	1,303,509	17,041,961	12,548,152	4,271,617		19,312,520	-	19,312,520
Amortization	60,596	175,351	960,611	812,652	130,398		1,400,445	-	1,400,445
Operating profit	2,712,148	10,807,569	15,850,986	13,595,004	2,226,626		29,240,672	-	29,240,672

¹ Includes other amounts not included in specific operating segments.

² Segment net revenue includes intersegment revenue.

³ CE is restated due to the reclassification of the Printing Solutions business segment, which has been changed from CE to Others.

(B) Regional information

The regional segment information provided to the Management Committee for the reportable segments as at and for the years ended December 31, 2017 and 2016 is as follows:

(1) For the year ended December 31, 2017

<i>(In millions of Korean won)</i>	Korea	America	Europe	Asia and Africa	China	Intercompany elimination within the Company	Consolidated
Net segment revenue	31,545,181	81,019,556	44,418,914	44,247,994	38,343,731	-	239,575,376
Non-current assets ¹	89,156,726	10,545,297	6,081,926	11,980,235	9,361,317	(699,370)	126,426,131

¹ Total of non-current assets other than financial instruments, deferred tax assets, and investments in associates and joint ventures.

(2) For the year ended December 31, 2016

<i>(In millions of Korean won)</i>	Korea	America	Europe	Asia and Africa	China	Intercompany elimination within the Company	Consolidated
Net segment revenue	20,201,828	68,728,575	38,253,185	39,099,991	35,583,166	-	201,866,745
Non-current assets ¹	68,978,040	7,041,731	730,490	9,626,711	11,132,720	(483,191)	97,026,501

¹ Total of non-current assets other than financial instruments, deferred tax assets, and investments in associates and joint ventures.

33. Related Party Transactions

(A) Sale and purchase transactions

Sale and purchase transactions with related parties for the years ended December 31, 2017 and 2016 are as follows:

(In millions of Korean won)	Name of Company ¹	2017					
		Sales		Disposal of fixed assets		Purchases	Purchases of fixed assets
Associates and joint ventures	Samsung SDS	₩	57,662	₩	-	₩ 1,971,449	₩ 308,297
	Samsung Electro-Mechanics		49,308		-	2,085,917	302
	Samsung SDI		62,122		1,600	974,496	63,655
	Cheil Worldwide		12,570		-	782,530	1,079
	Other		612,726		-	9,069,864	174,199
	Total (Associates and joint ventures)	₩	794,388	₩	1,600	₩14,884,256	₩ 547,532
Other related parties	Samsung C&T	₩	164,913	₩	68	₩ 379,500	₩ 4,131,961
	Other		163,023		35	934,452	329,063
	Total (Other related parties)	₩	327,936	₩	103	₩ 1,313,952	₩ 4,461,024
Others ²	Samsung Engineering	₩	8,980	₩	-	₩ 41,495	₩ 2,301,886
	S-1		42,355		-	362,748	84,281
	Other		73,878		3	205,680	406
	Total (Others)	₩	125,213	₩	3	₩ 609,923	₩ 2,386,573

¹ Transactions with separate entities that are related parties of the Company.

² Although these entities are not related parties of the Company in accordance with *Korean IFRS 1024*, they belong to the same large enterprise group in accordance with the Monopoly Regulation and Fair Trade Act.

(In millions of Korean won)	Name of Company ¹	2016					
		Sales		Disposal of fixed assets		Purchases	Purchases of fixed assets
Associates and joint ventures	Samsung SDS	₩	46,073	₩	-	₩ 1,585,089	₩ 199,728
	Samsung Electro-Mechanics		27,516		23	2,280,953	-
	Samsung SDI ²		59,322		397	1,072,830	32,576
	Cheil Worldwide ³		672		-	214,061	-
	Other		286,880		113	6,693,656	214,728
	Total (Associates and Joint ventures)	₩	420,463	₩	533	₩ 11,846,589	₩ 447,032
Other related parties	Samsung C&T	₩	42,905	₩	74	₩ 249,088	₩ 3,343,979
	Other		231,878		1,557,589	763,500	398,514
	Total (Other related parties)	₩	274,783	₩	1,557,663	₩ 1,012,588	₩ 3,742,493
Others ⁴	Samsung Engineering		15,677		-	17,627	2,485,027
	S-1		35,846		-	323,792	37,590
	Other		58,054		83,298	663,222	65
	Total (Others)	₩	109,577	₩	83,298	₩ 1,004,641	₩ 2,522,682

¹ Transactions with separate entities that are related parties of the Company.

² For the year ended December 31, 2016, Samsung SDI was included in associates as the Company's ownership of common outstanding stock was increased.

³ For the year ended December 31, 2016, Cheil Worldwide was included in associates due to acquisition of shares.

⁴ Although these entities are not related parties of the Company in accordance with *Korean IFRS 1024*, they belong to the same large enterprise group in accordance with the Monopoly Regulation and Fair Trade Act.

(B) Balances of receivables and payables

Balances of receivables and payables arising from sales and purchases of goods and services as at December 31, 2017 and December 31, 2016 are as follows:

(In millions of Korean won)		December 31, 2017			
		Name of Company ¹		Receivables	Payables
Associates and joint ventures	Samsung SDS	₩	14,182	₩	406,760
	Samsung Electro-Mechanics		7,052		197,364
	Samsung SDI		90,370		97,193
	Cheil Worldwide		369		500,706
	Other		250,059		954,537
	Total (Associates and joint ventures)	₩	362,032	₩	2,156,560
Other related parties	Samsung C&T	₩	242,506	₩	731,995
	Other ²		29,457		1,313,314
	Total (Other related parties)	₩	271,963	₩	2,045,309
Others ³	Samsung Engineering		2,518	₩	652,519
	S-1		2,589		58,309
	Other		3,369		35,774
	Total (Others)	₩	8,476	₩	746,602

¹ Balances due from and to separate entities that are related parties of the Company.

² Payables include the unsettled amount from purchasing cards with Samsung Card of ₩1,121,331 million. The Company has a purchasing card agreement with a limit of ₩2,343,000 million as at December 31, 2017. For the year ended December 31, 2017, the amounts used and reimbursed are ₩5,091,576 million and ₩5,279,430 million, respectively.

³ Although these entities are not related parties of the Company in accordance with *Korean IFRS 1024*, they belong to the same large enterprise group in accordance with the Monopoly Regulation and Fair Trade Act.

(In millions of Korean won)		December 31, 2016			
		Name of Company ¹		Receivables	Payables
Associates and joint ventures	Samsung SDS	₩	5,709	₩	362,062
	Samsung Electro-Mechanics		1,143		108,469
	Samsung SDI ²		89,721		76,211
	Cheil Worldwide ³		456		436,624
	Other		210,891		784,475
	Total (Associates and joint ventures)	₩	307,920	₩	1,767,841
Other related parties	Samsung C&T	₩	231,089	₩	435,505
	Other		31,752		1,932,924
	Total (Other related parties)	₩	262,841	₩	2,368,429
Others ⁴	Samsung Engineering	₩	10,664	₩	115,726
	S-1		4,160		47,098
	Other		3,058		28,841
	Total (Others)	₩	17,882	₩	191,665

¹ Balances due from and to separate entities that are related parties of the Company.

² For the year ended December 31, 2016, Samsung SDI was included in associates as the Company's ownership of common outstanding stock was increased.

³ For the year ended December 31, 2016, Cheil Worldwide was included in associates due to acquisition of shares.

⁴ Although these entities are not related parties of the Company in accordance with *Korean IFRS 1024*, they belong to the same large enterprise group in accordance with the Monopoly Regulation and Fair Trade Act.

(C) For the year ended December 31, 2017, the Company invested ₩25,293 million (for the year ended December 31, 2016,

₩84,306 million) in associates and joint ventures. And for the year ended December 31, 2017, the Company recovered ₩53,085 million (for the year ended December 31, 2016, ₩4,931) from associates and joint ventures. Also, for the year ended December 31, 2016, the Company invested 181,081 million to the entities that are not related parties of the Company in accordance with *Korean IFRS 1024*, but belong to the same conglomerate according to the Monopoly Regulation and Fair Trade Act.

(D) For the year ended December 31, 2017, the Company declared ₩1,165,659 million of dividends (for the year ended December 31, 2016, ₩505,296 million) to related parties. As at December 31, 2017 and 2016, there are no unpaid dividends to related parties. Also, for the year ended December 31, 2017, the Company declared ₩90,034 million of dividends (for the year ended December 31, 2016, ₩38,983 million) to the entities that are not related parties of the Company in accordance with *Korean IFRS 1024*, but belong to the same conglomerate according to the Monopoly Regulation and Fair Trade Act. As at December 31, 2017 and 2016, there are no unpaid dividends to these entities.

(E) Key management compensation

Key management includes directors (executive and non-executive) and members of the Executive Committee. The compensation paid or payable for employee services for the years ended December 31, 2017 and 2016 consists of:

<i>(In millions of Korean won)</i>	2017		2016	
Short-term employee benefits	₩	28,025	₩	16,822
Post-employment benefits		792		640
Other long-term employee benefits		8,617		8,671

34. Information about Non-Controlling Interests

(A) Changes in accumulated non-controlling interests

The profit or loss allocated to non-controlling interests and accumulated non-controlling interests of subsidiaries that are material to the Company for the years ended December 31, 2017 and 2016 are as follows:

2017	Percentage of	Balance as at	Net Income	Dividends	Other	Balance as at
<i>(In millions of Korean won)</i>	Non-Controlling	January 1				December 31
	Interests					
Samsung Display and its subsidiaries	15.2%	₩ 5,981,454	₩ 790,480	₩ (1,582)	₩ 6,116	₩ 6,776,468

2016	Percentage of	Balance as at	Net Income	Dividends	Other	Balance as at
<i>(In millions of Korean won)</i>	Non-Controlling	January 1				December 31
	Interests					
Samsung Display and its subsidiaries	15.2%	₩ 5,642,413	₩ 253,296	₩(13,472)	₩ 99,217	₩ 5,981,454

(B) The summarized financial information for each subsidiary with non-controlling interests that are material to the Company before intercompany eliminations for the years December 31, 2017 and 2016 are as follows:

(1) Summarized consolidated statements of financial position

<i>(In millions of Korean won)</i>	Samsung Display and its subsidiaries	
	December 31, 2017	December 31, 2016
Current assets	₩ 21,316,244	₩ 17,208,126
Non-current assets	37,884,399	30,421,181
Current liabilities	12,279,301	7,957,076
Non-current liabilities	5,828,551	3,191,759
Equity attributable to:	41,092,791	36,480,472
Owners of the parent	40,573,072	35,982,390
Non-controlling interests	519,719	498,082

(2) Summarized consolidated statements of comprehensive income

<i>(In millions of Korean won)</i>	Samsung Display and its subsidiaries	
	2017	2016
Sales	₩ 34,293,155	₩ 26,816,450
Net income	4,645,695	1,618,023
Other comprehensive income(loss)	(25,171)	721,849
Total comprehensive income attributable to:	4,620,524	2,339,872
Owners of the parent	4,597,147	2,343,120
Non-controlling interests	23,377	(3,248)

(3) Summarized consolidated statements of cash flows

<i>(In millions of Korean won)</i>	Samsung Display and its subsidiaries	
	2017	2016
Cash flows from operating activities	₩ 8,697,292	₩ 6,800,635
Cash flows from investing activities	(14,793,775)	(9,163,528)
Cash flows from financing activities	6,336,379	2,563,830
Effect of exchange rate changes on cash and cash equivalents	(18,642)	10,061
Increase in cash and cash equivalents	221,254	210,998
Cash and cash equivalents at beginning of period	596,861	385,863
Cash and cash equivalents at end of period	818,115	596,861

35. Business Combinations

Significant business combinations for the year ended December 31, 2017 are as follows:

To strengthen the Company's automotive electronics and audio businesses, Samsung Electronics America, the Company's subsidiary, acquired 100% of the equity shares of Harman and its subsidiaries on March 10, 2017.

(1) Overview of the acquired company

Name of the acquired company	Harman International Industries, Inc. and 109 subsidiaries
Headquarters location	Stamford, CT, USA
Representative director	Dinesh Paliwal
Industry	Design, development, manufacture and sale of audio and video system components

(2) Purchase price allocation

<i>(In millions of Korean Won)</i>	Amount
I. Consideration transferred	₩ 9,272,702
II. Identifiable assets and liabilities	
Cash and cash equivalents	647,729
Trade and other receivables	1,533,437
Inventory	1,068,865
Property, plant and equipment	858,790
Intangible assets	5,564,309
Other assets	902,824
Trade and other payables	3,436,020
Deferred tax liabilities	1,442,527
Other liabilities	873,637
Total net identifiable assets	4,823,770
III. Goodwill (I – II)	₩ 4,448,932

Had Harman and the subsidiaries been consolidated from January 1, 2017, the revenue and net income for the year ended December 31, 2017, would be ₩8,581,461 million and ₩213,287 million, respectively. The revenue and net income contributed by Harman and the subsidiaries since the acquisition amount to ₩7,103,437 million and ₩209,015 million, respectively.

36. Non-Current Assets and Liabilities Held-for-Sale (Assets of disposal group)

(A) Summary

- Sale of Printing Solutions business segment

During the year ended December 31, 2017, the Company sold its Printing Solutions business unit to HP, Inc. The transaction was completed on November 1, 2017.

(B) Details of assets and liabilities reclassified as held-for-sale, as at December 31, 2016 are as follows:

<i>(In millions of Korean Won)</i>	December 31, 2016	
Assets held-for-sale		
Trade receivables	₩	182,738
Inventories		270,642
Other current assets		115,037
Property, plant and equipment		84,869
Intangible assets		124,571
Other non-current assets		57,949
Total	₩	835,806
Liabilities held-for-sale		
Current liabilities	₩	272,726
Non-current liabilities		83,662
Total	₩	356,388

(C) Details of cumulative income or expense recognized in other comprehensive income related to the disposal group classified as held-for-sale as at December 31, 2016 are as follows:

<i>(In millions of Korean won)</i>	December 31, 2016	
Foreign currency translation	₩	(28,810)

37. Events After the Reporting Period

According to the resolution of the Board of Directors on October 31, 2017, the Company repurchased 0.89 million shares (ordinary shares 0.712 million, preference shares 0.178 million) at a cost of ₩2,228,135 million from November 1, 2017 to January 26, 2018. All repurchased shares were retired in January 30, 2018.

In accordance with the resolution of the Board of Directors on January 31, 2018, the Company decided to split their ordinary and preference shares by lowering the par value from ₩5,000 to ₩100. The stock split is scheduled for May 16, 2018. If the effect of the stock split is reflected after the retirement of the repurchased shares as of January 30, 2018, the Company expects to issue 6,419,324,700 ordinary shares and 903,620,000 preference shares.

3. Other Financial Information

A. Restatement of Financial Statements and Other Matters to Note

(1) Restatement of financial statements: N/A

(2) Acquisition, divestment, asset transfer and transfer of business (Separate only)

There are no major merger transactions in the past three business years on a parent basis(separate basis). For information on the business combinations and divestitures of subsidiaries, please refer to the notes on the business combination and assets held for sale (assets of disposal group) of the consolidated financial statements.

Details of divestiture

- Company Name: S-Printing Solution Corp.
- Location: 129 Samsung-ro, Yeongtong-gu, Suwon-si, Gyeonggi-do
- Purpose of split: To improve the competitiveness of S-Printing Solution
- Date of Approval: October 27, 2016 (at the extraordinary general meeting of shareholders)
- Date of split: November 1, 2016
-

On September 12, 2016, the Company entered into an agreement with HP Inc., to sell the Printing Solutions business including its shares of S-Printing for consideration of USD 1,050 million. This transaction was completed as of November 1, 2017.

The Company completed the deal to sell Printing Solutions business including shares of S-Printing Solution Corp on November 1, 2017.

Details of the above statement can be found in the “Important Matters Report“, published at DART (<http://dart.fss.or.kr/>).

The Split of S-Printing Solution business was reported on a separate basis.

(3) Information on the accounting treatment of the sales of assets and contingent liabilities relating to the asset backed securities

- Domestic: As of December 31, 2017, the Company provided a debt guarantee of KRW 32,664 million to its employees who took debt from financial institutions in order to finance employee housing rental. The Company's housing rental debt guarantee limit is KRW 49,937 million.

- Overseas :

(Unit: USD thousand)

Company	Relationship	Creditor	Guarantee Expiry Date	Transactions				Limit of Guarantee
				Beginning of period	Increase	Decrease	End of period	
SEA	Subsidiary	SMBC etc.	2018-12-16	1,000,000	0	1,000,000	0	1,423,000
SEM	Subsidiary	Santander etc.	2018-12-16	0	0	0	0	546,000
SAMCOL	Subsidiary	Citibank etc.	2018-12-16	83,585	0	16,646	66,939	168,000
SEDA	Subsidiary	HSBC etc.	2018-12-17	0	0	0	0	769,000
SECH	Subsidiary	Citibank etc.	2018-12-16	0	14,639	0	14,639	178,000
SEPR	Subsidiary	BBVA etc.	2018-12-16	68,739	0	28,589	40,150	180,000
SSA	Subsidiary	Citibank etc.	2018-12-16	34,327	0	34,327	0	323,000
SEMAG	Subsidiary	SocGen etc.	2018-12-16	0	0	0	0	110,000
SETK	Subsidiary	BTMU etc.	2018-12-16	187,867	54,478	0	242,345	755,000
SECE	Subsidiary	Citibank etc.	2018-12-16	0	0	0	0	94,768
SEEG	Subsidiary	HSBC	2018-06-13	0	0	0	0	50,000
SEIN	Subsidiary	BNP etc.	2018-11-08	0	0	0	0	186,000
SJC	Subsidiary	Mizuho Bank etc.	2018-12-16	0	0	0	0	885,782
SEUC	Subsidiary	Credit Agricole etc.	2018-12-16	0	0	0	0	150,000
SEDAM	Subsidiary	Citibank etc.	2018-12-16	101,326	0	101,326	0	371,000
SECA	Subsidiary	Nova Scotia	2017-10-10	0	0	0	0	0
SELA	Subsidiary	Citibank	2018-12-16	0	0	0	0	50,000
SEEH	Subsidiary	HSBC etc.	2018-12-16	0	0	0	0	647,000
SERK	Subsidiary	BNP etc.	2018-12-16	0	0	0	0	245,000
SELV	Subsidiary	Citibank	2018-12-16	0	0	0	0	10,000
SAPL	Subsidiary	BOA etc.	2018-12-16	0	0	0	0	411,000
SEV	Subsidiary	SCB	2018-11-08	0	0	0	0	15,000
SAVINA	Subsidiary	SCB etc.	2018-11-08	0	0	0	0	71,000
SET	Subsidiary	SCB	2018-11-08	0	0	0	0	30,000
SCIC	Subsidiary	HSBC etc.	2018-12-16	0	0	0	0	350,000
SME	Subsidiary	SCB	2018-11-08	0	0	0	0	110,000
SAMEX	Subsidiary	Citibank	2018-12-16	0	0	0	0	5,000
SEASA	Subsidiary	Citibank	2018-12-16	0	0	0	0	1,000
SSAP	Subsidiary	SCB	2018-11-08	10,282	0	10,282	0	30,000
Simpress	Subsidiary	BNP	2017-11-08	44,529	0	44,529	0	0
SEHK	Subsidiary	HSBC	2018-06-13	0	0	0	0	2,000
SEPM	Subsidiary	HSBC	2018-06-13	86,870	0	9,451	77,419	125,000
Adgear	Subsidiary	BOA	2018-11-08	0	0	0	0	2,000
Harman Finance International SCA	Subsidiary	JP Morgan etc.	2022-05-27	367,115	50,785	0	417,900	417,900
Total				1,984,640	119,902	1,245,150	859,392	8,711,450

※ SEC requires BOD approval for individual guarantees exceeding 2.5% of total equity. When the guarantee amount is between 0.1% and/or less than 2.5%, the approval decision is delegated to the Management Committee.

※ Simpress ceased to be affiliate(share sale in November 2017)

Refer to 3.Litigation (XI. Other Information Related to Investment Protection) or the notes to the consolidated financial statements for the information on contingent liabilities.

(4) Other matters requiring attention in relation to the use of the financial statements: N/A

B. Allowance for Bad Debt

The allowances for bad debt by account over the past 3 years are as follows:

(Unit: KRW million, %)

Period	Account	Receivables Amount	Allowance Amount	Allowance (%)
2017	Trade receivables	28,333,822	635,815	2.2%
	Short-term loans	7,258	81	1.1%
	Other receivables	4,135,935	26,975	0.7%
	Advances	1,758,547	4,874	0.3%
	Trade receivables (Long-term)	1,983,136	1,397	0.1%
	Other receivables (Long-term)	139,472	330	0.2%
	Advances (Long-term)	134,192	836	0.6%
	Long-term loans	130,550	467	0.4%
	Total	36,622,912	670,775	1.8%
2016	Trade receivables	24,699,961	420,750	1.7%
	Short-term loans	7,208	67	0.9%
	Other receivables	3,546,546	25,349	0.7%
	Advances	1,442,219	2,281	0.2%
	Trade receivables (Long-term)	4,563	139	3.0%
	Other receivables (Long-term)	35,683	153	0.4%
	Advances (Long-term)	175,211	1,369	0.8%
	Long-term loans	173,068	7,849	4.5%
	Total	30,084,459	457,957	1.5%
2015	Trade receivables	25,494,637	326,611	1.3%
	Short-term loans	8,570	80	0.9%
	Other receivables	3,400,227	47,564	1.4%
	Advances	1,709,840	3,837	0.2%
	Trade receivables (Long-term)	25,747	249	1.0%
	Other receivables (Long-term)	185,667	1,726	0.9%
	Advances (Long-term)	422,884	793	0.2%
	Long-term loans	180,839	2,334	1.3%
	Total	31,428,411	383,194	1.2%

※ Receivables amount represents the value after deducting present value discount account (Consolidated)

Allowance for bad debt by year

(Unit: KRW million)

	2017	2016	2015
1. Allowance for bad debt (beginning balance)	457,957	383,194	294,157
2. Net bad debt expense (① - ②±③)	3,685	29,843	11,373
① Bad debt expense, etc (Write-off)	38,584	31,334	14,149
② Bad debt recovered	559	1,491	2,776
③ Others	-34,340	-	-
3. Bad debts expense	216,503	104,606	100,410
4. Allowance for bad debts (Ending balance)	670,775	457,957	383,194

※ Based on consolidated financial statements.

The guideline for bad debt allowances for trade receivables

(1) Calculation of bad debt allowance

- Bad debt allowance is calculated based on write-off experience rate and future expected bad debt amount.

(2) Calculation basis of write-off experience rate and future expected bad debt:

- Experience rate of write-off is calculated based on the actual write-off rate for the average receivables balance of previous three years.
- Future expected bad debt is determined within the range of 1% ~ 100% of the receivable balance, taking into consideration cases of debtor bankruptcy, compulsory execution, death, disappearance, etc.

[Bad debt allowance by circumstance]

Circumstance	Rate of Allowance
Dispute or Conflict	25%
Receivable through utilization of third party collection agency	50%
Receivable under litigation	75%
Customer filed or in the process of filing bankruptcy	100%

(3) Instruction: Write-off is recorded when trade receivables contain the following characteristics:

- Objective proof of bad debts, such as bankruptcy, compulsory execution, discontinuance of business, debtor's death or disappearance
- Legal action has failed or the right of collection has been extinguished.
- When the collection agency gives as notice that collection is not possible.
- When the collateral is sold or insurance is received.
- The collection expenses exceed the amount of receivables.

The outstanding period of trade receivables

(Unit : KRW million)

	Less than 6 months	6 months ~ 1 year	1 year ~ 3 years	More than 3 years	Total
Amount	30,198,033	24,276	55,721	38,928	30,316,958
Ratio	99.6%	0.1%	0.2%	0.1%	100.0%

※ Receivables amount on net present value (consolidated)

C. Inventory

- The inventory status by Division over the last three years is as follows:

(Unit: KRW million)

Division		Category	2017	2016	2015	Note
CE		Finished Goods	1,853,446	2,045,068	1,679,255	
		Work In Process	119,337	109,934	73,060	
		Raw Material	2,570,348	1,826,473	1,671,254	
		Material In Transit	1,686,440	1,957,076	1,607,908	
		Total	6,229,571	5,938,551	5,031,477	
IM		Finished Goods	2,327,139	2,032,004	1,924,116	
		Work In Process	721,194	531,628	347,780	
		Raw Material	4,541,940	2,851,438	2,471,314	
		Material In Transit	867,103	594,955	492,149	
		Total	8,457,376	6,010,025	5,235,359	
DS	Semi-Conductor	Finished Goods	1,094,967	871,428	1,598,617	
		Work In Process	5,179,312	3,796,746	4,740,937	
		Raw Material	659,223	541,573	505,226	
		Material In Transit	39,312	62,078	109,333	
		Total	6,972,814	5,271,825	6,954,113	
	DP	Finished Goods	746,856	299,208	398,342	
		Work In Process	638,268	444,831	483,084	
		Raw Material	849,786	375,611	265,206	
		Material In Transit	133,038	69,782	49,957	
		Total	2,367,948	1,189,432	1,196,589	
	DS Total	Finished Goods	1,872,648	1,203,332	2,014,630	
		Work In Process	6,256,420	4,412,185	5,353,650	
		Raw Material	1,708,632	977,020	804,420	
		Material In Transit	64,934	135,230	162,897	
		Total	9,902,634	6,727,767	8,335,597	
Harman		Finished Goods	432,184	-	-	
		Work In Process	77,696	-	-	
		Raw Material	322,649	-	-	
		Material In Transit	140,939	-	-	
		Total	973,468	-	-	
Total		Finished Goods	7,304,437	5,905,339	5,769,460	
		Work In Process	7,113,901	5,017,384	5,779,303	
		Raw Material	9,413,217	6,494,166	5,859,262	
		Material In Transit	1,151,800	936,614	1,403,769	
		Total	24,983,355	18,353,503	18,811,794	

Inventory Ratio (%)– Total Assets [Inventory ÷ Total Assets]	8.3%	7.0%	7.8%	
Inventory Turn-over [Yearly CGS ÷ {(Beginning of Inventory + End of Inventory)÷2}]	6.0	6.5	6.8	

※ For CE division, the financial information was restated to exclude Printing Solution business.

Inventory Counts

(1) Inspection Date

- End of May and November (twice a year)
- For the difference in inventory values between the inspection date and at the end of the year, check the all warehouse record for the period in order to confirm the existence of the inventories as of the date of the financial statements.

(2) Inspection Method

- Internal warehouse: Closed & Total Inspection
- ※ Sample check in semiconductor and DP inventory, SVC materials

- Outside warehouse

Perform check for the possession confirmation documents and sample tests at the third party warehouse and Inventory in transit

- External auditors join and observe the inspection and carry out sample tests to check the existence and completeness.

Inventory Aging and Valuation

Inventories are stated at the lower of cost or net realizable value. Inventory as at December 31, 2017 is as follows:

(Unit : KRW million)

Category	Acquisition Cost	Inventory Valuation Reserve	Balance	Note
Finished Goods	8,201,526	-897,089	7,304,437	
Work In Process	7,331,394	-217,493	7,113,901	
Raw Material	10,196,123	-782,906	9,413,217	
Material In Transit	1,151,800	-	1,151,800	
Total	26,880,843	-1,897,488	24,983,355	

※ Prepared on a consolidated basis

D. Fair Value Estimation

Refer to 6. Financial Instruments by Category and 31. Financial Risk Management in 『III. Financial Affairs』.

E. List of Issued Debt Securities

Issued Debt Securities

(As of December 31, 2017)

(Unit : KRW Million, %)

Issuing Company	Type of Securities	Issuance Method	Date of Issuance	Total Nominal Amount	Interest Rate	Rating (Rating Institution)	Maturity Date	Payment Status	Management Company
Samsung Electronics	Corporate Bonds	Public Offering	1997.10.02	107,140	7.7	AA-(S&P), A1(Moody's)	2027.10.01	Partial Redemption	Goldman Sachs et al.
SEA	Corporate Bonds	Public Offering	2012.04.10	1,134,600	1.8	A+(S&P), A1(Moody's)	2017.04.10	Redemption	Goldman Sachs et al.
Harman International Industries, Inc	Corporate Bonds	Public Offering	2015.05.06	428,560	4.2	Baa2 (Moody's), BBB+ (S&P)	2025.05.15	Unredeemed	J.P.Morgan et al.
Harman Finance International SCA	Corporate Bonds	Public Offering	2015.05.20	447,738	2.0	Baa2 (Moody's), BBB+ (S&P)	2022.05.27	Unredeemed	HSBC et al.
Total	-	-	-	2,118,038	-	-	-	-	-

※ The exchange rate as of the base date has been applied.

※ SEA Corporate Bond was redeemed on April 2017.

Details and Compliance of the Bond Management Contract (Samsung Electronics)

(Base Date of Preparation: December 31, 2017)

(Unit : KRW Million, %)

Name of Security	Date of Issuance	Maturity Date	Issued Amount	Settlement Date of Bond Management Contract	Debenture Management Company
US\$ 100,000,000 7.7% Debenture	1997.10.02	2027.10.01	107,140	1997.10.02	The Bank of New York Mellon Trust Company, N.A.

Base Date of Implementation: December 31, 2017

Financial Ratios	Contract Details	Not applicable
	Implementation Status	Not applicable
Constraint on Collaterals	Contract Details	Less than 10% of net tangible assets
	Implementation Status	Compliant (there is no collateral for the relevant assets)
Constraint on Disposal of Assets	Contract Details	Certain requirements, such as transfer of obligations on the bond, must be satisfied to dispose all or most of an asset
	Implementation Status	Compliant (disposal of assets accounted for 0.3% of the total during '17)
Constraint on Governance	Contract Details	Not applicable
	Implementation Status	Not applicable
Submission of Implementation Report	Implementation Status	Not applicable

- ※ The date of the bond management contract was signed on the same day as the Fiscal Agency Agreement; accordingly, the Bank of New York Mellon Trust Company, N.A. is under the authority of the Fiscal Agent.
- ※ The exchange rate as of the base date has been applied.
- ※ The net tangible assets subjected to the limitation of collaterals are production facilities and stocks owned by the Company.
- ※ The base date of implementation is the date when most recent financial statements were prepared, which are used to determine the status of implementation.

Commercial Paper Balance

(Base Date : December 31, 2017)

(Unit : KRW Million)

Maturity		Under 10 Days	Above 10 Days / Under 30 Days	Above 30 Days / Under 90 Days	Above 90 Days / Under 180 Days	Above 180 Days / Under 1 Year	Above 1 Year / Under 2 Years	Above 2 Years / Under 3 Years	Above 3 Years	Total
Balance	Public	-	-	-	-	-	-	-	-	-
	Private	-	-	-	-	-	-	-	-	-
	Total	-	-	-	-	-	-	-	-	-

Asset-Backed Short Term Bond Unredeemed Balance

(Base Date : December 31, 2017)

Maturity		Under 10 Days	Above 10 Days / Under 30 Days	Above 30 Days / Under 90 Days	Above 90 Days / Under 180 Days	Above 180 Days / Under 1 Year	Total	Issue Limit	Balance Limit
Balance	Public	-	-	-	-	-	-	-	-
	Private	-	-	-	-	-	-	-	-
	Total	-	-	-	-	-	-	-	-

Corporate Bond Unredeemed Balance

(Base Date : December 31, 2017)

(Unit : KRW Million)

Maturity		Under 1 Year	Above 1 Year / Under 2 Years	Above 2 Years / Under 3 Years	Above 3 Years / Under 4 Years	Above 4 Years / Under 5 Years	Above 5 Years / Under 10 Years	Above 10 Years	Total
Balance	Public	5,357	5,357	5,357	5,357	453,095	455,345	-	929,868
	Private	-	-	-	-	-	-	-	-
	Total	5,357	5,357	5,357	5,357	453,095	455,345	-	929,868

※ The exchange rate as of the base date has been applied.

- Corporate Bond Unredeemed Balance (Samsung Electronics)

(Base Date : December 31, 2017)

(Unit : KRW Million)

Maturity		Under 1 Year	Above 1 Year / Under 2 Years	Above 2 Years / Under 3 Years	Above 3 Years / Under 4 Years	Above 4 Years / Under 5 Years	Above 5 Years / Under 10 Years	Above 10 Years	Total
Unredeemed	Public	5,357	5,357	5,357	5,357	5,357	26,785	-	53,570
Balance	Total	5,357	5,357	5,357	5,357	5,357	26,785	-	53,570

※ The exchange rate as of the base date has been applied.

- Corporate Bond Unredeemed Balance (Harman)

(Base Date : December 31, 2017)

(Unit : KRW Million)

Maturity		Under 1 Year	Above 1 Year / Under 2 Years	Above 2 Years / Under 3 Years	Above 3 Years / Under 4 Years	Above 4 Years / Under 5 Years	Above 5 Years / Under 10 Years	Above 10 Years	Total
Unredeemed	Public	-	-	-	-	447,738	428,560	-	876,298
Balance	Total	-	-	-	-	447,738	428,560	-	876,298

※ The exchange rate as of the base date has been applied.

Hybrid Bond Outstanding Balance

(Base Date : December 31, 2017)

(Unit : KRW Million)

Maturity		Under 1 Year	Above 1 Year / Under 5 Years	Above 5 Years / Under 10 Years	Above 10 Years / Under 15 Years	Above 15 Years / Under 20 Years	Above 20 Years / Under 30 Years	Above 30 Years	Total
Unredeemed Balance	Public	-	-	-	-	-	-	-	-
	Private	-	-	-	-	-	-	-	-
	Total	-	-	-	-	-	-	-	-

Contingent Convertible Bond Outstanding Balance

(Base Date : December 31, 2017)

(Unit : KRW Million)

Maturity		Under 1 Year	Above 1 Year / Under 2 Years	Above 2 Years / Under 3 Years	Above 3 Years / Under 4 Years	Above 4 Years / Under 5 Years	Above 5 Years / Under 6 Years	Above 10 Years / Under 20 Years	Above 20 Years / Under 30 Years	Above 30 Years	Total
Unredeemed Balance	Public	-	-	-	-	-	-	-	-	-	-
	Private	-	-	-	-	-	-	-	-	-	-
	Total	-	-	-	-	-	-	-	-	-	-

IV. Auditor's Report

1. Introduction

Samil PwC has audited the consolidated/separate financial statements of the Company for the fiscal year ended December 31, 2017, including statement of financial position as at December 31, 2017, statements of profit or loss, comprehensive income, changes in equity and cash flows for the year ended December 31, 2017. Nothing has come to attention that causes them to believe the accompanying year end consolidated/separate financial statements are not presented fairly, in all material respects, in accordance with K-IFRS. Samil PwC also conducted audits on the consolidated/separate financial statements of the Company for the fiscal year ended December 31, 2015 and 2016, and expressed an unqualified opinion on those financial statements.

There are 270 subsidiaries as at December 31, 2017. The subsidiary, Samsung Electronics Iberia(SES), has changed the external auditor to E&Y, nine subsidiaries including Samsung India Electronics(SIEL) have changed the external auditor to KPMG and the subsidiary, Mirero System, has changed the external auditor to PwC for the year ended December 31, 2017. The newly acquired two entities including Harman International Industries, Inc. appointed PwC as the external auditor and newly incorporated entity, Samsung Venture Capital Union #37 appointed KPMG as the external auditor at their own decision.

Year end	Company Auditor	Audit Opinion	Remarks
31 December 2017	Samil PwC	Unqualified	Not applicable
31 December 2016	Samil PwC	Unqualified	Not applicable
31 December 2015	Samil PwC	Unqualified	Not applicable

< Audit (Review) plan for the year ended December 31, 2017>

Review Period		Planned dates
2017 1Q	Pre-review	2017.03.06 ~ 2017.03.24
	Review	2017.04.07 ~ 2017.05.12
2017 2Q	Pre-review	2017.06.05 ~ 2017.06.23
	Review	2017.07.07 ~ 2017.08.11
2017 3Q	Pre-review	2017.09.06 ~ 2017.09.22
	Review	2017.10.11 ~ 2017.11.13
System and Internal Control Audit		2017.09.04 ~ 2017.09.29
Audit procedures		2017.12.11 ~ 2017.12.29
Completion of final audit procedures		2018.01.08 ~ 2018.02.27

[Audit Contract Description]

(In millions of Korean won)

Fiscal Year	Group Auditor	Description	Compensation	Total Hours
Year ended December 31, 2017	Samil PwC	- Review of the interim consolidated/separate financial statements - Audit on consolidated/separate financial statements	4,030	46,576
Year ended December 31, 2016	Samil PwC	- Review of the interim consolidated/separate financial statements - Audit on consolidated/separate financial statements	3,690	43,999
Year ended December 31, 2015	Samil PwC	- Review of the interim consolidated/separate financial statements - Audit on consolidated/separate financial statements	3,690	43,928

Samil PwC has performed non-audit services and has been paid KRW 1,006 million by the Company for the year ended December 31, 2017.

[Non-Audit Service Contract Description]

(In millions of Korean won)

Fiscal Year	Contract Date	Non-audit Service Description	Period for Service	Compensation	Note
Year ended December 31, 2017	January 2017	Tax Advisory	2017.1~2017.12	196	
	December 2016	Custom Tax Advisory	2017.1~2017.12	810	
	Subtotal			1,006	
Year ended December 31, 2016	December 2016	Tax Advisory	2016.12~2016.12	220	
	February 2016	Custom Tax Advisory	2016.01~2016.12	1,133	
	Subtotal			1,353	
Year ended December 31, 2015	June 2014	Incorporation, M&A and Divestiture Advisory	2015.01~2015.12	503	
	May 2014	Tax Advisory	2015.01~2015.12	47	
	Subtotal			550	

2. Regarding Internal Control

Samil PwC has reviewed management's report on the operations of the Internal Accounting Control System ("IACS") of the Company as of December 31, 2017 and nothing has come to attention that causes them to believe that management's report on the operations of the IACS is not presented fairly, in all material respects, in accordance with IACS standards.

V. Management Discussion and Analysis

1. Note on Forward-Looking Statements

This annual report includes forward-looking statements that relate to future events and can be generally identified by phrases containing words such as “believes,” “expects,” “anticipates,” “foresees,” “forecasts,” “estimates” or other words or phrases of similar meaning. These forward-looking statements are not guarantees of future performance and may involve known and unknown risks, uncertainties and other factors that may affect our actual results, performance, achievements or financial position, making them materially different from the actual future results, performance, achievements or financial position expressed or implied by these forward-looking statements. Unless otherwise specified, all data presented in this report is from our consolidated financial statements.

Uncertain events that could positively or negatively affect the Company’s management condition and financial performance include, but are not limited to:

- Trends of financial markets domestically and abroad, including changes in exchange rates and interest rates
- The Company’s strategic decision making, including disposals and purchases of businesses
- Unexpected sudden changes in core businesses such as CE, IM, Semiconductor, and DP
- Other changes domestically and abroad that can affect management condition and financial performance

The Company assumes no obligation to revise or update this report to reflect risks or uncertainties that arise after the reporting period.

2. Overview

In 2017, Samsung Electronics continued to face uncertainties caused by a surge in protectionism, mainly in the US, and escalating geopolitical risks, despite the global economy showing signs of stable growth.

Yet, we achieved record-high results thanks to the support of our shareholders and tireless efforts of our executives and employees, recording sales of KRW 240 trillion and an operating profit of KRW 54 trillion on a consolidated basis. On a parent basis, sales came in at KRW 162 trillion and operating profit set a new record at KRW 35 trillion.

In financial terms as of end-2017, the debt-to-equity ratio was 40.7%, capital adequacy ratio was 71.1%, and ROE was 20.7% on a consolidated basis. On a parent basis, debt-to-equity ratio was 30.8%, capital adequacy ratio was 76.5%, and ROE was 19.9%. These show our financial structure remained solid. Samsung brand ranked as the world’s sixth with an estimated value of USD 56.2 billion, the figures up one spot and 9% from the previous year (October 2017, *Interbrand*).

On the business side, we solidified our leadership and maintained dominant market shares. In the component business, we enhanced cost competitiveness by expanding cutting-edge processes to produce the industry’s first 10nm-class DRAM and our 4th generation V-NAND. In the set business, we launched innovative products such as the Galaxy S8 and Note8 featuring Infinity Display, QLED TVs equipped with quantum dot technology, and the Frame TV that boasts a unique and innovative design.

In 2018, protectionism and geopolitical uncertainties are projected to keep weighing on the global economy. In addition, rising competition in our main business areas and rapidly advancing technologies in fields such as AI and big data, which seem certain to shift the paradigm in the IT industry, pose additional challenges.

We will not dwell on our success, but rather use such an environment to seize new opportunities. We will continue to create innovation and remain steadfast in our efforts to growing our established businesses and developing new ones.

3. Financial Position and Performance (Consolidated)

A. Financial Position

(Unit: KRW million)

Classification	2017	2016	Increase/decrease	% change
Total assets	301,752,090	262,174,324	39,577,766	15.1%
Current assets	146,982,464	141,429,704	5,552,760	3.9%
- Cash and cash equivalents	30,545,130	32,111,442	-1,566,312	-4.9%
- Short-term financial instruments	49,447,696	52,432,411	-2,984,715	-5.7%
- Available-for-sale financial assets	3,191,375	3,638,460	-447,085	-12.3%
- Trade and other receivables	27,695,995	24,279,211	3,416,784	14.1%
- Inventories	24,983,355	18,353,503	6,629,852	36.1%
- Other current assets	11,118,913	10,614,677	504,236	4.8%
Non-current assets	154,769,626	120,744,620	34,025,006	28.2%
- Available-for-sale financial assets	7,752,180	6,804,276	947,904	13.9%
- Associates and joint ventures	6,802,351	5,837,884	964,467	16.5%
- Property, plant and equipment	111,665,648	91,473,041	20,192,607	22.1%
- Intangible assets	14,760,483	5,344,020	9,416,463	176.2%
- Other non-current assets	13,788,964	11,285,399	2,503,565	22.2%
Total liabilities	87,260,662	69,211,291	18,049,371	26.1%
Current liabilities	67,175,114	54,704,095	12,471,019	22.8%
Non-current liabilities	20,085,548	14,507,196	5,578,352	38.5%
Total equity	214,491,428	192,963,033	21,528,395	11.2%
Total liabilities and equity	301,752,090	262,174,324	39,577,766	15.1%
Capital adequacy ratio	71.1%	73.6%	-2.5%	-
Debt-to-equity ratio	40.7%	35.9%	4.8%	-
Inventory turnover	6.0	6.5	-0.5	-

※ Presented in accordance with Korean International Financial Reporting Standards ("K-IFRS")

- Samsung Electronics' total assets in 2017 were KRW 301.7521 trillion, an increase of KRW 39.5778 trillion (15.1%) from the previous year. This includes an increase of KRW 20.2 trillion in tangible assets and KRW 9.4 trillion in intangible assets caused by large-scale facility investment and M&A activities.
- Total liabilities were KRW 87.2607 trillion, an increase of KRW 18.0494 trillion (26.1%) from the previous year. This includes an increase of KRW 12.4710 trillion (22.8%) in current liabilities and an increase of KRW 5.5784 trillion (38.5%) in non-current liabilities.
- Total equity was KRW 214.4914 trillion, an increase of KRW 21.5284 trillion (11.2%) from the previous year. Retained earnings increased by KRW 22.7249 trillion primarily from net income of KRW 42.1867 trillion.
- In terms of financial ratios, the Company maintained the financial structure of a world-class company as the capital adequacy ratio decreased by 2.5%p from the previous year to 71.1% and the debt-to-equity ratio increased by 4.8%p from the previous year to 40.7%.

B. Performance

- In 2017, revenue increased by 18.7% to KRW 240 trillion. This strong result was driven by growing demand for components such as memory.
- Operating profit increased by 83.5% to KRW 53.6450 trillion, income before income tax expense increased by 83.0% to KRW 56.1960 trillion, and net income increased by 85.6% to KRW 42.1867 trillion.

(Unit: KRW million)

Classification	2017	2016	Increase/decrease	% change
Sales	239,575,376	201,866,745	37,708,631	18.7%
Cost of sales	129,290,661	120,277,715	9,012,946	7.5%
Gross profit	110,284,715	81,589,030	28,695,685	35.2%
Selling and administrative expenses	56,639,677	52,348,358	4,291,319	8.2%
Operating profit	53,645,038	29,240,672	24,404,366	83.5%
Other revenue	3,010,657	3,238,261	-227,604	-7.0%
Other expenses	1,419,648	2,463,814	-1,044,166	-42.4%
Gain on valuation using the equity method of accounting	201,442	19,501	181,941	933.0%
Financial revenue	9,737,391	11,385,645	-1,648,254	-14.5%
Financial expenses	8,978,913	10,706,613	-1,727,700	-16.1%
Income before income tax expense	56,195,967	30,713,652	25,482,315	83.0%
Income tax expense	14,009,220	7,987,560	6,021,660	75.4%

Classification	2017	2016	Increase/decrease	% change
Net income	42,186,747	22,726,092	19,460,655	85.6%

※ Presented in accordance with Korean International Financial Reporting Standards (“K-IFRS”)

Performance by division

[Unit: KRW million]

Classification	Division	Business	2017		2016		% change	
			Amount	Share	Amount	Share	Amount	(%)
Revenue	CE		45,108,919	18.8%	45,102,762	22.3%	6,157	0.0%
	IM		106,668,318	44.5%	100,302,112	49.7%	6,366,206	6.3%
	DS	Semiconductor	74,255,595	31.0%	51,157,002	25.3%	23,098,593	45.2%
		DP	34,465,363	14.4%	26,928,644	13.3%	7,536,719	28.0%
		Total	108,167,480	45.1%	78,148,203	38.7%	30,019,277	38.4%
	Harman		7,102,629	3.0%	-	-	7,102,629	-
	Overall revenue		239,575,376	100.0%	201,866,745	100.0%	37,708,631	18.7%
Operating profit	CE		1,653,300	3.1%	2,712,148	9.3%	-1,058,848	-39.0%
	IM		11,827,324	22.0%	10,807,569	37.0%	1,019,755	9.4%
	DS	Semiconductor	35,204,143	65.6%	13,595,004	46.5%	21,609,139	158.9%
		DP	5,398,401	10.1%	2,226,626	7.6%	3,171,775	142.4%
		Total	40,327,933	75.2%	15,850,986	54.2%	24,476,947	154.4%
	Harman		57,421	0.1%	-	-	57,421	-
	Overall operating profit		53,645,038	100.0%	29,240,672	100.0%	24,404,366	83.5%

※ For the CE division, the financial information was restated to exclude Printing Solution business

※ For Harman, the financial information was consolidated after the date of Acquisition.

[CE Division]

- The CE division's revenue in 2017 increased by 0.01% from the previous year to KRW 45.1089 trillion. Consolidated operating profit, however, decreased by KRW 1.0588 trillion to KRW 1.6533 trillion due to intense competition.
- The CE division is growing steadily, led mainly by its premium products that boast industry-leading technology such as ultra-definition QLED TVs, Family Hub refrigerators, Flexwash washing machines, and Breeze-free air conditioners.

Our TV business maintained the top position in the global TV market for a twelfth straight year on the back of outstanding technology and strength in design. We widened the gap with competitors once again, solidifying our position as market leader.

In 2018, we will continue to meet the high expectations of our consumers with products such as our QLED TVs featuring superior image quality and ultra-large TVs, which are leading the premium TV market.

The Home appliance business stepped up as a global leader by investing in and developing innovative products that make users' lives more convenient. Such efforts, for example, have kept us atop the global refrigerator market for six straight years.

In 2018, we will solidify our global reputation by rolling out products with superior usability and efficiency, such as the Family Hub 3.0 refrigerator and QuickDrive washing machine. Utilizing our leadership in the smart home appliance market, we plan to enhance both the internal and external ecosystems of our products and continue to invest in the B2B business, which includes products such as built-in air conditioner systems.

We will create new demand and strengthen our reputation in the global home appliance market. To accomplish this, we will develop smart home appliances tailored to different lifestyles in each region with improved functionality, user-centered designs, and IoT connectivity.

[IM Division]

- The IM division's revenue in 2017 was KRW 106.6683 trillion, an increase of 6.3% compared to the previous year. Operating profit as well increased by 9.4% to KRW 11.8273 trillion.
- The IM division, building on lessons learned from the Note7 quality issues, achieved solid earnings by launching innovative flagship products. Introduced in the first half of 2017, the Galaxy S8 was highly regarded for its unique features, such as Infinity Display, Bixby, and Samsung Connect. In the second half of the same year, the Galaxy Note8 launched, with its high-quality dual camera and improved S-pen functionality garnering much attention from loyal fans of the Note Series.
In the mass-market segment, Galaxy A and J Series models launched equipped with proven and popular features from our premium lineup, including a metal design, water and dust proofing, high-speed charging, and Samsung Pay.

In 2018, we plan to fortify our top position in the premium market with our differentiated products that satisfy consumer needs, such as a camera capable of recording super-slow-motion video and speakers boasting stereo sound. For our low-end range, we will focus on profitability by streamlining our lineups and improving yield. Meanwhile, we will continue to offer wearable devices and accessories that enhance usability and enrich user experience.

Alongside our achievements in hardware, we will expand the open ecosystem centered on Bixby. We intend to take full advantage of cutting-edge technologies like IoT to realize a multi-device experience on a Samsung-based platform, as expressed in our mid- to long-term roadmap.

We will stay at the forefront of technological innovation by utilizing our unparalleled R&D capabilities and investing in AI, cloud, 5G, and any other technology with potential to drive future growth.

[Semiconductor]

- The semiconductor business's 2017 revenue increased by 45.2% from the previous year to KRW 74.2556 trillion. Operating profit increased by 158.9% to KRW 35.2041 trillion.
- The memory business set new record by enhancing process technology, reducing costs, and expanding shipments of more profitable products thanks to stable demand from major mobile and server customers. We invested in next-generation technologies to build on our technological leadership, and the operation of the Pyeongtaek fab and investments in additional semiconductor lines enable us to manage demand fluctuations.
- In the DRAM business, a supply shortage continued as overall demand of mobile and server chips increased and other suppliers were slow in developing high-performance, high-quality products. Under these circumstances, we expect demand to keep concentrating on SEC. In November 2017, we unveiled the industry's first 10nm-class DRAM products and secured a competitive position at least a year ahead of our rivals in terms of development capability. Furthermore, we are nimbly responding to market changes by executing strategic decisions—e.g., we expanded the supply of HBM2, the second generation of high bandwidth memory, which supports AI and supercomputing.

For NAND, data-center, SSD, and mobile-device needs are behind a steady increase in demand, while supply is also growing as companies compete to boost production capacity. Under these circumstances, we achieved earnings growth in this area by improving cost competitiveness via expanding our advanced manufacturing processes to our 4th generation V-NAND and by increasing supply of premium products such as high-density, high-capacity SSDs.

- For the System LSI business, most of our growth came from mobile devices, although we expect smartphone and tablet markets to stall over time. That said, emerging markets for new products and services in areas such as IoT, wearable devices, automotive parts, and health care should keep overall demand from declining.

For SOC products, even though the overall mobile market is slowing, rising demand for low-end and mid-range devices is projected to drive growth in demand for integrated chips, which combine modem, AP, and connectivity solutions. We use our advanced manufacturing processes and supply of APs and SOCs to boost competitiveness in not only the premium market but also the growing market for low-end and mid-range products.

Meanwhile, we are witnessing increasing demand for high-resolution, high-quality image sensors from companies seeking to differentiate their mobile devices. Our CIS technology for mobile devices tops the industry thanks to our capabilities in areas like ISOCELL, FDTI, and slim design along with various solutions such as dual cameras. We target diversifying the business to include the automotive and security industries, to name a few.

The Foundry Business forms partnerships with major fabless companies based on its advanced manufacturing processes. Building on our 2018 achievement of launching the world's first 10nm products, we seek to maintain our leading status by being the first to develop the 7nm process.

[DP]

- The DP division's revenue in 2017 increased by 28.0% from the previous year to KRW 34.4654 trillion. Operating profit increased by 142.4% compared to the previous year to KRW 5.3984 trillion.
- We focused on solidifying our leadership in OLED by strengthening our product competitiveness, manufacturing efficiency, and consumer base. In 2018, we expect intensifying competition between OLED panels and LTPS LCD coupled with a decline in panel demand in the first half of 2018 to keep uncertainties high. In response, we will proactively address fluctuating demand from major smartphone companies and develop technologies that outperform those for LTPS LCD. We will also strive to enhance capabilities of new applications to capitalize on future growth drivers.

Amid growing competition in the LCD business, we prioritized improving yield and reducing costs alongside our efforts to expand sales of value-added products. In 2018, although increased production in China will present challenges, we will continue our transition to qualitative growth by offering cutting-edge products and raising our premium-product portion of sales.

[Harman]

- Since its acquisition in 2017, the Harman Division contributed revenue and operating profit of KRW 7.1026 trillion and 57.4 billion, respectively.
- The Harman Division is focused on maintaining its outstanding performance in the automotive parts and professional/consumer audio markets. To that end, it aims to keep growing by pursuing innovation, conducting strategic M&A activities, and empowering its popular brands. Harman is a recognized leader in the automotive parts industry and boasts a wide range of brands encompassing both mass-market and premium products. We will fully utilize the wide consumer appeal while maintaining the quality that is synonymous with Harman.

Our innovations have enabled us to establish a strong reputation in the audio equipment market. We will continue to

venture into new areas to attract new customers by launching promising products like wireless smart speakers and keep building on brand reputation to help improve our results.

We are committed to delivering another successful year in 2018, backed by our thorough preparation and ability to capitalize on new opportunities. We see rapid advancements in areas such as AI, IoT, autonomous vehicles, and big data—and the inevitable paradigm shift in the IT industry—as an opportunity to secure the technological strength to keep us at the forefront of the fourth industrial revolution.

We are steadfast in our goal to improve our performance, knowing that strong results contribute significantly to shareholder value. Political and economic uncertainties are likely to persist both at home and abroad, while industry competition to develop cutting-edge technology will keep the business environment challenging. We will continue to strengthen our quality assurance, develop innovative technologies to remain competitive, and devote resources to research new markets and launch new products. Through these efforts, we will remain a premier global firm that offers new value to both our customers and society as a whole.

-

C. New businesses

N/A

D. Changes in Organizational Structure

The Company in December 2009 reorganized from a two-division system (DMC and DS) to a system of seven independent business units. Additionally, the total number of regional headquarters increased to 10, as the Africa regional headquarters was newly established.

In April 2010, the Digital Imaging Business Unit was added through a merger with Samsung Digital Imaging, increasing the total number of business units to eight. In December 2010, an overseas semiconductor division was changed from a subsidiary to a regional headquarters.

In July 2011, the Company established the DS Business Headquarters to reinforce synergies in the component business. In December 2011, following an organizational change, the DMC division was re-established and the Company returned to a two-divisional system (DMC and DS).

In April 2012, the LED business unit was added through a merger with Samsung LED.

In May 2012, the Japan regional headquarters was established under the DS division.

In December 2012, following an organizational change, the CE and IM businesses were upgraded to individual division status, and the IT Solution business was divided into the Printing Solution and Computer businesses. The Printing Solution business was incorporated into the CE division, and the Computer business was integrated into the Mobile business under the IM division. The Medical Device business team was upgraded to a business unit within the CE division.

In December 2013, the Digital Imaging business was changed to a team unit, and then was integrated with the Mobile business.

In December 2015, the LED business was reorganized and converted into a team unit.

In June 2017, the System LSI Business was divided into Foundry and System LSI Businesses.

(Refer to 『2. Company History』 of 『I. Corporate Overview』 for more details about changes in the organizational structure.)

E. Foreign Exchange Risk

The Company is exposed to foreign exchange risks of currencies, including USD, in carrying out business activities. Uncertainty related to exchange rates could influence business transactions, assets and liabilities in the future. In particular, as the Company exports more than it imports, won cash flow can be negatively impacted when the won is strong. Therefore, the Company focuses on minimizing foreign exchange risk by matching the level of assets with liabilities in each currency through selling foreign currency bonds or other measures.

F. Recognition of Asset Impairment Loss

The Company conducts an annual review on asset impairment of CGUs (cash generating unit). The recoverable amount of a CGU is based on the calculation of value in use, and the value in use is calculated using a BTCF (before tax cash flow) assumption that based on the financial budget for the next five years approved by the management committee. In calculating terminal-year cash flow for period that exceeds the term, assumption of fixed average growth (does not exceed industry average) was used.

Impairment losses are recognized to the extent the carrying amount exceeds the recoverable amount of a CGU. The

recoverable amount is determined by choosing the greater figure between the use-value and fair-value, which deducts the disposal amount per cost. Recognized impairment losses in 2017 reached KRW 147.4 billion for tangible assets and KRW 143.1 billion for intangible assets

(Refer to 『2. Notes to Consolidated Financial Statements』 of 『III. Financial Affairs』 for more details about asset impairment loss)

4. Liquidity, Financing, and Expenditure

As of the end of 2017, the Company had KRW 83.1842 trillion of liquid funds.

These liquid funds include i) cash and cash equivalents, ii) short-term financial instruments, and iii) short-term available-for-sale securities. This balance decreased by KRW 4.9981 trillion from KRW 88.1823 trillion at the end of the previous period.

The Company's short-term borrowings (including alternatives for liquidity) are KRW 16.0462 trillion, an increase of KRW 2.0666 trillion from KRW 13.9796 trillion at the end of the previous year. Long-term borrowings are KRW 2.7678 trillion, an increase of KRW 1.4650 trillion from KRW 1.3028 trillion at the end of the previous year.

The Company's net cash (liquid funds - borrowings) is KRW 64.3702 trillion, which is a decrease of KRW 8.5297 trillion from KRW 72.8999 trillion at the end of the previous period.

The Company has ample liquidity according to the liquidity assessment index of the global credit rating agency Moody's. The Company's cash coverage (liquid funds/borrowings), which is a major liquidity index that Moody's uses, corresponds to the highest level, Aaa

	2017	2016	Note
Cash Coverage (Liquid funds/borrowings)	442%	577%	Moody's rating Aaa : >400%

5. Other Information for Investment Decision Making

A. Significant accounting policies and estimation

- Refer to 『2. Notes to Consolidated Financial Statements』 in 『III. Financial Affairs』 for significant accounting policies and estimates.

B. Environment and employees

- Refer to 『5. Sanctions and others』 in 『XI. Other information』 for environmental sanctions or administrative actions.
- No significant changes in employee positions during this period.

C. Legal regulations

- Refer to 『5. Sanctions and others』 of 『XI. Other information』 for major legal regulations on the Company's

businesses.

D. Derivatives and risk management policy

- To manage exchange rate risk, overseas companies enter into currency forwards which are denominated in the trading currency of a foreign currency position as opposed to the companies' reporting currencies. Overseas offices buy or sell currency forwards with less than one year maturity via a bank to avert risk.
- Refer to 『8. Derivative Instruments and Put Options』 in 『II. Businesses Overview』 and 『2. Notes to Financial Statements』 in 『III. Financial Affairs』 for Derivatives and risk management policy.

VI. Corporate Governance

1. Board of Directors

A. Overview of Board of Directors

As of December 31, 2017, the Board of Directors (BOD) consists of nine (9) directors, four (4) of whom are executive directors (Oh-Hyun Kwon, Boo-Keun Yoon, Jong-Kyun Shin, and Jae-Yong Lee) and five (5) independent directors (In-Ho Lee, Han-Joong Kim, Kwang-Soo Song, Byeong-Gi Lee, and Jae-Wan Bahk).

The Board has six (6) committees as follows: Management Committee, Audit Committee, Independent Director Recommendation Committee, Related Party Transactions Committee, Compensation Committee, and Governance Committee.

* On March 23, 2018, Executive Director Oh-Hyun Kwon and Independent Directors Han-Joong Kim and Byeong-Gi Lee retired. Executive Director Boo-Keun Yoon and Jong-Kyun Shin resigned.

* On March 23, 2018, Executive Directors Sang-Hoon Lee, Ki-Nam Kim, Hyun-Suk Kim, and Dong-Jin Koh and Independent Directors Jeong Kim, Sun-Uk Kim, and Byung-gook Park were newly appointed.

* As of April 2, 2018, (the "Reporting Date"), SEC's BOD is comprised of five Executive Directors (Jae-Yong Lee, Sang-Hoon Lee, Ki-Nam Kim, Hyun-suk Kim, Dong-Jin Koh) and six Independent Directors (In-Ho Lee, Kwang-Soo Song, Sun-Uk Kim, Jae-Wan Bahk, Byung-Gook Park, and Jeong Kim).

B. The BOD Agendas considered and voting results

Date	Agenda	Results	Independent directors				
			In-Ho Lee	Han-Joong Kim	Kwang-Soo Song	Byeong-Gi Lee	Jae-Wan Bahk
Jan 24, 2017	① Approval of FY2016 financial statements and annual business report	Approved	For	For	For	For	For
	② Buyback and cancellation of treasury stocks	Approved	For	For	For	For	For
Jan 24, 2017	③ Panel development contract with Samsung Display	Approved	For	For	For	For	For
	✳ <i>Reported Item</i>						
Jan 24, 2017	① Report on the internal accounting management system						
	② Assessment of the internal accounting management system						
Feb 24, 2017	① Decision to convene the 48 th (FY2017) AGM	Approved	For	For	For	For	For
	② Decision of the 48 th AGM agenda items	Approved	For	For	For	For	For
Feb 24, 2017	- report items:						
	1) FY2016 annual audit report						
Feb 24, 2017	2) FY2016 annual business report						
	- Item 1: Approval of FY2016 financial statements, including balance sheet, income statement, and the statement of appropriation of retained earnings, etc.						
Feb 24, 2017	- Item 2: Approval of remuneration limit for Directors						
	③ Guidelines for external contributions	Approved	For	For	For	For	For
Feb 24, 2017	④ Application for product liability insurance	Approved	For	For	For	For	For
	✳ <i>Reported Item</i>						
Feb 24, 2017	① Progress on initiatives to enhance shareholder value						

Mar 24, 2017	① Setting the remuneration of directors ② Approval of management plans for Social Contribution Fund for 2017 ③ Sponsoring the WorldSkills Competition <i>✱ Reported Item</i> ① Establishment and operation (draft) of Governance Committee	Approved Approved Approved	For For For	For For For	For For For	For For For	Absent Absent Absent
Apr 27, 2017	① Approval of the 49 th 1Q17 financial statements, business report and March quarterly dividend ② Buyback and cancellation of treasury stocks ③ Cancellation of shares held in treasury ④ Revision to CSR committee regulations ⑤ Donation ⑥ Property leasing contract with Samsung Display and compensation for the disposal of a property ⑦ Facility development contract with Samsung Display <i>✱ Reported Item</i> ① Result of the Holding company review	Approved Approved Approved Approved Approved Approved Approved	For For For For For For For	For For For For For For For	For For For For For For For	For For For For For For For	For For For For For For For
Jun 12, 2017	① Property leasing contract with Samsung Display ② Approval of test facility supply transactions with Samsung Display	Approved Approved	For For	For For	For For	For For	For For
Jul 27, 2017	① Approval of the 1H FY17 financial statements, half-year business report and June quarterly dividend ② Buyback and cancellation of treasury stocks ③ Application for property insurance ④ Panel development contract with Samsung Display	Approved Approved Approved Approved	For For For For	For For For For	For For For For	For For For For	For For For For
Oct 31, 2017	① Approval of the 3Q17 financial statements, business report and September quarterly dividend ② Buyback and cancellation of treasury stocks ③ Approval of shareholder return policy covering 2018-2020	Approved Approved Approved	For For For	For For For	For For For	For For For	For For For
Nov 24, 2017	① Approval of transactions with affiliate person(s) ② Payment of retirement pensions ③ Contribution for Special Disaster Area in Pohang ④ Contribution for 2018 Winter Fundraising Campaign	Approved Approved Approved Approved	For For For For	For For For For	For For For For	For For For For	For For For For

C. The Committees of BOD

(1) Composition of the BOD Committees as of December 31, 2017

Committee Name	Members	Member Name	Responsibility and Authority	Note
Management	3 Executive Directors	Oh-Hyun Kwon, Boo-Keun Yoon, Jong-Kyun Shin	See description below	Director Oh-Hyun Kwon retired and Director Boo-Keun Yoon and Jong-Kyun Shin resigned. Director Ki-Nam Kim, Hyun-Suk Kim, and Dong-Jin Koh were appointed. (March 23, 2018)
Related Party Transactions	3 Independent Directors	In-Ho Lee, Han-Joong Kim, Kwang-Soo Song		Director Han-Joong Kim retired and Director Sun-Uk Kim were appointed (March 23, 2018)
Compensation	3 Independent Directors	Kwang-Soo Song, In-Ho Lee, Byeong-Gi Lee		Director Byeong-Gi Lee retired and Director Jeong

				Kim was appointed (March 23, 2018)
Governance	5 Independent Directors	Byeong-Gi Lee, In-Ho Lee, Han-Joong Kim, Kwang-Soo Song, Jae-Wan Bahk		Director Byeong-Gi Lee and Hang-Joong Kim retired and Director Sun- Uk Kim, Byung-Gook Park, and Jeong Kim were appointed (March, 23, 2018)

※ The Audit Committee and Independent Director Recommendation Committee were excluded according to *Corporate Disclosure Guidelines* set forth by the *Financial Supervisory Service Authority* of South Korea.

i) The Management Committee

- **Responsibility:** The Management Committee deliberates and decides on matters specified by the BOD regulations and resolutions or specifically delegated by the BOD. The composition and operation of the management committee are determined by the BOD.

- **Authority**

The Management Committee deliberates and decides on the following matters:

(A) General management

1. Annual or mid to long-term management policy and strategy
2. Key management strategy
3. Business planning and restructuring
4. Establishment, relocation, and withdrawal of overseas branch and corporation
5. Initiating cooperation such as strategic partnership with foreign companies
6. Acquisition or disposal of domestic and overseas subsidiaries
(Provided that the value of transaction exceeds 0.1% of total equity)
7. Other major management matters
8. Establishment, relocation, and withdrawal of branches and operations
9. Appointment or dismissal of supervisors
10. Suspension or shutdown of production over 5% of the total production in the recent year
11. Licensing agreements and technology transfer, partnership in regards to technology of over 0.5% of the total equity
12. Acquisitions and transfers of patents related to new material and new technology of over 0.5% of the total equity
13. Collection and destruction of products corresponding to over 5% of the total sales in the recent business year
14. Contracting for over 5% of the total sales in the recent business year
15. Contracting or canceling of single sales agency and suppliers over 5% of the total sales in the recent business year
16. Set basic principles of organization management
17. Decision making on basic principles and any changes related to wages, bonus, and fringe benefits
18. Appointments, dismissals and changes of transfer agent
19. Closing of shareholder register and setting reference date
20. Establishing corporate guidelines for business and management related activities

(B) Matters related to Finance

1. Acquisitions or disposals of equity investments with a value no less than 0.1% and less than 2.5% of the total equity
2. Direct overseas investments with a value no less than 0.1% and less than 2.5% of the total equity
3. New debt guarantee (excluding extension of period) or collateral issuance with a value no less than 0.1% and less than 2.5% of the total equity
 - A. Collateral: Only in the case of providing collateral for others

B. Guarantee: Excludes performance guarantee (e.g., bid, contract, defect, difference guarantee) and tax payment guarantee

4. New credit agreement contract (excluding extension of period) with a value no less than 0.1% and less than 5% of the total equity
5. Approval of related party transactions:
Related party transactions with affiliates involving cash (loans and payments), equity (stocks and bonds) or assets (such as real estate and intangible property rights) with a value between KRW 3 billion and/or less than KRW 5 billion, as defined in the Monopoly Regulation and Fair Trade Act
※ Excluding cases where existing contracts are renewed without significant change.
6. Issuance of Corporate Bond
7. Acquisition and disposal of real estate with a value of over 0.1% of the total equity, provided that the transaction is with the third party
8. Any matters that the CEO deems necessary and important for the business (e.g., CAPEX)

(C) All other matters except for those delegated to the BOD and other committees in accordance with BOD regulations

ii) Related Party Transactions Committee

- *Responsibility*: Improve management transparency through establishment of a voluntary compliance system

- *Authority*

- 1) Right to receive reports on related party transactions:
The Committee can request reports on related party transactions with affiliates
※ According to the Monopoly Regulation and Fair Trade Act, the Related Party Transactions committee is permitted to conduct preliminary reviews and approves on related party transactions over KRW 5 billion and other transactions that are deemed as significant.
- 2) Right to order ex officio investigation of related party transactions
- 3) Right to propose corrective measures for related party transactions

iii) Compensation Committee

- *Responsibility*: Provide transparent and objective decision making process related to remuneration of directors

- *Authority*

- 1) Propose remuneration limit of directors at annual general shareholders' meeting
- 2) Establish the director remuneration system
- 3) Other matters delegated by the Board

iv) Governance Committee

- *Responsibility*: Fulfill corporate social responsibility and enhance shareholder value

- *Authority*

- 1) Matters related to corporate social responsibility
- 2) Matters related to shareholder value enhancement
 - Preliminary review on shareholder return policies
 - Activities for improving shareholders' rights and interests
 - Other important management matters that can significantly impact shareholder value
- 3) Establishment, composition, operation of organizations under the committee including research groups, councils, etc.
- 4) Other matters delegated by the Board

(2) The BOD Committee Activities

※ The Audit Committee and Independent Director Recommendation Committee were excluded according to *Corporate Disclosure Guidelines* set forth by the *Financial Supervisory Service Authority* of South Korea.

□ The Management Committee

Date	Agenda	Voting Results	Attendance/Voting Results of Independent Directors
Jan 24, 2017	① Memory investment ② S.LSI investment	Approved Approved	N/A (All comprised of executive directors)
Feb 24, 2017	① Memory investment ② Liquidation of an overseas subsidiary	Approved Approved	
Mar 8, 2017	① Closing of shareholder register for quarterly dividend	Approved	
Apr 24, 2017	① S.LSI investment ② Investment in Pyeongtaek complex ③ Capital increase for regional office ④ Rental contract	Approved Approved Approved Approved	
Jun 12, 2017	① Memory investment ② Investment in TP center ③ Closing of shareholder register for quarterly dividend ④ Withdrawal of a subsidiary liquidation plan	Approved Approved Approved Approved	
Jul 27, 2017	① Memory investment ② Construction of parking facilities in Hwasung campus ③ Establishment of a regional office ④ Establishment of a regional office	Approved Approved Approved Approved	
Aug 28, 2017	① Investment in a regional office ② Lifting of limits on account receivable collateral loans	Approved Approved	
Sep 12, 2017	① Closing of shareholder register for quarterly dividend	Approved	
Oct 26, 2017	① Memory investment	Approved	
Dec 20, 2017	① Investment in Pyeongtaek complex	Approved	

□ The Related Party Transactions Committee

Date	Agenda	Voting Results	Name of independent directors		
			In-Ho Lee (Attendance: 100%)	Han-Joong Kim (Attendance: 100%)	Kwang-Soo Song (Attendance: 100%)
			For/against		
Jan 23, 2017	- Report on 4Q16 related party transaction	-	-	-	-
Feb 22, 2017	- Prior review on large-scale related party transactions 1) Application for product liability insurance	-	-	-	-
Apr 26, 2017	- Prior review on large-scale related party transactions 1) Donation 2) Property leasing contract with Samsung Display and compensation for the disposal of a property - Report on 1Q17 related party transactions	- - -	- - -	- - -	- - -

Jun 12, 2017	- Prior review on large-scale related party transactions 1) Property leasing contract with Samsung Display	-	-	-	-
Jul 26, 2017	- Prior review on large-scale related party transactions 1) Application for property insurance - Report on 2Q17 related party transactions	-	-	-	-
Oct 30, 2017	- Report on 3Q17 related party transactions	-	-	-	-
Nov 24, 2017	- Prior review on large-scale related party transactions 1) Approval of commodity and service deal in 2018 2) The payment of retirement pensions	-	-	-	-

□ The Compensation Committee

Date	Agenda	Voting Results	Name of Independent Directors		
			Kwang-Soo Song (Attendance: 100%)	In-Ho Lee (Attendance: 100%)	Byeong-Gi Lee (Attendance: 100%)
			For/Against		
Feb 22, 2017	① Appointment of the head of compensation committee	Approved	For	For	For
	② Preliminary review on remuneration limit for Directors	Approved	For	For	For

□ The Governance Committee

Date	Agenda	Voting Results	Name of Independent Directors				
			Byeong-Gi Lee (Attendance: 100%)	In-Ho Lee (Attendance: 100%)	Han-Joong Kim (Attendance: 100%)	Kwang-Soo Song (Attendance: 100%)	Jae-Wan Bahk (Attendance: 100%)
			For/Against				
Feb 24, 2017	<i>✖ Reported Item</i> ① Report on 1Q17 results of CSR risk management council	-	-	-	-	-	-
Apr 24, 2017	<i>✖ Reported Item</i> ① Report on updates of IR activities	-	-	-	-	-	-
	② Report on 2Q17 results of CSR risk management council	-	-	-	-	-	-
Jun 12, 2017	① Review on establishment of corporate governance charter	Approved	For	For	For	For	For
	② Review on Governance Committee operation	Approved	For	For	For	For	For
Jul 27, 2017	<i>✖ Reported Item</i> ① Report on updates of IR activities	-	-	-	-	-	-
	② Report on 3Q17 results of CSR risk management council	-	-	-	-	-	-
Oct 31, 2017	① Review on shareholder return program for 2018-2020 <i>✖ Reported Item</i>	Approved	For	For	For	For	For

	① Report on 4Q17 results of CSR risk management council	-	-	-	-	-	-
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※ According to the resolution of the board of directors on April 27, 2017, the Governance Committee was established, merging the old CSR committee.

D. Director Independence

(1) Appointment of Directors

Directors are appointed by shareholders at the general meeting of shareholders. The Board is responsible for nominating candidates for Executive Directors, and the Independent Director Recommendation Committee nominates independent director candidates, who are presented to shareholders as separate agendas and voted on at the general meeting.

The Board submits shareholders' proposal for appointment of Directors at the general shareholder meeting as a separate agenda if the proposal satisfies related laws and regulations.

The composition of the Board of Directors, as appointed by shareholders, is as follows:

[As of December 31, 2017]

Position	Name	Nominated by	Responsibility	Transactions with the Company	Relation with the controlling shareholder
Executive Director	Oh-Hyun Kwon	BOD	Chairman of BOD President of Samsung Advanced Institute of Technology	N/A	N/A
Executive Director	Boo-Keun Yoon	BOD	Corporate Relations		
Executive Director	Jong-Kyun Shin	BOD	Talent Development		
Executive Director	Jae-Yong Lee	BOD	Overall management of the Company operation		Affiliate
Independent Director	In-Ho Lee	Independent Director Recommendation Committee	Overall management of the Company		N/A
Independent Director	Han-Joong Kim	Independent Director Recommendation Committee	Overall management of the Company		
Independent Director	Kwang-Soo Song	Independent Director Recommendation Committee	Overall management of the Company		
Independent Director	Byeong-Gi Lee	Independent Director Recommendation Committee	Overall management of the Company		
Independent Director	Jae-Wan Bahk	Independent Director Recommendation Committee	Overall management of the Company		

* On March 23, 2018, Executive Director Oh-Hyun Kwon and Independent Directors Han-Joong Kim and Byeong-Gi Lee retired. Executive Director Boo-Keun Yoon and Jong-Kyun Shin resigned.

* At the annual meeting on March 23, 2018, Executive Directors Sang-Hoon Lee, Ki-Nam Kim, Hyun-Suk Kim, Dong-Jin Koh and Independent Directors Jeong Kim, Sun-Uk Kim, and Byung-gook Park were newly appointed.

* On March 23, 2018, Executive Directors Ki-Nam Kim, Hyun-Suk Kim, Dong-Jin Koh were newly appointed as CEOs

(2) Independent Director Recommendation Committee

The Independent Director Recommendation Committee assesses the independence and ability of potential candidates and recommends selected candidates to be appointed at the annual general meeting of shareholders. The committee is composed of three Independent Directors (Han-Joong Kim, Byeong-Gi Lee, and Jae-Wan Bahk) and one Executive Director (Oh-Hyun Kwon) based on related laws that states more than a majority of members should be independent directors in accordance with clause 4 of Article 542-8 of the Commercial Law.

* As of December 31, 2017

* On March 23, 2018, Executive Director Oh-Hyun Kwon and Independent Directors Han-Joong Kim and Byeong-Gi Lee retired. Independent Directors Jeong Kim and Byung-gook Park were newly appointed.

* As of April 2, 2018, (the “Reporting Date”), the committee is comprised of three Independent Directors.

(3) Professionalism of Independent Directors

1) Support team for independent directors

- Support team: HR team of Corporate Management Office
- Major tasks: Supports the Board of Directors’ operation and independent directors’ performance of their duties
- Responsible Staff: five persons

2) Education programs to provide them with updates on key management issues

(1) Inspection of domestic business sites

- Period: Every year in May (on a regular basis)
- Organizer: HR team and management team of the region where the inspection takes place
- Content: On-site inspection to check the status of the business operation

(2) Inspection of overseas business sites

- Period: Every year in August (on a regular basis)

※ Was not conducted in 2017

- Organizer: HR team and management team of the region where the inspection takes place
- Content: On-site inspection to check the status of the business operation

2. Audit System

A. Profile of the member of the Audit Committee

As of December 31, 2017, SEC operates an Audit Committee consisting of three (3) Independent Directors.

Name	Career	Concurrent position in another company	Note
In-Ho Lee	- Corporate Advisor, Shinhan Bank (2009-2011) - Chief Executive Officer, Shinhan Financial Group (2005-2009) - President & CEO, Shinhan Bank (1999-2003)	-	Independent director
Han-Joong Kim	- Professor Emeritus, Yonsei University (2012-Present) - President, Yonsei University (2008-2012) - Professor, Dept. of Preventive Medicine & Public Health (1982-2012)	-	Independent director
Kwang-Soo Song	- Advisor, Kim & Chang Law Office (2007-Present) - Prosecutor General, Supreme Prosecutors' Office (2003-2005) - Chief Prosecutor, Daegu High Prosecutors' Office (2002-2003)	Independent Director of Doosan Corporation (2013 -present)	Independent director

* On March 23, 2018, Independent Director Han-Joong Kim retired and Independent Director Sun-Uk Kim was newly appointed.

B. Audit Committee Independence

Pursuant to related laws and the Articles of Incorporation, SEC has prepared internal audit regulations which encompass composition, operation, authority, and responsibility of the Audit Committees to carry out audit related activities.

The Audit Committee is exclusively composed of Independent Directors appointed at the general shareholder meeting and includes audit and financial experts. Also, the Audit Committee satisfies requirements of relevant laws.

(e.g., The stipulation that the role of head of the committee be held by an Independent Director.)

Requirements	Satisfied the requirement	Applicable Acts, etc.
- The audit committee shall consist of at least three directors.	Satisfied (three directors)	Article 415-2 (2) of the Commercial Act, Article 2 of the Company's Audit Committee Regulation
- The ratio of independent directors shall exceed two thirds of the total number of members.	Satisfied (all members are independent directors)	
- At least one member of the committee shall be an accounting or financing expert.	Satisfied (In-ho Lee)	Article 542-11 (2) of the Commercial Act, Article 3 of the Company's Audit Committee Regulation
- The representative of the committee shall be an independent director.	Satisfied	
- Other conditions (affiliates of majority shareholder, etc.)	Satisfied (n/a)	Article 542-11 (3) of the Commercial Act

The Audit Committee reviews accounting documents including financial statements and audit process related documents presented by the external auditor, and if needed, requests the external auditor to perform additional reviews on accounting books and records. To ensure the reliability of accounting related disclosures, the audit committee receives and reviews a report on internal accounting control systems prepared by internal accounting managers. In addition, the Committee attends the BOD meetings and other significant meetings. The Committee receives reports on deliberations of the Management Committee and on business performance from Directors and requests additional reviews and supplementation of data as needed.

C. Major Activities of the Audit Committee

Date	Agenda	Voting Results	Name of independent directors		
			In-Ho Lee (Attendance: 100%)	Han-Joong Kim (Attendance: 100%)	Kwang-Soo Song (Attendance: 100%)
			For/Against		
Jan 23, 2017	- Report on 2016 financial statement and business report - Report on 2016 internal accounting management system activities - Report on 4Q16 non-audit activities - Report on 2016 audit activities - Report on plans to appoint external auditor	- - - - -	- - - - -	- - - - -	- - - - -
Feb 22, 2017	- Approval of appointment of external auditor	Approved	For	For	For
Feb 24, 2017	- Review on the 48th AGM agenda - Assessment of 2016 activities of internal compliance system	- -	- -	- -	- -
Apr 26, 2017	- Report on the 49 th 1Q17 financial statement and business report - Report on 1Q17 non-audit activities - Report on 1Q17 external contributions	- - -	- - -	- - -	- - -
Jul 26, 2017	- Report on 1H FY17 financial statements and half-year business report - Report on Q2 FY17 non-audit activities - Report on Q2 FY17 external contributions - Report on 1H FY17 audit activities	- - - -	- - - -	- - - -	- - - -
Oct 30, 2017	- Report on the 49 th 3Q17 financial statement and business report - Report on Q3 FY17 non-audit activities - Report on Q3 FY17 external contributions	- - -	- - -	- - -	- - -

D. Compliance Officer

[As of December 31, 2017]

1. Compliance Officer	Name	Sang-Joo Lee
	Age	47
	Academic Background	Doctor of Laws (Korea University College of Law) / Master of Public Administration (Graduate School of Public Administration, Harvard University)
	Current Position	Corporate Compliance Team leader at Samsung Electronics Legal Affairs Office
	Career	- Dec '12: Senior vice president/Team leader of Samsung Electronics Global Legal Affairs Team - Dec '15: Senior vice president/Team leader of Samsung Electronics Compliance Team - Feb '16: Senior vice president/Concurrently Chief privacy officer of Samsung Electronics Global Privacy Office
2. Date of BOD resolution		Jan 28, 2016
3. Disqualifications		N/A
4. Others		N/A

※ On March 23, 2018 Officer Sang-Joo Lee resigned and Officer Young-Soo Kim was appointed with the resolution of the Board.

3. Voting Rights of Shareholders

A. Voting Method

The Company has not adopted a concentrated voting system, voting in writing, or an electronic voting system.

B. Minority Shareholder Right

No minority shareholder right was exercised during the reporting period.

C. Competition over Management Right

No competition over management rights occurred during the reporting period.

VII. Information on Shareholders

1. Ownership of Shares by the Major Shareholder and Related Parties

The following table presents changes in shares held by the major shareholder and related parties as of the beginning and end of the reporting period.

[As of December 31, 2017]

(Unit: Shares, %)

Name	Relationship	Type of share	Number of Shares owned				Note
			Beginning of period		End of period		
			# shares owned	%	# shares owned	%	
Kun-Hee Lee	Major shareholder	Common	4,985,464	3.54	4,985,464	3.86	-
Kun-Hee Lee	Major shareholder	Preferred	12,398	0.06	12,398	0.07	-
Samsung C&T	Affiliate	Common	5,976,362	4.25	5,976,362	4.63	-
Samsung Welfare Foundation	Affiliate	Common	89,683	0.06	89,683	0.07	-
Samsung Foundation Of Culture	Affiliate	Common	37,615	0.03	37,615	0.03	-
Ra-Hee Hong	Affiliate	Common	1,083,072	0.77	1,083,072	0.84	-
Jae-Yong Lee	Affiliate	Common	840,403	0.60	840,403	0.65	-
Samsung Life Insurance	Affiliate	Common	10,622,814	7.55	10,622,814	8.23	-
Samsung Life Insurance	Affiliate	Preferred	879	0.00	879	0.00	-
Samsung Life Insurance (special accounts)	Affiliate	Common	467,320	0.33	466,715	0.36	Open market transactions
Samsung Life Insurance (special accounts)	Affiliate	Preferred	51,614	0.25	30,778	0.17	Open market transactions
Samsung Fire & Marine Insurance	Affiliate	Common	1,856,370	1.32	1,856,370	1.44	-
Oh-Hyun Kwon	Affiliate	Common	1,300	0.00	500	0.00	Open market transactions
Total		Common	25,960,403	18.45	25,958,998	20.11	-
		Preferred	64,891	0.32	44,055	0.24	-

※ The definition of 'major shareholder' is as defined in the Monopoly Regulation and Fair Trade Act of Korea.

※ The changes in share ratio, excluding open market transactions, is due to the cancellation of treasury shares.

[The major shareholder]

- Name of the major shareholder: Kun-Hee Lee
- Career (recent 5 years): Chairman of Samsung Electronics (March 2010 ~)
 Honorary member of the International Olympic Committee (2017 ~)
 Member of the International Olympic Committee (1996 ~ 2017)

- The major shareholder of record has not changed during the reporting period.

2. Ownership of Shares

As of December 31, 2017, the date of Suspension of Entry in the Register of Shareholders, shareholders with over 5% voting shares, excluding the major shareholder, are as follows:

[As of December 31, 2017]

(Unit: Shares, %)

Classification	Name	Number of shares owned	Share ratio	Note
Shareholders with over 5% voting shares	National Pension Service	12,369,186	9.58	-
	Samsung Life Insurance	11,089,529	8.59	-
Employee Stock Ownership Association		-	-	-

※ Note: According to the shareholder register, as of December 31, 2017.

※ The number of shares owned and share ratio of Samsung Life Insurance includes special accounts.

3. Minority Shareholders

[As of December 31, 2017]

(Unit: Person, Shares, %)

Classification	Shareholder		Shares owned		Note
	# of shareholders	Portion	# of shares	Portion	
Minority shareholder	144,283	99.94	75,381,275	58.39	Shareholders with shares less than 1/100 of total issued stocks
Total	144,374	100.00	129,098,494	100.00	

※ Note: According to the shareholder register, as of December 31, 2017.

※ Based on the number of common stock with voting rights.

4. Stock Affairs

Preemptive Rights in the Articles of Incorporation	1. New shares to be issued by the Company shall be allocated to the shareholders in proportion to their respective shareholdings in accordance with Paragraph 6, Article 8. If shares are not subscribed for as a result of waiver or loss of the preemptive right of the shareholders to subscribe for new shares, or if fractional shares remain at the time of allocation of new shares, such shares may be disposed of by a resolution of the Board of Directors in accordance with applicable laws and regulations. 2. Notwithstanding the above Paragraph 1, new shares may be allocated to persons other than shareholders in the following cases: a. If the Company issues new shares or causes underwriters to underwrite new shares by a resolution of the Board of Directors in accordance with applicable laws and regulations including the Financial Investment Services and Capital Markets Act; b. If the Company allocates new shares preferentially to members of the Employee Stock Ownership Association by a resolution of the Board of Directors in accordance with applicable laws and regulations including the Financial Investment Services and Capital Markets Act; c. If the Company issues new shares for the issuance of depositary receipts (DR) by a resolution of the Board of Directors in accordance with the applicable laws and regulations including the Financial Investment Services and Capital Markets Act; d. If the Company issues new shares by public offering in accordance with Article 11-3; e. If new shares are issued by the exercise of stock options in accordance with Article 11-4; f. If the Company issues new shares to the extent of 20% of the total issued and outstanding shares to domestic or foreign financial institutions for the purpose of obtaining financing urgently or to the relevant partner company for the purpose of inducing technology therefrom, etc., by a resolution of the Board of Directors; provided that the issue price of the new shares shall not be less than the price prescribed by the Financial Investment Services and Capital Markets Act and other applicable laws and regulations. ☞ (Note) Paragraph 6, Article 8 In the case of rights issue, bonus issue or stock dividend, the holders of common shares shall be entitled to common shares, and the holders of preferred shares shall be entitled to preferred shares, in proportion to their respective shareholdings; provided that in the case of rights issue and stock dividend the Company may, if necessary, issue only one kind of shares, where all the shareholders shall be entitled to such kind of shares to be issued. ☞ (Note) Article 11-3 (Public Offering) 1. The Company may issue new shares by public offering to the extent that the new shares do not exceed 20% of the total number of issued and outstanding shares by a resolution of the Board of Directors, pursuant to Article 165-6, Paragraph 1, Item 3 of the Financial Investment Services and Capital Markets Act. 2. If the Company issues new shares by public offering, the type, quantity and issue price of the shares to be newly issued shall be determined by a resolution of the Board of Directors; provided that the issue price of such new shares shall not be less than the price prescribed by the Financial Investment Services and Capital Markets Act and other applicable laws and regulations. ☞ (Note) Article 11-4 (Stock Options) 1. The Company may grant stock options to its officers and employees (including officers and employees of the
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Preemptive Rights in the Articles of Incorporation	related companies as set forth in Article 542-3, Paragraph 1 of the Commercial Code; the same shall apply for the purpose of this Article) by a special resolution of the general meeting of shareholders pursuant to Article 542-3 of the Commercial Code, to the extent permitted by the Commercial Code; provided, however, that the Company may grant the stock options to its officers and employees (except for the directors of the Company) by a resolution of the Board of Directors to the extent determined by the relevant laws and regulations.		
	2. The person to whom stock options may be granted are the officers and employees who have contributed or have the capacity to contribute to the establishment, management, overseas business, technical innovation, etc. of the Company; provided, however, that the officers and employees who may not be entitled to stock options under the relevant laws and regulations shall be excluded.		
	3. The shares to be issued to the officers or employees by the exercise of their stock options (in case the Company pays, either in cash or treasury shares, the difference between the exercise price of stock options and the market price, they refer to the shares which are the basis for such calculation) shall be common shares in registered form or preferred shares in registered form.		
	4. Total number of shares to be delivered in accordance with the exercise of stock options shall be up to the extent permitted by the relevant laws and regulations.		
	5. The stock options may be exercised until an expiry date determined by a resolution of the general meeting of shareholders or the Board of Directors and such expiry date shall be determined within a period not exceeding eight (8) years from the date when two (2) years have elapsed from the date of the general meeting of shareholders or the date of the Board of Directors' meeting at which a resolution to grant such stock options is adopted; provided, however, that the person to whom a stock option is granted should serve the Company for at least two (2) years after the date of such resolution in order to exercise such stock option, unless otherwise set forth by relevant laws and regulations.		
	6. The terms and conditions for stock options, such as the contents and exercise price thereof shall be determined by a special resolution of the general meeting of shareholders or by a resolution of the Board of Directors in accordance with the relevant laws and regulations and the Articles of Incorporation; provided, however, that such matters which are not provided for as matters reserved for resolutions of the general meeting of shareholders or the Board of Directors' meeting under the relevant laws and regulations or the Articles of Incorporation may be determined by the Board of Directors or a committee authorized by the Board of Directors.		
	7. The Company may cancel the grant of stock options by a resolution of the Board of Directors in any of the following cases: a. In case the relevant officer or employee voluntarily retires from his/her office or leaves the Company after the grant of stock options; b. In case the relevant officer or employee causes substantial damages to the Company due to his/her willful misconduct or negligence; c. In case any of the causes for cancellation set forth in the stock option agreement occurs.		
Settlement date	December 31	Annual General Meetings of Shareholders	Within 3 months after the end of every business year
Shareholder register closing period			One month from January 1
Stock type	1,5,10,50,100,500,1000,10000(8 types)		
Transfer agent	Korea Securities Depository(T: +82-2-3774-3000): 23, Yeouinaru-ro 4-gil, Yeongdeungpo-gu, Seoul, Korea		
Shareholder benefit	n/a	Published on	Joongang Daily

5. Stock Price and Stock Transactions

A. Domestic Stock Market

(Unit: KRW, Shares)

Type			July '17	August	September	October	November	December
Common stock	Stock price	Highest	2,560,000	2,450,000	2,681,000	2,754,000	2,861,000	2,605,000
		Lowest	2,350,000	2,231,000	2,302,000	2,620,000	2,540,000	2,410,000
		Average	2,469,667	2,342,909	2,512,667	2,698,063	2,769,091	2,537,053
	Volume	Highest (day)	523,366	508,146	341,144	410,396	592,683	406,208
		Lowest (day)	136,111	102,069	158,870	116,121	90,724	147,005
		Monthly	4,444	5,361	4,679	3,401	4,621	4,515
Preferred stock	Stock price	Highest	2,056,000	1,959,000	2,152,000	2,240,000	2,335,000	2,096,000
		Lowest	1,849,000	1,800,000	1,876,000	2,067,000	2,096,000	1,956,000
		Average	1,956,381	1,898,273	2,033,619	2,155,188	2,233,818	2,041,526
	Volume	Highest (day)	84,712	72,665	78,594	72,304	117,674	55,848
		Lowest (day)	15,456	11,999	14,011	18,315	19,388	18,431
		Monthly	766	689	735	601	971	682

※ Monthly trading volume in 1,000 shares

B. Overseas Stock Market

☐ Name of Stock Exchange: London Stock Exchange (Common stock)

(Unit: USD, KRW, DR)

Type			July '17	August	September	October	November	December
Common stock	Stock price	Highest	1,141.00	1,101.00	1,194.00	1,233.00	1,287.00	1,199.00
		(Converted to KRW)	1,279,974	1,230,918	1,356,742	1,387,125	1,412,225	1,310,987
		Lowest	1,016.00	983.00	1,022.00	1,153.00	1,175.00	1,147.00
		(Converted to KRW)	1,163,320	1,121,701	1,147,399	1,322,145	1,271,820	1,246,445
		Average	1,088.14	1,038.64	1,110.81	1,186.09	1,253.09	1,172.32
	Volume	Highest (day)	76,483	95,469	48,751	82,433	58,894	50,503
		Lowest (day)	9,348	15,333	10,698	9,017	9,954	1,259
		Monthly	606	658	567	656	583	425

※ Monthly volume in 1,000 Depository Receipt (DR) shares.

※ Stock price converted to KRW is based on the closing price of the KRW/USD exchange rate of the applicable trading day.

Ratio: Common one (1) share equals DR two (2) shares

☐ Name of Stock Exchange: Luxembourg Stock Exchange (Preferred stock)

(Unit: USD, KRW, DR)

Type			July '17	August	September	October	November	December
Preferred stock	Stock price	Highest	911.00	871.00	945.00	1,010.00	1,055.00	986.00
		(Converted to KRW)	1,024,784	985,014	1,073,804	1,136,250	1,182,866	1,056,400
		Lowest	805.00	794.00	831.00	918.00	966.00	917.00
		(Converted to KRW)	921,725	906,033	940,277	1,052,671	1,045,598	993,019
		Average	863.67	839.36	895.38	949.86	1,011.77	946.58
	Volume	Highest (day)	12,349	11,484	11,762	17,333	7,836	10,494
		Lowest (day)	503	608	531	829	466	596
		Monthly	69	73	88	78	66	68

※ Monthly volume in 1,000 DR shares.

※ Stock price converted to KRW is based on KRW/USD exchange rate of closing price of trading day.

Ratio: Preferred one (1) share equals DR two (2) shares

VIII. Executives and Employees

1. Executives and Employees

A. Registered Executives

[As of December 31, 2017]

(Unit: # of shares)

Name	Gender	Date of birth	Position	Registered executive	Full/part-time	Responsibility	Major career	Number of shares owned		Length of Service	Term expiration
								Common	Preferred		
Oh-Hyun Kwon	M	1952.10	CEO	Registered executive	Full time	Chairman of BOD President of Samsung Advanced Institute of Technology	Head of DS Division	500	0	70 mo.	2018.03.23
Boo-Keun Yoon	M	1953.02	CEO	Registered executive	Full time	Corporate Relations	Head of CE Division	0	0	58 mo.	2019.03.14
Jong-Kyun Shin	M	1956.01	CEO	Registered executive	Full time	Talent Development	Head of IM Division	0	0	58 mo.	2019.03.14
Jae-Yong Lee	M	1968.06	Executive Director	Registered executive	Full time	Overall management of company operation	Vice Chairman	840,403	0	15 mo.	2019.10.26
In-Ho Lee	M	1943.11	Independent Director	Registered executive	Part time	Overall management of the Company	Former President & CEO of Shinhan Bank	0	0	94 mo.	2019.03.18
Han-Joong Kim	M	1948.11	Independent Director	Registered executive	Part time	Overall management of the Company	Former President of Yonsei University	0	0	70 mo.	2018.03.23
Kwang-Soo Song	M	1950.01	Independent Director	Registered executive	Part time	Overall management of the Company	Advisor at Kim & Chang Law Office	0	0	58 mo.	2019.03.14
Byeong-Gi Lee	M	1951.05	Independent Director	Registered executive	Part time	Overall management of the Company	Professor Emeritus at Seoul National University	0	0	70 mo.	2018.03.23
Jae-Wan Bahk	M	1955.01	Independent Director	Registered executive	Part time	Overall management of the Company	Professor of Public Administration at Sungkyunkwan University	0	0	22 mo.	2019.03.10

* On March 23, 2018, Executive Director Oh-Hyun Kwon and Independent Directors Han-Joong Kim and Byeong-Gi Lee retired. Executive Director Boo-Keun Yoon and Jong-Kyun Shin resigned.

* At the annual meeting on March 23, 2018, Executive Directors Sang-Hoon Lee, Ki-Nam Kim, Hyun-Suk Kim and Dong-Jin Koh and Independent Directors Jeong Kim, Sun-Uk Kim, and Byung-gook Park were newly appointed.

* On March 23, 2018, Executive Officers Ki-Nam Kim, Hyun-Suk Kim, Dong-Jin Koh were newly appointed as CEOs.

B. Concurrent Position with Other Companies

[As of December 31, 2017]

Concurrent office holder		Company	
Name	Position	Name of company	Position
Kwang-Soo Song	Independent director	Doosan Corp.	Independent director
Jae-Wan Bahk	Independent director	Lotte Shopping co., Ltd	Independent director
Oh-Hyun Kwon	Executive director, CEO	Samsung Display	Executive director, CEO

- ※ Persons above do not hold shares of the companies with which they hold a concurrent position.
- ※ Executive director Jae-Yong Lee resigned his post as an independent director on the board of Exor N.V. on April 5, 2017.
- ※ Executive director Oh-Hyun Kwon resigned his post as an executive director and CEO of Samsung Display on March 22, 2018.

C. Employees

[As of December 31, 2017]

(Unit: KRW million)

Division	Gender	No. of employees				Average length of service (years)	Total Compensation	Average Compensation per employee
		Regular	Contract-based	Others	Total			
CE	M	10,454	82	-	10,536	13.7	-	-
CE	F	2,193	21	-	2,214	8.9	-	-
IM	M	19,624	207	-	19,831	11.8	-	-
IM	F	7,672	38	-	7,710	9.6	-	-
DS	M	34,387	165	-	34,552	10.4	-	-
DS	F	14,539	15	-	14,554	10.1	-	-
Others	M	8,032	155	-	8,187	13.5	-	-
Others	F	2,162	38	-	2,200	9.4	-	-
Gender Total	M	72,497	609	-	73,106	11.6	9,013,438	127
Gender Total	F	26,566	112	-	26,678	9.8	2,120,076	88
Total		99,063	721	-	99,784	11.0	11,133,514	117

- ※ Total compensation and average compensation per employee were calculated before income tax and other deductions based on the earned income payment record submitted to the district tax office in accordance with Article 20 of the Income Tax Law.
- ※ The number of employees represent domestic employees resident in Korea, excluding four executive directors. According to the updated reporting standards, the number of employees includes employees on leave.
- ※ Average compensation per employee was calculated based on the annual average number of employees— 95,158 employees (male: 71,172, female: 23,986).

2. Remuneration for Directors

A. Summary on Total Remuneration

(1) Remuneration approved at the annual general meetings of shareholders

(Unit: KRW million)

	Number of Persons	Amount Approved at Shareholders' Meeting	Note
Executive Directors	4	-	-
Independent Directors	2	-	-
Audit Committee member or Auditor	3	-	-
Total	9	55,000	-

※ Number of persons is as of the Reporting Date.

※ The amount approved at shareholders' meeting is the remuneration limit for directors registered pursuant to Article 388 of the Korean Commercial Act ("**Registered Directors**"). It includes compensation for Registered Directors who retired this fiscal year and in past years.

(2) Remuneration Paid

[Remuneration paid for directors and auditors]

(Unit: KRW million)

Number of Persons	Total amount of remuneration	Average remuneration per Director	Note
9	41,757	4,640	-

※ Number of persons is as of the Reporting Date.

※ The total amount is the income based on the Income Tax Law that incumbent or retired executive directors, independent directors, and members of audit committee in the fiscal year received as a Registered Director in accordance with Article 159 of the Financial Investment Services and Capital Markets Act and Article 168 of the enforcement decree of the same law.

※ Average remuneration per Director was calculated by dividing total remuneration by annual average number of persons.

※ There is no recognized expenses (compensation expenses) in this term from granting stock options.

[Remuneration paid by type of director]

(Unit: KRW million)

	Number of Persons	Total amount of remuneration	Average remuneration per Director	Note
Executive Director (not including independent directors and audit committee members)	4	41,348	10,337	-
Independent Director (not including audit committee members)	2	156	78	-
Audit Committee member	3	253	84	-
Auditor	-	-	-	-

- ※ Number of persons is as of the Reporting Date.
- ※ The total amount is the income based on the Income Tax Law that incumbent or retired executive directors, independent directors, and members of audit committee in the fiscal year received as a Registered Director in accordance with Article 159 of the Financial Investment Services and Capital Markets Act and Article 168 of the enforcement decree of the same law.
- ※ Average remuneration per Director was calculated by dividing total remuneration by annual average number of persons.
- ※ There is no recognized expenses (compensation expenses) in this term from granting stock options.

3. Remuneration for Individual Executive Directors

Remuneration for individual Executive directors is disclosed on a semi-annual basis, in Half-year report and Business report, in accordance with the Financial Investment Services and Capital Markets Act (revised on March 2016).

(1) Summary on Remuneration by Executive

(Unit: KRW million)

Name	position	Total amount of remuneration	Remuneration not included in total amount
Oh-Hyun Kwon	CEO	24,381	-
Boo-Keun Yoon	CEO	7,669	-
Jong-Kyun Shin	CEO	8,427	-
Jae-Yong Lee	Executive Director	871	-

(2) Criteria and Methodology

(Unit: KRW million)

Name	Type of remuneration		Total amount	Criteria and methodology
Oh-Hyun Kwon (CEO)	Earned Income	Salary	1,840	In accordance with internal regulation for executive treatment (Board resolution), Remuneration is determined based on position (chairman), responsibilities, and performance results. He was paid KRW 156 million per month in January-October, KRW 174 million in November, and KRW 104 million in December.
		Bonus	7,719	- Traditional holiday bonus: 100% of monthly wage - Target Achievement Incentive: Paid biannual basis within the range of 0~200% of monthly

				wage, and CEO decides the amount according to the level of target achieved. (Adjusted according to division performance) - Performance Incentive: When performance exceeds the Company's goal; CEO decides the amount based on 20% profit. Paid once a year within 0~50% of salary. (Adjusted according to individual performance) - Long-term incentive: Calculated within the director remuneration limit, based on the average salary of 3 years, using ROE, stock performance, and EBIT margin as criteria. Paid over the course of 3 years. ※ Bonus was awarded based on his contribution in increasing operating profits to KRW 50 trillion, by achieving considerable earnings growth in the semiconductor business.
		Profit from exercising stock option	-	n/a
		Other Income	14,821	- Fringe benefits are provided in line with the internal regulation for executive treatment (Board resolution) - Special bonus (one-time)
	Retirement Income		-	n/a
	Other Income		-	n/a

(Unit: KRW million)

Name	Type of remuneration		Total amount	Criteria and methodology
Boo-Keun Yoon (CEO)	Earned Income	Salary	1,670	In accordance with internal regulation for executive treatment (Board resolution), Remuneration is determined based on his position (vice chairman), responsibilities, and performance results. He was paid KRW 144 million per month in January-November and KRW 86 million in December.
		Bonus	5,862	- Traditional holiday bonus: 100% of monthly wage - Target Achievement Incentive: Paid biannual basis within the range of 0~200% of monthly wage, and CEO decides the amount according to the level of target achieved. (Adjusted according to division performance) - Performance Incentive: When performance exceeds the Company's goal; CEO decides the amount based on 20% profit. Paid once a year within 0~50% of salary. (Adjusted according to individual performance) - Long-term incentive: Calculated within the director remuneration limit, based on the average salary of 3 years, using ROE, stock performance, and EBIT margin as criteria. Paid over the course of 3 years. ※ Bonus was awarded based on his leadership in achieving remarkable performance of the CE division, by delivering innovative products and improving the soundness of business.
		Profit from exercising stock option	-	n/a
		Other Income	137	- Fringe benefits are provided in line with the internal regulation for executive treatment (Board resolution)
	Retirement Income		-	n/a
	Other Income		-	n/a

(Unit: KRW million)

Name	Type of remuneration		Total amount	Criteria and methodology
Jong-Kyun Shin (CEO)	Earned Income	Salary	1,670	In accordance with internal regulation for executive treatment (Board resolution), Remuneration is determined based on his position (vice chairman), responsibilities, and performance results. He was paid KRW 144 million per month in January-November and KRW 86 million in December.
		Bonus	6,624	- Traditional holiday bonus: 100% of monthly wage - Target Achievement Incentive: Paid biannual basis within the range of 0~200% of monthly wage, and CEO decides the amount according to the level of target achieved. (Adjusted according to division performance) - Performance Incentive: When performance exceeds the Company's goal; CEO decides the amount based on 20% profit. Paid once a year within 0~50% of salary. (Adjusted according to individual performance) - Long-term incentive: Calculated within the director remuneration limit, based on the average salary of 3 years, using ROE, stock performance, and EBIT margin as criteria. Paid over the course of 3 years. ※ Bonus was awarded based on his efforts to strengthen business performance by expanding sales of premium products and providing next-generation 5G solutions
		Profit from exercising stock option	-	n/a
		Other Income	133	- Fringe benefits are provided in line with the internal regulation for executive treatment (Board resolution)
	Retirement Income		-	n/a
	Other Income		-	n/a

(Unit: KRW million)

Name	Type of remuneration		Total amount	Criteria and methodology
Jae-Yong Lee (Executive Director)	Earned Income	Salary	318	In accordance with internal regulation for executive treatment (Board resolution), Remuneration is determined based on his position (vice chairman), responsibilities, and performance results. He was paid KRW 159 million per month.
		Bonus	529	- Traditional holiday bonus: 100% of monthly wage - Target Achievement Incentive: Paid biannual basis within the range of 0~200% of monthly wage, and CEO decides the amount according to the level of target achieved. (Adjusted according to division performance) - Performance Incentive: When performance exceeds the Company's goal; CEO decides the amount based on 20% profit. Paid once a year within 0~50% of salary. (Adjusted according to individual performance) - Long-term incentive: Calculated within the director remuneration limit, based on the average salary of 3 years, using ROE, stock performance, and EBIT margin as criteria. Paid over the course of 3 years. ※ Bonus was awarded based on his remarkable management capabilities and leadership, demonstrated in the preemptive M&A plan conducted to respond to uncertain business conditions.
		Profit from exercising stock option	-	n/a
		Other Income	24	- Fringe benefits are provided in line with the internal regulation for executive treatment (Board resolution)
	Retirement Income		-	n/a

	Other Income	-	n/a
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A. Stock Options Granted and Exercised

(1) Status of Stock Options Granted to Registered Directors

There have been no stock options granted to, or exercised by, the executive BOD or Audit Committee members during the reporting period.

(2) Status of Stock Options Granted to Non-Registered Executives

As of December 31, 2017, there is no accumulated-basis unexercised stock option.

IX. Affiliates and Subsidiaries

1. Affiliates & Subsidiaries

A. Affiliates

- Name of affiliated group: Samsung

As at December 31, 2017, Samsung Group had a total of sixty-three (63) domestic affiliates, where five (5) affiliates (Samsung Active Asset Management, Samsung Hedge Asset Management, Harman International Korea, Red Bend Software Korea, SBTM) were added. Among the Samsung Group's sixty-three (63) domestic affiliates, sixteen (16) affiliates including Samsung Electronics are listed, and forty-seven (47) affiliates are unlisted.

[As of December 31, 2017]

	No. of affiliates	Name of affiliates
Listed	16	Samsung C&T, Samsung Electronics, Samsung SDI, SEMCO, Samsung Fire & Marine Insurance, Samsung Heavy Industries, Samsung Life Insurance, MULTICAMPUS, Samsung Securities, Samsung SDS, Samsung Card, Samsung Engineering, S1, Cheil Worldwide, Hotel Shilla, Samsung Biologics
Unlisted	47	Seoul Lakeside CC, Samwoo Architects & Engineers, CVnet Corporation, Samsung Bioepis, Samsung Display, Samsung Corning Advanced Glass, SU Materials, STECO, SEMES, Samsung Electronics Service, Samsung Electronics Sales, Samsung Electronics Logitech, Suwon Samsung Bluewings FC, Samsung Medison, Samsung Claim Adjustment Service, Samsung Fire & Marine Insurance Service, Samsung Fire & Marine Financial Service, Daejung Offshore Wind Power Co., Ltd., Samsung Futures, Samsung Asset Management, Saengbo, Samsung Life Service, Samsung SRA Asset Management, Samsung Life Financial Service, SD Flex, Cheil Fashion Retail, Natural9, Samsung Welstory, S-Printing Solution, SECUI, STM, S-Core, OpenHands, Miracom, Samsung Card Customer Service, Human TSS, S-1CRM, Shilla Stay, HDC Shilla Duty Free LTD, SERI, Samsung Lions, Samsung Venture Investment Corporation, Samsung Active Asset Management, Samsung Hedge Asset Management, Harman International Korea, Red Bend Software Korea, SBTM
Total	63	

※ S-Printing Solution ceased to be affiliate in February 2018 (share sale in November 2017)

B. Ownership Status of Affiliates and Subsidiaries

1) Domestic

Investee Investor	Samsung C&T Corp	Samsung Electronics	Samsung SDI	Samsung Electro-Mechanics	Samsung Heavy Industries	Hotel Shilla	Samsung Engineering	Cheil Worldwide	S1 Corporation	Samsung SDS	Samsung Lions	SERI	STECO
Samsung C&T Corporation		4.6			0.1		7.0			17.1		1.0	
Samsung Electronics			19.6	23.7	16.9	5.1		25.2		22.6		29.8	70.0
Samsung SDI	2.1				0.4	0.1	11.7		11.0			29.6	
Samsung Electro-mechanics	2.6				2.3							23.8	
Samsung Heavy Industries												1.0	
Cheil Worldwide					0.1						67.5		
Hotel Shilla													
S1 Corporation													
SERI													
Samsung SDS													
Samsung Life Insurance	0.1	8.6	0.1	0.3	3.3	7.9	0.1	0.1	5.4	0.1		14.8	
Samsung Fire & Marine Insurance	1.4	1.4					0.2		1.0				
Samsung Securities.						3.1			1.3				
Samsung Card						1.3		3.0	1.9				
SDC													
Samsung BioLogics													
Samsung Asset Management													
Miracom													
Harman International Industries													
Red Bend Software													
Total	6.3	14.7	19.7	24.0	23.1	17.5	19.0	28.4	20.6	39.7	67.5	100.0	70.0

※ (% of Ownership; As of December 31, 2017; Based on Common shares)

Investee Investor	SEMES	Samsung Electronic s Service	Samsung Electronics Sales	Suwon Samsung Bluewings FC	Samsung Electronics Logitech	Samsung Display	Samsung Medison	Samsung BioLogics	Samsung BioEpis	Samsung Corning Advance d Glass	SU Materials	CVnet	Seoul Lakeside CC
Samsung C&T Corporation								43.4				40.1	100.0
Samsung Electronics	91.5	99.3	100.0		100.0	84.8	68.5	31.5					
Samsung SDI						15.2							
Samsung Electro- mechanics													
Samsung Heavy Industries													
Cheil Worldwide				100.0									
Hotel Shilla													
S1 Corporation													
SERI													
Samsung SDS												9.4	
Samsung Life Insurance								0.1					
Samsung Fire & Marine Insurance													
Samsung Securities.													
Samsung Card													
SDC										50.0	50.0		
Samsung BioLogics									94.6				
Samsung Asset Management													
Miracom													
Harman International Industries													
Red Bend Software													
Total	91.5	99.3	100.0	100.0	100.0	100.0	68.5	75.0	94.6	50.0	50.0	49.5	100.0

※ (% of Ownership; As of December 31, 2017; Based on Common shares)

Investee Investor	Samwoo Architects & Engineers	SD Flex	Cheil Fashion Retail	Natural9	Samsung Welstory	Daejung Offshore Wind Power	SECUI	Human TSS	S-1CRM	STM	MULTI CAMPUS	S-Core	Open Hands
Samsung C&T Corporation	100.0		100.0	51.0	100.0		8.7						
Samsung Electronics													
Samsung SDI		50.0								100.0			
Samsung Electro-mechanics													
Samsung Heavy Industries						50.1							
Cheil Worldwide												5.2	
Hotel Shilla													
S1 Corporation								100.0	100.0			0.6	
SERI											15.2		
Samsung SDS							56.5				47.2	81.8	100.0
Samsung Life Insurance											0.0		
Samsung Fire & Marine Insurance													
Samsung Securities.													
Samsung Card													
SDC													
Samsung BioLogics													
Samsung Asset Management													
Miracom												0.5	
Harman International Industries													
Red Bend Software													
Total	100.0	50.0	100.0	51.0	100.0	50.1	65.2	100.0	100.0	100.0	62.4	88.1	100.0

※ (% of Ownership; As of December 31, 2017; Based on Common shares)

Investee Investor	Miracom	Shilla Stay	HDC Shilla Duty Free LTD	Samsung Life Insurance	Saengbo	Samsung Life Service	Samsung SRA Asset Management	Samsung Life Financial Service	Samsung Fire & Marine Insurance	Samsung Claim Adjustment Service	Samsung Fire & Marine Insurance Service	Samsung Fire & Marine Financial Service	Samsung Securities
Samsung C&T Corporation					19.3								
Samsung Electronics													
Samsung SDI													
Samsung Electro-mechanics													
Samsung Heavy Industries													
Cheil Worldwide	5.4												
Hotel Shilla		100.0	50.0	100.0									
S1 Corporation	0.6												
SERI													
Samsung SDS	83.6												
Samsung Life Insurance						50.0	99.8	100.0	100.0	15.0			
Samsung Fire & Marine Insurance											100.0	100.0	100.0
Samsung Securities.													
Samsung Card													
SDC													
Samsung BioLogics													
Samsung Asset Management													
Miracom													
Harman International Industries													
Red Bend Software													
Total	89.6	100.0	50.0	100.0	19.3	50.0	99.8	100.0	100.0	15.0	100.0	100.0	100.0

※ (% of Ownership; As of December 31, 2017; Based on Common shares)

Investee Investor	Samsung Securities	Samsung Card	S-Printing Solution	Samsung Card Customer Service	Samsung Asset Management	Samsung Futures	Samsung Venture Investment	Samsung Active Asset Management	Samsung Hedge Asset Management	Harman International Korea	Red Bend Software Korea
Samsung C&T Corporation							16.7				
Samsung Electronics							16.3				
Samsung SDI							16.3				
Samsung Electro-mechanics							17.0				
Samsung Heavy Industries							17.0				
Cheil Worldwide											
Hotel Shilla											
S1 Corporation											
SERI											
Samsung SDS											
Samsung Life Insurance	29.4	71.9			100.0						
Samsung Fire & Marine Insurance											
Samsung Securities.						100.0	16.7				
Samsung Card				100.0							
SDC											
Samsung BioLogics											
Samsung Asset Management								100.0	100.0		
Miracom											
Harman International Industries										100.0	
Red Bend Software											100.0
Total	29.4	71.9	-	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

※ (% of Ownership; As of December 31, 2017; Based on Common shares)

2) Overseas

Investor	Investee	Ownership
Samwoo Architects & Engineers	SAMOO DESIGNERS & ENGINEERS INDIA PRIVATE LIMITED	100.0
Samsung C&T America Inc.	Meadowland Distribution	100.0
Samsung C&T America Inc.	SAMSUNG OIL & GAS USA CORP	10.0
Samsung C&T America Inc.	Samsung Green repower, LLC	100.0
Samsung C&T America Inc.	Samsung Solar Construction Inc.	100.0
Samsung C&T America Inc.	QSSC, S.A. de C.V.	20.0
Samsung C&T America Inc.	Samsung Solar Energy LLC	100.0
Samsung C&T America Inc.	S-sprint Inc	24.0
Samsung Renewable Energy Inc.	SRE GRW EPC GP Inc.	100.0
Samsung Renewable Energy Inc.	SRE GRW EPC LP	100.0
Samsung Renewable Energy Inc.	SRE SKW EPC GP Inc.	100.0
Samsung Renewable Energy Inc.	SRE SKW EPC LP	100.0
Samsung Renewable Energy Inc.	SRE GRW LP Holdings LP	100.0
Samsung Renewable Energy Inc.	SRE SKW LP Holdings LP	100.0
Samsung Renewable Energy Inc.	SRE WIND PA GP INC.	100.0
Samsung Renewable Energy Inc.	SRE WIND PA LP	100.0
Samsung Renewable Energy Inc.	SRE GRS Holdings GP Inc.	100.0
Samsung Renewable Energy Inc.	SRE GRS Holdings LP	100.0
Samsung Renewable Energy Inc.	SRE K2 EPC GP Inc.	100.0
Samsung Renewable Energy Inc.	SRE K2 EPC LP	100.0
Samsung Renewable Energy Inc.	SRE KS HOLDINGS GP INC.	100.0
Samsung Renewable Energy Inc.	SRE KS HOLDINGS LP	100.0
Samsung Renewable Energy Inc.	SRE Belle River LP Holdings LP	100.0
Samsung Renewable Energy Inc.	SRE Armow EPC GP Inc.	100.0
Samsung Renewable Energy Inc.	SRE Armow EPC LP	100.0
Samsung Renewable Energy Inc.	SRE Armow LP Holdings LP	100.0
Samsung Renewable Energy Inc.	SRE North Kent 1 LP H.LP	100.0
Samsung Renewable Energy Inc.	SRE Wind GP Holding Inc.	100.0
Samsung Renewable Energy Inc.	SRE North Kent 2 LP Holdings LP	100.0
Samsung Renewable Energy Inc.	SRE Solar Development GP Inc.	100.0
Samsung Renewable Energy Inc.	SRE Solar Development LP	100.0
Samsung Renewable Energy Inc.	SRE Windsor Holdings GP Inc.	100.0
Samsung Renewable Energy Inc.	SRE Southgate Holdings GP Inc.	100.0
Samsung Renewable Energy Inc.	SRE Solar Construction Management GP Inc.	100.0
Samsung Renewable Energy Inc.	SRE Solar Construction Management LP	100.0
Samsung Renewable Energy Inc.	SRE DEVELOPMENT GP INC.	100.0
Samsung Renewable Energy Inc.	SRE DEVELOPMENT LP	100.0
Samsung Renewable Energy Inc.	SRE BRW EPC GP INC.	100.0
Samsung Renewable Energy Inc.	SRE BRW EPC LP	100.0
Samsung Renewable Energy Inc.	SRE North Kent 1 GP Holdings Inc	100.0
Samsung Renewable Energy Inc.	SRE North Kent 2 GP Holdings Inc	100.0
Samsung Renewable Energy Inc.	SRE Belle River GP Holdings Inc	100.0
Samsung Renewable Energy Inc.	SRE NK1 EPC GP Inc	100.0
Samsung Renewable Energy Inc.	SRE NK1 EPC LP	100.0
Samsung Renewable Energy Inc.	SRE Summerside Construction GP Inc.	100.0
Samsung Renewable Energy Inc.	SRE Summerside Construction LP	100.0
Samsung Green repower, LLC	SOLAR PROJECTS SOLUTIONS, LLC	50.0
SP Armow Wind Ontario GP Inc	SP Armow Wind Ontario LP	0.0
Samsung C&T Oil & Gas Parallel Corp.	PLL Holdings LLC	83.6
Samsung C&T Oil & Gas Parallel Corp.	PLL E&P LLC	90.0
SRE GRW EPC GP Inc.	SRE GRW EPC LP	0.0
SRE SKW EPC GP Inc.	SRE SKW EPC LP	0.0
PLL Holdings LLC	Parallel Petroleum LLC	61.0
SRE GRW LP Holdings LP	Grand Renewable Wind LP Inc.	45.0
SRE SKW LP Holdings LP	South Kent Wind LP Inc.	50.0
SRE WIND PA GP INC.	SRE WIND PA LP	0.0
SRE GRS Holdings GP Inc.	Grand Renewable Solar GP Inc.	50.0
SRE GRS Holdings GP Inc.	SRE GRS Holdings LP	0.0
SRE K2 EPC GP Inc.	SRE K2 EPC LP	0.0
SRE KS HOLDINGS GP INC.	KINGSTON SOLAR GP INC.	50.0

Investor	Investee	Ownership
SRE KS HOLDINGS GP INC.	SRE KS HOLDINGS LP	0.0
SRE Belle River LP Holdings LP	SP Belle River Wind LP	42.5
SP Belle River Wind GP Inc	SP Belle River Wind LP	0.0
SRE Armow EPC GP Inc.	SRE Armow EPC LP	0.0
SRE Armow LP Holdings LP	SP Armow Wind Ontario LP	50.0
SRE North Kent 1 LP H.LP	North Kent Wind 1 LP	35.0
SRE Wind GP Holding Inc.	SP Armow Wind Ontario GP Inc	50.0
SRE Wind GP Holding Inc.	SRE GRW LP Holdings LP	0.0
SRE Wind GP Holding Inc.	SRE SKW LP Holdings LP	0.0
SRE Wind GP Holding Inc.	SRE Armow LP Holdings LP	0.0
SRE Wind GP Holding Inc.	South Kent Wind GP Inc.	50.0
SRE Wind GP Holding Inc.	Grand Renewable Wind GP Inc.	50.0
South Kent Wind GP Inc.	South Kent Wind LP Inc.	0.0
Grand Renewable Wind GP Inc.	Grand Renewable Wind LP Inc.	0.0
North Kent Wind 1 GP Inc	North Kent Wind 1 LP	0.0
SRE North Kent 2 LP Holdings LP	North Kent Wind 2 LP	50.0
North Kent Wind 2 GP Inc	North Kent Wind 2 LP	0.0
SRE Solar Development GP Inc.	SRE Solar Development LP	0.0
SRE Solar Construction Management GP Inc.	SRE Solar Construction Management LP	0.0
SRE DEVELOPMENT GP INC.	SRE DEVELOPMENT LP	0.0
SRE BRW EPC GP INC.	SRE BRW EPC LP	0.0
SRE North Kent 1 GP Holdings Inc	SRE North Kent 1 LP H.LP	0.0
SRE North Kent 1 GP Holdings Inc	North Kent Wind 1 GP Inc	50.0
SRE North Kent 2 GP Holdings Inc	SRE North Kent 2 LP Holdings LP	0.0
SRE North Kent 2 GP Holdings Inc	North Kent Wind 2 GP Inc	50.0
SRE Belle River GP Holdings Inc	SRE Belle River LP Holdings LP	0.0
SRE Belle River GP Holdings Inc	SP Belle River Wind GP Inc	50.0
SRE NK1 EPC GP Inc	SRE NK1 EPC LP	0.0
SRE Summerside Construction GP Inc.	SRE Summerside Construction LP	0.0
Samsung Solar Energy LLC	Samsung Solar Energy 1 LLC	100.0
Samsung Solar Energy 1 LLC	CS SOLAR LLC	50.0
Samsung C&T Deutschland GmbH	SCNT Investment Atlantic SPRL	0.0
Samsung C&T Deutschland GmbH	POSS-SLPC, s.r.o	20.0
Samsung C&T Deutschland GmbH	Solluce Romania 1 B.V.	20.0
Samsung C&T Deutschland GmbH	Solluce Slovenia 1 B.V.	20.0
Samsung C&T Deutschland GmbH	S.C. Otelix S.A	94.3
Samsung C&T U.K. Ltd.	Samsung Nigeria Co., Ltd.	0.1
Solluce Romania 1 B.V.	LJG GREEN SOURCE ENERGY ALPHA S.R.L.	78.0
Solluce Slovenia 1 B.V.	ZE Solar 1 D.O.O.	70.0
Cassava Investment Korea Pte. Ltd.	PT. Cahaya Borneo Sukses Agrosindo	49.0
Cassava Investment Korea Pte. Ltd.	PT. Cassava Borneo Sukses Plantation	49.0
Samsung C&T Thailand Co., Ltd	Samsung Development (Thailand) Co., Ltd.	67.0
Cheil Holding Inc.	Samsung Const. Co. Phils.,Inc.	75.0
Samsung C&T Singapore Pte., Ltd.	Samsung Chemtech Vina LLC	48.3
Samsung C&T Singapore Pte., Ltd.	S-print Inc	16.0
Samsung C&T Singapore Pte., Ltd.	Cassava Investment Korea Pte. Ltd.	12.7
Samsung C&T Singapore Pte., Ltd.	PT. INSAM BATUBARA ENERGY	10.0
Samsung C&T Singapore Pte., Ltd.	Malaysia Samsung Steel Center Sdn.Bhd	30.0
Samsung C&T Singapore Pte., Ltd.	S&G Biofuel PTE.LTD	12.6
S&G Biofuel PTE.LTD	PT. Gandaerah Hendana	95.0
S&G Biofuel PTE.LTD	PT. Inecda	95.0
Samsung C&T Hongkong Ltd.	Samsung C&T Thailand Co., Ltd	13.2
Samsung C&T Hongkong Ltd.	SAMSUNG (TIANJIN) INTERNATIONAL TRADING CO., LTD	100.0
Samsung C&T Hongkong Ltd.	SAMSUNG TRADING (SHANGHAI) CO., LTD	100.0
Samsung C&T Hongkong Ltd.	Samsung Precision Stainless Steel(pinghu) Co.,Ltd.	45.0
Samsung C&T Hongkong Ltd.	Samsung Corporation (Guangzhou) Limited.	100.0
Samsung Electronics	Samsung Japan Corporation	100.0
Samsung Electronics	Samsung R&D Institute Japan Co. Ltd.	100.0
Samsung Electronics	Samsung Electronics America, Inc.	100.0
Samsung Electronics	Samsung Electronics Canada, Inc.	100.0
Samsung Electronics	Samsung Electronics Mexico S.A. De C.V.	63.6

Investor	Investee	Ownership
Samsung Electronics	Samsung Electronics (UK) Ltd.	100.0
Samsung Electronics	Samsung Electronics(London) Ltd.	100.0
Samsung Electronics	Samsung Electronics Holding GmbH	100.0
Samsung Electronics	Samsung Electronics Iberia, S.A.	100.0
Samsung Electronics	Samsung Electronics France S.A.S	100.0
Samsung Electronics	Samsung Electronics Hungarian Private Co. Ltd.	100.0
Samsung Electronics	Samsung Electronics Italia S.P.A.	100.0
Samsung Electronics	Samsung Electronics Europe Logistics B.V.	100.0
Samsung Electronics	Samsung Electronics Benelux B.V.	100.0
Samsung Electronics	Samsung Electronics Overseas B.V.	100.0
Samsung Electronics	Samsung Electronics Polska, SP.Zo.o	100.0
Samsung Electronics	Samsung Electronics Portuguesa S.A.	100.0
Samsung Electronics	Samsung Electronics Nordic Aktiebolag	100.0
Samsung Electronics	Samsung Electronics Austria GmbH	100.0
Samsung Electronics	Samsung Electronics Slovakia s.r.o	55.7
Samsung Electronics	Samsung Electronics Europe Holding Cooperatief U.A.	100.0
Samsung Electronics	Samsung Electronics Display (M) Sdn. Bhd.	75.0
Samsung Electronics	Samsung Electronics (M) Sdn. Bhd.	100.0
Samsung Electronics	Samsung Vina Electronics Co., Ltd.	100.0
Samsung Electronics	Samsung Asia Private Ltd.	100.0
Samsung Electronics	Samsung India Electronics Private Ltd.	100.0
Samsung Electronics	Samsung R&D Institute India-Bangalore Private Limited	100.0
Samsung Electronics	Samsung Electronics Australia Pty. Ltd.	100.0
Samsung Electronics	PT Samsung Electronics Indonesia	100.0
Samsung Electronics	Thai Samsung Electronics Co., Ltd.	91.8
Samsung Electronics	Samsung Malaysia Electronics (SME) Sdn. Bhd.	100.0
Samsung Electronics	Samsung Electronics Hong Kong Co., Ltd.	100.0
Samsung Electronics	Suzhou Samsung Electronics Co., Ltd.	69.1
Samsung Electronics	Samsung (CHINA) Investment Co., Ltd.	100.0
Samsung Electronics	Samsung Electronics Suzhou Semiconductor Co., Ltd.	100.0
Samsung Electronics	Samsung Electronics Huizhou Co., Ltd.	89.6
Samsung Electronics	Tianjin Samsung Electronics Co., Ltd.	48.2
Samsung Electronics	Samsung Electronics Taiwan Co., Ltd.	100.0
Samsung Electronics	Tianjin Samsung Telecom Technology Co., Ltd.	90.0
Samsung Electronics	Shanghai Samsung Semiconductor Co., Ltd.	100.0
Samsung Electronics	Samsung Electronics Suzhou Computer Co., Ltd.	73.7
Samsung Electronics	Shenzhen Samsung Electronics Telecommunication Co., Ltd.	95.0
Samsung Electronics	Samsung Semiconductor (China) R&D Co., Ltd.	100.0
Samsung Electronics	Samsung (China) Semiconductor Co., Ltd.	100.0
Samsung Electronics	Samsung SemiConductor Xian	100.0
Samsung Electronics	Samsung Gulf Electronics Co., Ltd.	100.0
Samsung Electronics	Samsung Electronics Egypt S.A.E	0.1
Samsung Electronics	Samsung Electronics South Africa(Pty) Ltd.	100.0
Samsung Electronics	Samsung Electronics Latinoamerica (Zona Libre)	100.0
Samsung Electronics	Samsung Electronica da Amazonia Ltda.	87.0
Samsung Electronics	Samsung Electronics Argentina S.A.	98.0
Samsung Electronics	Samsung Electronics Chile Limitada	4.1
Samsung Electronics	Samsung Electronics Rus Company LLC	100.0
Samsung Electronics	Samsung Electronics Rus Kaluga LLC	100.0
Samsung Electronics	Tianjin Samsung LED Co., Ltd.	100.0
Samsung BioEpis	SAMSUNG BIOEPIS UK LIMITED	100.0
Samsung BioEpis	Samsung Bioepis CH GmbH	100.0
Samsung BioEpis	SAMSUNG BIOEPIS AU PTY LTD	100.0
Samsung BioEpis	Samsung Bioepis TR Pharmaceutical Distributor LLC	100.0
Samsung BioEpis	SAMSUNG BIOEPIS BR PHARMACEUTICAL LTDA	100.0
Samsung Display	Intellectual Keystone Technology LLC	41.9
Samsung Display	Samsung Display Slovakia s.r.o.	100.0
Samsung Display	Samsung Display Vietnam Co., Ltd.	100.0
Samsung Display	Samsung Suzhou Module Co., Ltd.	100.0
Samsung Display	Samsung Suzhou LCD Co., Ltd.	60.0
Samsung Display	Samsung Display Dongguan Co., Ltd.	100.0

Investor	Investee	Ownership
Samsung Display	Samsung Display Tianjin Co., Ltd.	95.0
Samsung Display	NovaLED GmbH	9.9
SEMES	SEMES America, Inc.	100.0
SEMES	SEMES (XIAN) Co., Ltd.	100.0
Samsung Medison	Samsung Medison India Private Ltd.	100.0
Studer Japan Ltd.	Harman International Japan Co., Ltd.	100.0
Samsung Electronics America, Inc.	NexusDX, Inc.	100.0
Samsung Electronics America, Inc.	NeuroLogica Corp.	100.0
Samsung Electronics America, Inc.	Dacor Holdings, Inc.	100.0
Samsung Electronics America, Inc.	Samsung HVAC America, LLC	100.0
Samsung Electronics America, Inc.	SmartThings, Inc.	100.0
Samsung Electronics America, Inc.	Samsung Oak Holdings, Inc.	100.0
Samsung Electronics America, Inc.	Joyent, Inc.	100.0
Samsung Electronics America, Inc.	Samsung Pay, Inc.	100.0
Samsung Electronics America, Inc.	Prismview, LLC	100.0
Samsung Electronics America, Inc.	Samsung Semiconductor, Inc.	100.0
Samsung Electronics America, Inc.	Samsung Research America, Inc.	100.0
Samsung Electronics America, Inc.	Samsung Electronics Home Appliances America, LLC	100.0
Samsung Electronics America, Inc.	Samsung International, Inc.	100.0
Samsung Electronics America, Inc.	Harman International Industries, Inc.	100.0
Dacor Holdings, Inc.	Dacor	100.0
Dacor Holdings, Inc.	EverythingDacor.com, Inc.	100.0
Dacor Holdings, Inc.	Distinctive Appliances of California, Inc.	100.0
Dacor	Dacor Canada Co.	100.0
Samsung Oak Holdings, Inc.	Stellus Technologies, Inc.	100.0
Joyent, Inc.	Joyent Ltd.	100.0
Samsung Semiconductor, Inc.	Samsung Austin Semiconductor LLC.	100.0
Samsung Electronics Canada, Inc.	AdGear Technologies Inc.	100.0
Samsung Electronics Canada, Inc.	SigMast Communications Inc.	100.0
Samsung Research America, Inc.	SAMSUNG NEXT LLC	100.0
Samsung Research America, Inc.	Viv Labs, Inc.	100.0
SAMSUNG NEXT LLC	SAMSUNG NEXT FUND LLC	100.0
Samsung International, Inc.	Samsung Mexicana S.A. de C.V.	100.0
Samsung Electronics Mexico S.A. De C.V.	Samsung Electronics Digital Appliance Mexico, SA de CV	100.0
Harman International Industries, Inc.	Studer Japan Ltd.	100.0
Harman International Industries, Inc.	Harman International Industries Canada Ltd.	100.0
Harman International Industries, Inc.	Harman Becker Automotive Systems, Inc.	100.0
Harman International Industries, Inc.	Harman Professional, Inc.	100.0
Harman International Industries, Inc.	Red Bend Software Inc.	100.0
Harman International Industries, Inc.	S1NN USA, Inc.	100.0
Harman International Industries, Inc.	Harman Connected Services Holding Corp.	100.0
Harman International Industries, Inc.	AMX Holding Corporation	100.0
Harman International Industries, Inc.	Harman Financial Group LLC	100.0
Harman International Industries, Inc.	Harman Belgium SA	100.0
Harman International Industries, Inc.	Harman Consumer Division Nordic A/S	100.0
Harman International Industries, Inc.	Harman Consumer Finland OY	100.0
Harman International Industries, Inc.	Harman France SNC	100.0
Harman International Industries, Inc.	Harman International SNC	0.0
Harman International Industries, Inc.	Harman Inc. & Co. KG	66.0
Harman International Industries, Inc.	Harman KG Holding, LLC	100.0
Harman International Industries, Inc.	Harman Becker Automotive Systems Italy S.R.L.	100.0
Harman International Industries, Inc.	Harman Finance International, SCA	100.0
Harman International Industries, Inc.	Harman Finance International GP S.a.r.l	100.0
Harman International Industries, Inc.	Harman Malaysia Sdn. Bhd.	100.0
Harman International Industries, Inc.	Harman International Singapore Pte. Ltd.	100.0
Harman International Industries, Inc.	Harman Industries Holdings Mauritius Ltd.	100.0
Harman International Industries, Inc.	Harman International Mexico, S. de R.L. de C.V.	99.9
AMX LLC	AMX UK Limited	100.0
AMX LLC	Harman Professional Singapore Pte. Ltd	100.0
Harman Becker Automotive Systems, Inc.	Harman International Estonia OU	100.0
Harman Investment Group, LLC	Harman Becker Automotive Systems Manufacturing Kft	5.0

Investor	Investee	Ownership
Harman Investment Group, LLC	Harman do Brasil Industria Electronica e Participacoes Ltda.	100.0
Harman Professional, Inc.	AMX LLC	35.5
Harman Professional, Inc.	Southern Vision Systems, Inc	100.0
Harman Professional, Inc.	Harman da Amazonia Industria Electronica e Participacoes Ltda.	0.0
Harman Professional, Inc.	Harman do Brasil Industria Electronica e Participacoes Ltda.	0.0
Red Bend Software Inc.	Red Bend Software SAS	100.0
Harman Connected Services, Inc.	Harman Connected Services Engineering Corp.	100.0
Harman Connected Services, Inc.	Triple Play Integration LLC	100.0
Harman Connected Services, Inc.	Harman Connected Services South America S.R.L.	100.0
Harman Connected Services, Inc.	Harman Connected Services UK Ltd.	100.0
Harman Connected Services, Inc.	Harman Connected Services Corp. India Pvt. Ltd.	1.6
Harman Connected Services, Inc.	Global Symphony Technology Group Private Ltd.	100.0
Harman Connected Services Holding Corp.	Harman Connected Services, Inc.	100.0
Harman Connected Services Holding Corp.	Harman Connected Services AB.	100.0
AMX Holding Corporation	AMX LLC	64.5
Harman Financial Group LLC	Harman International (India) Private Limited	0.0
Harman Financial Group LLC	Harman International Mexico, S. de R.L. de C.V.	0.1
Harman Financial Group LLC	Harman de Mexico, S. de R.L. de C.V.	0.0
Samsung Electronics (UK) Ltd.	Samsung Semiconductor Europe Limited	100.0
Samsung Electronics Holding GmbH	Samsung Semiconductor Europe GmbH	100.0
Samsung Electronics Holding GmbH	Samsung Electronics GmbH	100.0
Samsung Electronics Hungarian Private Co. Ltd.	Samsung Electronics Czech and Slovak s.r.o.	31.4
Samsung Electronics Hungarian Private Co. Ltd.	Samsung Electronics Slovakia s.r.o	44.3
Samsung Electronics Benelux B.V.	Samsung Electronics Mexico S.A. De C.V.	36.4
Samsung Electronics Benelux B.V.	SAMSUNG ELECTRONICS BALTICS SIA	100.0
Samsung Electronics Benelux B.V.	Samsung Electronics West Africa	100.0
Samsung Electronics Benelux B.V.	Samsung Electronics East Africa	100.0
Samsung Electronics Benelux B.V.	Samsung Electronics Egypt S.A.E	99.9
Samsung Electronics Benelux B.V.	Samsung Electronics Israel Ltd.	100.0
Samsung Electronics Benelux B.V.	Samsung Electronics Tunisia S.A.R.L.	99.0
Samsung Electronics Benelux B.V.	Samsung Electronics Pakistan(Private) Ltd.	100.0
Samsung Electronics Benelux B.V.	Samsung Electronics South Africa Production (pty) Ltd.	100.0
Samsung Electronics Benelux B.V.	Samsung Electronics Turkey	100.0
Samsung Electronics Benelux B.V.	Samsung Semiconductor Israel R&D Center,Ltd.	100.0
Samsung Electronics Benelux B.V.	Samsung Electronics Levant Co.,Ltd.	100.0
Samsung Electronics Benelux B.V.	Samsung Electronics Maghreb Arab	100.0
Samsung Electronics Benelux B.V.	Samsung Electronics Venezuela, C.A.	100.0
Samsung Electronics Benelux B.V.	Samsung Electronica da Amazonia Ltda.	13.0
Samsung Electronics Benelux B.V.	Samsung Electronics Chile Limitada	95.9
Samsung Electronics Benelux B.V.	Samsung Electronics Peru S.A.C.	100.0
Samsung Electronics Benelux B.V.	Samsung Electronics Ukraine Company LLC	100.0
Samsung Electronics Benelux B.V.	Samsung R&D Institute Rus LLC	100.0
Samsung Electronics Benelux B.V.	Samsung Electronics Central Eurasia LLP	100.0
Samsung Electronics Benelux B.V.	Samsung Electronics Caucasus Co. Ltd	100.0
Samsung Electronics Nordic Aktiebolag	Samsung Nanoradio Design Center	100.0
AKG Acoustics GmbH	Harman Professional Denmark ApS	100.0
AKG Acoustics GmbH	Studer Professional Audio GmbH	100.0
Harman Professional Denmark ApS	Martin Professional Japan Ltd.	40.0
Harman Professional Denmark ApS	R&D International BVBA	100.0
Harman Professional Denmark ApS	Harman International s.r.o	100.0
Harman Professional Denmark ApS	Harman Professional France SAS	100.0
Harman Professional Denmark ApS	Harman Professional Germany GmbH	100.0
Harman Professional Denmark ApS	Martin Manufacturing (UK) Ltd	100.0
Harman Professional Denmark ApS	Martin Professional Pte. Ltd.	100.0
Harman France SNC	Harman International SNC	100.0
Harman International SNC	Harman France SNC	0.0
Harman Becker Automotive Systems GmbH	Harman International Romania SRL	0.0
Harman Becker Automotive Systems GmbH	iOnRoad Technologies Ltd	100.0
Harman Holding GmbH & Co. Kg	Harman Becker Automotive Systems GmbH	100.0
Harman Holding GmbH & Co. Kg	Harman Deutschland GmbH	100.0
Harman Holding GmbH & Co. Kg	Harman Becker Automotive Systems Manufacturing Kft	95.0

Investor	Investee	Ownership
Harman Holding Gmbh & Co. Kg	Duran Audio B.V.	100.0
Harman Holding Gmbh & Co. Kg	Harman RUS CIS LLC	100.0
Harman Inc. & Co. KG	Harman Holding Gmbh & Co. Kg	100.0
Harman Inc. & Co. KG	Harman Management Gmbh	100.0
Harman Inc. & Co. KG	Harman Hungary Financing Ltd.	100.0
Harman Connected Services GmbH	Harman Connected Services OOO	100.0
Harman KG Holding, LLC	Harman Inc. & Co. KG	34.0
Harman Becker Automotive Systems Manufacturing Kft	Harman Professional Kft	100.0
Harman Becker Automotive Systems Manufacturing Kft	Harman Consumer Nederland B.V.	100.0
Harman Becker Automotive Systems Manufacturing Kft	Harman International Romania SRL	100.0
Harman Becker Automotive Systems Manufacturing Kft	Red Bend Ltd.	100.0
Harman Becker Automotive Systems Manufacturing Kft	Towersec Ltd.	100.0
Harman Hungary Financing Ltd.	Harman International Industries Limited	100.0
Harman Finance International GP S.a.r.l	Harman Finance International, SCA	0.0
Harman Consumer Nederland B.V.	AKG Acoustics Gmbh	100.0
Harman Consumer Nederland B.V.	Harman Holding Limited	100.0
Duran Audio B.V.	Harman Investment Group, LLC	100.0
Duran Audio B.V.	Duran Audio Iberia Espana S.L.	100.0
Harman Connected Services AB.	Harman Connected Services Japan Co., Ltd.	100.0
Harman Connected Services AB.	Harman Connected Services Finland OY	100.0
Harman Connected Services AB.	Harman Connected Services GmbH	100.0
Harman Connected Services AB.	Harman Connected Services Poland Sp.zoo	100.0
Harman Connected Services AB.	Harman Connected Services Limited	100.0
Harman Connected Services AB.	Harman Connected Services Solutions (Beijing) Co., Ltd.	100.0
Harman Connected Services AB.	Harman Connected Services Solutions (Chengdu) Co., Ltd.	100.0
Harman Connected Services AB.	Harman Connected Services Taiwan Inc.	100.0
AMX UK Limited	AMX Gmbh	100.0
AMX UK Limited	Knight Image Limited	100.0
AMX UK Limited	Inspiration Matters Limited	100.0
AMX UK Limited	Endeleo Limited	100.0
Harman Automotive UK Limited	Harman de Mexico, S. de R.L. de C.V.	100.0
Harman International Industries Limited	Harman Automotive UK Limited	100.0
Harman International Industries Limited	Arcam Limited	100.0
Harman International Industries Limited	Harman International Industries PTY Ltd.	100.0
Harman Connected Services UK Ltd.	Harman Connected Services Morocco	100.0
Arcam Limited	A&R Cambridge Limited	100.0
Samsung Electronics Austria GmbH	Samsung Electronics Switzerland GmbH	100.0
Samsung Electronics Europe Holding Cooperatief U.A.	Samsung Electronics Czech and Slovak s.r.o.	68.6
Samsung Electronics Europe Holding Cooperatief U.A.	Samsung Electronics Romania LLC	100.0
Samsung Electronics Europe Holding Cooperatief U.A.	Porta Nuova Varesine Building 2 S.r.l.	49.0
Samsung Electronics Europe Holding Cooperatief U.A.	Samsung Electronics Poland Manufacturing SP.Zo.o	100.0
Samsung Electronics Europe Holding Cooperatief U.A.	Samsung Electronics Greece S.A.	100.0
Samsung Electronics Europe Holding Cooperatief U.A.	Samsung Electronics Air Conditioner Europe B.V.	100.0
Samsung Electronics Europe Holding Cooperatief U.A.	Innoetics E.P.E.	100.0
Samsung Electronics Europe Holding Cooperatief U.A.	Samsung Denmark Research Center ApS	100.0
Samsung Electronics Europe Holding Cooperatief U.A.	Samsung France Research Center SARL	100.0
Samsung Electronics Europe Holding Cooperatief U.A.	Samsung Cambridge Solution Centre Limited	100.0
Samsung Electronics Europe Holding Cooperatief U.A.	NovaLED GmbH	40.0
Samsung Asia Private Ltd.	Samsung Electronics Japan Co., Ltd.	100.0
Samsung Asia Private Ltd.	Samsung Electronics Display (M) Sdn. Bhd.	25.0
Samsung Asia Private Ltd.	Samsung Electronics New Zealand Limited	100.0
Samsung Asia Private Ltd.	Samsung Electronics Philippines Corporation	100.0
Samsung Asia Private Ltd.	Samsung R&D Institute BanglaDesh	100.0
Samsung Asia Private Ltd.	Samsung Electronics Vietnam Co., Ltd.	100.0
Samsung Asia Private Ltd.	Samsung Electronics Vietnam THAINGUYEN Co., Ltd.	100.0
Samsung Asia Private Ltd.	Samsung Electronics HCMC CE Complex Co. Ltd.,	100.0
Samsung Asia Private Ltd.	Samsung SDI(Malaysia) Sdn. Bhd.	25.0
Samsung Asia Private Ltd.	Samsung Electro-Mechanics(Thailand) Co., Ltd.	25.0
Samsung Asia Private Ltd.	iMarket Asia Co., Ltd.	11.4
Samsung India Electronics Private Ltd.	Samsung Nepal Services Pvt, Ltd	100.0
PT Samsung Electronics Indonesia	PT Samsung Telecommunications Indonesia	100.0

Investor	Investee	Ownership
Thai Samsung Electronics Co., Ltd.	LAOS Samsung Electronics Sole Co., Ltd	100.0
Harman Professional Singapore Pte. Ltd	AMX Products And Solutions Private Limited	100.0
Harman Connected Services Technologies Pvt. Ltd.	Aditi Technologies Europe GmbH	100.0
Harman Connected Services Technologies Pvt. Ltd.	INSP India Software Development Pvt. Ltd.	99.0
Harman Connected Services Corp. India Pvt. Ltd.	Harman Connected Services Technologies Pvt. Ltd.	100.0
Harman Connected Services Corp. India Pvt. Ltd.	INSP India Software Development Pvt. Ltd.	1.0
Samsung Electronics Hong Kong Co., Ltd.	iMarket Asia Co., Ltd.	11.3
Suzhou Samsung Electronics Co., Ltd.	Samsung Suzhou Electronics Export Co., Ltd.	100.0
Samsung (CHINA) Investment Co., Ltd.	Suzhou Samsung Electronics Co., Ltd.	19.2
Samsung (CHINA) Investment Co., Ltd.	Samsung Mobile R&D Center China-Guangzhou	100.0
Samsung (CHINA) Investment Co., Ltd.	Samsung Tianjin Mobile Development Center	100.0
Samsung (CHINA) Investment Co., Ltd.	Samsung R&D Institute China-Shenzhen	100.0
Samsung (CHINA) Investment Co., Ltd.	Samsung Electronics Huizhou Co., Ltd.	10.3
Samsung (CHINA) Investment Co., Ltd.	Tianjin Samsung Electronics Co., Ltd.	43.1
Samsung (CHINA) Investment Co., Ltd.	Beijing Samsung Telecom R&D Center	100.0
Samsung (CHINA) Investment Co., Ltd.	Samsung Electronics Suzhou Computer Co., Ltd.	26.3
Samsung (CHINA) Investment Co., Ltd.	Samsung Electronics China R&D Center	100.0
Samsung (CHINA) Investment Co., Ltd.	Samsung Electronics (Beijing) Service Company Limited	100.0
Harman International (China) Holdings Co., Ltd.	Harman Automotive InfoTech (Dalian) Co., Ltd.	100.0
Harman International (China) Holdings Co., Ltd.	Harman (Suzhou) Audio and Infotainment Systems Co., Ltd.	100.0
Harman International (China) Holdings Co., Ltd.	Harman Technology (Shenzhen) Co., Ltd.	100.0
Harman International (China) Holdings Co., Ltd.	Harman (China) Technologies Co., Ltd.	100.0
Harman Holding Limited	Harman Commercial (Shanghai) Co., Ltd.	100.0
Harman Holding Limited	Harman International (China) Holdings Co., Ltd.	100.0
Harman Holding Limited	Harman Automotive Electronic Systems (Suzhou) Co., Ltd.	100.0
Samsung Gulf Electronics Co., Ltd.	Samsung Electronics Egypt S.A.E	0.1
Samsung Electronics Maghreb Arab	Samsung Electronics Tunisia S.A.R.L	1.0
Harman Industries Holdings Mauritius Ltd.	Harman International (India) Private Limited	100.0
Global Symphony Technology Group Private Ltd.	Harman Connected Services Corp. India Pvt. Ltd.	98.4
iOnRoad Technologies Ltd	iOnRoad Ltd	100.0
Red Bend Ltd.	Red Bend Software Japan Co., Ltd.	100.0
Red Bend Ltd.	Red Bend Software Ltd.	100.0
Red Bend Ltd.	Broadsense Ltd.	100.0
Samsung Electronics Latinoamerica (Zona Libre)	Samsung Electronics Latinoamerica Miami, Inc.	100.0
Samsung Electronics Latinoamerica (Zona Libre)	Samsung Electronica Colombia S.A.	100.0
Samsung Electronics Latinoamerica (Zona Libre)	Samsung Electronics Panama. S.A.	100.0
Samsung Electronica da Amazonia Ltda.	Samsung Electronics Argentina S.A.	2.0
Harman do Brasil Industria Electronica e Participacoes Ltda.	Harman da Amazonia Industria Electronica e Participacoes Ltda.	100.0
Samsung Electronics Central Eurasia LLP	Samsung Electronics Caucasus Co. Ltd	0.0
Samsung SDI	Intellectual Keystone Technology LLC	41.0
Samsung SDI	Samsung SDI Japan Co., Ltd.	89.2
Samsung SDI	Samsung SDI America, Inc.	91.7
Samsung SDI	Samsung SDI Hungary Rt.	100.0
Samsung SDI	Samsung SDI Europe GmbH	100.0
Samsung SDI	Samsung SDI Battery Systems GmbH	100.0
Samsung SDI	Samsung SDI(Malaysia) Sdn, Bhd.	68.6
Samsung SDI	Samsung SDI Vietnam Co., Ltd.	100.0
Samsung SDI	Samsung SDI Energy Malaysia Sdn, Bhd.	100.0
Samsung SDI	Samsung SDI(Hong Kong) Ltd.	97.6
Samsung SDI	Samsung SDI China Co., Ltd.	100.0
Samsung SDI	Samsung SDI-ARN (Xi'An) Power Battery Co., Ltd.	50.0
Samsung SDI	Samsung SDI-Sungrow Energy Storage Battery Co., Lt	65.0
Samsung SDI	Samsung SDI (Changchun) Power Battery Co., Ltd.	50.0
Samsung SDI	Samsung SDI (Tianjin) Battery Co., Ltd.	50.0
Samsung SDI	Samsung SDI Brasil Ltda.	45.0
Samsung SDI	Novaled GmbH	50.1
Samsung SDI	SAMSUNG SDI WUXI CO.,LTD.	100.0
Samsung SDI	Samsung Chemical Electronic Materials (SuZhou) Co., Ltd.	100.0
Samsung SDI	Samsung SDI(Wuxi) Battery Systems Co., Ltd.	50.0
Samsung SDI	iMarket Asia Co., Ltd.	8.7
Samsung SDI America, Inc.	Samsung SDI Brasil Ltda.	40.4

Investor	Investee	Ownership
Samsung SDI(Hong Kong) Ltd.	Tianjin Samsung SDI Co., Ltd.	80.0
Samsung SDI(Hong Kong) Ltd.	Samsung SDI Brasil Ltda.	14.6
Samsung Electro-Mechanics	Samsung Electro-Machanics Japan Co., Ltd.	100.0
Samsung Electro-Mechanics	Samsung Electro-Mechanics America, Inc.	100.0
Samsung Electro-Mechanics	SAMSUNG ELECTRO-MECHANICS GMBH	100.0
Samsung Electro-Mechanics	Samsung Electro-Mechanics(Thailand) Co., Ltd.	75.0
Samsung Electro-Mechanics	Samsung Electro-Mechanics Philippines, Corp.	100.0
Samsung Electro-Mechanics	Calamba Premier Realty Corporation	39.8
Samsung Electro-Mechanics	Samsung Electro-Mechanics Pte Ltd.	100.0
Samsung Electro-Mechanics	Samsung Electro-Mechanics Vietnam Co., Ltd.	100.0
Samsung Electro-Mechanics	Dongguan Samsung Electro-Mechanics Co., Ltd.	100.0
Samsung Electro-Mechanics	Tianjin Samsung Electro-Mechanics Co., Ltd.	81.8
Samsung Electro-Mechanics	Samsung High-Tech Electro-Mechanics(Tianjin) Co., Ltd.	95.0
Samsung Electro-Mechanics	Samsung Electro-Mechanics (Shenzhen) Co., Ltd.	100.0
Samsung Electro-Mechanics	Kunshan Samsung Electro-Mechanics Co., Ltd.	100.0
Samsung Electro-Mechanics	iMarket Asia Co., Ltd.	8.7
Samsung Electro-Mechanics America, Inc.	Samsung Electro-Mechanics do Brasil Intermediacoes de Negocios Ltda.	100.0
Calamba Premier Realty Corporation	Batino Realty Corporation	100.0
Samsung Fire & Marine Insurance	SAMSUNG FIRE & MARINE MANAGEMENT CORPORATION	100.0
Samsung Fire & Marine Insurance	SAMSUNG FIRE & MARINE INSURANCE COMPANY OF EUROPE	100.0
Samsung Fire & Marine Insurance	P.T. Asuransi Samsung Tugu	70.0
Samsung Fire & Marine Insurance	Samsung Vina Insurance Co., Ltd	75.0
Samsung Fire & Marine Insurance	SAMSUNG REINSURANCE PTE. LTD	100.0
Samsung Fire & Marine Insurance	Samsung Property & Casualty Insurance Company(China), Ltd	100.0
Samsung Fire & Marine Insurance	Samsung Fire & Marine Insurance Management Middle East	100.0
Samsung Fire & Marine Insurance	SAMSUNG FIRE & MARINE CONSULTORIA EM SEGUROS LTDA.	100.0
Samsung Heavy Industries Co.,Ltd.	Camellia Consulting Corporation	100.0
Samsung Heavy Industries Co.,Ltd.	Samsung Heavy Industries India Pvt.Ltd.	100.0
Samsung Heavy Industries Co.,Ltd.	SAMSUNG HEAVY INDUSTRIES (M) SDN.BHD	100.0
Samsung Heavy Industries Co.,Ltd.	Samsung Heavy Industries(Ningbo) Co., Ltd	100.0
Samsung Heavy Industries Co.,Ltd.	Samsung Heavy Industries(Rongcheng) Co., Ltd	100.0
Samsung Heavy Industries Co.,Ltd.	Rongcheng Gaya Heavy Industries Co., Ltd	100.0
Samsung Heavy Industries Co.,Ltd.	SAMSUNG HEAVY INDUSTRIES NIGERIA LIMITED	100.0
Samsung Heavy Industries Co.,Ltd.	Samsung Heavy Industries Mozambique LDA	100.0
Samsung Heavy Industries Co.,Ltd.	SAMSUNG HEAVY INDUSTRIES BRASIL ASSESSORIA EM PROJETOS EMPRESARIAIS LTDA.	100.0
Samsung Heavy Industries Co.,Ltd.	SHI BRAZIL CONSTRUCTION	100.0
SAMSUNG HEAVY INDUSTRIES NIGERIA LIMITED	SHI - MCI FZE	70.0
Samsung Life Insurance	Porta Nuova Varesine Building 2 S.r.l.	51.0
Samsung Life Insurance	30 GRESHAM STREET(JERSEY) LIMITED	100.0
Samsung Life Insurance	THAI SAMSUNG LIFE INSURANCE CO., LTD.	48.9
Samsung Life Insurance	Beijing Samsung Real Estate Co., Ltd	90.0
Samsung Asset Management	Samsung Asset Management (New York), Inc.	100.0
Samsung Asset Management	Samsung Asset Management(London) Ltd.	100.0
Samsung Asset Management	Samsung Private Equity Manager I Co., Ltd	100.0
Samsung Asset Management	Samsung Asset Management (Hong Kong) Ltd.	100.0
30 GRESHAM STREET(JERSEY) LIMITED	30 GRESHAM STREET (SINGAPORE) LIMITED	100.0
Samsung Asset Management (Hong Kong) Ltd.	Samsung Asset Management (Beijing) Ltd.	100.0
CHEIL INDUSTRIES ITALY SRL	COLOMBO VIA DELLA SPIGA S.R.L	100.0
Samsung Fashion Trading Co., Ltd	Eight Seconds(Shanghai)Co., Ltd.	100.0
Samsung C&T Corporation	MYODO METAL CO., LTD.	100.0
Samsung C&T Corporation	Samsung C&T Japan Corporation	100.0
Samsung C&T Corporation	Samsung C&T America Inc.	100.0
Samsung C&T Corporation	Samsung Finance Corporation.	80.0
Samsung C&T Corporation	Samsung E&C America, INC.	100.0
Samsung C&T Corporation	SAMSUNG OIL & GAS USA CORP	90.0
Samsung C&T Corporation	Samsung Renewable Energy Inc.	100.0
Samsung C&T Corporation	SCNT Power Norte S. De R.L. de C.V.	100.0
Samsung C&T Corporation	QSSC, S.A. de C.V.	60.0
Samsung C&T Corporation	Samsung C&T Oil & Gas Parallel Corp.	100.0
Samsung C&T Corporation	Samsung C&T Canada Ltd.	100.0

Investor	Investee	Ownership
Samsung C&T Corporation	Samsung C&T Deutschland GmbH	100.0
Samsung C&T Corporation	Samsung C&T U.K. Ltd.	100.0
Samsung C&T Corporation	Samsung C&T ECUK Limited	100.0
Samsung C&T Corporation	Whessoe engineering Limited	100.0
Samsung C&T Corporation	SCNT Investment Atlantic SPRL	100.0
Samsung C&T Corporation	POSS-SLPC, s.r.o	50.0
Samsung C&T Corporation	Solluce Romania 1 B.V.	80.0
Samsung C&T Corporation	SAM investment Manzanilo.B.V	53.3
Samsung C&T Corporation	Solluce Slovenia 1 B.V.	80.0
Samsung C&T Corporation	Ecosolar OOD	100.0
Samsung C&T Corporation	Ecoenergy Solar OOD	100.0
Samsung C&T Corporation	Agriplam EOOD	100.0
Samsung C&T Corporation	Fishtrade EOOD	100.0
Samsung C&T Corporation	Manageproject EOOD	100.0
Samsung C&T Corporation	Solar Park EOOD	100.0
Samsung C&T Corporation	Veselinovo Energy OOD	100.0
Samsung C&T Corporation	Samsung C&T Construction Hungary Kft.	100.0
Samsung C&T Corporation	Samsung C&T (KL) Sdn.,Bhd.	100.0
Samsung C&T Corporation	Samsung C&T Malaysia SDN. BHD	100.0
Samsung C&T Corporation	Erdsam Co., Ltd.	100.0
Samsung C&T Corporation	Samsung Chemtech Vina LLC	51.7
Samsung C&T Corporation	S-print Inc	40.0
Samsung C&T Corporation	Cassava Investment Korea Pte. Ltd.	29.7
Samsung C&T Corporation	Samsung Development (Thailand) Co., Ltd.	33.0
Samsung C&T Corporation	Samsung C&T Thailand Co., Ltd	43.9
Samsung C&T Corporation	Cheil Holding Inc.	40.0
Samsung C&T Corporation	Samsung Const. Co. Phils.,Inc.	25.0
Samsung C&T Corporation	Samsung Design Philippines Inc	100.0
Samsung C&T Corporation	PT. INSAM BATUBARA ENERGY	90.0
Samsung C&T Corporation	Samsung C&T India Pte., Ltd.	100.0
Samsung C&T Corporation	Samsung C&T Corporation India Private Limited	100.0
Samsung C&T Corporation	Malaysia Samsung Steel Center Sdn.Bhd	70.0
Samsung C&T Corporation	Samsung C&T Singapore Pte., Ltd.	100.0
Samsung C&T Corporation	S&G Biofuel PTE.LTD	50.5
Samsung C&T Corporation	SAMSUNG C&T Mongolia LLC.	70.0
Samsung C&T Corporation	Samsung C&T Eng.&Const. Mongolia LLC.	100.0
Samsung C&T Corporation	S&WOO CONSTRUCTION PHILIPPINES,INC.	40.0
Samsung C&T Corporation	Samsung C&T Hongkong Ltd.	100.0
Samsung C&T Corporation	Samsung C&T Taiwan Co., Ltd.	100.0
Samsung C&T Corporation	Samsung Precision Stainless Steel(pinghu) Co.,Ltd.	55.0
Samsung C&T Corporation	SAMSUNG C&T (SHANGHAI) CO., LTD.	100.0
Samsung C&T Corporation	Samsung C&T (Xi'an) Co., Ltd.	100.0
Samsung C&T Corporation	Samsung Nigeria Co., Ltd.	99.9
Samsung C&T Corporation	SAMSUNG C&T CORPORATION SAUDI ARABIA	100.0
Samsung C&T Corporation	SAM Gulf Investment Limited	100.0
Samsung C&T Corporation	Samsung C&T Chile Copper SpA	100.0
Samsung C&T Corporation	SCNT Power Kelar Inversiones Limitada	100.0
Samsung C&T Corporation	Samsung C&T Corporation Rus LLC	100.0
Samsung C&T Corporation	JSC BALKHASH THERMAL POWER PLANT	50.0
Samsung C&T Corporation	Samsung SDI America, Inc.	8.3
Samsung C&T Corporation	Samsung SDI(Malaysia) Sdn. Bhd.	6.4
Samsung C&T Corporation	Samsung SDI(Hong Kong) Ltd.	2.4
Samsung C&T Corporation	Beijing Samsung Real Estate Co.. Ltd	10.0
Samsung C&T Corporation	Cheil Industries Corp., USA	100.0
Samsung C&T Corporation	CHEIL INDUSTRIES ITALY SRL	100.0
Samsung C&T Corporation	Samsung Fashion Trading Co. ,Ltd	100.0
Samsung C&T Corporation	CHEIL INDUSTRIES INC. VIETNAM COMPANY LIMITED	100.0
Samsung C&T Corporation	Samsung C&T Corporation UEM Construction JV Sdn Bhd	60.0
Samsung C&T Corporation	iMarket Asia Co., Ltd.	19.3
Samsung Welstory	WELSTORY VIETNAM COMPANY LIMITED	90.0
Samsung Welstory	Shanghai Ever-Hongjun Business Mgt Service Co.,LTD	85.0

Investor	Investee	Ownership
Samsung Welstory	Shanghai Welstory Food Company Limited	70.0
MULTICAMPUS	LANGUAGE TESTING INTERNATIONAL, INC	82.4
Eight Seconds(Shanghai)Co., Ltd.	Eight Seconds (Shanghai) Trading Co., Ltd.	100.0
PengTai Greater China Co., Ltd.	PengTai China Co., Ltd.	100.0
PengTai Greater China Co., Ltd.	PengTai Taiwan Co., Ltd.	100.0
PengTai Greater China Co., Ltd.	PengTai Interactive Advertising Co.,Ltd.	100.0
PengTai China Co., Ltd.	PengTai e-Commerce Co.,Ltd.	100.0
PengTai China Co., Ltd.	PengTai Marketing Service Co., Ltd.	100.0
PengTai Interactive Advertising Co.,Ltd.	MEDIALYTICS Inc.	51.0
PengTai Interactive Advertising Co.,Ltd.	Beijing Pengtai Baozun E-commerce Co., Ltd.	51.0
iMarket Asia Co., Ltd.	iMarket China Co., Ltd.	80.0
Samsung Securities	Samsung Securities (America), Inc.	100.0
Samsung Securities	Samsung Securities (Europe) Limited.	100.0
Samsung Securities	Samsung Securities (Asia) Limited.	100.0
Samsung SDS	iMarket Asia Co., Ltd.	40.6
Samsung SDS	Samsung SDS Global SCL America, Inc.	100.0
Samsung SDS	SAMSUNG SDS GSCL CANADA., LTD.	100.0
Samsung SDS	Samsung SDS America, Inc.	100.0
Samsung SDS	Neo EXpress Transportation (NEXT), Inc.	51.0
Samsung SDS	Samsung SDS Europe Ltd.	100.0
Samsung SDS	Samsung SDS Global SCL Hungary Kft.	100.0
Samsung SDS	Samsung SDS Global SCL Slovakia, s.r.o.	100.0
Samsung SDS	Samsung SDS Global SCL Poland Sp. Z o.o.	100.0
Samsung SDS	Samsung SDS Global SCL UK Ltd.	100.0
Samsung SDS	Samsung GSCL Sweden AB	100.0
Samsung SDS	Samsung SDS Global SCL France SAS	100.0
Samsung SDS	Samsung SDS Global SCL Greece Societe Anonyme	100.0
Samsung SDS	Samsung SDS Global SCL Baltics, SIA	100.0
Samsung SDS	Samsung SDS Global SCL Italy S.R.L. A Socio Unico	100.0
Samsung SDS	Samsung SDS Global Supply Chain Logistics Spain S.L.U	100.0
Samsung SDS	Samsung SDS Global SCL Netherlands Cooperatief U.A	100.0
Samsung SDS	Samsung SDS Global SCL Germany GmbH	100.0
Samsung SDS	Samsung GSCL Portugal, Sociedade Unipessoal Lda	100.0
Samsung SDS	Samsung SDS Global SCL Austria GmbH	100.0
Samsung SDS	Samsung SDS Global SCL Czech s.r.o.	100.0
Samsung SDS	Samsung SDS Global SCL Switzerland GmbH	100.0
Samsung SDS	Samsung SDS GSCL Romania SRL	100.0
Samsung SDS	Samsung SDS Asia Pacific Pte. Ltd.	100.0
Samsung SDS	Samsung Data Systems India Private Limited	100.0
Samsung SDS	SAMSUNG SDS VIETNAM CO., LTD.	100.0
Samsung SDS	Samsung SDS India Private Limited	100.0
Samsung SDS	SAMSUNG SDS GLOBAL SCL ASIA PACIFIC PTE. LTD.	100.0
Samsung SDS	Samsung SDS GSCL Vietnam Co Ltd	100.0
Samsung SDS	PT. Samsung SDS Global SCL Indonesia	100.0
Samsung SDS	Samsung SDS Global SCL Philippines Co., Ltd.	100.0
Samsung SDS	Samsung SDS Global SCL Thailand Co.,Ltd	100.0
Samsung SDS	Samsung SDS Global SCL Malaysia SDN.BHD.	100.0
Samsung SDS	SAMSUNG SDS GLOBAL SCL AUSTRALIA PTY., LTD.	100.0
Samsung SDS	SDS-ACUTECH CO., LTD	50.0
Samsung SDS	ALS SDS Joint Stock Company	51.0
Samsung SDS	Samsung SDS China, Ltd.	100.0
Samsung SDS	Samsung IT Services (Beijing) Co., Ltd.	100.0
Samsung SDS	Samsung SDS Global SCL Hong Kong Ltd	100.0
Samsung SDS	SDS Kerry (Shanghai) Supply Chain Solutions Limited	50.0
Samsung SDS	SAMSUNG SDS Global SCL Egypt Co. Ltd.	100.0
Samsung SDS	Samsung SDS Global SCL South Africa (PTY) Ltd.	100.0
Samsung SDS	Samsung SDS Global SCL Transport and Logistics Joint Stock Company	100.0
Samsung SDS	Samsung SDS Global Supply Chain Logistics Middle East DWC-LLC	100.0
Samsung SDS	Samsung SDS Latin America Solucoes Em Tecnologia Ltda	99.7
Samsung SDS	SAMSUNG SDS GLOBAL SCL LATIN AMERICA LOGISTICA LTD	99.7
Samsung SDS	Samsung SDS Global SCL Rus Limited Liability Company	100.0

Investor	Investee	Ownership
Miracom	MIRACOM INC ASIA PACIFIC LTD	100.0
Samsung SDS Global SCL America, Inc.	Samsung SDS Latin America Solucoes Em Tecnologia Ltda	0.3
Samsung SDS Global SCL America, Inc.	Samsung SDS Mexico, S.A. DE C.V.	99.0
Samsung SDS Global SCL America, Inc.	Samsung SDS Global SCL Panama S. A.	100.0
Samsung SDS Global SCL America, Inc.	Samsung SDS Global SCL Chile Limitada	100.0
Samsung SDS Global SCL America, Inc.	Samsung SDS Global SCL Peru S.A.C.	100.0
Samsung SDS Global SCL America, Inc.	Samsung SDS Global SCL Colombia S.A.S.	100.0
Samsung SDS Global SCL America, Inc.	SAMSUNG SDS GLOBAL SCL LATIN AMERICA LOGISTICA LTD	0.3
Samsung SDS Europe Ltd.	Samsung SDS Global SCL Netherlands Cooperatief U.A	0.0
Samsung SDS Global SCL Netherlands Cooperatief U.A	Samsung SDS Global SCL Poland Sp. Z.o.o.	0.0
Samsung SDS Global SCL Netherlands Cooperatief U.A	Samsung SDS Global SCL Greece Societe Anonyme	0.0
Samsung SDS Global SCL Netherlands Cooperatief U.A	Samsung SDS GSCL Romania SRL	0.0
Samsung SDS Global SCL Netherlands Cooperatief U.A	Samsung SDS Global SCL Rus Limited Liability Company	0.0
Samsung SDS China, Ltd.	Samsung SDS Global SCL Beijing Co., Ltd	100.0
Samsung IT Services (Beijing) Co., Ltd.	Samsung SDS Global Development Center Xi'an	100.0
MIRACOM INC ASIA PACIFIC LTD	MIRACOM INC CHINA LTD	100.0
Samsung Engineering	Offshore 1 consulting Corporation	49.0
Samsung Engineering	Samsung Engineering America Inc.	100.0
Samsung Engineering	Samsung Engineering Hungary Ltd.	100.0
Samsung Engineering	Samsung Engineering (Malaysia) SDN. BHD.	100.0
Samsung Engineering	PT Samsung Engineering Indonesia Co., Ltd.	100.0
Samsung Engineering	Samsung Engineering (Thailand) Co., Ltd.	81.0
Samsung Engineering	Samsung Engineering India Private Ltd.	100.0
Samsung Engineering	Samsung Engineering Vietnam	100.0
Samsung Engineering	Samsung Engineering Construction(Shanghai) Co., Lt	100.0
Samsung Engineering	SAMSUNG ENGINEERING CONSTRUCTION XIAN CO., LTD.	100.0
Samsung Engineering	Samsung Saudi Arabia Company Limited.	100.0
Samsung Engineering	Muharrag Wastewater Services Company W.L.L.	64.8
Samsung Engineering	Muharrag STP Company B.S.C.	4.6
Samsung Engineering	Muharrag Holding Company 1 Ltd.	45.0
Samsung Engineering	Samsung Ingenieria Mexico Construccion Y Operacion S.A. De C.V.	99.9
Samsung Engineering	Samsung Engineering Trinidad Co., Ltd.	100.0
Samsung Engineering	Samsung Ingenieria Manzanillo, S.A. De C.V.	99.9
Samsung Engineering	Grupo Samsung Ingenieria Mexico, S.A. De C.V.	100.0
Samsung Engineering	Samsung Ingenieria Energia S.A. De C.V.	100.0
Samsung Engineering	Samsung Engineering Bolivia S.A	100.0
Samsung Engineering	Samsung Ingenieria DUBA S.A. de C.V.	100.0
Samsung Engineering	Samsung Engineering Kazakhstan LLP	100.0
Samsung Engineering America Inc.	SEA Construction, LLC	100.0
Samsung Engineering America Inc.	SEA Louisiana Construction, L.L.C.	100.0
Samsung Engineering (Malaysia) SDN. BHD.	Muharrag Wastewater Services Company W.L.L.	0.3
Samsung Engineering India Private Ltd.	Samsung Saudi Arabia Company Limited.	0.0
Samsung Saudi Arabia Company Limited.	Samsung EPC Company Ltd.	75.0
Muharrag Holding Company 1 Ltd.	Muharrag Holding Company 2 Ltd.	100.0
Muharrag Holding Company 2 Ltd.	Muharrag STP Company B.S.C.	89.9
S1 Corporation	SOCM LLC	100.0
S1 Corporation	S-1 CORPORATION VIETNAM CO., LTD	100.0
S1 Corporation	Samsung Beijing Security Systems	100.0
Cheil Worldwide	Cheil USA Inc.	100.0
Cheil Worldwide	Cheil Central America Inc.	100.0
Cheil Worldwide	IRIS Worldwide Holdings Limited	85.2
Cheil Worldwide	Cheil Europe Ltd.	100.0
Cheil Worldwide	Cheil Germany GmbH	100.0
Cheil Worldwide	Cheil France SAS	100.0
Cheil Worldwide	Cheil Nordic AB	100.0
Cheil Worldwide	Cheil India Pvt. Ltd.	100.0
Cheil Worldwide	Cheil (Thailand) Ltd.	100.0
Cheil Worldwide	Cheil Singapore Pte. Ltd.	100.0
Cheil Worldwide	Cheil Vietnam Co. Ltd.	99.0
Cheil Worldwide	Cheil Integrated Marketing Philippines, Inc.	100.0
Cheil Worldwide	Cheil Malaysia SDN BHD	100.0

Investor	Investee	Ownership
Cheil Worldwide	Cheil China	100.0
Cheil Worldwide	Cheil Hong Kong Ltd.	100.0
Cheil Worldwide	Bravo Asia Limited	100.0
Cheil Worldwide	Bravo Asia-Shanghai	100.0
Cheil Worldwide	Cheil MEA FZ-LLC	100.0
Cheil Worldwide	Cheil South Africa Pty., Ltd.	100.0
Cheil Worldwide	CHEIL KENYA LIMITED	99.0
Cheil Worldwide	Cheil Communications Nigeria Ltd.	99.0
Cheil Worldwide	Cheil Jordan LLC	100.0
Cheil Worldwide	Cheil Ghana Limited	100.0
Cheil Worldwide	Cheil Brazil Communications Ltda.	100.0
Cheil Worldwide	Cheil Mexico Inc. SA de CV	98.0
Cheil Worldwide	Cheil Chile SpA.	100.0
Cheil Worldwide	Cheil Rus LLC	100.0
Cheil Worldwide	Cheil Ukraine LLC	100.0
Cheil Worldwide	Cheil Kazakhstan LLC	100.0
Hotel Shilla	Samsung Hospitality America Inc.	100.0
Hotel Shilla	Shilla Travel Retail Pte. Ltd.	100.0
Hotel Shilla	SHILLA LIMITED Macao	100.0
Hotel Shilla	Samsung Shilla Business Service Beijing Co., Ltd.	100.0
Hotel Shilla	SHILLA LIMITED Hong Kong	100.0
Hotel Shilla	Shilla Travel Retail Hong Kong Limited	100.0
Hotel Shilla	Shilla Travel Retail Taiwan Limited	64.0
SBTM Corporation	Sasmsung Hospitality U.K. Inc.	100.0
SBTM Corporation	SAMSUNG HOSPITALITY ROMANIA SRL	100.0
SBTM Corporation	Sasmsung Hospitality Vietnam Co., Ltd.	99.0
SBTM Corporation	Samsung Hospitality Philippines Inc.	100.0
SBTM Corporation	Samsung Hospitality India Private Limited	100.0
Iris Americas, Inc.	Iris USA, Inc.	100.0
Iris Americas, Inc.	Iris Atlanta, Inc.	100.0
Iris Americas, Inc.	Iris Experience, Inc.	100.0
Iris Americas, Inc.	Iris Latin America, Inc.	100.0
Iris Americas, Inc.	Iris Worldwide San Diego, Inc.	100.0
Iris Latin America, Inc.	Irisnation Latina No.2, S. de R.L. de C.V.	0.0
Iris Latin America, Inc.	Irisnation Latina, S. de R.L. de C.V.	0.0
Iris Canada Holdings Ltd	Pricing Solutions Ltd	75.1
Cheil USA Inc.	The Barbarian Group LLC	100.0
Cheil USA Inc.	McKinney Ventures LLC	100.0
Cheil USA Inc.	Cheil India Pvt. Ltd.	0.0
Cheil USA Inc.	Cheil Mexico Inc. SA de CV	2.0
Samsung Hospitality America Inc.	Samsung Hospitality Europe GmbH	100.0
IRIS Worldwide Holdings Limited	Iris Nation Worldwide Limited	100.0
IRIS Worldwide Holdings Limited	Josh & James Limited	100.0
Iris Nation Worldwide Limited	Iris Americas, Inc.	100.0
Iris Nation Worldwide Limited	Irisnation Latina No.2, S. de R.L. de C.V.	100.0
Iris Nation Worldwide Limited	Irisnation Latina, S. de R.L. de C.V.	100.0
Iris Nation Worldwide Limited	Iris Canada Holdings Ltd	100.0
Iris Nation Worldwide Limited	Iris London Limited	100.0
Iris Nation Worldwide Limited	Iris Promotional Marketing Ltd	100.0
Iris Nation Worldwide Limited	Iris Ventures 1 Limited	100.0
Iris Nation Worldwide Limited	Founded Partners Limited	76.0
Iris Nation Worldwide Limited	Iris Products (Worldwide) Limited	100.0
Iris Nation Worldwide Limited	Iris Korea Limited	100.0
Iris Nation Worldwide Limited	Iris PR Limited	100.0
Iris Nation Worldwide Limited	Iris Concise Limited	100.0
Iris Nation Worldwide Limited	Iris Digital Limited	100.0
Iris Nation Worldwide Limited	Iris Amsterdam B.V.	100.0
Iris Nation Worldwide Limited	Datalytics Limited	100.0
Iris Nation Worldwide Limited	Iris Ventures (Worldwide) Limited	100.0
Iris Nation Worldwide Limited	Iris Culture Limited	100.0
Iris Nation Worldwide Limited	Concise Consultants Limited	100.0

Investor	Investee	Ownership
Iris Nation Worldwide Limited	Atom42 Limited	64.0
Iris Nation Worldwide Limited	Pricing Solutions (UK) Limited	75.0
Iris Nation Worldwide Limited	Iris Services Limited Dooel Skopje	100.0
Iris Nation Worldwide Limited	Irisnation Singapore Pte Limited	100.0
Iris Nation Worldwide Limited	Iris Worldwide Integrated Marketing Pvt Limited	100.0
Iris Nation Worldwide Limited	Iris Sydney PTY Ltd	100.0
Iris Nation Worldwide Limited	Iris Worldwide (Thailand) Limited	100.0
Iris Nation Worldwide Limited	Iris (Beijing) Advertising Company Limited	100.0
Iris Nation Worldwide Limited	Irisnation Hong Kong Limited	100.0
Iris London Limited	Iris Partners LLP	100.0
Iris Promotional Marketing Ltd	Holdings BR185 Limited	100.0
Iris Ventures 1 Limited	Iris Germany GmbH	100.0
Founded Partners Limited	Founded Partners, Inc.	100.0
Iris Ventures (Worldwide) Limited	THE ELEPHANT ROOM LIMITED	49.0
Iris Germany GmbH	Pepper NA, Inc.	100.0
Iris Germany GmbH	Pepper Technologies Pte Limited	100.0
Cheil Europe Ltd.	BEATTIE MCGUINNESS BUNGAY LIMITED	100.0
Cheil Europe Ltd.	Cheil Italia S.r.l	100.0
Cheil Europe Ltd.	CHEIL SPAIN S.L	100.0
Cheil Europe Ltd.	CHEIL BENELUX B.V.	100.0
Cheil Germany GmbH	Cheil Austria GmbH	100.0
Cheil Singapore Pte. Ltd.	PengTai Greater China Co., Ltd.	95.0
Cheil Singapore Pte. Ltd.	PT. CHEIL WORLDWIDE INDONESIA	100.0
Cheil Integrated Marketing Philippines, Inc.	Cheil Philippines Inc.	30.0
Cheil Hong Kong Ltd.	PengTai Greater China Co., Ltd.	3.1
Samsung Shilla Business Service Beijing Co., Ltd.	Tianjin Samsung International Travel Service Co., Ltd	100.0
Cheil MEA FZ-LLC	One Agency FZ LLC	100.0
Cheil MEA FZ-LLC	One RX Project Management Design and Production Limited Company	0.0
Cheil South Africa Pty., Ltd.	CHEIL KENYA LIMITED	1.0
Cheil South Africa Pty., Ltd.	Cheil Communications Nigeria Ltd.	1.0
One Agency FZ LLC	One RX India PVT. LTD	100.0
One Agency FZ LLC	One RX Project Management Design and Production Limited Company	100.0
One Agency FZ LLC	One RX Interior Design LLC	100.0
One Agency FZ LLC	One RX Printing LLC	100.0
One Agency FZ LLC	One Agency South Africa Pty., Ltd.	100.0
One RX Interior Design LLC	One RX India PVT. LTD	0.0
Holdings BR185 Limited	Brazil 185 Participacoes Ltda	100.0
Brazil 185 Participacoes Ltda	Iris Router Marketing Ltda	100.0
Samsung Electro-Mechanics	SAMSUNG ELECTRO MECHANICS SOFTWARE INDIA BANGALORE PRIVATE LIMITED	99.9

※ (% of Ownership; As of December 31, 2017; Based on Common shares)

C. Related Statute & Regulation

Restrictions on cross-holdings within Samsung Group in accordance with the Monopoly Regulation and Fair Trade Act

(1) Date of announcement by authorities of companies deemed Samsung Group companies for the current year:
May 1, 2017

(2) Summary of Regulation

- Restriction on cross-holdings
- Restriction on debt guarantee between affiliates
- Restriction on affiliate voting rights of financial & insurance companies
- Decision making of board committee and public disclosure of large-scale inter-affiliate transactions
- Public disclosure of important matters of unlisted companies
- Other public disclosures related to Group status

D. Concurrent Position with Other Affiliates

[As of December 31, 2017]

Name	Concurrent Position		
	Name of company	Position	Full/Part Time
OH-HYUN KWON	Samsung Display	CEO	Full Time
DONGSOO JUN	Samsung Medison	CEO	Full Time
Hee Chan Roh	Samsung Corning Advanced Glass	Executive director	Part Time
Jong sung Kim	Samsung Corning Advanced Glass	Auditor	Part Time
Jaikyeong Shinn	STECO	CEO	Full Time
KWANGYOUNG CHUNG	Samsung Electro-mechanics	Executive director	Part Time
Honggyeong Kim	Samsung Display	Non-Executive director	Part Time
JEONGJUN CHOI	Samsung Electronics Logitech	Auditor	Part Time
	Samsung Electronics Sales	Auditor	Part Time
	Samsung Electronics Service	Auditor	Part Time
	Samsung Display	Auditor	Part Time
Taeyang Yoon	SEMES	Non-Executive director	Part Time
Kwang-Chae PARK	Samsung Medison	Executive director	Full Time
Sooncheol Park	Samsung Medison	Auditor	Part Time
	Samsung Venture Investment	Auditor	Part Time
GIHO SEO	SEMES	Non-Executive director	Part Time
Jason Oh	STECO	Auditor	Part Time
Wonjoon Lee	SERI	Auditor	Part Time
SE-YEON JANG	SEMES	Non-Executive director	Part Time
Joe Hur	STECO	Non-Executive director	Part Time

E. Equity Investments in other corporations

(1) The total book value of equity investments in other corporations as of December 31, 2017 is KRW 56,643 billion (for the purpose of business operations, etc.)

[As of December 31, 2017]

(Unit: 1,000 shares, KRW million, %)

Account	Name of Company	Acquisition Date	Objective	Acquisition Cost	Beginning Balance			Increase/Decrease			Ending Balance			Current Financial Stats	
					Qty	Owner ship (%)	Book Value	Acquisition/Disposal		Valuation	Qty	Owner ship (%)	Book Value	Total Asset	Net Income (Loss)
								Qty	Amount						
Listed	Samsung Electro-Mechanics	1977.01	Business	250	17,693	23.69	445,244				17,693	23.69	445,244	7,767,403	161,739
Unlisted	STECO	1995.06	Management	24,000	2,590	70	35,861				2,590	70	35,861	170,098	4,792
Unlisted	SEMES	1992.12	Management	1,000	2,173	91.54	71,906				2,173	91.54	71,906	1,043,351	141,341
Unlisted	SERI	1991.05	Business	320	3,576	29.8	24,942				3,576	29.8	24,942	119,761	226
Listed	Samsung SDS	1992.07	Business	6,160	17,472	22.58	560,827				17,472	22.58	560,827	7,277,761	530,348
Unlisted	Samsung Electronics Service	1998.01	Management	30,000	6,000	99.33	48,121				6,000	99.33	48,121	347,411	-907
Unlisted	Samsung Electronics Sales	2000.12	Management	3,100	1,767	100	247,523				1,767	100	247,523	590,110	2,329
Unlisted	Samsung Electronics Logitech	1999.04	Management	76	1,011	100	46,669				1,011	100	46,669	155,878	14,480
Unlisted	Samsung Display	2012.04	Management	16,009,547	221,969	84.78	18,509,307				221,969	84.78	18,509,307	50,723,199	3,352,802
Unlisted	SVIC #21	2011.11	Management	19,800	1	99	97,570	0	-11,484		1	99	86,086	85,857	-2,564
Unlisted	SVIC #22	2011.11	Management	19,800	1	99	147,282	0	-7,920		1	99	139,362	126,264	-2,146
Unlisted	SVIC #26	2014.11	Management	19,800	2	99	183,744	0	-11,613		2	99	172,131	192,444	25,885

※ Account noted here reflects approved for sales securities.

Account	Name of Company	Acquisition Date	Objective	Acquisition Cost	Beginning Balance			Increase/Decrease			Ending Balance			Current Financial Stats	
					Qty	Ownership (%)	Book Value	Acquisition/Disposal		Valuation	Qty	Ownership (%)	Book Value	Total Asset	Net Income (Loss)
								Qty	Amount						
Unlisted	SVIC #27	2014.09	Management	5,940	0	99	15,642	0	3,366		0	99	19,008	16,871	-1,660
Unlisted	SVIC #28	2015.02	Management	7,425	2	99	159,984	0	3,669		2	99	163,653	186,164	-3,778
Unlisted	SVIC #32	2016.08	Management	19,800	0	99	40,194	1	29,373		1	99	69,567	64,077	-5,534
Unlisted	SVIC #33	2016.11	Management	4,950	0	99	4,950	1	63,617		1	99	68,567	64,873	-4,959
Unlisted	Samsung Medison	2011.02	Management	286,384	87,350	68.45	477,648				87,350	68.45	477,648	329,064	10,199
Unlisted	S-Printing Solution	2016.11	Business	218,370	10,000	100	218,370	-10,000	-218,370		0	0	0	-	-
Listed	Samsung BioLogics	2011.04	Business	30,000	20,837	31.49	443,193				20,837	31.49	443,193	7,183,091	-96,972
Unlisted	Intellectual Discovery	2011.05	Business	5,000	1,784	15.71	5,241				1,784	15.71	5,241	37,234	-1,416
Listed	Samsung Heavy Industries	1977.09	Business	125	65,931	16.91	609,862			-126,588	65,931	16.91	483,274	13,818,113	-340,748
Listed	Hotel Shilla	1979.12	Business	252	2,005	5.11	96,527			73,673	2,005	5.11	170,200	2,249,913	25,286
Listed	Cheil Worldwide	1988.09	Business	185	29,038	25.24	491,599				29,038	25.24	491,599	2,238,293	127,153
Listed	A-Tech Solution	2009.11	Business	26,348	1,592	15.92	16,636			2,786	1,592	15.92	19,422	166,863	-468
Listed	iMarket Korea	2000.12	Business	1,900	647	1.8	6,732			-900	647	1.8	5,832	1,086,928	15,659
Listed	KT Skylife	2001.12	Business	3,344	240	0.5	4,152			-924	240	0.5	3,228	792,893	57,314
Listed	Samsung SDI	1977.01	Business	304	13,463	19.58	1,242,605				13,463	19.58	1,242,605	15,751,477	657,236

※ Account noted here reflects approved for sales securities

Account	Name of Company	Acquisition Date	Objective	Acquisition Cost	Beginning Balance			Increase/Decrease			Ending Balance			Current Financial Stats	
					Qty	Ownership (%)	Book Value	Acquisition/Disposal		Valuation	Qty	Ownership (%)	Book Value	Total Asset	Net Income (Loss)
								Qty	Amount						
Listed	Wonik IPS	2016.04	Business	16,214	1,851	4.48	48,495			13,326	1,851	4.48	61,821	409,618	95,365
Listed	Wonik Holdings	2013.12	Business	15,411	1,759	2.28	11,857			2,023	1,759	2.28	13,880	1,061,203	80,652
Listed	Dongjin Semichem	2017.11	Business	48,277	0	0	0	2,468	48,277	6,263	2,468	4.8	54,540	667,169	44,997
Listed	#N/A	2017.11	Business	55,618	0	0	0	835	55,618	752	835	4.8	56,370	895,950	43,086
Unlilisted	SSLM	2012.04	Business	52,296	220	0.39	0	-220	0		0	0	0	-	-
Unlilisted	Kihyup Technology Banking	1995.01	Business	5,000	1,000	17.24	5,000				1,000	17.24	5,000	98,120	1,107
Unlilisted	The Korea Economic Daily	1987.05	Business	150	72	0.39	365				72	0.39	365	348,436	31,324
Unlilisted	Samsung Venture Investment	1999.11	Business	4,900	980	16.33	7,515			269	980	16.33	7,784	100,809	7,583
Unlilisted	Cyber Bank	2000.12	Business	8,000	1,083	7.46	0				1,083	7.46	0	0	0
Unlilisted	FineChips	2001.12	Business	10	2	3.81	10				2	3.81	10	6,332	1,827
Unlilisted	SK Telink	2010.11	Business	4,357	15	1.13	4,357	-15	-4,357		0	0	0	-	-
Unlilisted	Inkel	2006.11	Business	130	0	0	0				0	0	0	103,217	-6,643
Listed	Yongpyong Resort	2007.05	Business	1,869	400	0.83	3,432			748	400	0.83	4,180	868,156	16,953
Unlilisted	Sambo Computer	2012.09	Withdrawal	0	0	0.01	0				0	0.01	0	30,787	3,590
Unlilisted	Icube #1	2009.12	Business	4,000	0	16.23	4,000		-2,386		0	16.23	1,614	1,837	23

※ Account noted here reflects approved for sales securities

Account	Name of Company	Acquisition Date	Objective	Acquisition Cost	Beginning Balance			Increase/Decrease			Ending Balance			Current Financial Stats	
					Qty	Owner ship (%)	Book Value	Acquisition/Disposal		Valuation	Qty	Owner ship (%)	Book Value	Total Asset	Net Income (Loss)
								Qty	Amount						
Unlisted	KT Wibro	2010.07	Business	60,000	600	24.2	48,500	-600	-48,500		0	0	0	-	-
Unlisted	Shinsung Eng & Construction	2010.07	Withdrawal	1	0	0.01	0				0	0.01	0	186,429	9,096
Unlisted	Woobang	2010.07	Withdrawal	0	1	0	0				1	0	0	451,452	48,233
Unlisted	Heehoon Design & Global	2010.07	Withdrawal	0	2	0.02	0	-2	0		0	0	0	-	-
Unlisted	Daewoo Development	2012.12	Withdrawal	0	0	0.02	0				0	0.02	0	197,601	25,144
Unlisted	Daewoo Songdo Development	2012.12	Withdrawal	0	9	0.02	0				9	0.02	0	19,367	-350
Unlisted	Zyle Motor Sales	2012.12	Withdrawal	0	1	0	0				1	0	0	277,101	19,463
Unlisted	Woojung Construction	2014.04	Withdrawal	0	5	0.35	0	-5	0		0	0	0	-	-
Unlisted	Hyunjin	2014.04	Withdrawal	0	23	0.25	0	-23	0		0	0	0	-	-
Unlisted	Sungwon Construction	2014.04	Withdrawal	0	1	0.03	0				1	0.03	0	27,744	-627
Unlisted	Inhee	2014.04	Withdrawal	0	2	0.17	0				2	0.17	0	9,566	-217
Unlisted	Samsung Solution	2014.04	Withdrawal	0	4	6.7	0				4	6.7	0	9,048	1,398
Unlisted	STX Construction	2014.04	Withdrawal	0	0	0.01	0	0	0		0	0	0	-	-
Unlisted	Poonglim	2014.05	Withdrawal	0	52	0.37	0				52	0.37	0	236,539	-37,906
Unlisted	Semiconductor Growth Fund	2017.03	Business	500	0	0	0	34,500,000	34,500		34,500,000	66.67	34,500	51,906	-470

※ Account noted here reflects approved for sales securities

Account	Name of Company	Acquisition Date	Objective	Acquisition Cost	Beginning Balance			Increase/Decrease			Ending Balance			Current Financial Stats	
					Qty	Ownership (%)	Book Value	Acquisition/Disposal		Valuation	Qty	Ownership (%)	Book Value	Total Asset	Net Income (Loss)
								Qty	Amount						
Unlilisted	JNT	2011.02	Business	1,800	0	24	2,496		-738		0	24	1,758	5,014	-113
Unlilisted	SV	2011.02	Business	1,850	0	14.85	2,292		-2,292		0	14.85	0	16,643	17,678
Unlilisted	Seoul Investment Partners	2011.1	Business	1,550	0	19.38	2,689		-221		0	19.38	2,468	10,666	214
Unlilisted	Daishin Aju IB	2011.08	Business	258	0	3	726				0	3	726	19,468	-4,924
Unlilisted	TS	2011.11	Business	1,700	0	20.48	1,204		-528		0	20.48	676	11,277	559
Unlilisted	IMM	2011.11	Business	760	0	7.6	708		-708		0	0	0	-	-
Unlilisted	L&S	2012.07	Business	848	0	7.46	1,756				0	7.46	1,756	23,481	-78
Unlilisted	Maltani Corp.	2012.04	Business	16,544	45	15	16,270			-1,133	45	15	15,137	67,049	1,212
Unlilisted	Pantech	2013.06	Business	53,000	53,000	10.03	0				53,000	10.03	0	59,660	485
Unlilisted	KTCNP-GC	2013.12	Business	960	0	3.56	5,255		-1,423		0	3.56	3,832	124,960	17,748
Unlilisted	Postech Technology Investment	2013.12	Business	600	0	10	600		-70		0	10	530	5,153	1,133
Unlilisted	AI Research Institute	2016.07	Business	3,000	600	14.29	3,000				600	14.29	3,000	17,754	-2,351
Unlilisted	SECA	1992.08	Position Security	3,823	0	100	90,922				0	100	90,922	762,677	32,908
Unlilisted	SEA	1978.07	Position Security	59,362	492	100	10,063,081		6,990,726		492	100	17,053,807	26,266,636	-70,198
Unlilisted	SELA	1989.04	Position Security	319	40	100	86,962				40	100	86,962	373,528	30,281

※ Account noted here reflects approved for sales securities

Account	Name of Company	Acquisition Date	Objective	Acquisition Cost	Beginning Balance			Increase/Decrease			Ending Balance			Current Financial Stats	
					Qty	Ownership (%)	Book Value	Acquisition/Disposal		Valuation	Qty	Ownership (%)	Book Value	Total Asset	Net Income (Loss)
								Qty	Amount						
Unlisted	SEM	1995.07	Position Security	3,032	3,837	63.58	165,638				3,837	63.58	165,638	1,049,929	42,174
Unlisted	SEASA	1996.06	Position Security	4,696	21,854	98	6,779				21,854	98	6,779	55,595	471
Unlisted	SEDA	1994.01	Position Security	13,224	77,205,709	87.04	647,620				77,205,709	87.04	647,620	5,838,533	1,449,426
Unlisted	SECH	2002.12	Position Security	597	0	4.1	597				0	4.1	597	445,411	14,837
Unlisted	SESA	1989.01	Position Security	3,276	8,021	100	142,091				8,021	100	142,091	734,838	37,303
Unlisted	SENA	1992.03	Position Security	392	1,000	100	69,372				1,000	100	69,372	701,024	32,492
Unlisted	SEH	1991.05	Position Security	1,954	753	100	650,157				753	100	650,157	1,823,278	459,848
Unlisted	SEP	1982.09	Position Security	204	1,751	100	37,616				1,751	100	37,616	159,301	7,254
Unlisted	SEF	1991.08	Position Security	230	2,700	100	234,115				2,700	100	234,115	1,262,849	79,144
Unlisted	SEUK	1995.07	Position Security	33,908	109,546	100	433,202				109,546	100	433,202	1,615,723	99,024
Unlisted	SEHG	1982.02	Position Security	28,042	0	100	354,846				0	100	354,846	711,919	130,726
Unlisted	SEAG	2002.01	Position Security	40	0	100	32,162				0	100	32,162	313,654	13,216
Unlisted	SEI	1993.05	Position Security	862	677	100	143,181				677	100	143,181	822,200	88,682
Unlisted	SEBN	1995.07	Position Security	236	539,138	100	914,751				539,138	100	914,751	1,484,026	64,009
Unlisted	SELS	1991.05	Position Security	18,314	1,306	100	24,288				1,306	100	24,288	1,927,321	6,718

※ Account noted here reflects approved for sales securities

Account	Name of Company	Acquisition Date	Objective	Acquisition Cost	Beginning Balance			Increase/Decrease			Ending Balance			Current Financial Stats	
					Qty	Ownership (%)	Book Value	Acquisition/Disposal		Valuation	Qty	Ownership (%)	Book Value	Total Asset	Net Income (Loss)
								Qty	Amount						
Unlilsted	SEPOL	1996.04	Position Security	5,462	106	100	78,267				106	100	78,267	528,044	43,468
Unlilsted	SSA	1998.12	Position Security	263	2,000	100	32,622				2,000	100	32,622	496,007	11,388
Unlilsted	SESK	2002.06	Position Security	8,976	0	55.68	263,767				0	55.68	263,767	2,019,572	100,617
Unlilsted	SEEH	2008.01	Position Security	4,214	0	100	1,369,992				0	100	1,369,992	9,474,960	18,294
Unlilsted	SEO	1997.01	Position Security	120	0	100	-10,043				0	100	-10,043	105,267	732
Unlilsted	SERC	2006.01	Position Security	24,877	0	100	188,290				0	100	188,290	1,189,176	87,648
Unlilsted	SERK	2007.07	Position Security	4,600	0	100	204,555				0	100	204,555	960,288	123,604
Unlilsted	SEAU	1987.11	Position Security	392	53,200	100	111,964				53,200	100	111,964	382,133	-45,978
Unlilsted	SEMA	1989.09	Position Security	4,378	16,247	100	153,602		-50,200		16,247	100	103,402	155,377	15,028
Unlilsted	SGE	1995.05	Position Security	827	0	100	32,836				0	100	32,836	743,222	71,314
Unlilsted	SEEG	2012.07	Position Security	23	0	0.05	39				0	0.05	39	492,092	4,346
Unlilsted	SEIN	1991.08	Position Security	7,463	46	99.99	118,909				46	99.99	118,909	1,562,037	77,149
Unlilsted	SDMA	1995.03	Position Security	21,876	71,400	75	244,382		-225,641		71,400	75	18,741	24,794	-458
Unlilsted	SIEL	1995.08	Position Security	5,414	216,787	100	75,263				216,787	100	75,263	6,200,351	654,408
Unlilsted	SRI-B	2005.05	Position Security	7,358	17	100	31,787				17	100	31,787	222,450	34,412

※ Account noted here reflects approved for sales securities

Account	Name of Company	Acquisition Date	Objective	Acquisition Cost	Beginning Balance			Increase/Decrease			Ending Balance			Current Financial Stats	
					Qty	Ownership (%)	Book Value	Acquisition/Disposal		Valuation	Qty	Ownership (%)	Book Value	Total Asset	Net Income (Loss)
								Qty	Amount						
Unlilsted	SAVINA	1995.01	Position Security	5,839	0	100	28,365				0	100	28,365	218,680	29,641
Unlilsted	TSE	1988.01	Position Security	1,390	11,020	91.83	279,163				11,020	91.83	279,163	2,191,519	164,627
Unlilsted	STE	1996.01	Position Security	4,206	2,499	49	0				2,499	49	0	6,426	0
Unlilsted	SME	2003.05	Position Security	4,796	17,100	100	7,644				17,100	100	7,644	337,274	23,742
Unlilsted	SAPL	2006.07	Position Security	793	877,133	100	981,483				877,133	100	981,483	6,138,652	1,293,924
Unlilsted	SEHK	1988.09	Position Security	349	274,250	100	79,033				274,250	100	79,033	1,157,142	26,691
Unlilsted	SET	1994.11	Position Security	456	27,270	100	112,949				27,270	100	112,949	1,447,907	35,193
Unlilsted	SESS	1994.12	Position Security	18,875	0	100	504,313				0	100	504,313	950,691	77,856
Unlilsted	SCIC	1996.03	Position Security	23,253	0	100	640,452				0	100	640,452	13,409,281	268,481
Unlilsted	SEHZ	1992.12	Position Security	792	0	89.56	255,535				0	89.56	255,535	6,440,564	689,594
Unlilsted	SSEC	1995.04	Position Security	32,128	0	69.06	130,551				0	69.06	130,551	558,610	-6,924
Unlilsted	TSEC	1993.04	Position Security	15,064	0	48.2	138,101				0	48.2	138,101	672,687	99,195
Unlilsted	SSDP	1993.08	Position Security	4,446	0	87.1	65,319		-65,319		0	0	0	-	-
Unlilsted	TSTC	2001.03	Position Security	10,813	0	90	490,041				0	90	490,041	1,019,144	186,454
Unlilsted	SSET	2002.02	Position Security	6,009	0	95	121,624				0	95	121,624	129,943	14,974

※ Account noted here reflects approved for sales securities

Account	Name of Company	Acquisition Date	Objective	Acquisition Cost	Beginning Balance			Increase/Decrease			Ending Balance			Current Financial Stats	
					Qty	Ownership (%)	Book Value	Acquisition/Disposal		Valuation	Qty	Ownership (%)	Book Value	Total Asset	Net Income (Loss)
								Qty	Amount						
Unlilsted	SESC	2002.09	Position Security	5,471	0	73.7	34,028				0	73.7	34,028	910,833	43,592
Unlilsted	SSS	2001.01	Position Security	1,200	0	100	19,189				0	100	19,189	5,169,438	236,410
Unlilsted	SSCR	2006.09	Position Security	3,405	0	100	9,332				0	100	9,332	27,417	2,689
Unlilsted	TSOE	2010.04	Position Security	33,837	0	82	73,893		-73,893		0	0	0	-	-
Unlilsted	TSLED	2012.04	Position Security	119,519	0	100	119,519				0	100	119,519	416,578	50,736
Unlilsted	SCS	2012.09	Position Security	111,770	0	100	3,888,196		351,484		0	100	4,239,680	8,076,107	1,359,007
Unlilsted	SSCX	2016.04	Position Security	1,141	0	100	1,141				0	100	1,141	796,942	40,268
Unlilsted	SJC	1975.12	Position Security	273	1,560	100	370,647		-117,539		1,560	100	253,108	934,227	2,415
Unlilsted	SRJ	1992.08	Position Security	3,120	122	100	117,257				122	100	117,257	138,096	7,767
Unlilsted	TSST Japan	2004.03	Business	1,639	30	49.03	0				30	49.03	0	304	0
Unlilsted	SDIB	1996.09	Business	3,110	586	0.08	0	-586	0		0	0	0	-	-
Unlilsted	Semiconductor Portal	2002.12	Business	38	0	1.21	10				0	1.21	10	1,474	52
Unlilsted	Nanosys, Inc	2010.08	Business	4,774	1,747	1.43	2,387				1,747	1.35	2,387	25,063	-2,772
Unlilsted	ONE BLUE LLC	2011.07	Business	1,766	0	16.67	1,766				0	16.67	1,766	30,714	540
Unlilsted	TidalScale	2013.08	Business	1,112	2,882	8.87	1,112				2,882	8.28	1,112	1,276	-6,584

※ Account noted here reflects approved for sales securities

Account	Name of Company	Acquisition Date	Objective	Acquisition Cost	Beginning Balance			Increase/Decrease			Ending Balance			Current Financial Stats	
					Qty	Ownership (%)	Book Value	Acquisition/Disposal		Valuation	Qty	Ownership (%)	Book Value	Total Asset	Net Income (Loss)
								Qty	Amount						
Unlisted	Sentiance	2012.12	Business	3,422	7	10.85	3,422				7	7.53	3,422	14,318	-6,277
Unlisted	Mantis Vision	2014.01	Business	1,594	355	2.93	1,980				355	2.87	1,980	5,355	-5,607
Unlisted	Argus Cyber Security	2015.09	Business	356	10	0.39	356	-10	-356		0	0	0	-	-
Unlisted	INEDA	2014.04	Business	3,181	6,100	7.42	5,397				6,100	2.88	5,397	12,631	-12,045
Unlisted	Leman	2014.08	Business	1,019	17	3.66	1,019				17	3.38	1,019	2,550	-3,518
Unlisted	Alces	2014.09	Business	4,832	421	18.5	4,832		-4,832		421	18.5	0	928	-2,347
Unlisted	Keyssa	2016.01	Business	3,332	1,235	2.58	3,332				1,235	2.12	3,332	18,899	-18,317
Unlisted	Zyomed	2016.01	Business	2,044	1,464	2.88	2,044				1,464	2.88	2,044	14,891	-562
Unlisted	SensiFree	2016.01	Business	2,111	490	17.88	2,111				490	17.88	2,111	2,000	-3,063
Unlisted	Bot Home Automation	2016.03	Business	2,307	540	0.89	2,307				540	0.89	2,307	439,274	-154,043
Unlisted	Unispectral	2016.02	Business	1,112	115	8.83	1,112	2,193	1,018		2,308	13.07	2,130	2,097	-4,084
Unlisted	Quobyte	2016.04	Business	2,865	729	11.83	2,865				729	11.83	2,865	1,748	-2,520
Unlisted	Afero	2016.05	Business	5,685	723	5.6	5,685				723	5.72	5,685	5,548	-13,835
Unlisted	Graphcore	2016.06	Business	3,494	3,000	7.73	3,494				3,000	4.45	3,494	99,882	-14,592
Unlisted	Soundhound	2016.12	Business	7,059	306	1.24	7,059				306	1.17	7,059	65,821	-22,412

※ Account noted here reflects approved for sales securities

Account	Name of Company	Acquisition Date	Objective	Acquisition Cost	Beginning Balance			Increase/Decrease			Ending Balance			Current Financial Stats	
					Qty	Ownership (%)	Book Value	Acquisition/Disposal		Valuation	Qty	Ownership (%)	Book Value	Total Asset	Net Income (Loss)
								Qty	Amount						
Unlisted	Aimotive	2017.12	Business	3,302	0	0	0	2	3,302		2	3.18	3,302	11,124	-2,700

※ Account noted here reflects approved for sales securities

※ Based on separate financial statements

※ Ownership percentage is calculated based on the number of common shares issued

※ Financial stats of Sungwon Construction and Mantis Vision are as of December 31, 2016 and September 30, 2017 due to difficulties of obtaining the most recent financial data.

※ Impairment loss of KRW 265,584 million was recognized for SEMA, SDMA, SJC, Alces using net asset valuation method.

X. Related Party Transactions

1. Credit Offerings for Affiliates and Subsidiaries

A. Details on Debt Guarantee (as of December 31, 2017)

- Domestic: N/A

- Overseas:

(Unit: USD thousand)

Company	Relationship	Creditor	Guarantee Expiry Date	Transactions				Limit of Guarantee
				Beginning of period	Increase	Decrease	End of period	
SEA	Subsidiary	SMBC, etc.	2018-12-16	1,000,000	0	1,000,000	0	1,423,000
SEM	Subsidiary	Santander, etc.	2018-12-16	0	0	0	0	546,000
SAMCOL	Subsidiary	Citibank, etc.	2018-12-16	83,585	0	16,646	66,939	168,000
SEDA	Subsidiary	HSBC, etc.	2018-12-17	0	0	0	0	769,000
SECH	Subsidiary	Citibank, etc.	2018-12-16	0	14,639	0	14,639	178,000
SEPR	Subsidiary	BBVA, etc.	2018-12-16	68,739	0	28,589	40,150	180,000
SSA	Subsidiary	Citibank, etc.	2018-12-16	34,327	0	34,327	0	323,000
SEMAG	Subsidiary	SocGen, etc.	2018-12-16	0	0	0	0	110,000
SETK	Subsidiary	BTMU, etc.	2018-12-16	187,867	54,478	0	242,345	755,000
SECE	Subsidiary	Citibank, etc.	2018-12-16	0	0	0	0	94,768
SEEG	Subsidiary	HSBC	2018-06-13	0	0	0	0	50,000
SEIN	Subsidiary	BNP, etc.	2018-11-08	0	0	0	0	186,000
SJC	Subsidiary	Mizuho Bank, etc.	2018-12-16	0	0	0	0	885,782
SEUC	Subsidiary	Credit Agricole, etc.	2018-12-16	0	0	0	0	150,000
SEDAM	Subsidiary	Citibank, etc.	2018-12-16	101,326	0	101,326	0	371,000
SECA	Subsidiary	Nova Scotia	2017-10-10	0	0	0	0	0
SELA	Subsidiary	Citibank	2018-12-16	0	0	0	0	50,000
SEEH	Subsidiary	HSBC, etc.	2018-12-16	0	0	0	0	647,000
SERK	Subsidiary	BNP, etc.	2018-12-16	0	0	0	0	245,000
SELV	Subsidiary	Citibank	2018-12-16	0	0	0	0	10,000
SAPL	Subsidiary	BOA, etc.	2018-12-16	0	0	0	0	411,000
SEV	Subsidiary	SCB	2018-11-08	0	0	0	0	15,000
SAVINA	Subsidiary	SCB, etc.	2018-11-08	0	0	0	0	71,000
SET	Subsidiary	SCB	2018-11-08	0	0	0	0	30,000
SCIC	Subsidiary	HSBC, etc.	2018-12-16	0	0	0	0	350,000
SME	Subsidiary	SCB	2018-11-08	0	0	0	0	110,000
SAMEX	Subsidiary	Citibank	2018-12-16	0	0	0	0	5,000
SEASA	Subsidiary	Citibank	2018-12-16	0	0	0	0	1,000
SSAP	Subsidiary	SCB	2018-11-08	10,282	0	10,282	0	30,000
Simpress	Subsidiary	BNP	2017-11-08	44,529	0	44,529	0	0
SEHK	Subsidiary	HSBC	2018-06-13	0	0	0	0	2,000
SEPM	Subsidiary	HSBC	2018-06-13	86,870	0	9,451	77,419	125,000
Adgear	Subsidiary	BOA	2018-11-08	0	0	0	0	2,000
Harman Finance International, S.C.A	Subsidiary	JP Morgan, etc.	2022-05-27	367,115	50,785	0	417,900	417,900
Total				1,984,640	119,902	1,245,150	859,392	8,711,450

- ※ SEC requires BOD approval for individual guarantees exceeding 2.5% of total equity. When the guarantee amount is between 0.1% and/or less than 2.5%, the approval decision is delegated to the Management Committee.
- ※ Simpress was sold in November, 2017 and has been excluded from affiliates.

B. Purchase and Sales of Securities

N/A

2. Acquisitions and Disposals of Assets with Affiliates and Subsidiaries

In 2017, the Company sold its assets to affiliates including SAS (Samsung Austin Semiconductor LLC.) to increase the production capacity of such affiliates. Also, the Company acquired assets including facility equipment from affiliates to improve efficiency of domestic production.

(Unit: KRW million)

Company	Relationship	Transaction Type	Transaction Date	Date based on	Asset	Amount	Gain or Loss on Disposal
SAS	Affiliates	Asset sale/purchase	2017.09.25	Date of sale/purchase	Machinery, etc.	71,874	58,136
SCS	Affiliates	Asset sale/purchase	2017.11.08	Date of sale/purchase	Machinery, etc.	57,809	5,121
SESS	Affiliates	Asset sale/purchase	2017.11.23	Date of sale/purchase	Machinery, etc.	11,856	2,531
SEV	Affiliates	Asset sale/purchase	2017.10.20	Date of sale/purchase	Machinery, etc.	4,933	1,491
Kngine	Affiliates	Asset purchase	2017.08.24	Date of purchase	Other intangible assets	4,821	-
SSEC	Affiliates	Asset purchase	2017.03.16	Date of purchase	Machinery, etc.	2,527	-
SSEC(E)	Affiliates	Asset purchase	2017.09.12	Date of purchase	Machinery, etc.	2,027	-
TSE	Affiliates	Asset sale	2017.11.22	Date of sale	Machinery, etc.	928	34
SEVT	Affiliates	Asset sale/purchase	2017.11.08	Date of sale/purchase	Machinery, etc.	821	-37
SEHC	Affiliates	Asset sale	2017.12.15	Date of sale	Machinery, etc.	562	85
SESK	Affiliates	Asset sale/purchase	2017.08.21	Date of sale/purchase	Machinery, etc.	504	13
SEUK	Affiliates	Asset purchase	2017.02.28	Date of purchase	Machinery, etc.	433	-
TSLED	Affiliates	Asset purchase	2017.11.08	Date of purchase	Machinery, etc.	382	-
SEH	Affiliates	Asset sale	2017.12.04	Date of sale	Machinery, etc.	368	49
SEHZ	Affiliates	Asset sale/purchase	2017.09.12	Date of sale/purchase	Machinery, etc.	280	2
SEMES	Affiliates	Asset sale/purchase	2017.12.22	Date of sale/purchase	Machinery, etc.	235	95
TSTC	Affiliates	Asset purchase	2017.09.08	Date of purchase	Machinery, etc.	215	-
SEDA	Affiliates	Asset sale	2017.03.14	Date of sale	Machinery, etc.	203	125
S-Printing	Affiliates	Asset sale	2017.10.27	Date of sale	Machinery, etc.	139	139
SEIN	Affiliates	Asset sale/purchase	2017.08.01	Date of sale/purchase	Machinery, etc.	132	44
SII	Affiliates	Asset sale	2017.08.30	Date of sale	Machinery, etc.	128	19
Samsung Display	Affiliates	Asset sale/purchase	2017.12.01	Date of sale/purchase	Machinery, etc.	120	-31

※ Transaction amount was properly calculated by an appropriate method based upon market value assessment.

※ Acquisitions and disposals of assets with affiliates and subsidiaries listed above are not subject to BOD decision.

※ Transaction date: most recent transaction

※ S-Printing Solution was sold in November, 2017 and was excluded from affiliates in February, 2018.

3. Business Transactions with Affiliates and Subsidiaries

(Unit: KRW million)

Company	Relationship	Type of Transaction	Date	Details	Amount
SEA	Subsidiary	Sales/Purchases	2017.01 - 2017.12	Purchase/sales of HHP and Home Appliance, etc.	27,420,881
SSI	Subsidiary	Sales/Purchases	2017.01 - 2017.12	Purchase/sales of semiconductors, etc	25,191,705
SEVT	Subsidiary	Sales/Purchases	2017.01 - 2017.12	Purchase/sales of HHP, etc	18,775,338
SSS	Subsidiary	Sales/Purchases	2017.01 - 2017.12	Sales of semiconductors, etc	17,175,980
SEV	Subsidiary	Sales/Purchases	2017.01 - 2017.12	Purchase/sales of HHP and Home Appliance, etc.	13,022,856
SEHZ	Subsidiary	Sales/Purchases	2017.01 - 2017.12	Purchase/sales of HHP and TV, etc	12,465,959

※ In 2017, SEC conducted purchase and sales transactions with SEA (Samsung Electronics America, Inc.), etc.

4. Transactions with Stakeholders other than Affiliates and Subsidiaries

As of December 31, 2017, the Company's outstanding loan amount is KRW 128.5 billion related to support for supplier companies for product competitiveness and mutual development as well as home loan and educational expenses for employee welfare. In addition, the Company provides debt guarantees up to KRW 7.3 billion for employees' loans from financial institutions relating to housing rental.

(Unit: KRW million)

Company	Relationship	Loans				
		Account title	Transaction			
			Beginning of period (January 1, 2017)	Increase	Decrease	End of period (December 31, 2017)
SEST Co., Ltd., etc	Partner company	Short-term loans	31,271	3,135	2,194	32,212
Bumjin I&D, etc	Partner company and employee	Long-term loans	84,382	139,563	127,622	96,323
Total			115,653	142,698	129,816	128,535

※ The amount shown above reflects the discounted present value of debts before accounting for impaired debts.

XI. Other Information

1. Public Disclosure

A. Progress on major disclosed information

Report date	Title	Report	Progress
Sep 12, 2016	Sale of business operations of the Printing Solutions business	The sale of the Business is expected to be completed within 1 year, upon completion of the conditions set out in the Agreement (Schedule of the completion is subject to change). ※ Following the completion of the transaction, some details of the disclosed information have been updated, including schedules. The sale of the Business was completed on November 1, 2017, with the completion of the conditions set out in the Agreement which include the approvals of the combination of enterprises from countries concerned with this transaction (South Korea, U.S., EU, China, etc.).	Transaction completed (Nov 1, 2017)
Jan 31, 2018	Decision of Stock Split	Stock Split Schedule -Period for tendering existing shares: April 26, 2018 -Period of trading suspension: April 25-One day before the listing of new share certificates -Period for closing shareholders' registry April 27-May 14 -Listing date of new share certificates: May 16 ※ There have been modifications to the schedule. -Period for tendering existing shares: May 2, 2018 -Period of trading suspension: April 30, May 2, May 3 (3 business days. May 1 is a non-trading day) -Period for closing shareholders' registry: May 3-May 10 -Listing date of new share certificates: May 4	The schedule has been modified (March 16, 2018)

2. Summary of Results of Shareholder Meetings

[As of December 31, 2017]

Date	Agenda	Results
The 48th annual general meeting of shareholders ('17.3.24)	1. Approval of Balance Sheet, Income Statement, and Statement of Appropriation of Retained Earnings (Draft), etc. for FY2016	Approved
	2. Approval of the limit on the remuneration for Directors	Approved
The 48th extraordinary general meeting of shareholders ('16.10.27)	1. Approval of Printing solutions business Spin-off	Approved
	2. Appointment of Executive director - Nominee : Jae-Yong Lee	Approved
The 47th annual general meeting of shareholders ('16.3.11)	1. Approval of Balance Sheet, Income Statement, and Statement of Appropriation of Retained Earnings (Draft), etc. for FY2015	Approved
	2. Election of Directors	
	2.1: Appointment of Independent Directors	
	2.1.1: In-Ho Lee	Approved
	2.1.2: Kwang-Soo Song	Approved
	2.1.3: Jae-Wan Bahk	Approved
	2.2: Appointment of Executive Director	
	2.2.1: Boo-Keun Yoon	Approved
	2.2.2: Jong-Kyun Shin	Approved
	2.2.3: Sang-Hoon Lee	Approved
	2.3: Appointment of Members of Audit Committee	
	2.3.1: In-Ho Lee	Approved
	2.3.2: Kwang-Soo Song	Approved

	3. Approval of the Remuneration Limit for the Directors	Approved
	4. Approval of amendments to the Articles of Incorporation	Approved
The 46th annual general meeting of shareholders ('15.3.13)	1. Approval of Balance Sheet, Income Statement, and Statement of Appropriation of Retained Earnings (Draft), etc. for FY2014 2. Appointment of Directors 2.1: Appointment of Independent Directors 2.1.1: Han-Joong Kim 2.1.2: Byeong-Gi Lee 2.2: Appointment of Executive Director (Oh-Hyun Kwon) 2.3: Appointment of Member of Audit Committee (Han-Joong Kim) 3. Approval of the limit on the remuneration for Directors	Approved Approved Approved Approved Approved Approved

3. *Litigation*

(1) The litigation with Apple Inc. (“Apple”) is ongoing in the United States as at the reporting date. On August 24, 2012, the jury at the District Court for the Northern District of California determined that the Company partially infringed Apple’s design and utility patent and should pay damages to Apple. However, on March 1, 2013, the Judge ordered a new trial for a certain portion of the damages, ruling that it was originally miscalculated. On November 21, 2013 a jury verdict was rendered on the recalculated damages amount, and on March 6, 2014, the Judge made a final judgement to confirm the total damages and deny Apple’s bid for a permanent injunction against the Company. The Company appealed the decision on the damages amount on March 7, 2014, and a hearing on the appeal was held on December 4, 2014. On May 18, 2015, the appeals court affirmed in part and reversed in part a previous decision, and remanded it. On June 17, 2015, the Company petitioned for an en banc rehearing regarding the design infringement, and on August 13, 2015, the federal court dismissed the Company’s request. After the remand procedure, the Court of First Trial announced a partial final judgment on the appeals on September 18, 2015. On October 13, 2015, the immediate appeal was dismissed and on November 19, 2015, the Federal Circuit Court denied an en banc rehearing request. On December 11, 2015, the Company made payment for the damages. On December 14, 2015, the Company filed an appeal to the Supreme Court regarding the design patent infringement ruling. Thereafter, the two parties have submitted in writing to the District court details of supplemental damages incurred in connection with the ruling. On March 21, 2016, the Supreme Court granted the design-related appeals filed by the Company and on March 22, 2016, the Court of First Trial ordered all proceedings for review of damages scheduled to commence March 28, 2016 suspended until the sentence rendered by the Supreme Court was confirmed. On June 1, 2016, the Company submitted the draft document in the design-related appeal, and on June 8, 2016, several companies and organizations presented the document in support of the Company. Apple filed a dissenting document on July 29, 2016, and on August 5, 2016, several companies and organizations presented an advocative document. On August 29, 2016, the Company submitted a rebuttal letter. Oral statements for the appeal were held at the Supreme Court on October 11, 2016. On December 6, 2016, the Supreme Court issued a ruling citing the Company appeal and returned the case to the Court of Appeals. On February 7, 2017, the Court of Appeals reversed the case to the Court of First Trial. On October 12, 2017, the Court of First Instance held a court hearing on the necessity of retrial and decided to proceed with a design retrial on October 22, 2017.

Additionally, on May 5, 2014, the jury at the District Court for the Northern District of California in another ongoing patent lawsuit determined that the Company partially infringed Apple’s utility patent and should pay damages to Apple. On November 25, 2014, the first trial judgment was pronounced to confirm the jury’s verdict. The Company appealed on November 25, 2014 and the rehearing was held on January 5, 2016. On August 27, 2014, the Judge denied Apple’s request for a permanent injunction on the Company’s product. However, on September 17, 2015, the appellate court reversed and remanded a previous decision and on December 16, 2015, the Federal Circuit Court denied an en banc hearing request of the Company. On January 18, 2016, the Court of First instance ordered a permanent injunction on the Company’s product. On February 26, 2016, a panel of appellate court reversed the first trial judgment of November 25, 2014 and ruled that the Company had not infringed Apple’s some patents and its remaining patents were invalid. On March 30, 2016, Apple filed a petition for an en banc rehearing about the rulings of the panel. On October 7, 2016, the en banc appellate court reversed the panel rulings, affirming the first trial judgment that the Company had infringed Apple’s patent. On March 10, 2017, the Company appealed to the Supreme Court against the rulings of the en banc appellate Court. On October 4, 2017, the US Department of Justice filed with the Supreme Court a legal advice letter against the company’s appeal. On November 6, 2017, the Supreme Court dismissed the appeal. On January 11, 2018, the Court of First Trial held a court hearing on whether to recognize additional damages.

The final conclusion and the effect of the patent lawsuits with Apple are uncertain as at the reporting date.

(2) The Company is involved in claims, disputes, and investigations conducted by regulatory bodies at the reporting date, including civil claims from some overseas buyers for price-fixing related to the sale of TFT-LCD. Although the outflow of resources and timing of these matters are uncertain, the Company believes the outcome will not have a material impact on the financial condition of the Company.

(3) In addition, during the normal course of business with numerous companies, the Company has been involved in various claims, disputes, and investigations conducted by regulatory bodies. Although the outflow of resources and timing of these matters are uncertain, the Company believes the outcome will not have a material impact on the financial condition of the Company.

4. Guarantees

- Domestic: As of December 31, 2017, the Company provided a debt guarantee of KRW 32,664 million to its employees who took debt from financial institutions in order to finance employee housing rental. The Company's housing rental debt guarantee limit is KRW 49,937 million.

- Overseas:

(Unit: USD thousand)

Company	Relationship	Creditor	Guarantee Expiry Date	Transactions				Limit of Guarantee
				Beginning of period	Increase	Decrease	End of period	
SEA	Subsidiary	SMBC etc.	2018-12-16	1,000,000	0	1,000,000	0	1,423,000
SEM	Subsidiary	Santander etc.	2018-12-16	0	0	0	0	546,000
SAMCOL	Subsidiary	Citibank etc.	2018-12-16	83,585	0	16,646	66,939	168,000
SEDA	Subsidiary	HSBC etc.	2018-12-17	0	0	0	0	769,000
SECH	Subsidiary	Citibank etc.	2018-12-16	0	14,639	0	14,639	178,000
SEPR	Subsidiary	BBVA etc.	2018-12-16	68,739	0	28,589	40,150	180,000
SSA	Subsidiary	Citibank etc.	2018-12-16	34,327	0	34,327	0	323,000
SEMAG	Subsidiary	SocGen etc.	2018-12-16	0	0	0	0	110,000
SETK	Subsidiary	BTMU etc.	2018-12-16	187,867	54,478	0	242,345	755,000
SECE	Subsidiary	Citibank etc.	2018-12-16	0	0	0	0	94,768
SEEG	Subsidiary	HSBC	2018-06-13	0	0	0	0	50,000
SEIN	Subsidiary	BNP etc.	2018-11-08	0	0	0	0	186,000
SJC	Subsidiary	Mizuho Bank etc.	2018-12-16	0	0	0	0	885,782
SEUC	Subsidiary	Credit Agricole etc.	2018-12-16	0	0	0	0	150,000
SEDAM	Subsidiary	Citibank etc.	2018-12-16	101,326	0	101,326	0	371,000
SECA	Subsidiary	Nova Scotia	2017-10-10	0	0	0	0	0
SELA	Subsidiary	Citibank	2018-12-16	0	0	0	0	50,000
SEEH	Subsidiary	HSBC etc.	2018-12-16	0	0	0	0	647,000
SERK	Subsidiary	BNP etc.	2018-12-16	0	0	0	0	245,000
SELV	Subsidiary	Citibank	2018-12-16	0	0	0	0	10,000
SAPL	Subsidiary	BOA etc.	2018-12-16	0	0	0	0	411,000
SEV	Subsidiary	SCB	2018-11-08	0	0	0	0	15,000
SAVINA	Subsidiary	SCB etc.	2018-11-08	0	0	0	0	71,000
SET	Subsidiary	SCB	2018-11-08	0	0	0	0	30,000
SCIC	Subsidiary	HSBC etc.	2018-12-16	0	0	0	0	350,000
SME	Subsidiary	SCB	2018-11-08	0	0	0	0	110,000

SAMEX	Subsidiary	Citibank	2018-12-16	0	0	0	0	5,000
SEASA	Subsidiary	Citibank	2018-12-16	0	0	0	0	1,000
SSAP	Subsidiary	SCB	2018-11-08	10,282	0	10,282	0	30,000
Simpres	Subsidiary	BNP	2017-11-08	44,529	0	44,529	0	0
SEHK	Subsidiary	HSBC	2018-06-13	0	0	0	0	2,000
SEPM	Subsidiary	HSBC	2018-06-13	86,870	0	9,451	77,419	125,000
Adgear	Subsidiary	BOA	2018-11-08	0	0	0	0	2,000
Harman Finance International SCA	Subsidiary	JP Morgan etc.	2022-05-27	367,115	50,785	0	417,900	417,900
Total				1,984,640	119,902	1,245,150	859,392	8,711,450

※ SEC requires BOD approval for individual guarantees exceeding 2.5% of total equity. When the guarantee amount is between 0.1% and/or less than 2.5%, the approval decision is delegated to the Management Committee.

※ Simpres ceased to be affiliate (share sale in November 2017)

5. Sanctions and Others

The Company is engaged in a lawsuit after being issued a corrective order and fine (KRW 14,126 million) on July 10, 2012 for violating Article 23 (Prohibition of Unfair Trade Practices) of the Monopoly Regulation and Fair Trade Act. In order to comply with relevant laws on fair trade, the Company has strengthened internal oversight and provides education to executives and employees for the prevention of unfair trade practices.

The Company was ordered to pay a fine (KRW 26 million) on January 13, 2014 from the Fair Trade Commission for violating Article 4 (Announcement of Critical Information and Integrated Notification) of the Fair Labeling and Advertising Act, and has subsequently paid the fine. To comply with the Fair Labeling and Advertising Act, the Company offers education and training to executives and employees regarding these laws.

The Company was ordered to pay a fine (KRW 267 million) on March 7, 2013, for violating the Article 266 of Occupational Safety and Health Act (regarding the installment of shut-off valve) in connection with a hydrofluoric acid gas leak that occurred on January 28, 2013, and has subsequently paid the fine. The Company was issued an improvement order and ordered to pay a fine (KRW 2.2 million) on April 1, 2013, for violating the Article 46 of Toxic Chemicals Control Act (Recording and Preservation of Documents). The Company reported compliance with the improvement order and subsequently paid the fine. As a result of this incident, three employees and executives (a director of chemicals supply management on Career Level 4, with 20 years of continuous employment, a manager in charge of toxic chemicals on Career Level 3 with 8 years of continuous employment, retired, and a manager of chemicals supply management on Career Level 2 with 11 years of continuous employment) of SEC were ordered to pay a fine (KRW 15 million) at the first trial (in Suwon District Court) on October 31, 2014. In order to improve the capabilities of safety managers, the Company has implemented a relevant education program, and is strengthening inspection on maintenance and management of dangerous facilities.

For the hydrofluoric acid gas leak occurred on May 2, 2013, SEC was ordered to pay a fine (KRW 4.5 million) on February 27, 2014, for violating the Occupational Safety and Health Act and the Article 24 of Toxic Chemicals Control Act (Standards for Control of Poisonous Substances) and not wearing safety devices, and has subsequently paid the fine. The Company has completely revamped the Health & Safety Management Program, including creating a new organization devoted to the prevention of leakage to avoid such accidents from reoccurring in the future.

The Company was ordered and paid a fine of KRW 166.7 million in August 2013 for violating the Article 9 of Toxic Chemicals Control Act (Confirmation of Chemicals) and not reporting the importation of certain chemicals, following a joint inspection by the Ministry of Environment and Gyeong-Gi Province. The Company has improved the reporting process for imported chemicals to ensure proper notification of such chemicals.

Pursuant to the results of a joint inspection by the prosecutor's office and the Ministry of Employment and Labor conducted from June 16 through June 20 (2014), a current executive of SEC (Head director of safety & healthy management, and also advisor, with 32 years of continuous employment) and SEC were ordered and respectively paid a fine of KRW 2 million for violating the obligation of taking safety measures in accordance with the Article 24 of Occupational Safety and Health Act (Health Measures) on July 29, 2015 (in Suwon District Court). To prevent recurrence, the Company has implemented its own Environment & Safety Preliminary Evaluation System and conducts regular inspections.

Pursuant to the results of a regular inspection by the Ministry of Employment and Labor conducted from December 14 through December 16 (2015), SEC paid a fine of KRW 18.76 million for violating the obligation of implementing safety measures in accordance with the Article 49-2 (7) of Occupational Safety and Health Act (Submission of Process Safety Reports) on December 31, 2015 (The original amount was KRW 23.45 million, but was reduced by 20%, as the company voluntarily paid the fine by the due date). Regarding this matter, the Company has implemented its own safety inspection program for Hazardous Machineries and equipment to reinforce compliance management.

Pursuant to the results of a Process Safety Management (PSM) inspection by the Ministry of Employment and Labor conducted from December 5 through the December 9 (2016), the Company was ordered to pay a fine of KRW 3.52 million for violating the obligation of implementing safety measures in accordance with the Article 49-2 (7) of Occupational Safety and Health Act (Submission of Process Safety Reports) on December 14, 2016 and has subsequently paid the fine. In order to comply with relevant laws, the Company is training process safety experts on site and conducting its own process safety evaluation.

The years of continuous employment described herein is based on the date of order.

Special prosecutor investigating an influence-peddling scandal involving South Korea's president has indicted the Company's five executives (Jae-Yong Lee, Vice Chairman, Gee-Sung Choi, former Vice Chairman, Choong-Ki Chang, former President, Sang-Jin Park, former President, and Seong-Soo Hwang, former Senior Vice President) on allegations of violating the Act on the Aggravated Punishment, etc. of Specific Economic Crimes (embezzlement) on February 28, 2017. The Seoul Central District Court found them partially guilty on the charges on August 25, 2017, and sentenced Vice Chairman Jae-Yong Lee to five years in jail, former Vice Chairman Gee-Sung Choi to four years in jail, former President Choong-Ki Chang to four years in jail. Former President Sang-Jin Park received a three-year jail term, suspended for five years, and former Senior Vice President Seong-Soo Hwang was sentenced to 2 and half years, suspended for four years. On February 5, 2018, the Seoul High Court reversed the first trial judgment and Vice Chairman Jae-Yong Lee was suspended for four years and former Vice Chairman Gee-Sung Choi, former President Choong-Ki Chang, former President Sang-Jin Park, and former Senior Vice President Seong-Soo Hwang were suspended for two years each. The appeal against the decision has been filed and is under trial. The Company will check on future developments of the court proceedings and relevant matters.

The Company (including Samsung Display) has joint responsibility for reimbursement of liabilities that Samsung Display accrued before the separation.

On October 23, 2013, the Company signed a basic agreement on comprehensive business cooperation with Corning Incorporated, etc. Inflow or outflow of resources can occur hereafter, as the agreement includes conditions for compensation of loss. Projected disbursement as of the end of reporting term was counted as liabilities.

For information on the Company's commitments and contingencies, refer to Commitments and Contingencies notes in the Consolidated Financial Statements.

6. Return of Short-Swing Profits

The Company has not received any notification of short-swing profits from the Securities & Futures Commission (the Governor of the Financial Supervisory Service) in the past three years and until the date of this report.

7. External Contributions

Contribution	Amount	Details	Others
Management plans for the Social Contribution Fund for 2017	KRW 12.59 billion	- The Social Contribution Fund consists of voluntary contributions of employees and matching funds raised by SEC. SEC plans to contribute KRW 12.59 billion in matching funds in 2017. - The contributions will be spent on sponsoring domestic and overseas volunteer programs as well as supporting local social contribution activities.	The BoD approved this action at the meeting held on March 24, 2017.
Sponsoring WorldSkills Competition	EUR 1.35 million (approximately KRW 1.63 billion)	- The Company will be sponsoring the WorldSkills Competition Abu Dhabi 2017 as a Global Premium Partner and WorldSkills International, the organizing committee of WorldSkills Competition, from 2017 to 2018.	
Ho-Am Foundation	KRW 4 billion	- The donation was made to support the foundation's activities such as 'Ho-Am prize'.	The BoD approved this action at the meeting held on April 27, 2017.
Samsung Press Foundation	KRW 1.7 billion	- The donation was made to support the foundation's activities to promote the development of the press infrastructure, such as Samsung Journalism Award.	
Samsung Welfare Foundation	KRW 24 billion	- The donation was made to support the learning of low-income middle school students.	
Samsung Life Public Welfare Foundation	KRW 37.7 billion	- The donation was made to support the operation of Samsung Medical Center.	
Sungkyunkwan University	KRW 15 billion	- The donation was made to support the Samsung Scholarship program.	The BoD approved this action at the meeting held on November 24, 2017.
Korea Disaster Relief Association	KRW 3 billion	-The donation was made to support and help Pohang residents recover from the earthquake.	
Community Chest Of Korea	KRW 20.1 billion	-The donation was made to fulfil corporate social responsibility by participating in the annual campaign designed to help the marginalized.	

8. Subsequent Events

Samsung Electronics' plan to conduct a 50:1 stock split, announced on January 31, 2018, was approved at the Company's Annual General Meeting on March 23. The listing of new share certificates is scheduled at May 4, 2018.

Based on totals at the close of trading on January 30, the stock split will increase the total number of common shares outstanding to 6,419,324,700 and preferred shares outstanding to 903,629,000.

1. Details of Stock Split	Classification		Before Stock Split	After Stock Split
	Face Value per Share (KRW)		5,000	100
	Total Number of Shares Outstanding	Common Shares	128,386,494	6,419,324,700
		Different Classes of Shares	18,072,580	903,629,000
2. Stock Split Schedule	Scheduled Date of Shareholders' Meeting		March 23, 2018	
	Period for Tendering Existing Shares	From	March 26, 2018	
		To	May 2, 2018	
	Scheduled Period of Trading Suspension		April 30, May 2, May 3, 2018 (3 business days, May 1 is a non-trading day)	
	Period for Closing Shareholders' Registry	From	May 3, 2018	
		To	May 10, 2018	

	Scheduled Listing Date of New Share Certificates	May 4, 2018
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9. Acquisition and Divestments

[S-Printing Solution]

(1) Details on the Split

- Company Name: S-Printing Solution Corp.
- Location: 129 Samsung-ro, Yeongtong-gu, Suwon-si, Gyeonggi-do
- Purpose of split: S-Printing Solution competitiveness improvement
- Date of approval: October 27, 2016 (temporary shareholders' meeting)
- Date of split: November 1, 2016

(2) Details on the selling

On September 12, 2016, the Company made a deal to sell Printing Solutions business including shares of S-Printing Solution Corp to HP Inc. at USD 1.05 billion and completed the deal on November 1, 2017.

Details of the above statement can be found in the “Important Matters Report“, published in DART (<http://dart.fss.or.kr/>).

Please refer to the Management Combination and Non-current Assets Held-for-Sale (Assets of disposal group) notes in the Consolidated Financial Statements for acquisition and divestments on the consolidation basis.

10. Green Management

The Company is actively securing “Green Technology Certifications” to follow the Low Carbon Green Growth policy of the Korean government.

(Green Technology Certification)

The Company is acquiring Green Technology Certifications in accordance with Clause 2, Article 32 of the Framework Act on Low Carbon Green Growth. Green technology development is a part of our “Planet First” strategy, which strives to conduct business activities in a manner that respect people and nature. Since the certification system was introduced in 2010, the Company has secured 18 valid Green Technology Certifications as of December 31, 2017. In addition, the Company has acquired 80 ‘Green Technology’ product certificates for 642 models of commercialized products.

Certified Green Technologies as of December 31, 2017 are as follows:

Division	Business	Name of green technology and green projects	No.
CE		Low-power consumption SOC design technology for digital TVs, etc.	15
IM		Pentile display power-saving technology that adapts to level of illumination of the surroundings, etc.	3
Total			18

※ The data reported above is with respect to Samsung Electronics Co., Ltd. only.

※ Please refer to the 『11. Other Information (B. Environmental regulations)』 in 『II. Businesses Overview』 for greenhouse gas emission and energy use.